
ELMFIRE Verification Guide

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About This Guide

This document aggregates the standalone reports for every runnable verification case currently discovered under `cases/Verification`. The opening summary records environment and execution configuration and reports each case's scientific decision from its case-local `outputs/metrics.json`; command completion and PDF availability are not substituted for verification evidence.

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1 How to Read This Guide

Start with the verification summary for the aggregate-build environment, case-declared execution configuration, and scientific decision table. The case identifiers in the summary link to the corresponding case-report entries. The complete standalone reports follow in the stable numerical order defined by `cases/Verification/CASE_REGISTRY.md`; each retains its own scientific context, equations, metrics, citations, acceptance criteria, and limitations.

Interpret PASS and FAIL as the decision made by the case-local postprocessor against that case's declared acceptance criteria. A missing result, an unrun or incomplete workflow, or an unsupported capability is shown as NOT EVALUABLE. Command completion and PDF availability alone do not establish a scientific decision.

The aggregate-build environment describes how this guide was assembled; it is not necessarily the execution environment of every included case. Aggregating the compiled PDFs keeps case-local macros, labels, bibliographies, and packages isolated from one another. After standalone reports change, rebuild this guide with `make verification-report` or rebuild both suite reports with `make reports`.

2 Verification summary

2.1 Environment captured for this aggregate

This environment was captured when the aggregate report was generated. It describes the report-build process and must not be interpreted as the execution environment of every case unless the cases were run and aggregated in the same job. ELMFIRE is not executed to obtain this information.

Configuration	Recorded value
Captured at (UTC)	2026-09-09T22:26:08+00:00
Repository revision	7769fc9e1181
Repository state	dirty
Host platform	macOS-15.7.7-arm64-arm-64bit
Python runtime	CPython 3.12.9 (/opt/homebrew/anaconda3/bin/python3)
Conda environment	not active
ELMFIRE command	elmfire
ELMFIRE resolved path	not found on PATH
ELMFIRE version declaration	2025.0717
ELMFIRE executable	not available
SHA-256	
MPI launcher	mpirun (Open MPI) 5.0.6
Slurm allocation	not a Slurm build job
Loaded modules	not recorded
Python packages	NumPy 1.26.4; Matplotlib 3.9.2; PyYAML 6.0.1; Rasterio 1.5.0; GDAL not installed; Pandas 2.2.2; GeoPandas not installed; Shapely not installed

2.2 Case-declared execution configuration

These values come only from each case's `case.yaml`. They summarize the declared executable command, namelist, MPI allocation, and namelist contract. A missing declaration is reported explicitly rather than inferred.

Case	ELMFIRE command	Input namelist	MPI ranks	Schema
CASE01_BET	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE02_FGR	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE03_ETC	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE04_STI	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE05_TDWB	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE06_F2D	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE07_IPC	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE08_FBC	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE09_MMD	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE10_WSR	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE11_WSC	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE12_TRS	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE13_PRM	not applicable	not applicable	not declared	1
CASE14_SRS	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE15_CRO	\${ELMFIRE_BIN:-elmfire}	data/inputs/elmfire.data	not declared	1
CASE16_NSP	\${ELMFIRE_BIN:-elmfire}	data/inputs/elmfire.data	not declared	1
CASE17_SCV	\${ELMFIRE_BIN:-elmfire}	data/128/inputs/elmfire.data	not declared	1
CASE18_TCV	\${ELMFIRE_BIN:-elmfire}	data/inputs/elmfire_180.data	not declared	1
CASE19_WTH	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE20_PIG	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE21_WDE	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE22_VSL	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE23_FQD	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE24_CXI	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE25_MQD	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1

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Case	ELMFIRE command	Input namelist	MPI ranks	Schema
CASE26_CNF	\${ELMFIRE_BIN:-elmfire}	variants/1mph/elmfire.data.in	not declared	1
CASE27_FBT	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE28_OVA	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE29_SUP	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE30_MIT	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE31_PFT	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE32_RCV	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE33_SDO	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE34_FMS	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE35_WSS	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE36_SLS	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE37_DMS	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE38_LMS	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE39_DHC	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE40_WSD	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE41_CFP	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE42_WAF	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1

2.3 Scientific decision summary

This table is generated from each case's case-local `outputs/metrics.json`. A successful run script or report build is not treated as scientific evidence of PASS. Missing, incomplete, blocked, or non-decisional results are reported as NOT EVALUABLE.

Total cases: 42 PASS: 30 FAIL: 8 CHARACTERIZED: 0 NOT EVALUABLE: 4

Case	Category	Title	Decision
CASE01_BET	coupling tests	Biomass Emission-Time Resolution Independence	PASS
CASE02_FGR	coupling tests	Finite-Duration Firebrand Generation	PASS
CASE03_ETC	coupling tests	Eulerian Firebrand Transport Convergence	FAIL
CASE04_STI	coupling tests	Steady Transport and Immediate Ignition	PASS
CASE05_TDW	coupling tests	Firebrand Transport in Time-Dependent Wind	PASS
CASE06_F2D	coupling tests	Two-Dimensional Firebrand-Driven Spread	PASS
CASE07_IPC	coupling tests	Ignition Probability and SFT Convergence	FAIL
CASE08_FBC	coupling tests	Deposited Firebrand Consumption	FAIL
CASE09_MMD	coupling tests	Mixed-Mode Ignition-Delay Effects	PASS
CASE10_WSR	coupling tests	WUI Structural Resolution Sensitivity	NOT EVALUABLE
CASE11_WSC	coupling tests	WUI Surface-Coupling Synchronization	NOT EVALUABLE
CASE12_TRS	coupling tests	WUI Temporal-Resolution Sensitivity	FAIL
CASE13_PRM	coupling tests	WUI Parametric Response Maps	NOT EVALUABLE
CASE14_SRS	coupling tests	Two-Dimensional Spatial-Resolution Sensitivity	NOT EVALUABLE
CASE15_CRO	unit tests	Case 15: Constant rate of spread	PASS
CASE16_NSP	unit tests	Case 16: No-spread limiting behavior	PASS
CASE17_SCV	unit tests	Case 17: Spatial convergence	FAIL
CASE18_TCV	unit tests	Case 18: Temporal convergence	FAIL
CASE19_WTH	coupling tests	Case 19: Transient WUI heat flux	PASS
CASE20_PIG	coupling tests	Case 20: Point ignition and isotropic spread	PASS
CASE21_WDE	coupling tests	Case 21: Wind-driven elliptical spread	PASS
CASE22_VSL	coupling tests	Case 22: Spread across opposing valley slopes	PASS

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Case	Category	Title	Decision
CASE23_FQD	coupling tests	Case 23: Spatially varying fuel quadrants	PASS
CASE24_CXI	coupling tests	Case 24: Combined fuel, slope, and wind interactions	PASS
CASE25_MQD	coupling tests	Case 25: Spatially varying moisture quadrants	PASS
CASE26_CNF	coupling tests	Case 26: Surface, passive-crown, and active-crown fire	PASS
CASE27_FBT	coupling tests	Case 27: Lagrangian firebrand generation and transport	FAIL
CASE28_OVA	coupling tests	Case 28: Overnight spread-rate adjustment	PASS
CASE29_SUP	coupling tests	Case 29: Initial- and extended-attack suppression	FAIL
CASE30_MIT	coupling tests	Case 30: Multiple-ignition topology	PASS
CASE31_PFT	coupling tests	Case 31: Planar-front transport	PASS
CASE32_RCV	coupling tests	Case 32: Raster-orientation covariance	PASS
CASE33_SDO	coupling tests	Case 33: Slope-driven directional spread	PASS
CASE34_FMS	unit tests	Case 34: Standard fuel-model Rothermel sweep	PASS
CASE35_WSS	unit tests	Case 35: Midflame wind-speed Rothermel sweep	PASS
CASE36_SLS	unit tests	Case 36: Slope-factor Rothermel sweep	PASS
CASE37_DMS	unit tests	Case 37: Dead-fuel moisture and extinction sweep	PASS
CASE38_LMS	unit tests	Case 38: Static live-fuel moisture sweep	PASS
CASE39_DHC	unit tests	Case 39: Dynamic herbaceous curing sweep	PASS
CASE40_WSD	coupling tests	Case 40: Wind-slope vector and direction sweep	PASS
CASE41_CFP	unit tests	Case 41: Custom fuel-parameter Rothermel sweep	PASS
CASE42_WAF	coupling tests	Case 42: Wind-adjustment-factor coupling sweep	PASS

Decision vocabulary. PASS and FAIL are case-defined scientific decisions. CHARACTERIZED denotes descriptive validation metrics without a justified acceptance threshold. NOT EVALUABLE denotes missing, unrun, incomplete, unsupported, or otherwise non-decisional evidence.

Case 01: Biomass Emission-Time Resolution Independence

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE01_BET

This case verifies that biomass-emission duration is represented as a physical residence time rather than a grid-dependent number of solver steps. Changing cell size must preserve accumulated firebrand density for the same unit-width source and forcing.

The test exposes incorrect conversion between per-cell and unit-width generation rates, timestep-dependent source integration, and unintended cell-width scaling. Success requires overlapping downwind profiles and mesh-independent far-field accumulation within tolerance.

2 Mathematical formulation and numerical implementation

A surface-fire cell emits firebrands while the flame front crosses that cell. For a square cell of size $\Delta x = \Delta y$ and surface-fire spread rate V_s , the emission duration is

$$t_{\text{emission}} = \frac{\Delta x}{V_s}.$$

Without this duration scaling, changing the resolution would change the time for which a burning cell acts as a firebrand source and would therefore change the accumulated deposition solely as a numerical artifact.

This case represents a one-dimensional firebrand-transport problem with a 1 m-wide strip perpendicular to the transport direction. The unit-width firebrand generation rate is

$$\text{GR}_{1\text{m}} = 33.3 \text{ pcs s}^{-1} \text{ MW}^{-1}.$$

ELMFIRE applies `EMBER_GR_PER_MW_VEGE` to the total fire power of a computational pixel. A pixel with fireline intensity I_B and cross-stream width Δy has fire power $I_B \Delta y$. Its firebrand generation rate is therefore

$$\dot{N}_{\text{pixel}} = I_B \Delta y \text{GR}_{\text{ELMFIRE}}.$$

To keep the represented source equal to the 1 m-wide source at every grid resolution, preprocessing sets

$$\text{GR}_{\text{ELMFIRE}} = \frac{\text{GR}_{1\text{m}}}{\Delta y} = \frac{33.3}{\Delta x} \text{ [pcs s}^{-1} \text{ MW}^{-1}].$$

The configured `EMBER_GR_PER_MW_VEGE` values are consequently 6.66, 3.33, and 1.11 pcs/s/MW for the 5, 10, and 30 m variants. In all three variants, $\text{GR}_{\text{ELMFIRE}} \Delta y = \text{GR}_{1\text{m}}$, so the unit-width source strength is independent of cell size.

2.1 Implementation path

The documented residence-time formulation requires a burning cell to emit only while the flame resides in that cell [1]. In the current implementation, `READ_SPOTTING` reads the PER-MW source parameters, `SPOTTING` calls `EMBER_TO_EMIT_PER_CELL` to convert fireline intensity and cell width to an ember rate, and `CALC_SPOTTING_DURATION` determines the source lifetime from the cell fire state. Accumulated deposition, fireline intensity, and spread-rate rasters are then written for postprocessing. The report checks both halves of the discretization:

preprocessing keeps the represented one-metre source strength intensive by scaling the configured rate with cross-stream cell size, and postprocessing derives the expected density from the simulated FLIN and VS at the same physical state. A source-duration or pixel-power scaling defect would therefore create a systematic resolution trend. This is a controlled code-verification test; it does not validate the empirical generation constant or the transport distribution against experiments.

3 Derivation of expected behavior

Combining the pixel generation rate with the emission duration gives the number emitted while the front crosses one cell. Dividing that number by the represented bin area eliminates Δx and gives the expected accumulated density:

$$N''_{\text{ref}} = \frac{I_B \Delta y \text{GR}_{\text{ELMFIRE}}(\Delta x/V_s)}{\Delta x (1 \text{ m})} = \text{GR}_{1\text{m}} \frac{I_B}{V_s} \quad [\text{pcs m}^{-2}].$$

Thus, the reference is calculated separately from the fire behavior actually produced by each simulation; it is not a fixed nominal number. The postprocessor finds the maximum positive fireline intensity in the usable interior of the latest `flin*.tif` and reads the spread rate from the corresponding cell of `vs*.tif`. ELMFIRE writes these fields in kW/m and ft/min for this configuration, so the implemented conversion is

$$N''_{\text{ref}} = 33.3 \frac{I_{B,\text{sim}}/1000}{V_{s,\text{sim}}(0.3048/60)} \quad [\text{pcs m}^{-2}].$$

For the current outputs, the mean values across the variants are $I_{B,\text{sim}} = 2.200042 \text{ MW/m}$ and $V_{s,\text{sim}} = 0.577015 \text{ m/s}$. These values give $N''_{\text{ref}} = 126.966 \text{ pcs/m}^2$. The per-variant values and selected raster cells are retained in `../outputs/metrics.json`.

The generated numerical substitutions make the cancellation visible. With $CFL_w = 0.5$ and $U = 6.71 \text{ m s}^{-1}$,

$\Delta x \text{ (m)}$	5	10	30
$\Delta t_{\text{generated}} \text{ (s)}$	0.372578	0.745156	2.000000
$GR_{\text{ELM}} = 33.3/\Delta x$	6.66	3.33	1.11
$GR_{\text{ELM}} \Delta y$	33.3	33.3	33.3

in $\text{pcs s}^{-1} \text{ MW}^{-1}$ for the last two rows. The 5- and 10-m CFL steps divide the requested 500-s duration exactly. At 30 m, the nominal CFL step is 2.235469 s; preprocessing selects the conservative whole-second divisor 2 s, giving 250 steps and preserving the 500-s stop while keeping $CFL_w = 0.4473 < 0.5$. Thus the source-scaling reference is exactly 33.3 in all three variants, and its relative error is

$$e_{GR} = \max_i \frac{|GR_i \Delta y_i - 33.3|}{33.3}, \quad e_{GR,\text{ref}} = 0.$$

The run-derived FLIN and VS values enter only as controlled fire-behavior normalizers; the ember-flux field being tested does not determine them. With $0.3048/60 = 0.00508$,

$$N''_{\text{ref}} = 33.3 \frac{FLIN/1000}{VS(0.00508)} = 6.555118 \frac{FLIN [\text{kW m}^{-1}]}{VS [\text{ft min}^{-1}]} \quad \text{pcs m}^{-2}.$$

For each generated mesh the expected far-field error is therefore zero, and the substituted decision requires $\text{mean}_{150 < x \leq 200} |N'' - N''_{\text{ref}}|/N''_{\text{ref}} \leq 0.05$.

4 Verification metrics and acceptance criteria

For each resolution, the centre-row accumulated count N_i is converted to an areal density for the represented one-metre strip,

$$n''_i = \frac{N_i}{\Delta x (1 \text{ m})}.$$

The independent reference uses the maximum-FLIN valid interior cell from the same run,

$$n''_{\text{ref}} = GR_{1\text{m}} \frac{FLIN/1000}{VS(0.3048/60)},$$

where FLIN is read in kW m^{-1} , VS in ft min^{-1} , and $GR_{1\text{m}} = 33.3 \text{ pcs s}^{-1} \text{ MW}^{-1}$. The comparison mask contains finite physical cells with $150 < x \leq 200 \text{ m}$; the two-cell halo is excluded.

4.1 Justification of metric quantification

The source-scaling check is an algebraic invariant rather than an empirical model tolerance. Preprocessing computes $G_{R,\Delta x}\Delta x$ from the same double-precision inputs used to write the namelist, so a relative allowance of 10^{-12} admits only floating-point roundoff and still detects any omitted or duplicated width factor. The 5% density-error limit is applied to a far-field mean because the derived expectation is a developed plateau, not a pointwise subcell solution. It permits the cell integration, discrete selection of the run-derived FLIN/VS reference cell, and finite-window averaging to differ slightly while remaining far below the order-one, resolution-proportional error caused by incorrect extensive/intensive scaling. Requiring the limit at every resolution makes the criterion a grid-consistency test rather than an agreement obtained by cancellation in one variant. Completeness and the conjunctive decision are logical requirements: without every source and reference raster, neither the normalization nor the claimed resolution independence is testable.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Final <code>ember_flux</code> , <code>flin</code> , and <code>vs</code> rasters for each manifest variant; shape, cell size, nodata, and halo are validated.	3 of 3 variants usable
Unit-width source scaling	Configured rate multiplied by cross-stream cell width, $G_{R,\Delta x}\Delta x$.	$33.3 \text{ pcs s}^{-1} \text{ MW}^{-1}$ to 10^{-12} relative tolerance
Far-field mean relative error	$E_r = \text{mean}_{150 < x \leq 200} n''_i - n''_{\text{ref}} / n''_{\text{ref}}$, evaluated separately for every resolution.	$E_r \leq 0.05$ for every variant
Overall decision	Conjunction of completeness, source scaling, and all per-resolution error decisions.	PASS only if every component passes

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `dx10`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

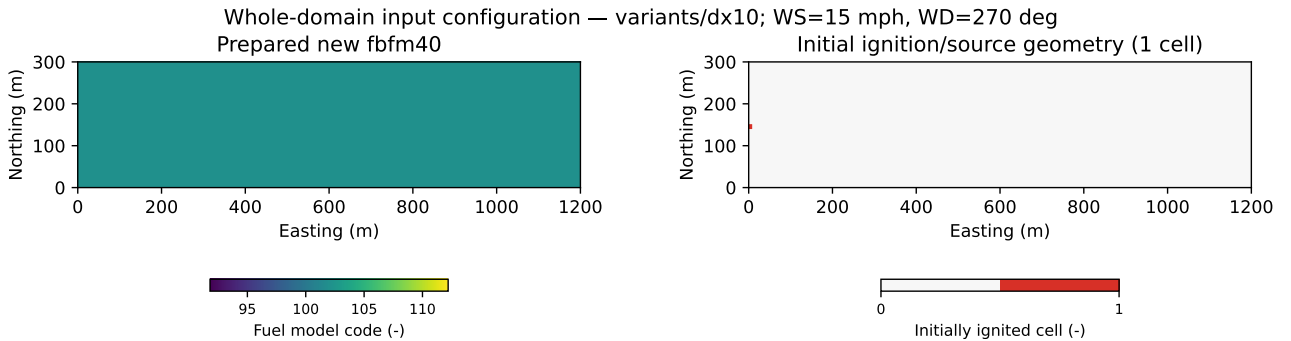


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The preprocessor creates flat, uniform FBFM102 domains at $\Delta x = \Delta y = 5, 10$, and 30 m. The physical simulation domain is the same for every resolution. Each raster also includes a two-cell numerical buffer outside each side of that physical domain. The ignition is placed in the first usable column rather than in the buffer.

Crosswind transport and secondary firebrand ignition are disabled so the accumulated deposition can be evaluated as a one-dimensional profile.

ELMFIRE writes the accumulated number of deposited firebrands in each target pixel. The area represented by a streamwise bin in the unit-width problem is $\Delta x \times 1$ m, so the postprocessor converts the pixel count N_i to density using

$$N''(x_i) = \frac{N_i}{\Delta x(1 \text{ m})} = \frac{N_i}{\Delta x} \quad [\text{pcs m}^{-2}].$$

The two buffer columns on each end are excluded from this profile.

For each completed variant, the mean accumulated density over $150 < x \leq 200$ m is compared with that variant's FLIN/VS-derived reference. The mean relative error must not exceed 5%. The case passes only when all three variants have valid outputs and pass independently. Agreement across the three resolutions demonstrates that both the emission duration and the unit-width generation-rate scaling are resolution independent.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from **dx10**. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. Firebrand counts are normalized by the cell length and the represented 1-m cross-stream strip, giving units of pcs m^{-2} .

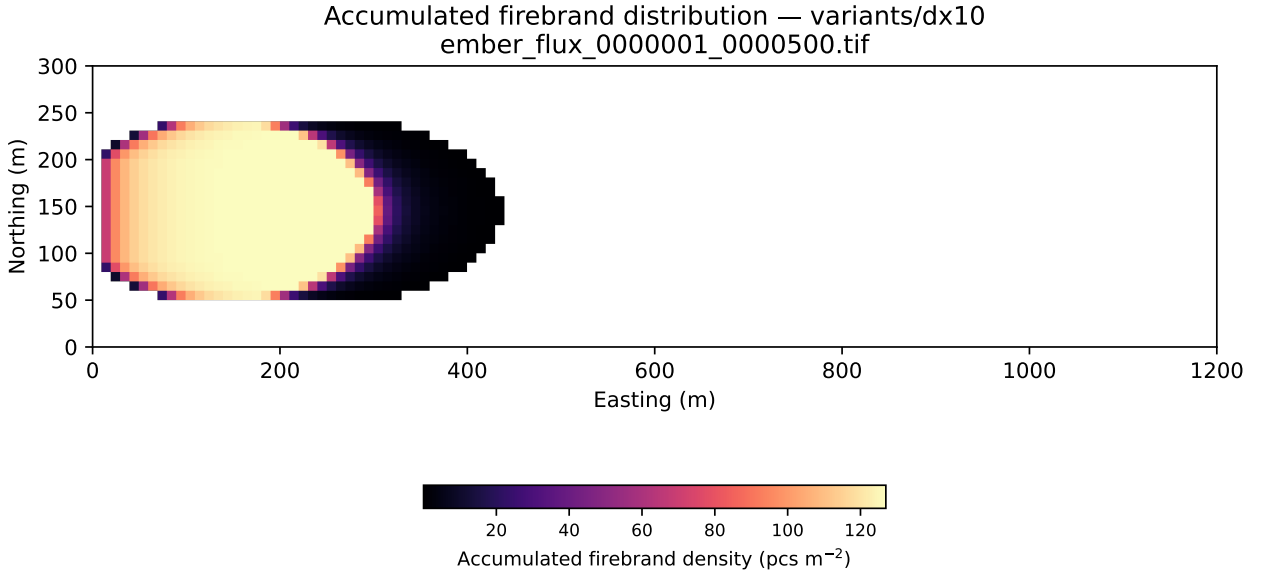


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figure 3 first presents the physical observable in the same three-resolution arrangement used to define the test. All bars are calculated from final ELMFIRE ember-flux rasters, and each dashed line uses the FLIN and spread rate extracted from that same variant.

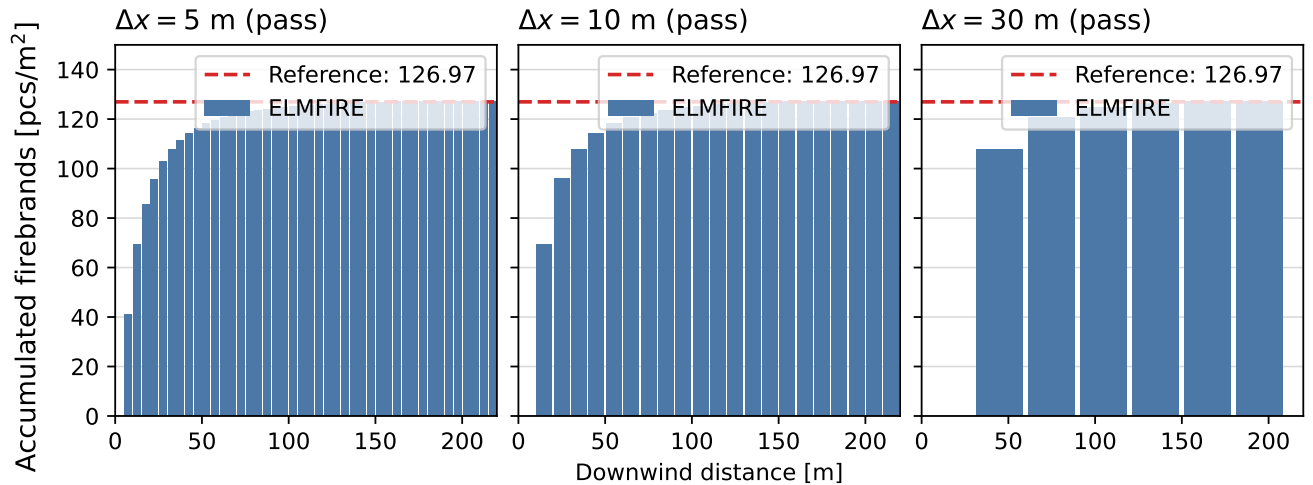


Figure 3: Accumulated firebrand density for the 5, 10, and 30 m grids. The dashed red line in each panel is the run-derived reference. The two-cell numerical halo is excluded from every profile.

7 Calculated metrics and verification decision

Quantity	Value
Overall status	pass
Mean reference fireline intensity	2.200042 MW/m
Mean reference spread rate	0.577015 m/s
Calculated accumulation reference	126.966 pcs/m ²
Variants configured	3
Variants computed	3

Table 2: Biomass emission-time verification summary.

The overall decision is sourced from the generated metric macros and is **pass**.

8 Technical interpretation

Agreement between the accumulated deposition density and the independently calculated FLIN/VS reference verifies the complete chain from biomass consumption through emission duration, per-unit-area source scaling, transport, and raster accumulation. Consistent agreement across the three grids further indicates that the implemented source normalization is not tied to cell area.

9 Scope and limitations

This case isolates a uniform, one-dimensional transport problem with crosswind dispersion, delayed ignition, and secondary spotting disabled. It verifies emission-time scaling and accumulated deposition, but does not by itself validate stochastic trajectories or ignition physics. The preprocessing parameters, variant manifest, calculated raster samples, and metric values are retained with the case so that the reported decision can be reproduced.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 02: Finite-Duration Firebrand Generation

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE02_FGR

This case verifies finite-duration firebrand generation when the source lifetime is not commensurate with the solver timestep. It compares timesteps shorter and longer than the residence time while holding the source, grid, wind, and transport distribution fixed.

The test exposes whole-step over-generation, truncation of the last active interval, and timestep-dependent deposition. Success requires both variants to recover the same integrated source count and downwind profile within tolerance.

2 Mathematical formulation and numerical implementation

Let a burning computational cell generate firebrands at the constant rate GR from ignition time t_{ign} until $t_{\text{ign}} + t_{\text{res}}$. The number generated during an ELMFIRE step $[t, t + \Delta t]$ must use the overlap between that step and the active source interval:

$$\Delta t_{\text{active}} = \max(0, \min(t + \Delta t, t_{\text{ign}} + t_{\text{res}}) - \max(t, t_{\text{ign}})),$$
$$\Delta N = \text{GR} \Delta t_{\text{active}}.$$

This formulation handles the first and final partial steps and also handles a single timestep that spans the entire residence time. Summing over all steps must give

$$N_{\text{total}} = \text{GR} t_{\text{res}}.$$

The case sets $\text{GR} = 10.0$ pcs/s and $t_{\text{res}} = 10.0$ s, so the expected total is 100.0 firebrands.

2.1 Implementation path and verification rationale

The documented finite-duration source formulation identifies a timestep-integration error when the solver step is comparable to or longer than the fireline residence time [1]. In current ELMFIRE, `READ_SPOTTING` selects the `PER-AREA` source, `SPOTTING` computes the active-cell emission, and the source timing state maintained by `CALC.SPOTTING.DURATION` bounds that emission. The accumulated Eulerian field is advanced by `EMBER.TRAJECTORY_EULERIAN`. Comparing a fine step with a step longer than the complete source lifetime directly tests whether partial-step overlap, rather than the nominal solver step, controls the emitted amount. The total-count metric detects temporal over- or under-integration; the profile additionally detects correct totals deposited in wrong cells. Surface spread, ignition, consumption, and crosswind dispersion are disabled, so those processes cannot compensate for a source-timing error. The test verifies numerical accounting and the configured transport kernel, not the physical realism of the prescribed lognormal distribution.

3 Derivation of expected behavior

Generated firebrands are distributed with a lognormal distance model using $\mu = 2.18$ and $\sigma = 1.23$. The distribution is truncated at its 99th percentile. ELMFIRE rounds that distance to an integer number K of cells and normalizes the retained probability:

$$P_i = \frac{F(i\Delta x) - F((i-1)\Delta x)}{F(K\Delta x) - F(10^{-6})}, \quad i = 1, \dots, K,$$

where F is the lognormal cumulative distribution. The analytical count in cell i is $N_i = 100P_i$. The postprocessor reproduces this cell-integrated reference and compares it directly with the final `ember_flux` raster.

The two time integrations can be reproduced explicitly. For $\Delta t = 0.13$ s, 76 complete steps cover $76(0.13) = 9.88$ s and the final overlap is $10 - 9.88 = 0.12$ s, so

$$N_{\text{fine}} = 10[76(0.13) + 0.12] = 100.$$

For the $\Delta t = 12.7$ s member, the first step spans the complete source interval and

$$\Delta t_{\text{active}} = \min(12.7, 10) - \max(0, 0) = 10 \text{ s}, \quad N_{\text{coarse}} = 10(10) = 100.$$

For the spatial reference, $z_{0.99} = 2.3263479$ gives

$$x_{0.99} = \exp[2.18 + 1.23(2.3263479)] = 154.6876 \text{ m}, \quad K = \text{NINT}(154.6876/10) = 15.$$

The retained CDF mass is $F(150) - F(10^{-6}) = 0.9893135$. Substitution in the cell formula gives $N_1 = 54.5524$, $N_2 = 20.8937$, $N_3 = 9.4216$ particles, with all 15 bins summing to 100 after normalization. Hence the exact metric references are $E_N = 0$ and $E_{L_1} = 0$; the numerical tests substitute $E_N \leq 0.005$ and $E_{L_1} \leq 0.01$ for both timesteps.

4 Verification metrics and acceptance criteria

The integrated reference count is $N_{\text{ref}} = G_R t_{\text{res}} = 10 \times 10 = 100$ particles. The analytical spatial reference integrates the configured lognormal landing distribution over each downwind cell, truncates it at the 99th percentile, and renormalizes the cell probabilities to N_{ref} . The ignition row and physical columns are read from each manifest; all halo cells and nodata values are excluded.

4.1 Justification of metric quantification

Both limits have a direct particle-count interpretation. Since $N_{\text{ref}} = 100$, the integrated-error bound $E_N \leq 0.005$ permits an absolute imbalance of only 0.5 particle. This is large enough for raster roundoff and percentile-edge binning, but too small to hide a one-particle source-duration error. The normalized L_1 bound of 0.01 permits at most one particle's worth of total spatial redistribution after the analytical lognormal probabilities have been integrated over the same cells. Unlike a signed error, this norm cannot conceal compensating upstream and downstream errors. Applying both conditions to the short- and long-timestep variants separates conservation of the total from correct overlap-limited placement. The exact variant count is required because the intended assertion concerns both sides of the residence-time boundary.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	One final <code>ember_flux</code> raster with current manifest geometry for each timestep.	2 of 2 variants usable
Integrated-count error	$E_N = \sum_{ij} N_{ij} - N_{\text{ref}} /N_{\text{ref}}$, using the full valid output raster.	$E_N \leq 0.005$ for each timestep
Profile normalized L_1 error	$E_{L_1} = \sum_k N_k - N_{k,\text{ref}} /N_{\text{ref}}$ on matched physical downwind cells.	$E_{L_1} \leq 0.01$ for each timestep
Overall decision	Conjunction of output completeness and both errors for both timestep variants.	PASS only if every component passes

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `dt0p13`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

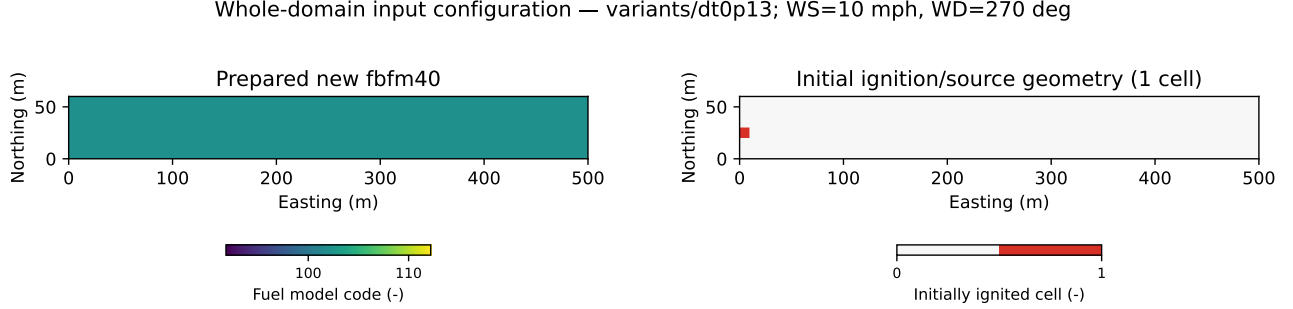


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The preprocessor creates two variants with $\Delta t = 0.13$ s and $\Delta t = 12.7$ s. Because timestep variation is the quantity under test, both fractional values are retained. The requested 120-s stop is rounded upward to an exact multiple: 924 steps give 120.12 s for the 0.13-s member, and 10 steps give 127.0 s for the 12.7-s member. The dump interval is set to the same aligned stop, while the verified 10-s source duration is unchanged. Both use a 10 m grid and the same 500 m by 60 m physical domain. This gives 50 by 6 usable cells and a 54 by 10 raster after two numerical buffer cells are added outside all four sides. A single ignition is placed in the first usable column and the middle usable row.

The source is isolated with `NO_SURFACE_FIRE=.TRUE.`, so it does not create additional burning cells. The current `PER-AREA` generation model calculates

$$GR_{\text{pixel}} = \text{EMBER_GR} \Delta x \Delta y.$$

Using `EMBER_GR=0.1` pcs/s/m² and a 100 m² cell gives the required 10 pcs/s source. The fixed source duration is selected with `USE_PHYSICAL_SPOTTING_DURATION=.FALSE.` and `TAU_EMBERGEN=10` s. Firebrand ignition, consumption, and crosswind dispersion are disabled so that they cannot change the accumulated count.

Each variant must satisfy both conditions:

- total-count relative error no greater than 0.5%;
- profile L_1 relative error no greater than 1%.

The overall case passes only if outputs exist and both timestep variants pass. The two-cell buffers are excluded from profile extraction, while the total count is checked over the complete raster to detect misplaced firebrands.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from `dt0p13`. It is read from the actual `ELMFIRE` output named in the panel, masks nodata, and excludes the two-cell numerical buffer. Firebrand counts are normalized by the cell length and the represented 1-m cross-stream strip, giving units of pcs m⁻².

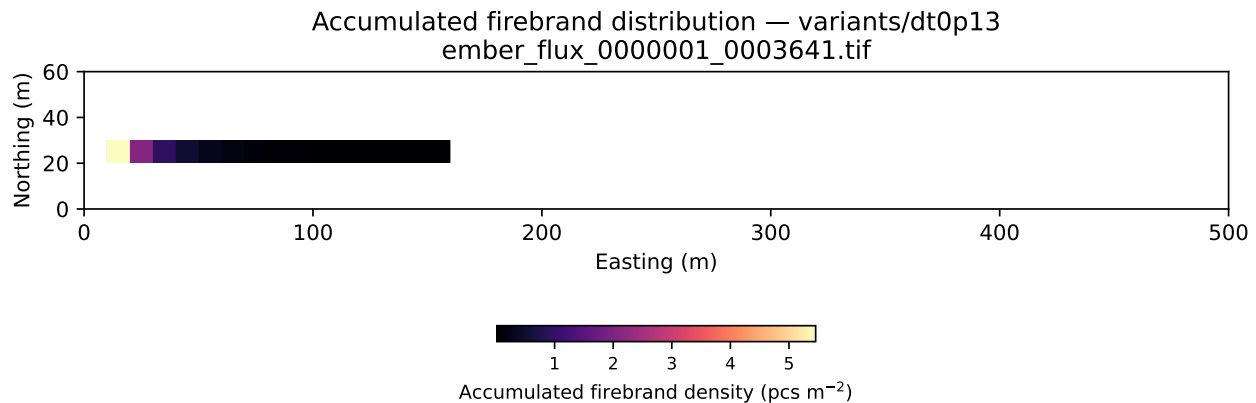


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Overall status: **pass**. Figure 3 first compares each actual ELMFIRE profile with the cell-integrated analytical distribution. Agreement for the 12.7 s variant is the critical result because that timestep exceeds the complete 10 s source duration.

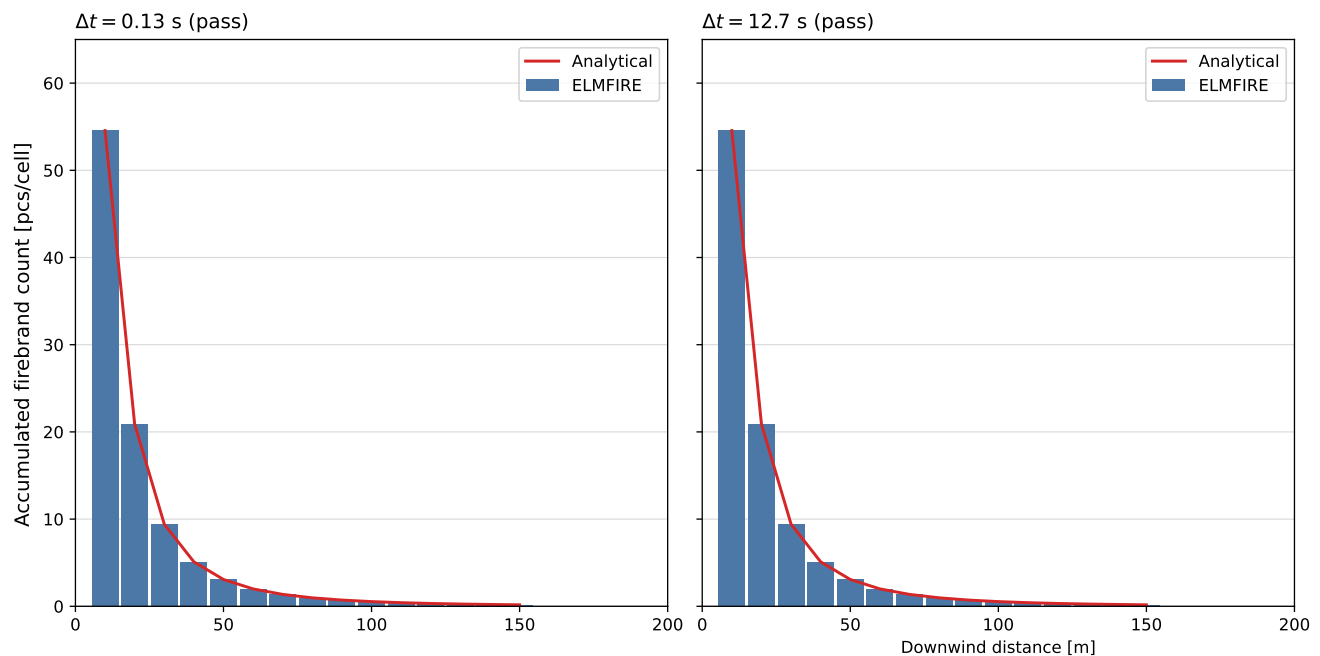


Figure 3: Accumulated firebrands per downwind cell for the 0.13 and 12.7 s timestep variants. Blue bars are ELMFIRE output and the red curve is the cell-integrated, truncated-lognormal reference for 100 generated firebrands.

7 Calculated metrics and verification decision

The overall decision is **pass** and is taken directly from postprocessing.

8 Technical interpretation

Agreement of both totals and profiles across a timestep longer than the complete source duration demonstrates correct partial-step accounting; a discrepancy isolates temporal source integration or deposition indexing.

Timestep	Observed total	Total relative error	Profile L_1 error
0.13 s	99.9999	5.198e-07	8.965e-07
12.7 s	100	2.399e-08	3.449e-07

Table 2: Firebrand-count and spatial-distribution errors for the two timestep variants.

9 Scope and limitations

The generated variant namelists, raster dimensions, buffer width, source location, rates, duration, and analytical parameters are recorded in `../variants/manifest.json`. Per-variant raster filenames and numerical errors are recorded in `../outputs/metrics.json`.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 03: Eulerian Firebrand Transport Convergence

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE03_ETC

This case verifies convergence of Eulerian firebrand transport in a uniform one-dimensional wind. Surface propagation is suppressed and deposited firebrands ignite immediately, so the leading edge isolates generation, transport, deposition, and level-set ignition updates.

The test exposes grid- or CFL-dependent advection speed, lost transient output, and inconsistent front extraction. Success requires the measured leading-edge rate to approach the prescribed wind speed over the accepted grid and CFL range.

2 Mathematical formulation and numerical implementation

This sweep reconstructs the wind-CFL convergence experiment [1]. `READ_SPOTTING` selects `PER-AREA`, `EULERIAN`, and `DIRECT`; `SPOTTING` creates the source, `EMBER_TRAJECTORY_EULERIAN` advances it, and `EULERIAN_SPOTTING_MAIN` converts the arriving ember state into a new ignition while updating `TIME_OF_ARRIVAL` and the level set. Because surface spread is suppressed, sustained propagation must traverse that complete code path. The wind-CFL sweep changes only the dimensionless ratio of wind displacement per step to cell size while the physical domain and areal source remain fixed. Consequently, a transport-stencil, timestep, direct-ignition, or TOA update error appears as resolution- or CFL-dependent leading-edge speed. The comparison to the wind speed is an idealized numerical reference and does not claim that real firebrands travel deterministically at the ambient wind speed.

3 Derivation of expected behavior

With surface spread disabled and direct ignition enabled, the transported ember field is the sole propagation mechanism. Uniform advection therefore gives the analytical leading-edge trajectory $x_{LE}(t) = u_{wind}t$ and the normalized propagation rate $ROS/u_{wind} = 1$. Grid and wind-CFL refinement should reduce numerical transport error without changing this limiting value.

The generated timestep is not an independent free parameter:

$$\Delta t = CFL_w \frac{\Delta x}{6.71}.$$

For the five required CFL values $\{0.2, 0.4, 0.5, 0.6, 0.8\}$, the generated timesteps are

Δx (m) \ CFL_w	0.2	0.4	0.5	0.6	0.8	
5	0.149031	0.298063	0.372578	0.447094	0.596125	
10	0.298063	0.596125	0.745156	0.894188	1.192250	s.
20	0.596125	1.192250	1.490313	1.788376	2.384501	
30	0.894188	1.788376	2.235469	2.682563	3.576751	

These $4 \times 5 = 20$ members form the acceptance set. Uniform advection over the 300–1500 m comparison interval predicts

$$\Delta t_{\text{ref}} = \frac{1500 - 300}{6.71} = 178.8376 \text{ s}, \quad \overline{ROS}_{\text{ref}} = \frac{1200}{178.8376} = 6.71 \text{ m s}^{-1}.$$

At the 260-s stop the characteristic displacement is $6.71(260) = 1744.6$ m, plus the generated half-cell initial coordinate where absolute position is needed. Thus every required variant has the exact normalized reference $\overline{ROS}/U = 1$ and $E_U = |1 - 1| = 0$; the implemented allowance is $E_U \leq 0.10$.

4 Verification metrics and acceptance criteria

ELMFIRE writes the level-set field ϕ after every solver time step. For every dump, the postprocessor reads the physical centerline, locates all sign changes, and linearly interpolates between adjacent cell centers to find the subcell locations where $\phi = 0$. The most downwind zero crossing is the leading edge $x_{LE}(t)$. Consecutive dumps give the stepwise rate

$$ROS_i = \frac{x_{LE}(t_{i+1}) - x_{LE}(t_i)}{t_{i+1} - t_i}. \quad (1)$$

The reported average is the time-weighted mean of these stepwise rates while both front observations lie between 300 m and 1500 m. Thus every solver step in the comparison interval contributes, and the result equals the total leading-edge displacement divided by elapsed time. This interval excludes startup behavior, the maximum downwind flight scale near the source, and the downstream boundary. Raster shape, cell size, buffer exclusion, dump-to-time matching, front-observation count, rate-interval count, and a diagnostic least-squares coefficient of determination are recorded for every result. A CSV trajectory preserves each dump time, selected ϕ raster, leading-edge location, and stepwise ROS.

Every variant with $CFL_{wind} \leq 0.8$ passes when

$$|ROS/u_{wind} - 1| \leq 0.10. \quad (2)$$

The higher-CFL points are retained to expose the onset of temporal transport error but do not determine overall pass/fail status.

4.1 Justification of metric quantification

The required range $CFL_{wind} \leq 0.8$ stays below the unit-CFL transport limit and leaves a 20% margin for the coupled source, transport, and direct- ignition updates. Thus the acceptance set probes the intended stable regime; larger CFL values diagnose degradation without being used to redefine that regime after results are known. The 10% speed band is deliberately wider than the largest single-cell position uncertainty divided by the 1200 m comparison length, but much narrower than the order-one error produced by a stalled front or by advancing trackers for only a fraction of a solver step. Time-weighting the rates over 300 m–1500 m further prevents an isolated grid crossing from controlling the decision. Requiring all 20 stable-regime variants guards against compensating resolution and timestep errors; incomplete trajectories cannot establish convergence and are therefore non-passing.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Required-output completeness	Transient phi rasters, dump times, and at least one valid front-rate interval between 300 and 1500 m.	All 20 variants with $CFL_{wind} \leq 0.8$
Normalized mean-ROS error	$E_U = \overline{ROS}/u_{wind} - 1 $, where $\overline{ROS}$ is displacement divided by elapsed time over the comparison interval.	$E_U \leq 0.10$ for every required variant
Overall decision	All required variants must be complete and satisfy the normalized-speed limit; higher-CFL variants are diagnostic only.	PASS iff all required components pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `dx10_cf10p2`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

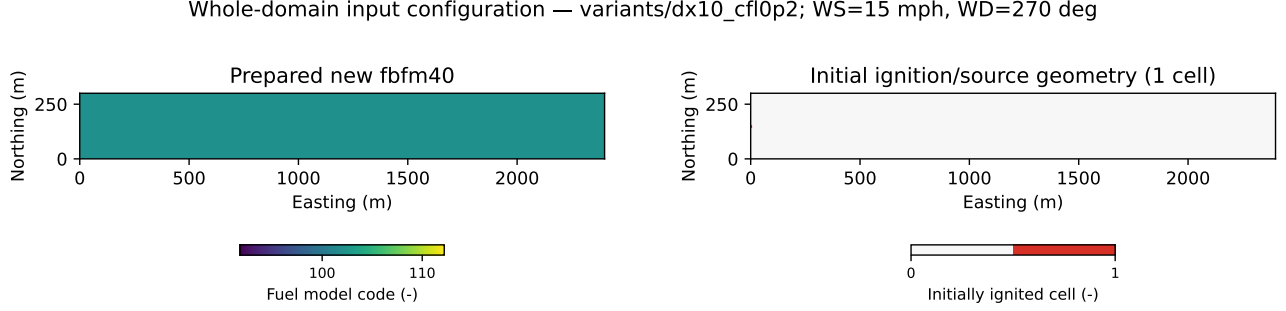


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The active physical domain is 2400 m long by 300 m wide. Two additional numerical cells are placed outside every physical boundary for each resolution. The initial burning cell is the first usable downwind cell and the center row; it is therefore never in the buffer. Fuel model 102, flat terrain, a uniform 15 mph wind from west to east, and deterministic input rasters are used. Crosswind dispersion and ember consumption are disabled.

The current spotting interface uses the following explicit model selections:

- generation: `PER-AREA`;
- spotting distance: `EMPIRICAL`;
- accumulation: `EULERIAN`;
- ignition: `DIRECT`.

The customized downwind distribution has $\mu = 2.18$, $\sigma = 1.23$, and $P_\epsilon = 0.01$. Every ignited cell emits for 10 s. The configured areal generation rate is resolution-independent:

$$GR_A = 10 [\text{pcs m}^{-2} \text{s}^{-1}]. \quad (3)$$

ELMFIRE therefore emits $N = GR_A \Delta x^2 \Delta t$ firebrands from one burning square cell during one solver step. This is the appropriate source definition because surface-fire spread and its fireline-intensity-dependent power are disabled.

5.1 Resolution sweep and predicted trend

Four grid spacings, $\Delta x = 5, 10, 20$, and 30 m, are crossed with wind-based CFL values 0.2, 0.4, 0.5, 0.6, 0.8, 1.0, and 1.2, for 28 variants. Each time step is prescribed by

$$\Delta t = CFL_{wind} \frac{\Delta x}{u_{wind}}. \quad (4)$$

Because timestep variation is the verification variable, these fractional values are not rounded. For each member preprocessing uses $N = \lceil 260/\Delta t \rceil$ and writes $t_{stop} = N\Delta t$, recording the requested 260 s, aligned stop, and integer step count in the manifest. All variants retain the same physical domain; only the number of active and buffer cells changes.

6 Actual simulation results

Figure 2 shows the representative transported level-set field from `dx10_cf10p2`. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. The zero contour identifies the transported front, while the signed field exposes domain-wide numerical structure away from the extracted centerline.

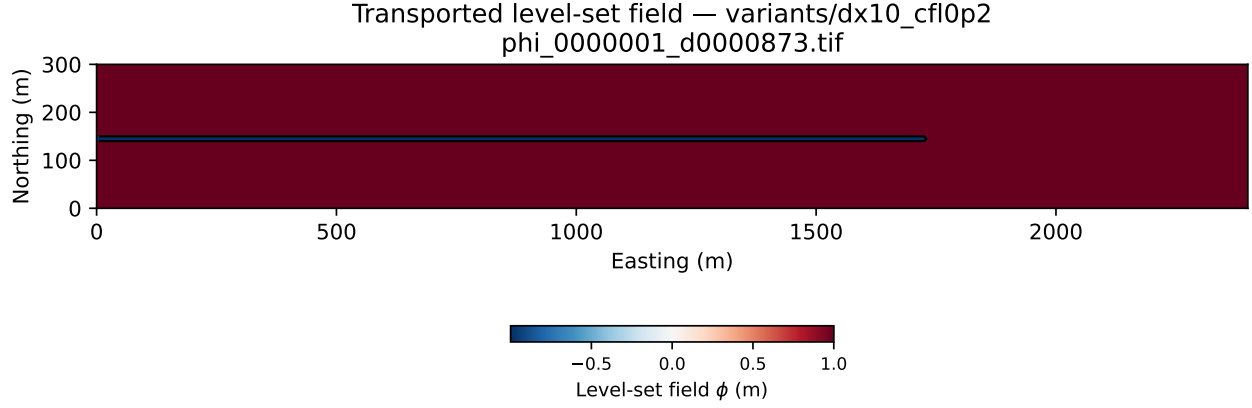


Figure 2: Whole-domain transported level-set field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figure 3 first presents the complete wind-CFL and grid-resolution sweep in the reference coordinate system. Each point is the normalized speed calculated from a complete transient level-set history.

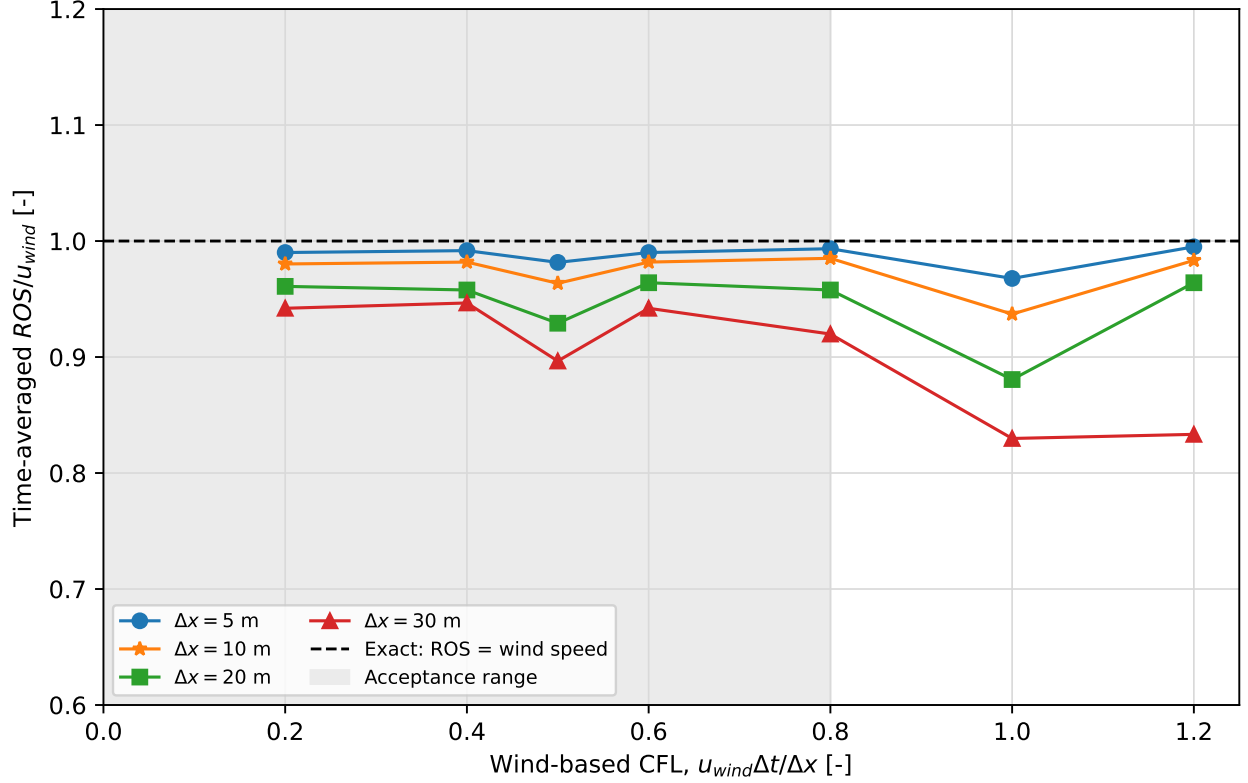


Figure 3: Time-averaged Eulerian firebrand-driven spread speed normalized by the uniform wind speed as a function of wind-based CFL number. The shaded region marks the CFL range subject to the 10% acceptance criterion.

Figure 4 is the supplemental diagnostic that traces one representative point from the level-set contour through the reported mean. It expands one member of the sweep, `dx10_cf10p5`, to show the complete path from the level-set field to the scalar convergence metric. Panel (a) shows one actual centerline profile, zoomed around the extracted front. In this direct-ignition, zero-surface-spread test, ELMFIRE’s level set is nearly binary away from the interface: burned and unburned cell centers remain close to -1 and $+1$. This is expected and is not a corrupted field. Adjacent cell-center values ϕ_j and ϕ_{j+1} with opposite signs bracket the front. Linear interpolation gives

$$x_{LE} = x_j - \phi_j \frac{x_{j+1} - x_j}{\phi_{j+1} - \phi_j}. \quad (5)$$

All centerline sign changes are evaluated, and the most downwind crossing is selected. This matters after spotting because a line can contain several separate burned regions and therefore several fire fronts.

Panel (b) assembles those independently extracted locations into $x_{LE}(t)$. The gray interval is the prescribed 300 m to 1500 m comparison window expressed in time. The dashed least-squares line is a trajectory-linearity diagnostic; its slope is 6.4957 m s^{-1} , but it is not the reported acceptance metric.

Panels (c) and (d) show how that metric is calculated. Panel (c) displays a 20-second detail from the start of the averaging interval so the alternating peak and trough values are individually visible. Each pair of consecutive dumps within the comparison window supplies $ROS_i = \Delta x_{LE,i} / \Delta t_i$. Because the final solver step can have a different duration, the average is explicitly time weighted:

$$\overline{ROS} = \frac{\sum_i ROS_i \Delta t_i}{\sum_i \Delta t_i} = \frac{x_{LE}(t_N) - x_{LE}(t_0)}{t_N - t_0}. \quad (6)$$

The representative result is $\overline{ROS} = 6.4655 \text{ m s}^{-1}$, or $\overline{ROS}/u_{wind} = 0.96356$. Thus the oscillatory instantaneous values caused by cell-by-cell front motion are not mistaken for the effective transport speed.

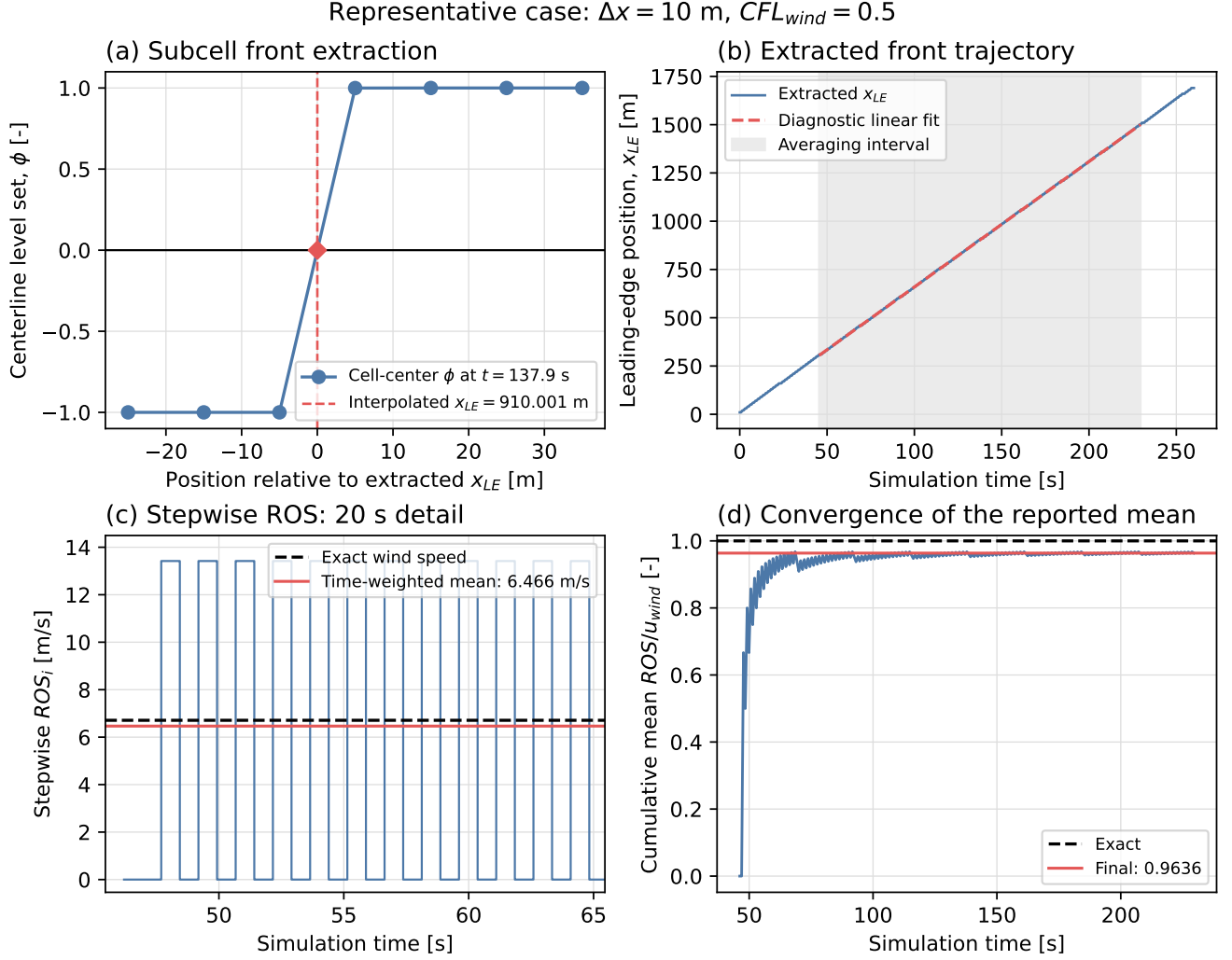


Figure 4: Detailed extraction for the representative $\Delta x = 10$ m, $CFL_{wind} = 0.5$ simulation. (a) Linear interpolation of centerline $\phi = 0$ crossings; (b) resulting leading-edge trajectory and diagnostic fit; (c) a 20-second zoom of consecutive-dump finite-difference ROS and its time-weighted mean; (d) convergence of the cumulative normalized mean to its reported value.

7 Calculated metrics and verification decision

Quantity	Value
Overall status	fail
Prepared variants	28
Variants with simulation results	28
Required variants	20
Required variants evaluated	20
Reference wind speed [m/s]	6.71
Maximum relative error	0.10
Acceptance CFL limit	0.8

Table 2: Verification status after processing all transient level-set outputs.

8 Technical interpretation

The expanded sweep fails its aggregate criterion because the $\Delta x = 30$ m, $CFL_{wind} = 0.5$ member gives

$$ROS/u_{wind} = 0.89655$$

corresponding to a relative error of 0.10345. This is slightly larger than the allowed 0.10. The $\Delta x = 10$ m representative case remains within tolerance and is used above to explain the extraction procedure. All 28 current-fingerprint variants completed and the generated controls match the declared wind, grid, and CFL sweep. The isolated coarse-grid excess is therefore a numerical transport/level-set discrepancy in the current source behavior, not an incomplete run or an admissible configuration correction.

9 Scope and limitations

From this case directory, configure `ELMFIRE_BIN` and run `./run_case.sh`. The runner preprocesses the manifest, executes every variant sequentially with isolated outputs and logs, postprocesses the results, and compiles the standalone report.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 04: Steady Transport and Immediate Ignition

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE04_STI

This case verifies the limiting mechanism shared by steady Eulerian transport and zero-delay wildland ignition. The `transport_impulse` variant disables surface spread and uses an areal source; `wildland_no_delay` uses a fire-power-based source and additionally checks accumulated population.

The paired design exposes failures in generation, transport, deposition, direct ignition, and time-of-arrival updates. Success requires both variants to recover wind-speed arrival behavior and the wildland variant to recover its one-metre-strip accumulation reference.

2 Mathematical formulation and numerical implementation

For constant wind speed $U = 6.71$ m/s and zero ignition delay, an ember-deposited cell becomes an active source immediately. Translation invariance then gives

$$t_{\text{ref}}(x) = \frac{x - x_0}{U}, \quad x_{\text{LE,ref}}(t) = x_{\text{LE},0} + U(t - t_0), \quad \overline{ROS}_{\text{ref}} = U, \quad (1)$$

where x_0 is the first usable cell centre. The final arrival-time raster stores the complete cell-crossing history used for the TOA profile metric. Leading-edge history is evaluated independently from every dumped instantaneous level-set field. At the physical centre row, the postprocessor linearly interpolates every adjacent pair of cell-centred ϕ values that changes sign and retains the rightmost $\phi = 0$ crossing. Here $x_{\text{LE},0}$ is the first extracted crossing at dump time t_0 . Once the zero contour has left the physical domain, that dump has no observable in-domain leading edge and is excluded from the trajectory fit rather than being replaced by a stationary domain-edge value. Mean ROS is the least-squares slope through all observable dump samples.

For immediate-ignition variant, a burning cell emits during its surface-fire residence time. Using the FLIN and spread rate written by the same executed cell gives

$$t_{\text{emission}} = \frac{\Delta x}{VS}, \quad N_{\text{ref}} = GR_{1m} \frac{FLIN}{1000} \frac{\Delta x}{VS}. \quad (2)$$

Here FLIN is in kW m⁻¹ and VS is converted from ft min⁻¹ to m s⁻¹. The documented nominal FLIN and VS give 1497.2 particles for this illustrative state [1]; the metric uses run-derived values so it remains consistent with the active fuel calculation. ELMFIRE forms pixel fire power from fireline intensity times cross-stream width Δy . The configured coefficient $GR_{\text{ELM}} = 33.3/\Delta y$ pcs/s/MW therefore makes the raw pixel count represent a one-metre strip. For the one-dimensional unit-width profile, each centre-row pixel count is converted to the density of the represented one-metre strip,

$$n_i = \frac{N_i}{\Delta x(1 \text{ m})} \quad [\text{pcs m}^{-2}]. \quad (3)$$

The two-cell numerical halo is excluded. The complete physical raster is still inspected, and its all-row deposited total is retained as a conservation diagnostic, but it is not mixed into the one-dimensional plotted observable. For the PER-AREA variant, the configured source gives $n_{\infty} = GR\Delta x\tau_{\text{ember}}/(1 \text{ m})$. For the PER-MW variant, Equation 2 divided by $\Delta x(1 \text{ m})$ gives the run-derived density reference.

The discrete algorithm is: generate embers in every active source cell; distribute them among downwind bins; accumulate them in landing cells; apply DIRECT ignition with $P_{\text{ign}} = 1$ and zero local/cell delay; update ember-ignition, level-set, and TOA states together; then advance the level-set clock. The two variants differ only where required by their source experiment.

3 Derivation of expected behavior

A deposition distribution with finite support changes which upstream cells contribute near x_0 , but not the propagation characteristic in Equation 1. Consequently TOA should be linear, and the leading edge extracted from ϕ should oscillate by at most one cell around $x_{LE,0} + U(t - t_0)$. Its fitted slope should approach U . Every cell marked by the ember-ignition raster must have finite nonnegative TOA.

In both variants, locations farther than the maximum travel distance receive contributions from a complete set of upstream emitters. Conservation then produces the configuration-consistent plateaus. The PER-AREA source uses $GR = 10 \text{ pcs m}^{-2} \text{ s}^{-1}$, $\Delta x = 10 \text{ m}$, and $\tau_{\text{ember}} = 6 \text{ s}$, giving 600 pcs m^{-2} under the represented-strip normalization. The PER-MW reference is calculated from co-located final FLIN and VS. The 155–195 m interval contains cells with complete upstream support.

The wind-CFL upper-bound step is

$$\Delta t_{\text{max}} = 0.5(10)/6.71 = 0.745156 \text{ s.}$$

To retain the requested 100-s stop as an exact multiple, preprocessing selects $N = \lceil 100/0.745156 \rceil = 135$ and $\Delta t = 100/135 = 0.740741 \text{ s}$. The resulting wind CFL is 0.4970, below the declared 0.5 bound. With $x_0 = 5 \text{ m}$, representative exact TOA substitutions in the declared plateau interval are

$$t_{\text{ref}}(155) = \frac{155 - 5}{6.71} = 22.3547 \text{ s}, \quad t_{\text{ref}}(195) = \frac{195 - 5}{6.71} = 28.3159 \text{ s.}$$

Likewise $x_{LE,\text{ref}}(50) = 5 + 6.71(50) = 340.5 \text{ m}$, and the front reaches the 500-m boundary after $(500 - 5)/6.71 = 73.7705 \text{ s}$. These substitutions give exact reference values of zero for TOA relative error, leading-edge NRMSE, and ROS relative error before raster quantization.

For the PER-AREA member, the complete-support density is

$$n''_{\infty} = GR \Delta x \tau_{\text{ember}} / (1 \text{ m}) = 10(10)(6) = 600 \text{ pcs m}^{-2}.$$

For the PER-MW member, the reference evaluated at each co-located FLIN/VS cell is

$$n''_{\text{ref}} = 33.3 \frac{FLIN/1000}{VS(0.3048/60)} = 6.555118 \frac{FLIN}{VS} \text{ pcs m}^{-2},$$

with FLIN in kW m^{-1} and VS in ft min^{-1} . The historical nominal substitution gives 1497.2 particles; current acceptance uses the generated co-located fields rather than that rounded historical value. Both plateau references therefore have expected relative error zero and are tested with a 0.10 allowance.

4 Verification metrics and acceptance criteria

For valid centre-row cells, define $e_i = |t_i - t_{\text{ref},i}|/t_{\text{ref},i}$. Each variant requires at least two cells, $\bar{e} \leq 0.10$, and $\max e_i \leq 0.25$. Leading-edge error is the RMSE relative to $x_{LE,0} + U(t - t_0)$ divided by $U(t_{\text{last}} - t_0)$ and must not exceed 0.10; t_{last} is the last dump with an in-domain $\phi = 0$ crossing. Every scheduled time must have exactly one matching ϕ raster. Fitted mean-ROS relative error must not exceed 0.10. All ember-ignited physical cells must have finite, nonnegative TOA. Each variant additionally requires $|\tilde{n} - n_{\text{ref}}|/n_{\text{ref}} \leq 0.10$, where $\tilde{n}$ is the median centre-row density from Equation 3 over the fully developed 155–195 m plateau. The corresponding analytical or run-derived reference is calculated independently of the ember-flux values. Domain-wide deposited totals are also recorded as diagnostics. Exactly two manifest variants and one final raster of every required kind are mandatory. The overall decision is the conjunction of completeness and all component decisions; missing or stale products are non-passing.

4.1 Justification of metric quantification

The 10% limits on mean TOA, leading-edge NRMSE, fitted ROS, and plateau density are engineering verification bands for a first-order, cell-centred transport/ignition calculation. They are broad enough to accommodate grid-crossing time quantization and the use of a localized run-derived FLIN/VS reference, yet they remain small compared with the order-one signature of a nonpropagating or solver-step-limited implementation. The 25% pointwise TOA limit is intentionally looser than the mean limit because a single cell next to the source or

front can carry a disproportionately large relative timing error; requiring at least two cells and the 10% mean prevents that allowance from masking a systematic bias. The plateau median is used because the expectation is spatially constant and the median is insensitive to one raster edge cell. Ignition-state consistency has zero physical tolerance: a cell cannot be classified as ember ignited without a finite, nonnegative arrival time. Both variants and all components are conjunctive because transport, ignition, and accumulation are separate claims of this combined test.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Two manifest variants with one final TOA, ignition-state, and accumulated-firebrand product of each required kind, plus one ϕ raster for every scheduled dump time.	2 of 2 variants complete; no missing ϕ dumps
TOA profile error	Mean and maximum of $ t_i - t_{ref,i} /t_{ref,i}$ on valid centre-row physical cells.	Mean ≤ 0.10 ; maximum ≤ 0.25
Leading-edge error	RMSE of the rightmost interpolated $\phi = 0$ crossing relative to $x_{LE,0} + U(t - t_0)$, normalized by the observable reference displacement.	NRMSE ≤ 0.10
Mean-ROS error	Relative error of fitted leading-edge ROS against $U = 6.71 \text{ m s}^{-1}$.	≤ 0.10
Ignition-state consistency	Every ember-ignited physical cell has a finite, nonnegative TOA.	No inconsistent cells
Accumulation plateau error	Relative error of median one-metre-strip density over 155–195 m against the independent reference.	≤ 0.10 for both variants
Overall decision	Conjunction of completeness and every component for both variants.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `transport_impulse`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

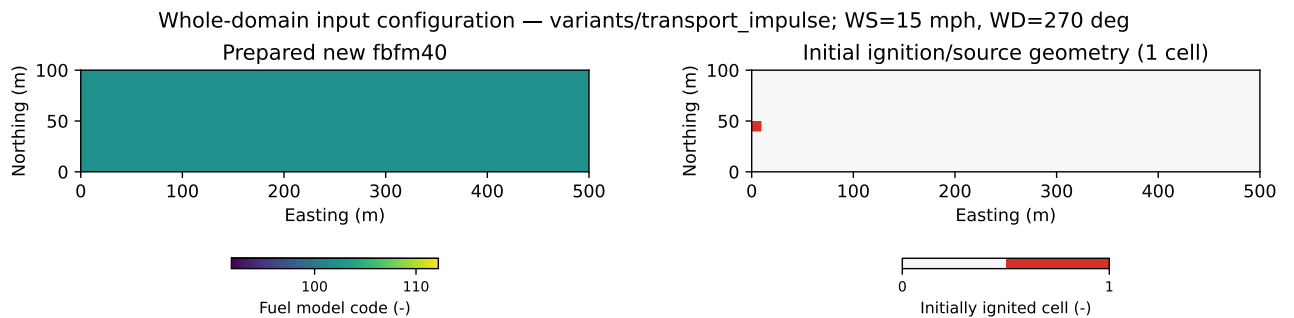


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Parameter	Configured value	Units	Verification role
Domain	500×100	m	common physical extent
Grid and halo	10; two cells/side	m	prescribed resolution; excluded halo
Time step	$100/135 = 0.740741$	s	wind CFL $0.4970 < 0.5$
Wind	6.71, from 270°	m/s	constant transport velocity
Fuel/terrain	FBFM 102; flat	—	uniform wildland control
Transport variant	PER-AREA; surface off; 100 s	—	isolates transport-impulse variant transport
Wildland variant	PER-MW; surface on; 100 s	—	immediate-ignition variant residence source
Transport/ignition	EMPIRICAL; EULERIAN; DIRECT	—	current spotting path
Ignition response	$P_{ign} = 1$; delays zero	—	immediate limiting regime
Crosswind/consumption	disabled/disabled	—	one-dimensional conservation

Table 2: Configuration shared by and distinguishing the two combined variants.

Before inspecting output, both TOA curves should overlay $(x - x_0)/U$. Both dumped- ϕ leading-edge histories should follow $x_{LE,0} + U(t - t_0)$ with grid-scale steps, and their mean ROS should be near 6.71 m/s. The two density bar profiles should rise near the inlet and approach their horizontal references: 600 pcs m^{-2} for the PER-AREA transport variant and the FLIN/VS-derived value for the wildland variant.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from `transport_impulse`. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. Firebrand counts are normalized by the cell length and the represented 1-m cross-stream strip, giving units of pcs m^{-2} .

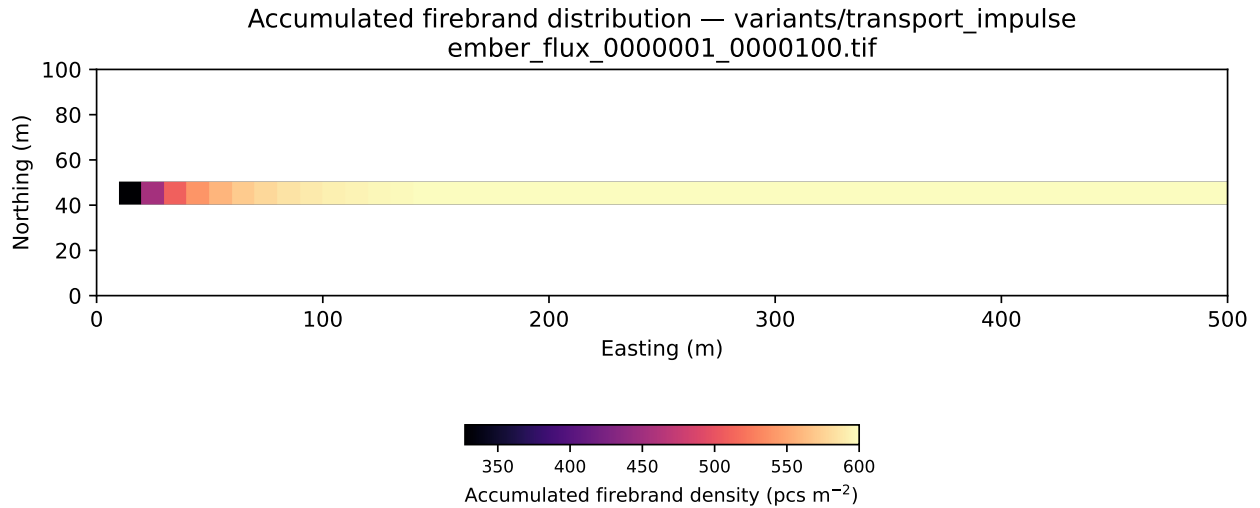


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The runner regenerated both input stacks and postprocessed only the current ELMFIRE products recorded in `variants/manifest.json`. Figure 3 presents the four physical observables used in the steady-transport reference experiment: accumulated density, arrival time, leading-edge position, and instantaneous leading-edge rate of spread.

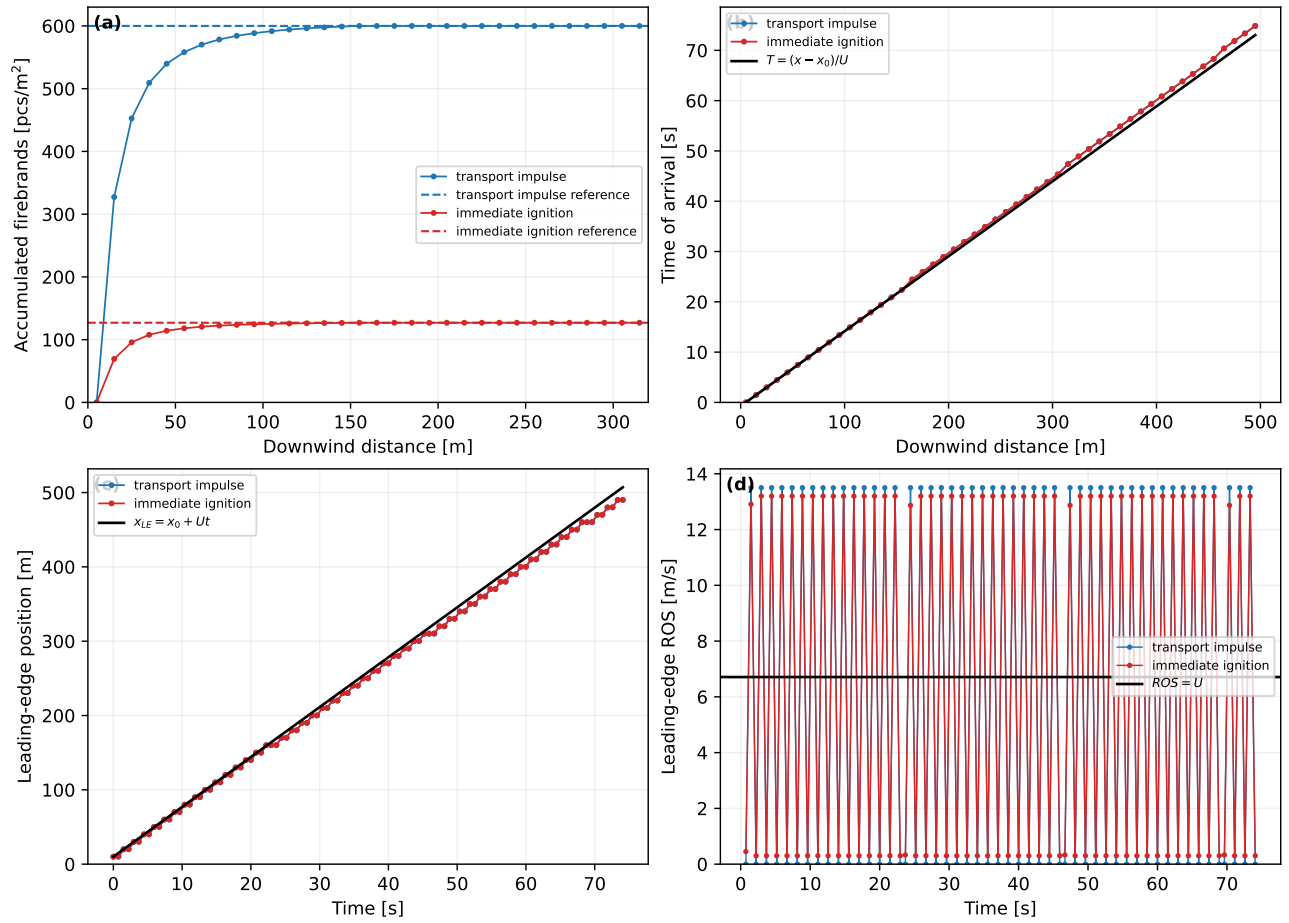


Figure 3: Steady-transport and immediate-ignition observables from actual ELMFIRE outputs. Solid colored curves identify the two controlled variants; black or color-matched reference curves show the derived analytical behavior.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
Output completeness	2 variants	2/2	PASS
Transport ϕ history	all scheduled dumps	137/137 dumps; 101 contours	PASS
Transport TOA mean error	≤ 0.10	0.01476	PASS
Transport leading-edge NRMSE	≤ 0.10	0.0165	PASS
Transport mean-ROS error	≤ 0.10	0.02537	PASS
Wildland ϕ history	all scheduled dumps	137/137 dumps; 101 contours	PASS
Wildland TOA mean error	≤ 0.10	0.01476	PASS
Wildland leading-edge NRMSE	≤ 0.10	0.01639	PASS
Wildland mean-ROS error	≤ 0.10	0.02536	PASS
Transport accumulation error	≤ 0.10	8.138e-08	PASS
Wildland accumulation error	≤ 0.10	5.625e-08	PASS
Transport deposited total	diagnostic	2.869e+05 pcs	—
Wildland deposited total	diagnostic	2.099e+05 pcs	—
Overall decision	all components pass	PASS	pass

Table 3: Metrics read directly from `outputs/metrics.json`.

8 Technical interpretation

The combined decision distinguishes a transport defect from a source-duration or normalization defect. Agreement in transport-impulse variant but not immediate-ignition variant implicates the residence-time PER-MW path; disagreement in both implicates shared transport, direct ignition, or TOA state handling. A full-area total of $2.099\text{e}+05$ particles was deposited across 5 active rows. A plateau-only failure cannot be hidden by a plausible TOA curve, and an average ROS cannot hide local TOA or product inconsistency. The numerical values in the table and figure provide the executed source revision’s decision rather than a qualitative visual judgment.

9 Scope and limitations

This is deterministic code verification of steady one-dimensional Eulerian transport and immediate ignition. It does not validate the empirical spotting PDF against experiments, time-varying wind, crosswind transport, stochastic ignition, delayed ignition, Lagrangian tracking, or consumption. Surface spread is present in the immediate-ignition variant source calculation but the wind-speed leading edge is expected to dominate. Run `./run_case.sh` from any working directory to regenerate both variants and the standalone report.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 05: Firebrand Transport in Time-Dependent Wind

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE05_TDW

This case verifies Eulerian firebrand transport when wind varies in space and time. Surface spread, crosswind dispersion, consumption, and ignition delay are disabled so the leading edge follows a controlled characteristic equation.

The test exposes incorrect meteorology interpolation, direction conversion, tracker lifetime, or transport/level-set timestep coupling. Success requires leading-edge position and running-average spread rate to agree with an independently integrated characteristic solution.

2 Mathematical formulation and numerical implementation

The imposed one-dimensional velocity is

$$u(x, t) = \bar{u} + A \sin\left(\frac{2\pi x}{\lambda_x}\right) \sin\left(\frac{2\pi t}{T}\right), \quad (\bar{u}, A, \lambda_x, T) = (6.71, 3.355, 100, 60) \quad (1)$$

in SI units [1]. The exact leading edge is the characteristic satisfying

$$\frac{dx_{\text{LE}}}{dt} = u(x_{\text{LE}}, t), \quad x_{\text{LE}}(0) = 5 \text{ m}. \quad (2)$$

The postprocessor integrates this ODE with fourth-order Runge–Kutta and a 0.005 s substep.

The preprocessor samples the analytical wind every 1 s into 121 bands of `ws.tif`; `wd.tif` holds a constant 270-degree meteorological direction. ELMFIRE linearly interpolates adjacent weather bands using `DT_METEOROLOGY`. The namelist explicitly selects `PER-AREA` generation, `EULERIAN` accumulation, and `DIRECT` ignition. Because surface fire is disabled, the per-area source avoids any dependence on fireline intensity. Each ignited cell emits for 10 s.

3 Derivation of expected behavior

The characteristic velocity is strictly positive: $3.355 \leq u \leq 10.065 \text{ m s}^{-1}$, so the reference leading edge is monotone. At each reporting time the simulated position is reconstructed as the furthest centre-row cell whose final ELMFIRE arrival time is no later than that time. Grid crossing makes the discrete instantaneous ROS alternate between zero and approximately $\Delta x / \Delta t$. For the reference-aligned diagnostic, a centred nine-point average should recover the smooth reference $u(x_{\text{LE}}(t), t)$.

The maximum prescribed speed used to select the generated solver step is $u_{\text{max}} = 6.71 + 3.355 = 10.065 \text{ m s}^{-1}$, so

$$\Delta t = 0.5(10) / 10.065 = 0.496771 \text{ s}.$$

Independent RK4 integration with the stated 0.005-s substep and $x(0) = 5 \text{ m}$ gives

$t \text{ (s)}$	0	30	60	90	120
$x_{\text{ref}} \text{ (m)}$	5.000	192.328	377.023	566.967	752.389

from the generated analytical wind, not from ELMFIRE output. Although $\sin(2\pi t/60) = 0$ makes the instantaneous reference velocity 6.71 m s^{-1} at these particular reporting times, the integrated position retains the intervening space-time oscillations; for example the 120-s displacement is $752.389 - 5 = 747.389 \text{ m}$, not a constant-wind substitution.

For matched sample i , the explicit references are $e_{x,i} = x_{\text{ELM},i} - x_{\text{ref},i} = 0$ and $e_{u,i} = \overline{ROS}_{\text{ELM},i} - u(x_{\text{ref},i}, t_i) = 0$. Hence the ideal metric values are $\text{mean } |e_x| = 0$, $\max |e_x| = 0$, $\text{mean } |e_u| = 0$, and $\text{coverage} = 1$; the substituted allowances are 10 m, 20 m, 2 m s^{-1} , and 0.90, respectively.

4 Verification metrics and acceptance criteria

For all finite samples, the postprocessor calculates mean and maximum absolute position error, the mean absolute error of the nine-point running-average ROS, and temporal coverage. The case passes only if

$$\text{MAE}_x \leq 10 \text{ m}, \quad \max |e_x| \leq 20 \text{ m}, \quad \text{MAE}_{\overline{ROS}} \leq 2 \text{ m s}^{-1}, \quad f_{\text{coverage}} \geq 0.90.$$

Missing or unusable output is explicitly non-passing.

4.1 Justification of metric quantification

The 10 m position-MAE limit equals one grid cell, and the 20 m maximum limit equals two cells. These bounds recognize that a continuous characteristic is reconstructed from cell-centred arrival events while still rejecting persistent phase errors. Differentiation magnifies that position quantization, so ROS is assessed after a centred nine-point average; its 2 m s^{-1} allowance is smaller than the configured 3.355 m s^{-1} velocity amplitude. A run that merely reproduces the mean wind but misses the oscillatory response therefore cannot satisfy the ROS criterion. The 90% coverage requirement allows a small number of unmatched samples near raster or temporal boundaries but prevents selective omission of a substantial high-error portion of either wind cycle. The four conditions measure bias, worst local error, derivative fidelity, and sampling support, respectively, so all are required for acceptance.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Final TOA raster with valid physical centre-row coverage; two-cell halo and nodata are excluded.	Usable output required
Position MAE	Mean $ x_{LE,ELMFIRE}(t) - x_{ODE}(t) $ over finite matched samples.	$\leq 10 \text{ m}$
Maximum position error	Maximum absolute leading-edge difference over the same samples.	$\leq 20 \text{ m}$
Running-ROS MAE	Mean absolute error of the centred nine-point ELMFIRE ROS against the characteristic solution.	$\leq 2 \text{ m s}^{-1}$
Temporal coverage	Fraction of required characteristic times having finite ELMFIRE leading-edge observations.	≥ 0.90
Overall decision	Conjunction of completeness and all four quantitative components.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the base time-dependent-wind run. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel-set or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

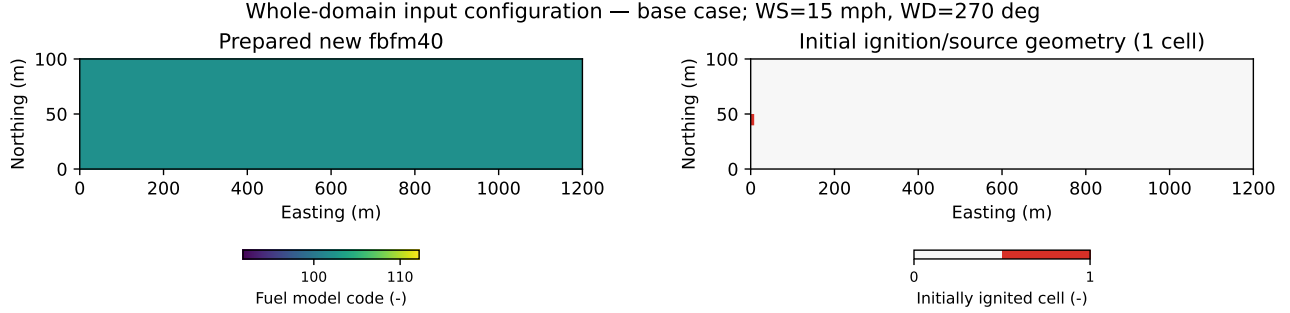


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Configuration of the reconstructed time-dependent-wind case.

Parameter	Value	Role
Physical domain	1200 m by 100 m	contains the 120-s trajectory
Cell size	10 m	streamwise discretization
Numerical halo	2 cells per side	excluded from observations
Duration	120 s	two wind periods
Wind bands	121 at 1 s	time-varying forcing
Wind CFL	0.5	resolves wind transport
Generation	PER-AREA, $1 \text{ pcs m}^{-2} \text{ s}^{-1}$	no-surface source
Emission duration	10 s	fixed finite-duration source
Source selection	100% deterministic	removes sampling variability
Accumulation/ignition	EULERIAN/DIRECT	tested transport and ignition path

The predicted position is a smooth monotone curve. Instantaneous numerical ROS is grid-quantized, while its nine-point mean should follow the oscillatory ODE velocity.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from the base time-dependent-wind run. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. The field provides the spatial counterpart to the leading-edge history and reveals whether the changing wind produces a coherent transported front across the complete domain.

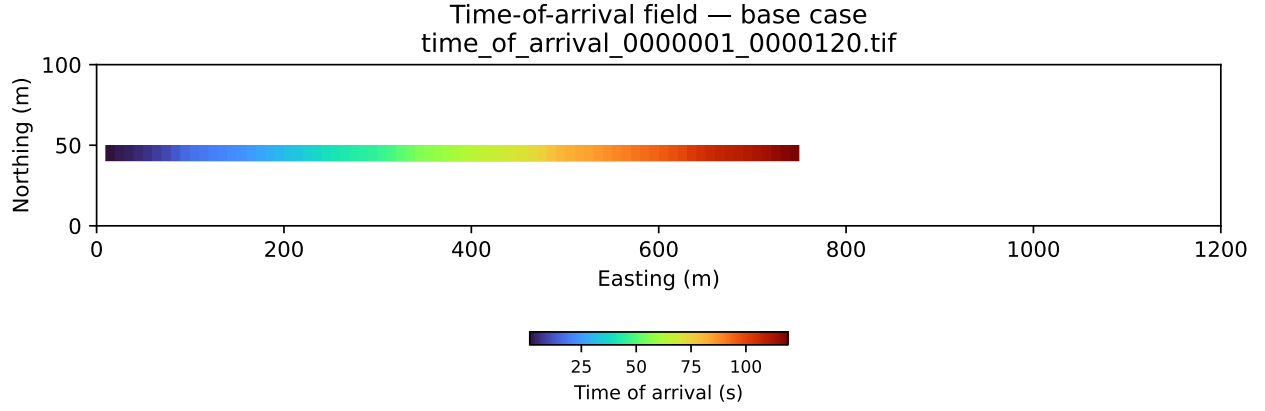


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figure 3 first reproduces the two principal time-dependent-wind observables. The numerical curves are re-constructed from the final ELMFIRE arrival-time raster; the smooth curves are obtained by integrating the prescribed leading-edge ODE.

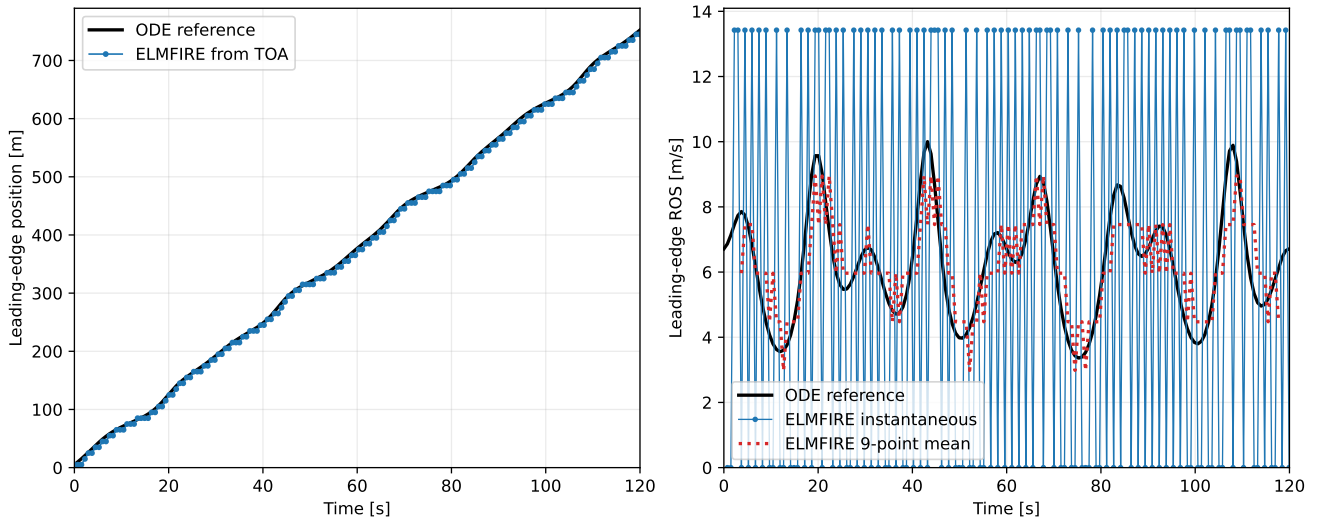


Figure 3: Time-dependent-wind transport comparison. Left: ELMFIRE and ODE leading-edge position. Right: ODE reference, grid-scale instantaneous ROS, and the nine-point running-average ELMFIRE ROS.

Figure 4 adds the time-resolved residuals and acceptance limits. It exposes localized phase errors that can be hidden by a single mean value.

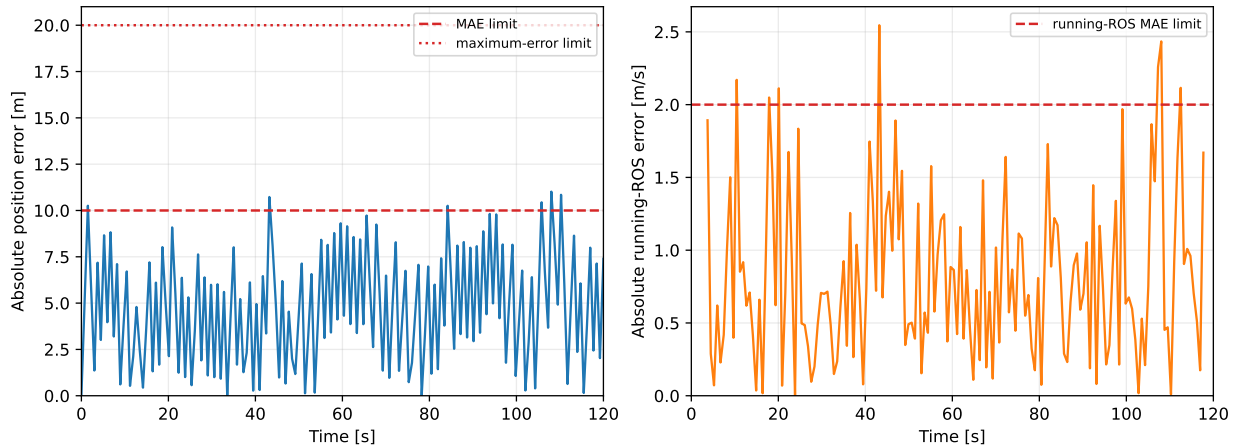


Figure 4: Absolute leading-edge-position and running-ROS errors through the simulation. Horizontal lines identify the applicable acceptance limits.

7 Calculated metrics and verification decision

Table 3: Acceptance metrics calculated by postprocessing.

Metric	Limit	Calculated	Pass
Position MAE	≤ 10 m	4.761474516898714	True
Position maximum error	≤ 20 m	11.01743438309552	True
Running ROS MAE	≤ 2 m/s	0.8128946040373389	True
Time coverage	≥ 0.90	1.0	True

The decision read directly from `outputs/metrics.json` is **pass** (`verification_passed = True`).

8 Technical interpretation

The current execution passes all four predeclared checks. The ELMFIRE leading edge reaches 745.0 m at 120 s, compared with the ODE reference value 752.3894781520776 m. The position MAE is 4.761474516898714 m, the maximum position error is 11.01743438309552 m, the nine-point running-ROS MAE is 0.8128946040373389 m/s, and the time-coverage fraction is 1.0; each lies within its declared limit. Input inspection confirms 121 one-second wind bands spanning the prescribed forcing history and an explicit deterministic source-selection percentage. The agreement therefore verifies the configured time-indexed-wind transport observable at the stated spatial and temporal resolution. Instantaneous ROS remains grid-quantized, which is why the smoother nine-point mean is the acceptance observable.

9 Scope and limitations

This is deterministic code verification of the Eulerian algorithm. It does not validate the empirical landing-distance distribution, crosswind dispersion, Lagrangian sampling, or physical ignition probability. One-second weather bands approximate the continuous analytical forcing by ELMFIRE’s native linear interpolation.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 06: Two-Dimensional Firebrand-Driven Spread

ELMFIRE Spotting Verification Guide

September 2, 2026

1 Verification purpose

Case ID: CASE06_F2D

This case verifies that Eulerian spotting produces a genuinely two-dimensional, firebrand-driven fireline under a manufactured two-component, space-time wind. It exercises emission, vector transport, deposition, direct spot ignition, and level-set propagation together.

The test exposes loss of either wind component, incorrect direction conversion, one-dimensional source geometry, missing secondary ignitions, non-monotone burned area, and stale transient output. Success requires nested perimeters near 30, 60, 90, and 120 s that advance in both directions and exceed the surface-only reach.

2 Mathematical formulation and numerical implementation

The prescribed horizontal wind is

$$u(x, t) = \bar{u} + A \sin(2\pi x/\lambda_x) \sin(2\pi t/T), \quad (1)$$

$$v(y, t) = \bar{v} + B \sin(2\pi y/\lambda_y) \sin(2\pi t/T), \quad (2)$$

with $\bar{u} = 6.71 \text{ m s}^{-1}$, $A = 3.355 \text{ m s}^{-1}$, $\bar{v} = 1.0 \text{ m s}^{-1}$, $B = 0.5 \text{ m s}^{-1}$, $\lambda_x = \lambda_y = 100 \text{ m}$, and $T = 60 \text{ s}$. The preprocessor evaluates these equations at every cell centre and every one-second meteorology time. It writes $|\mathbf{u}|$ in mph and converts the transport-to angle to ELMFIRE meteorological wind-from convention.

For each Eulerian source tracker, the current spotting implementation advances

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{u}(\mathbf{x}_k, t_k) \Delta t_k, \quad \Delta t_k = \Delta x / \max(|\mathbf{u}|, 0.01),$$

and distributes the source ember count through the configured downwind lognormal landing distribution. Direct ignition updates the level-set state, after which both the primary headfire and spot fires create new burning cells. The configured surface reference speed is $R_s = 0.71 \text{ m s}^{-1}$ [1], and the corresponding emission residence time is $t_e = \Delta x / R_s = 14.08 \text{ s}$.

The prescribed u and v components determine an individual transport characteristic, but they do not uniquely determine the maximum coordinates of the coupled fireline in this configuration. The observed $\phi = 0$ perimeter is the union of the surface-propagated primary fire and surface-propagated spot fires created after nonlocal, probability-distributed deposition. Consequently, $x_{\max}$ and $y_{\max}$ do not generally satisfy the uncoupled equations $dx_{\max}/dt = u(x_{\max}, t)$ and $dy_{\max}/dt = v(y_{\max}, t)$. An exact endpoint metric would require an independent solution of the complete coupled problem.

3 Derivation of expected behavior

Because the wind has positive mean components, successive fronts must progress downwind and develop a nonzero crosswind span. The level-set burned set is irreversible, so for burned masks $B_j = \{(x, y) : \phi(x, y, t_j) \leq 0\}$ the expected inclusion is $B_j \subseteq B_{j+1}$. A conservative surface-only leading-edge bound is

$$x_{\text{surf}}(120) = x_0 + R_s(120) = 90.2 \text{ m}.$$

A front substantially beyond this location, together with nonzero ember-ignition cells, demonstrates that firebrands rather than the surface headfire alone control the long-range advance. Accordingly, the principal expected result is a family of nested two-dimensional contours rather than a one-dimensional TOA curve.

The largest generated transport speed is

$$|\mathbf{u}|_{\max} = \sqrt{(6.71 + 3.355)^2 + (1.0 + 0.5)^2} = 10.1762 \text{ m s}^{-1},$$

so preprocessing sets $\Delta t = 0.5(10)/10.1762 = 0.491344 \text{ s}$. The surface residence time is independently

$$t_e = 10/0.71 = 14.0845 \text{ s}.$$

Starting from $x_0 = 5 \text{ m}$, a surface-only head fire would be at

$t \text{ (s)}$	30	60	90	120
$x_{\text{surf}} \text{ (m)}$	26.3	47.6	68.9	90.2

and would advance $0.71(30) = 21.3 \text{ m}$ between successive target times. The declared 20-m interval threshold is therefore slightly below the surface-only analytical increment but still requires two resolved cells. At 120 s, the firebrand-excess test $x_{\max} - 90.2 \geq 30 \text{ m}$ is equivalently $x_{\max} \geq 120.2 \text{ m}$, together with a positive ember-ignition count. Exact irreversible evolution gives $|B_j \setminus B_{j+1}|/|B_j| = 0$; the numerical allowance is 0.02. The final span reference is qualitative nonzero two-dimensional growth, made quantitative as at least $40 \text{ m} = 4\Delta x$.

4 Verification metrics and acceptance criteria

The postprocessor selects transient **phi** rasters within 2 s of all four target times and excludes the two-cell halo. It calculates: (i) completeness of all four contours; (ii) the fraction $|B_j \setminus B_{j+1}|/|B_j|$, limited to 0.02; (iii) downwind leading-edge advance, required to be at least 20 m in every interval; (iv) final crosswind span, required to be at least 40 m; and (v) firebrand excess advance $x_{\max}(120) - x_{\text{surf}}(120)$, required to be at least 30 m with at least one ember-ignition cell. The overall result is PASS only when every component passes.

4.1 Justification of metric quantification

The thresholds are expressed in grid-resolvable units. Selecting a contour within 2 s limits time mismatch to less than 7% of the earliest 30 s target and only a few solver steps. Exact level-set evolution is nested, so the 2% burned-area allowance is reserved for zero-contour classification along a cell-wide perimeter; a larger loss would indicate physical unburning or inconsistent dump selection. The minimum interval advance of 20 m, final crosswind span of 40 m, and firebrand excess of 30 m equal two, four, and three cells, respectively. These multi-cell thresholds cannot be satisfied by a stationary front, a single misclassified boundary cell, or the surface-only reference, yet they do not prescribe a detailed perimeter shape that the manufactured problem does not supply. At least one ember-ignition cell is an exact mechanism check, and all four times are required to support the claimed two-dimensional evolution.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Perimeter completeness	Unique transient phi raster selected within 2 s of 30, 60, 90, and 120 s.	4 of 4 target times
Nesting violation	Maximum $ B_j \setminus B_{j+1} / B_j $ over consecutive physical burned masks.	≤ 0.02
Downwind progression	Increase in maximum burned-cell x between consecutive target times.	$\geq 20 \text{ m}$ every interval
Final crosswind span	$\max(y) - \min(y)$ of the final physical burned mask.	$\geq 40 \text{ m}$
Firebrand-driven excess	$x_{\max}(120) - [x_0 + R_s(120)]$, together with the final ember-ignition map.	$\geq 30 \text{ m}$ and at least one ember-ignited cell
Overall decision	Conjunction of completeness, nesting, progression, span, and excess-advance checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the base two-dimensional run. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

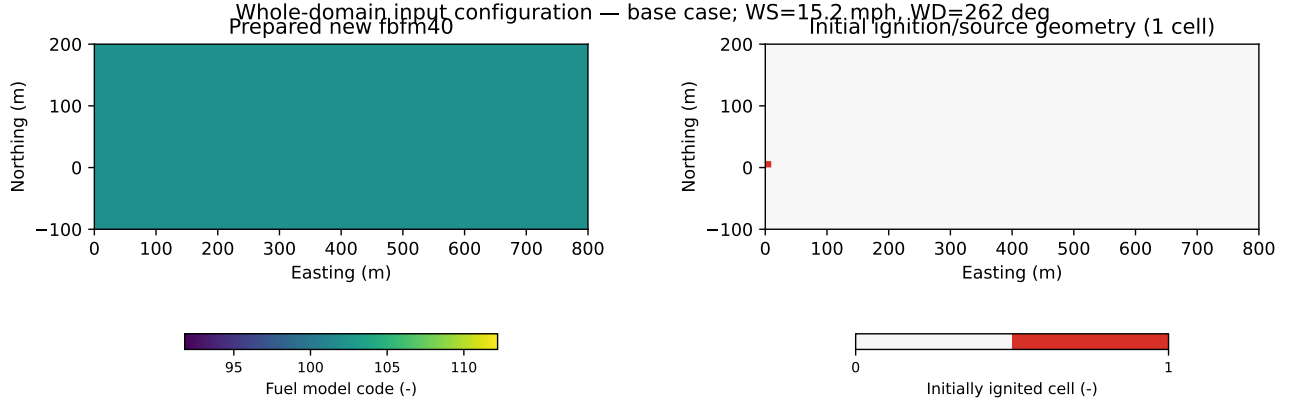


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Configured two-dimensional transport case experiment.

Parameter	Value	Role
Physical domain	800×300 m	two-dimensional propagation
Cell size	10 m	reference spatial resolution
Raster grid	84×34 cells	includes two-cell halo
Meteorology	121 bands at 1 s	space-time wind
Simulation step	0.491 026 s	wind CFL approximately 0.5
Initial ignition	(5, 5) m	localized upwind source
Surface headfire	enabled	primary moving source
Generation	PER-AREA, $10 \text{ pcs m}^{-2} \text{ s}^{-1}$	deterministic ember supply
Accumulation / ignition	Eulerian / direct	tested implementation path
Crosswind PDF	$\mu_y = 0, \sigma_y = 0.1$ m	narrow $k_{\max} = 0$ limit
Output times	30, 60, 90, 120 s	transient perimeter comparison

Before inspecting results, the predicted output is four nested contours elongated in positive x , displaced and broadened in y , with many secondary ember ignitions and a final leading edge far beyond 90.2 m.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from the base two-dimensional run. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. Counts are divided by physical cell area, giving pcs m^{-2} and exposing the full lateral as well as downwind distribution.

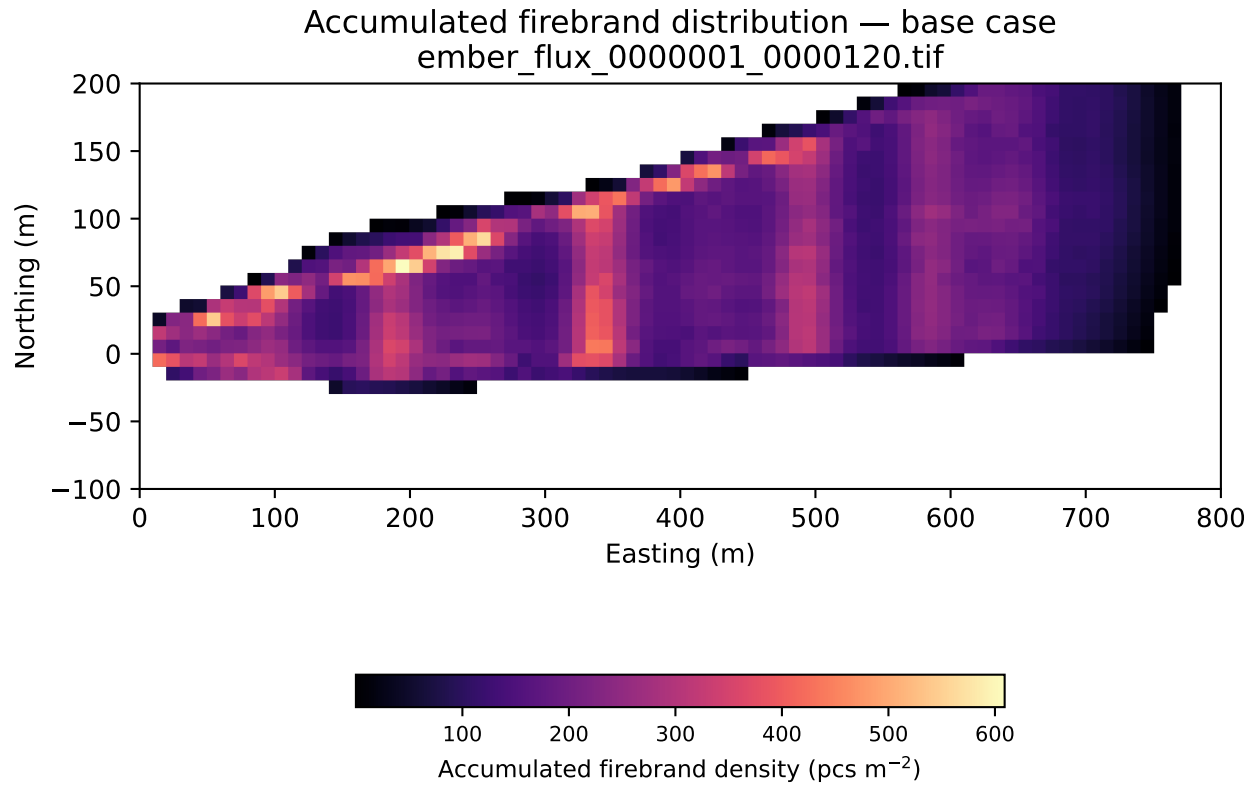


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The selected transient perimeter times were 30.44, 60.4, 90.35, 120 s. Exact filenames are recorded in `outputs/metrics.json`. Figure 3 first presents the evolving zero-level contours, which are the principal two-dimensional reference observable.

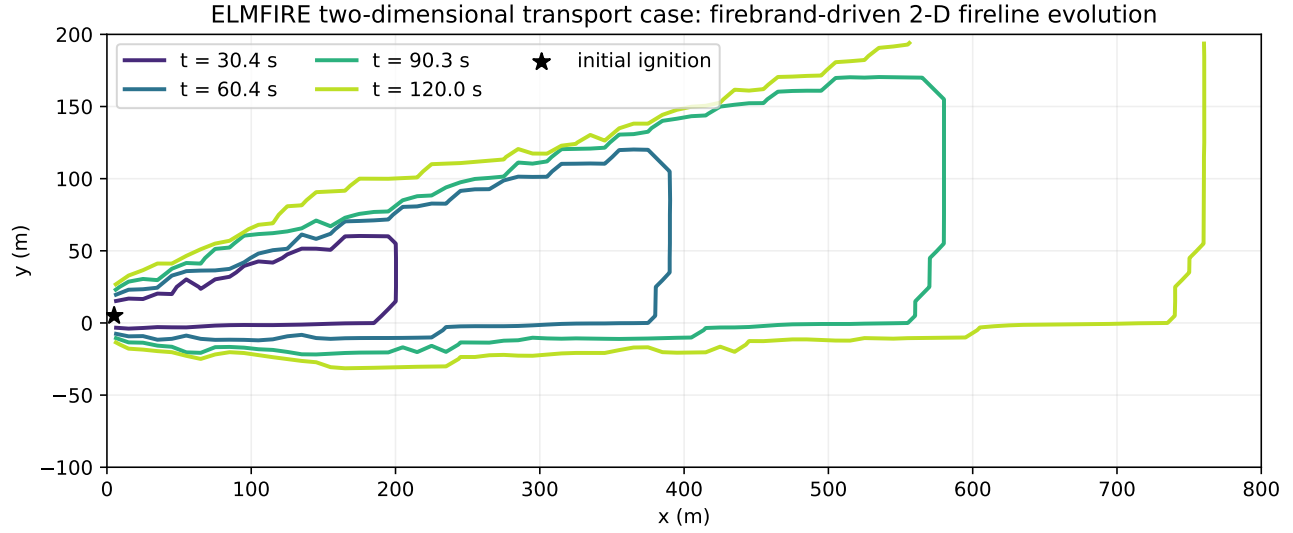


Figure 3: Actual ELMFIRE fire perimeters at four output times under the prescribed two-component, space-time wind. The star marks the initial ignition.

Figure 4 identifies the cells actually ignited by deposited firebrands. The accumulated-firebrand field is not repeated here because its whole-domain distribution is already presented in Figure 2.

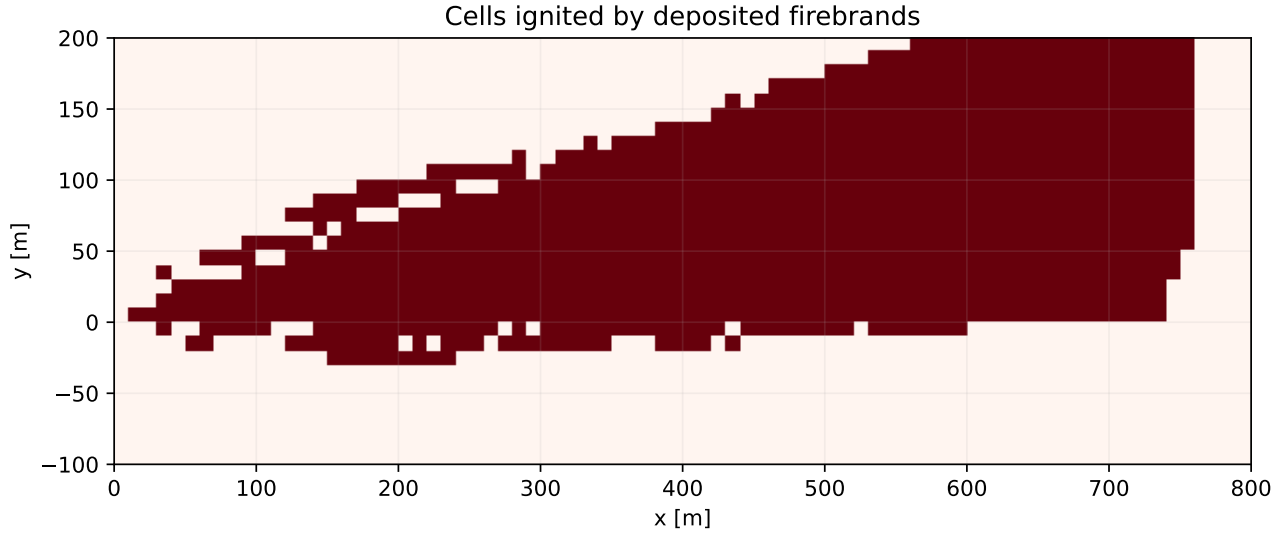


Figure 4: Whole-domain cells marked by ELMFIRE as ignitions caused by deposited firebrands. The numerical halo is excluded.

7 Calculated metrics and verification decision

8 Technical interpretation

The result is **pass**. The maximum nesting violation is 0.0, while the leading edge advances by 190, 190, 180 m over the three intervals. The final crosswind span is 220.0 m, confirming that the calculation is not a replicated one-dimensional transect. The final front exceeds the surface-only bound by 664.8 m and the ember-ignition raster

Table 3: Acceptance metrics calculated from actual ELMFIRE outputs.

Metric	Acceptance limit	Calculated value	Status
Required contours	4 within 2 s	30.44, 60.4, 90.35, 120	PASS
Nesting violation	≤ 0.02	0.0	PASS
Interval advances	each ≥ 20 m	190, 190, 180 m	PASS
Final crosswind span	≥ 40 m	220.0 m	PASS
Firebrand excess	≥ 30 m and ignitions	664.8 m; 1097 cells	PASS
Overall	all components true	PASS	pass

contains 1097 positive cells; together these observations attribute the long-range evolving fireline to secondary ignitions from transported firebrands.

9 Scope and limitations

This test verifies quantified mechanism and evolution invariants of the current Eulerian 2-D coupling and re-constructs the intended contour diagnostic. The known wind components alone are not treated as an exact outer-fireline solution because surface spread and nonlocal secondary ignitions remain active. The test does not reproduce the comparison solution cell-for-cell, compare Eulerian and Lagrangian contours, validate the landing PDF against experiments, or prove grid convergence. The configured flat terrain isolates horizontal transport; additional terrain-slope cases would be required to verify topographic coupling.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 07: Ignition Probability and SFT Convergence

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE07_IPC

This case verifies statistical convergence of the stochastic smoldering-to-flaming (SFT) model and the resulting ignition-probability response. Pooled ignition-delay samples must converge to the configured SFT scale, while ensemble-mean firebrand-driven spread must respond consistently to ignition probability.

The test exposes biased sampling, insufficient ensemble size, incorrect delay extraction, and grid-dependent probability application. Success is determined from confidence-aware delay statistics and declared ensemble metrics, never from a single realization.

2 Mathematical formulation and numerical implementation

For accumulated firebrands in a cell, ELMFIRE’s SIMPLE model uses

$$F(t) = 1 - (1 - P)^{t/\tau}, \quad P_{\Delta t} = 1 - (1 - P)^{\Delta t/\tau}, \quad (1)$$

where $P = P_{\text{ign,small}}$, $\tau = \tau_{\text{ign}}$, and $P_{\Delta t}$ is the Bernoulli probability evaluated during one solver step. Consequently $F(\tau) = P$: at $P = 0.9$, the theoretical 90th percentile is $q_{0.9} = \tau = 10$ s. A successful local transition is followed by `CELL_IGNITION_DELAY=0`, so the final TOA records the full ignition time without an additional development lag.

For the dedicated delay members, transient ember-flux output is written every wind-CFL step. If a cell’s first nonzero transient flux occurs at dump k , its first deposition lies in $(t_{k-1}, t_k]$. Postprocessing uses the interval midpoint $t_{d,i} = (t_{k-1} + t_k)/2$ and forms the actual-output sample

$$d_i = t_{\text{ign},i} - t_{d,i}. \quad (2)$$

Only centerline cells marked in the final ember-ignition raster, outside the two-cell halo and in the 155–2800 m complete-support interval, are admitted. The dump spacing Δt is retained as the temporal-censoring resolution.

For the response sweep,

$$CFL_w = \frac{u_w \Delta t}{\Delta x} \leq 0.5, \quad r_m = \frac{R_m}{u_w}, \quad (3)$$

and each member ROS R_m is the least-squares slope of centerline distance against final TOA over 100–2800 m. Ten explicit seeds are paired across $\Delta x = 5, 10, 20, 30$ m and $P = 0, 0.25, 0.5, 0.75, 1$.

3 Derivation of expected behavior

Equation 1 gives the exact event-level expectation $q_{0.9} = 10$ s. A finite sample quantile fluctuates, but its bootstrap confidence interval should contract as independently seeded members add cell samples. The final pooled estimate should therefore approach 10 s rather than an ROS-derived surrogate. Timestep-resolved deposition limits the deterministic timing uncertainty to the deposition interval.

At $P = 0$, the ignition hazard is zero, and no spotting-driven front should propagate. Increasing P increases $F(t)$ for every $t > 0$ and therefore cannot reduce population-mean ROS. At $P = 1$, ignition is immediate in the continuous model and $\bar{\tau}$ should approach one. Because physical extent is fixed and the generated CFL_w values remain conservatively bounded near 0.5, the four resolution curves should nearly coincide.

For the delay member, $P = 0.9$ and $\tau = 10$ s, hence

$$F(5) = 1 - 0.1^{0.5} = 0.683772, \quad F(10) = 1 - 0.1 = 0.9, \quad F(20) = 1 - 0.1^2 = 0.99.$$

Solving $F(q_{0.9}) = 0.9$ gives $0.1^{q_{0.9}/10} = 0.1$ and therefore $q_{0.9} = 10$ s exactly. With $U = 15(0.44704) = 6.7056 \text{ ms}^{-1}$, preprocessing first computes the $CFL_w = 0.5$ upper bounds, then reduces each step to divide 240 s exactly. The actual generated steps and corresponding per-step ignition probabilities are

Δx (m)	5	10	20	30
Δt (s)	0.372671	0.745342	1.490683	2.222222
$1 - 0.1^{\Delta t/10}$	0.082232	0.157702	0.290534	0.400516

For each resolution the conservative CFL-derived timestep defines the probability transformation and cross-resolution control. Preprocessing selects $N = \lceil 240/\Delta t_{CFL} \rceil$, writes the conservative divisor $\Delta t = 240/N$, and keeps $t_{stop} = 240$ s exactly; the manifest records nominal and actual CFL values plus the integer step count. This shows why the step probability changes while the continuous delay CDF does not.

The response design contains 4 resolutions $\times$ 5 probabilities $\times$ 10 paired seeds = 200 required runs, plus 10 delay members. The exact limiting references are $R/U = 0$ at $P = 0$, monotone nondecrease with P , and $R/U \rightarrow 1$ at $P = 1$. Numerically these become a maximum cross-resolution range of 0.10, a $P = 1$ floor of 0.90, and adjacent decreases no larger than their pooled 95-percent confidence half-width. The delay substitution is $|\hat{q}_{0.9} - 10| \leq 1$ s, with at least 1000 samples and bootstrap half-width at most 1 s.

4 Verification metrics and acceptance criteria

For every response group with $N = 10$, the postprocessor calculates the mean, sample standard deviation, standard error, and two-sided Student- t 95% confidence interval. Adjacent probability means may decrease only within their pooled confidence half-widths. At each P , the spread of the four resolution means must not exceed 0.10, and every $P = 1$ mean must be at least 0.90.

For the delay ensemble, samples are pooled after each independent member. At the final member, all ten members must be usable, at least 1000 delays must be available, $|\hat{q}_{0.9} - 10 \text{ s}| \leq 1 \text{ s}$, and the 2000-replicate bootstrap 95% interval half-width must not exceed 1 s. A missing final TOA, ignition map, dump-time table, transient sequence, stale raster shape, or empty sample set makes that member incomplete. Overall PASS is the conjunction of all response and delay components; missing required output yields `insufficient_output`, never a pass.

4.1 Justification of metric quantification

A two-sided 95% Student- t interval is used because each response mean has only ten independent ensemble members and its variance is estimated from those members. Allowing an adjacent decrease only within the pooled confidence half-width distinguishes sampling noise from evidence against monotonicity without assuming equal variances. The 0.10 cross-resolution range and the 0.90 unit-probability floor each represent ten percentage points on the naturally normalized $[0, 1]$ response scale: they admit finite-ensemble and grid-crossing variability but reject a practically material resolution effect or failure to approach immediate ignition.

The target $q_{0.9} = 10$ s follows directly from the prescribed SFT distribution [1]. Requiring 1000 pooled delays gives substantial support in the upper decile rather than allowing a few extreme samples to set the quantile. The 1 s accuracy and bootstrap-half-width limits are each 10% of the model timescale, so both bias and uncertainty must be small on the scale being verified. Two thousand bootstrap replicates provide stable percentile endpoints for this engineering decision without treating the bootstrap interval as an exact physical confidence bound. Full member completeness is required because silently dropping failed stochastic runs can bias both response means and delay quantiles.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Response-ensemble completeness	Ten usable members for every resolution-probability group.	200 of 200 members
Probability monotonicity	Adjacent ensemble-mean normalized ROS differences, allowing decrease only within pooled 95% confidence half-widths.	Every adjacent check passes
Resolution spread	Range of the four resolution means at each ignition probability.	≤ 0.10
Unit-probability response	Minimum ensemble-mean normalized ROS at $P_{ign} = 1$.	≥ 0.90
SFT sample completeness	Ten usable delay members and pooled interval-censored ignition-delay samples.	10 members and at least 1000 samples
SFT 90th percentile	Absolute difference between pooled $\hat{q}_{0.9}$ and the configured 10 s scale.	≤ 1 s
SFT uncertainty	Half-width of the 2000-replicate bootstrap 95% interval for $\hat{q}_{0.9}$.	≤ 1 s
Overall decision	Conjunction of every response and SFT component; incomplete output is not evaluable.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `dx10_pign0p5_r01`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

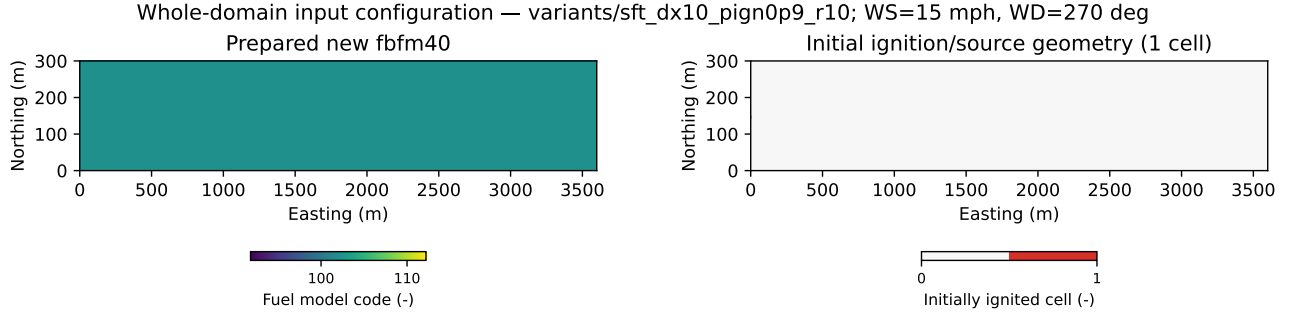


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Parameter	Configured value	Unit	Verification role
Physical domain	3600×300	m	Fixed physical extent
Buffer	2 cells per side	cells	Excluded numerical halo
Response grids	5, 10, 20, 30	m	Resolution comparison
Delay grid	10	m	delay-ensemble spatial baseline
Wind	15 (6.7056)	mph (m/s)	Uniform transport
Wind CFL	≤ 0.5	—	stable near-fixed $\Delta t / \Delta x$
Stop time	240	s	exact timestep multiple
Response probabilities	0, 0.25, 0.50, 0.75, 1	—	probability-response design
Delay model	$P = 0.9, \tau = 10$	—, s	delay-distribution design
Ensembles	20×10 response; 10 delay	runs	Sampling statistics
Generation	PER-AREA, 10	$\text{pcs m}^{-2} \text{ s}^{-1}$	No-surface source
Accumulation/ignition	EULERIAN/SIMPLE	—	Implementation under test
Surface/development delay	disabled/0	—, s	Isolate SFT waiting time
Delay output cadence	one wind-CFL step	s	Bracket first deposition

Table 2: Configuration of the merged ensemble-convergence case verification.

The predicted response plot contains four nondecreasing curves beginning at zero and approaching one. The delay plot should approach the horizontal 10 s line, with its sampling interval narrowing as ensemble members are pooled.

6 Actual simulation results

Figure 2 shows the representative ember-ignition field from `dx10_pign0p5_r01`. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. This representative ensemble member supplies spatial evidence alongside the ensemble probability and delay statistics; it is not used as a replacement for those statistical metrics.

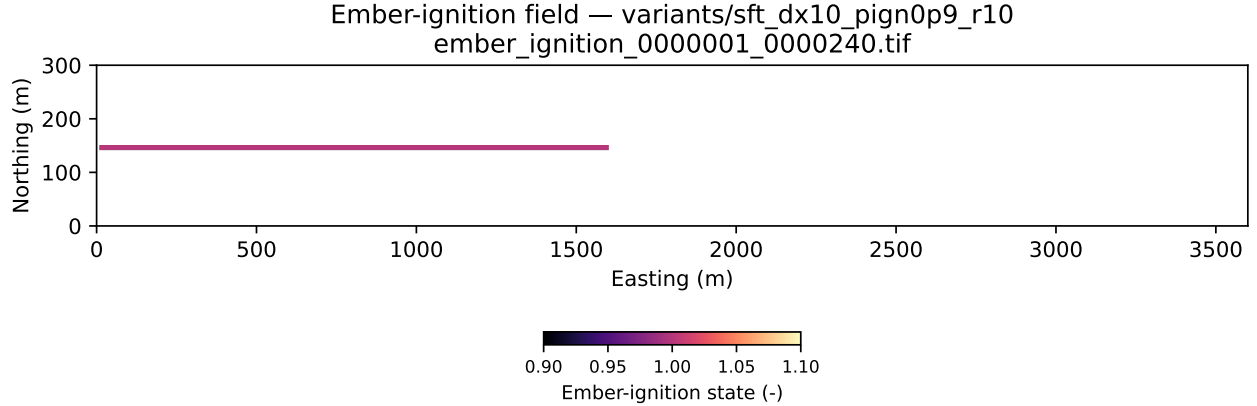


Figure 2: Whole-domain ember-ignition field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The distribution-level ignition-delay check is presented first because it compares the sampled stochastic variable directly with its prescribed model quantile. Figure 3 uses only delays reconstructed from actual arrival-time, ember-ignition, dump-time, and transient-flux outputs.

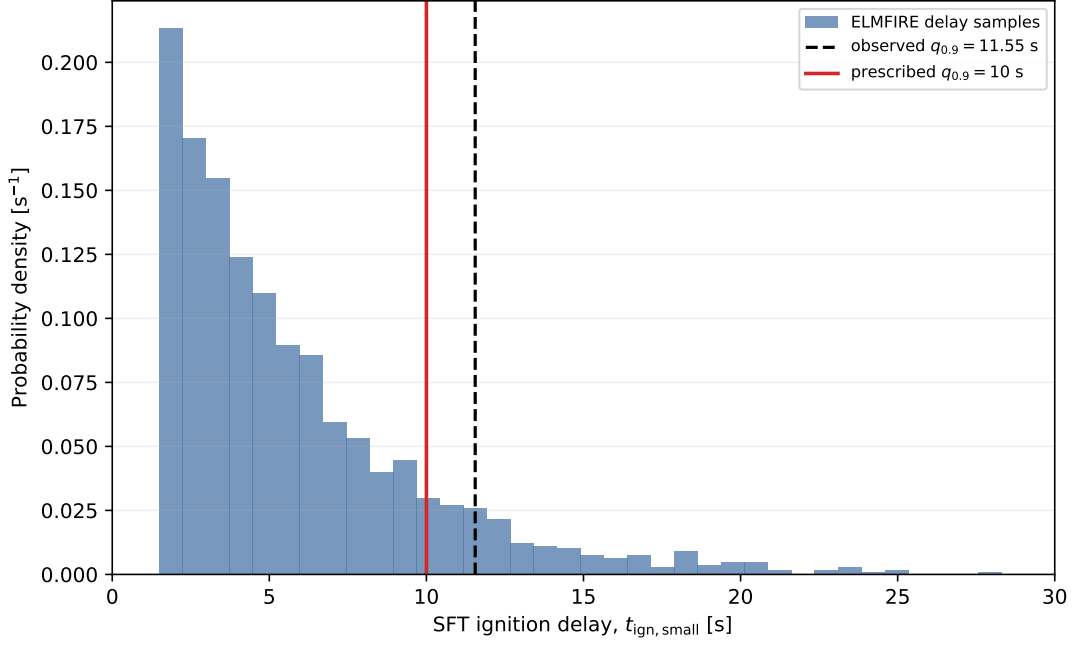


Figure 3: Probability density of sampled SFT ignition delays. The solid red line is the prescribed 90th percentile and the dashed black line is the observed pooled 90th percentile.

Figure 4 presents the second reference observable: the ensemble-mean normalized leading-edge ROS response to ignition probability and spatial resolution.

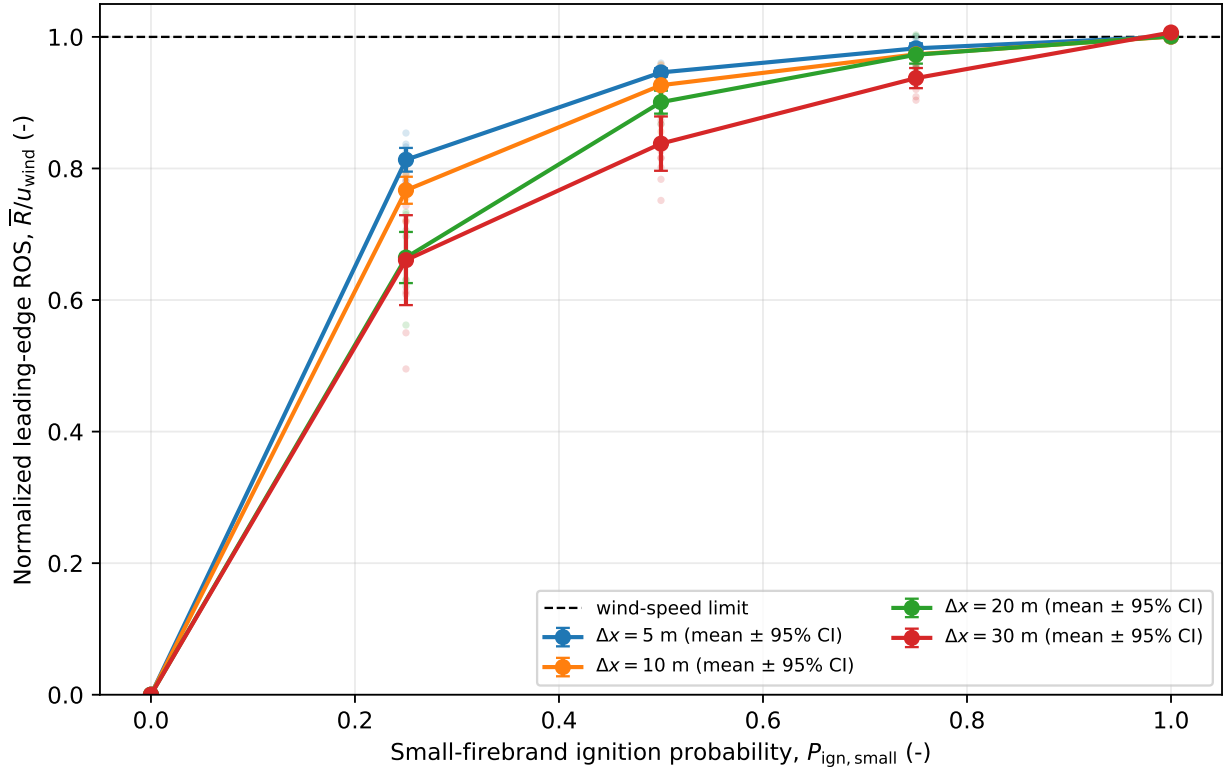


Figure 4: Ensemble-mean normalized leading-edge ROS versus $P_{\text{ign,small}}$. Error bars are 95% confidence intervals and faint points are independently seeded ELMFIRE members.

Figure 5 is an additional sample-size diagnostic. It shows how the pooled delay quantile and its bootstrap uncertainty change as independent members are added.

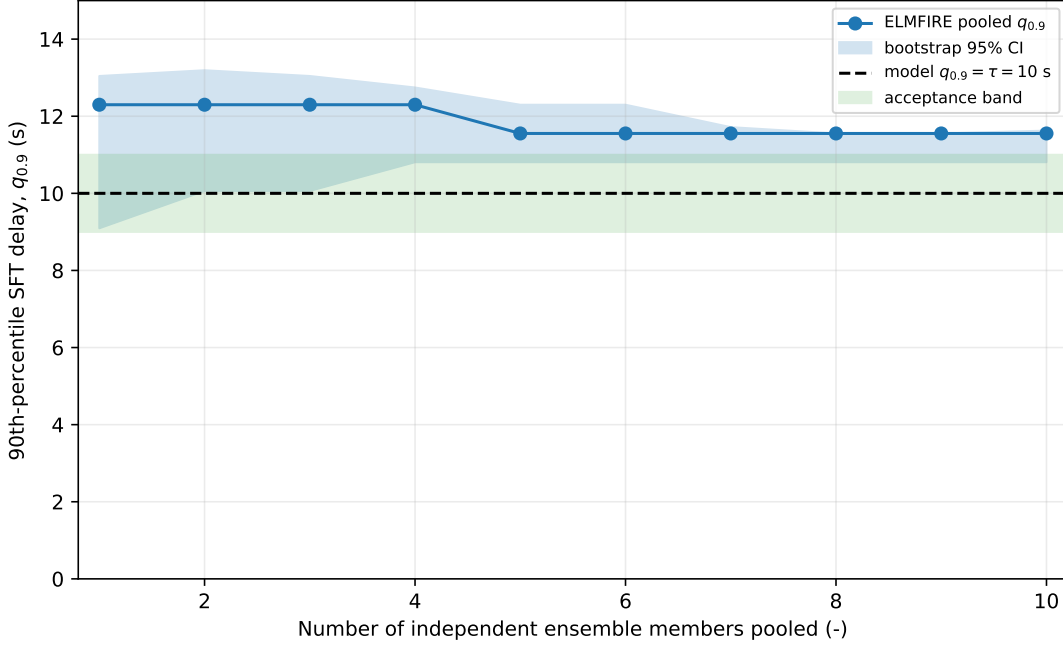


Figure 5: Cumulative convergence of the pooled ELMFIRE 90th-percentile SFT delay. The dashed line is the model expectation, blue shading is the bootstrap 95% interval, and the green band is the ± 1 s acceptance interval.

7 Calculated metrics and verification decision

Metric	Requirement	Calculated
Response members	200 usable	200/200
Probability response	monotone within 95% uncertainty	True
Maximum resolution-mean spread	≤ 0.10	0.1526
Minimum $P = 1$ mean	≥ 0.90	1.0004
SFT members	10 usable	10/10
SFT delay samples	≥ 1000	1440
SFT 90th percentile	10 ± 1 s	11.5528 s
SFT quantile absolute error	≤ 1 s	1.5528 s
SFT bootstrap half-width	≤ 1 s	0.4099 s
SFT component	all delay checks pass	False
Overall status	all components pass	fail

Table 3: Metrics read directly from `outputs/metrics.json`.

The machine-readable overall decision is **False**.

8 Technical interpretation

All 200/200 response members and 10/10 dedicated delay members are usable. The maximum cross-resolution response spread is 0.1526, exceeding the 0.10 limit, although the minimum unit-probability mean passes. The 1440 admitted SFT samples give a 90th-percentile delay of 11.5528 s; its 1.5528-s absolute error exceeds the 1-s limit, while the bootstrap half-width passes. Thus the result is a genuine **fail**, not an incomplete evaluation. The generated probability, seed, timestep, and ensemble controls are internally consistent. The source updates

probability as $1 - (1 - P)^{\Delta t/\tau}$ and advances a newly deposited cell through its staged ignition state on later invocations; the observed extra-step delay and grid dependence therefore identify current source timing/discretization behavior rather than a case-configuration defect.

9 Scope and limitations

The case verifies one SIMPLE SFT time scale, one delay probability, one uniform wind and fuel, Eulerian accumulation, zero full-development delay, and the specified grid/probability sweep. Midpoint reconstruction of first deposition is interval-censored at one wind-CFL step; that resolution is recorded but is not sampling uncertainty. Cell samples within a run share a propagating fire history, so independent run count and pooled cell count are both reported. The test does not validate the empirical choice of P or τ , physical ignition thresholds, surface-fire competition, PHYSICAL ignition, Lagrangian transport, or firebrand consumption.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 08: Deposited Firebrand Consumption

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE08_FBC

This case verifies removal of deposited firebrands by the consumption model. Paired consumption-off and consumption-on runs use identical forcing, seed, source model, and resolution so their active deposited populations can be compared directly.

The test exposes non-decaying stock, negative active counts, resolution-dependent consumption, and unintended feedback on surface arrival. Success requires the specified population reduction, invariant surface arrival, and convergence toward the finest-grid spatial profile.

2 Mathematical formulation and numerical implementation

Let N'' denote active deposited firebrands per unit area. With mean particle mass $m_{fb} = 0.2$ g, the pile mass loading is

$$\psi = \frac{m_{fb}N''}{10^4} \quad [\text{g cm}^{-2}]. \quad (1)$$

The source estimates the local air velocity from the 20-ft wind using a fuel- and canopy-dependent coefficient. It then evaluates the empirical pile heat- flux parameters

$$t_{rise} = \max(-1.02v_a^2 + 9.08v_a + 35, 0), \quad (2)$$

$$HF_{rise} = \max(-0.32v_a^2 + 1.59v_a + 0.1, 0) \tanh(12\psi), \quad (3)$$

$$HF_{decay} = \max(10^{-6}, -0.05v_a^2 + 0.02v_a - 0.01) \tanh(12\psi). \quad (4)$$

After applying the source's heat-flux scaling, the lifetime is the duration for which the modeled incident heat flux exceeds $HF_{fb,live} = 10 \text{ kW m}^{-2}$:

$$t_{fb,live} = \max(t_{half,2} - t_{half,1}, 0). \quad (5)$$

The intended explicit update is

$$N''^{n+1} = N''^n - \frac{N''^n}{\max(t_{fb,live}, t_{min})} \Delta t, \quad t_{min} = 10 \text{ s}. \quad (6)$$

ELMFIRE stores pixel counts, so the implementation multiplies the areal decay rate by pixel area before subtracting from `EMBER_FLUX`. Only the consumption-on member calls this routine. Final `ember_flux` therefore represents the active deposited population. The transient `ember_flux.transient` product records new deposition during an output interval and is reset after each dump; it is not used as an active-stock surrogate. Instead, independent consumption-on runs terminated at predeclared times provide the temporal samples from their final active-stock rasters.

The current source is also audited as part of this test. It presently assigns `MIN_LIFETIME=1E9` s, whereas the documented model requires 10 s [1], as expressed in Equation 6. The verification preserves this behavior and is expected to fail the consumption-effect metric until that implementation discrepancy is corrected.

3 Derivation of expected behavior

For otherwise identical runs, let $A_{on}(\Delta x)$ and $A_{off}(\Delta x)$ be the centerline sums of active deposited particles at 260 s. The retained fraction is

$$f_{ret}(\Delta x) = \frac{A_{on}(\Delta x)}{A_{off}(\Delta x)}. \quad (7)$$

The documented result that more than 96% is consumed gives $f_{ret} \leq 0.04$ [1]. Because the delayed-ignition baseline uses long ignition delays and is surface-fire dominated, removing deposited embers should not materially change TOA. The active-particle line density is recovered from a centerline pixel count as $n'(x_i) = N_i/\Delta x$ in pcs/m.

For grid convergence, the 5, 10, and 30 m consumption-on profiles are linearly interpolated to the 1 m cell centers. Their normalized integral error is

$$E_{\Delta x} = \frac{\int |n'_{\Delta x}(x) - n'_1(x)| dx}{\int |n'_1(x)| dx}. \quad (8)$$

Consistent refinement requires $E_{30} \geq E_{10} \geq E_5$, with $E_5 \leq 0.15$.

For a cell whose calculated lifetime is $T_n = \max(t_{fb, live, n}, 10)$, Equation 6 gives the explicit retained fraction

$$\frac{N''_{n+1}}{N''_n} = 1 - \frac{\Delta t_n}{T_n}, \quad \frac{N''_m}{N''_0} = \prod_{n=0}^{m-1} \left(1 - \frac{\Delta t_n}{T_n}\right).$$

This is the reference update that the consumption-on run must execute; the consumption-off reference has every factor equal to one. The published statement that more than 96 percent is consumed is converted without further fitting to

$$1 - f_{ret} \geq 0.96 \iff f_{ret} = \frac{A_{on}}{A_{off}} \leq 0.04.$$

The nominal surface-CFL upper bounds are $0.5\Delta x/0.71$. To keep every requested 10-s-multiple observation time exact, preprocessing selects the largest divisor no greater than that bound, preferring whole seconds when feasible:

$$\Delta t = \{0.666667, 2, 5, 10\} \text{ s}$$

for $\Delta x = \{1, 5, 10, 30\}$ m. Thus all 260-s paired stops and 50–500-s history stops are exact timestep multiples, and every actual surface CFL remains below 0.5. The intended lifetime is recalculated locally from v_a and $\psi = 0.2N''/10^4$, so no constant lifetime is invented for the report.

For the paired control, the exact expected TOA difference is zero and the active-stock lower bound is zero particles. The convergence reference is the generated 1-m profile: $E_1 = 0$, with expected refinement ordering $E_{30} \geq E_{10} \geq E_5 \geq E_1$. The decision substitutes $E_5 \leq 0.15$, paired TOA difference ≤ 10 s, and minimum active count $\geq -10^{-6}$, in addition to the 0.04 retained-fraction limit.

4 Verification metrics and acceptance criteria

All 48 manifest variants are required: eight paired 260-s metric runs and 40 consumption-on stop-time runs (four resolutions at ten sample times). Exactly one final active-flux and TOA raster per variant is required. The two-cell numerical halo is excluded. The components are:

1. $\max_{\Delta x} f_{ret} \leq 0.04$;
2. paired centerline TOA difference ≤ 10 s at every resolution;
3. every active count $\geq -10^{-6}$ particles;
4. $E_5 \leq 0.15$; and
5. $E_{30} \geq E_{10} \geq E_5$.

The overall decision is the conjunction of completeness and all five components. Missing, duplicated, dimensionally stale, or unreadable outputs produce an insufficient-output result rather than a pass.

4.1 Justification of metric quantification

The retained-fraction ceiling of 0.04 is not tuned to these outputs; it is the complement of the reported result that more than 96% of deposited firebrands are consumed by the evaluation time [1]. The paired TOA allowance of 10 s is a mechanism-isolation check at the scale of one output-time increment: consumption may alter the active stock but must not materially change the surface-dominated arrival field. The lower bound of -10^{-6} particle is only a floating-point allowance around the exact nonnegativity invariant and cannot conceal a physically meaningful negative population.

The convergence ordering is more informative than agreement at one grid. It requires every refinement to move toward the 1 m profile. The $E_5 \leq 0.15$ limit allows cell averaging of a steep transported-stock profile while requiring at least 85% agreement in normalized integral content; errors of that size remain clearly separated from the coarse-grid shape changes the sweep is intended to expose. All paired and stop-time runs are mandatory because omitting one would break either the on/off control or the temporal consumption history.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Eight paired metric runs plus 40 active-stock stop-time runs, with final-raster shape/nodata validation and halo exclusion.	48 of 48 variants
Retained active fraction	For each resolution, consumption-on active population divided by its consumption-off pair at 260 s.	Maximum ≤ 0.04
Surface-TOA invariance	Maximum paired centreline TOA difference on common finite physical cells.	≤ 10 s at every resolution
Active-count nonnegativity	Minimum value over every active-firebrand raster.	$\geq -10^{-6}$ particles
Fine-grid profile convergence	Normalized L_1 error of the 5-m consumption-on profile relative to the 1-m profile.	$E_5 \leq 0.15$
Ordered convergence	Normalized profile errors for 5, 10, and 30-m grids.	$E_{30} \geq E_{10} \geq E_5$
Overall decision	Conjunction of completeness and all five physical/numerical checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `dx10_consumption_on`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

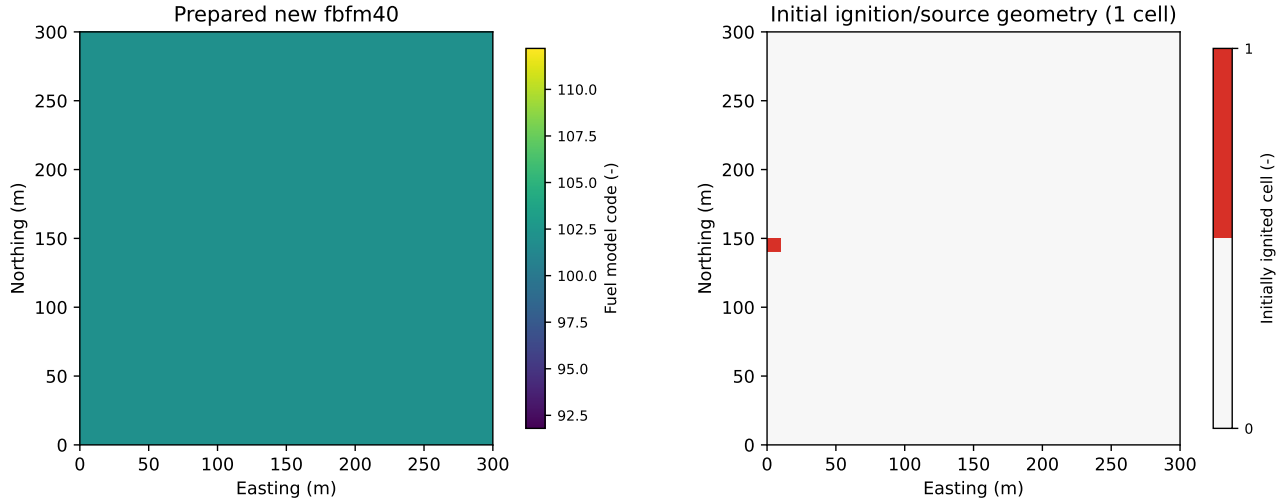


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Parameter	Configured value	Role
Physical domain	300 × 300 m	identical extent at all resolutions
Cell sizes	1, 5, 10, 30 m	active-stock convergence sweep
Numerical halo	2 cells per side	excluded from all metrics
Metric evaluation time	260 s	paired consumption-on/off comparison
History sample times	50–500 s in 50-s increments	final active-stock samples for the temporal curve
Temporal CFL	at most 0.5 based on 0.71 m/s	conservative exact-duration divisors
Wind	15 mph from 270 degrees	uniform downwind forcing
Generation	PER-MW, 33.3/ Δy pcs/s/MW	delayed-ignition baseline vegetation source
Transport/accumulation	EMPIRICAL/EULERIAN	active-stock implementation
Ignition	PHYSICAL; 300 s development	prevents spotting-dominated spread
Consumption	paired off/on	isolates finite lifetime
Critical live heat flux	10 kW/m ²	hard-coded lifetime threshold
Crosswind distribution	disabled	one-dimensional profile diagnostic

Table 2: Configuration used by every paired variant.

Before inspecting results, consumption-on curves should be far below the paired no-consumption curve upstream of 200 m, retained fractions should lie below the 4% line, and progressively finer active-stock profiles should agree. TOA should be nearly unchanged between paired runs.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from `dx10_consumption_on`. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. Counts are divided by physical cell area, giving pcs m⁻², so the retained and consumed portions of the deposit field can be interpreted spatially.

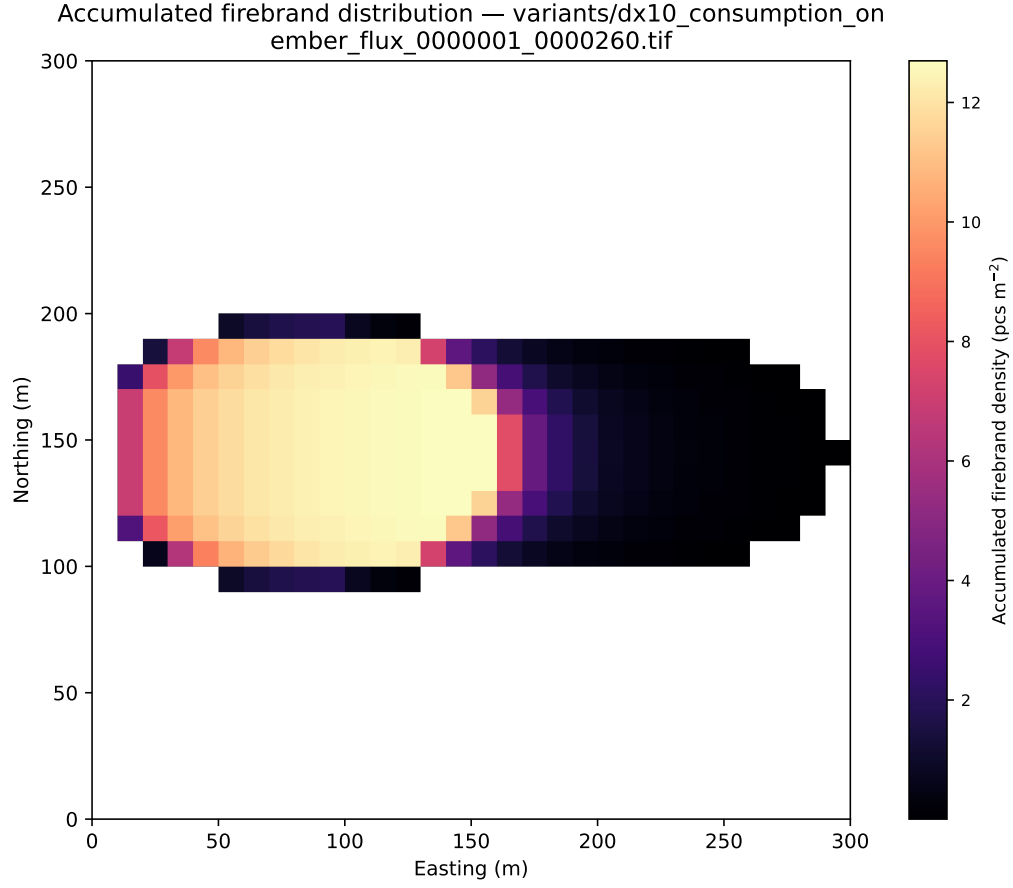


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

All 48 variants were rerun. The temporal curves use ten independent consumption-on stop-time runs per resolution, from 50 to 500 s in 50 s increments. Each plotted sample is therefore a final ELMFIRE active-stock field, not an interval-deposition raster. Figure 3 first presents the four physical observables for the finest consumption-on variant: active firebrands, arrival time, leading-edge position, and leading-edge ROS.

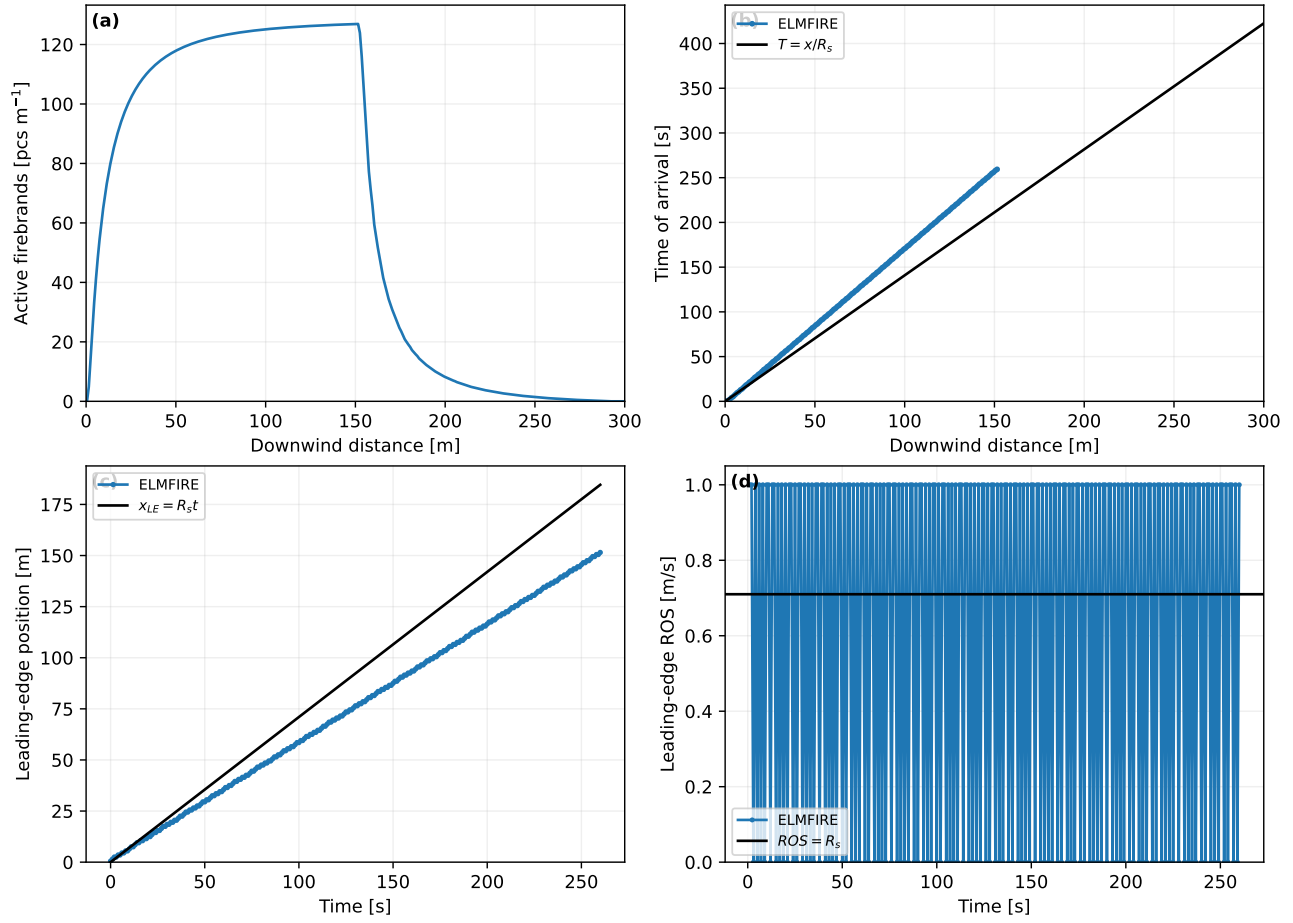


Figure 3: Consumption-on state at 1 m resolution. The top row gives the active firebrand and arrival-time profiles; the bottom row gives the reconstructed leading-edge trajectory and ROS with surface-spread references.

Figure 4 reproduces the temporal and spatial resolution views of the active population. The left panel samples the physical location nearest 200 m; the right panel selects the dump nearest 250 s.

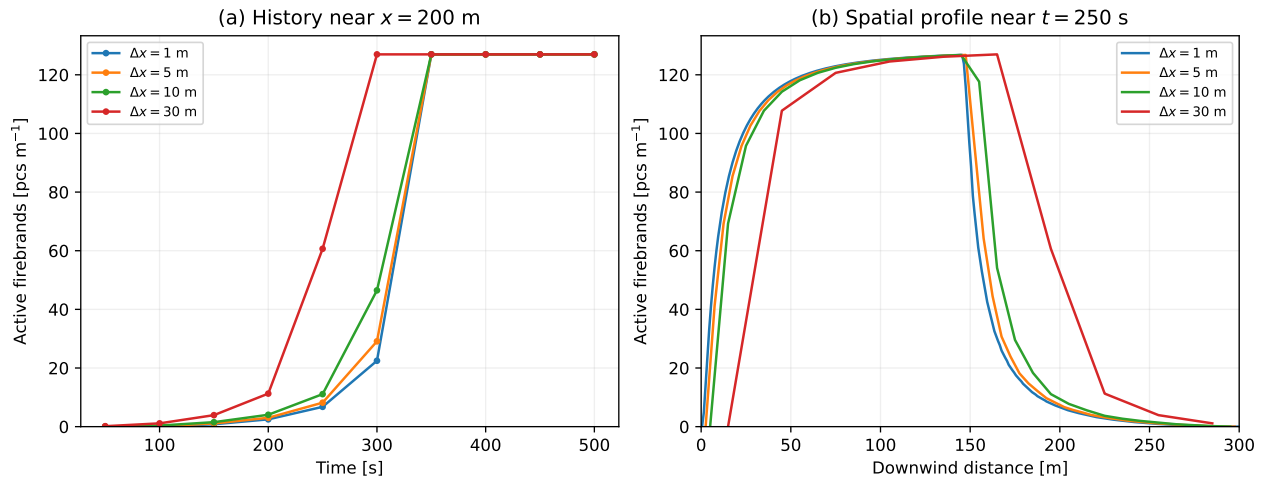


Figure 4: Consumption-model resolution sweep. Left: active-firebrand history near 200 m. Right: active-firebrand spatial profile near 250 s. Every curve is read from a predeclared stop-time run's final ELMFIRE raster.

Figure 5 is the supplemental paired diagnostic that makes the consumption-on/off effect and the 4% retention

criterion explicit.

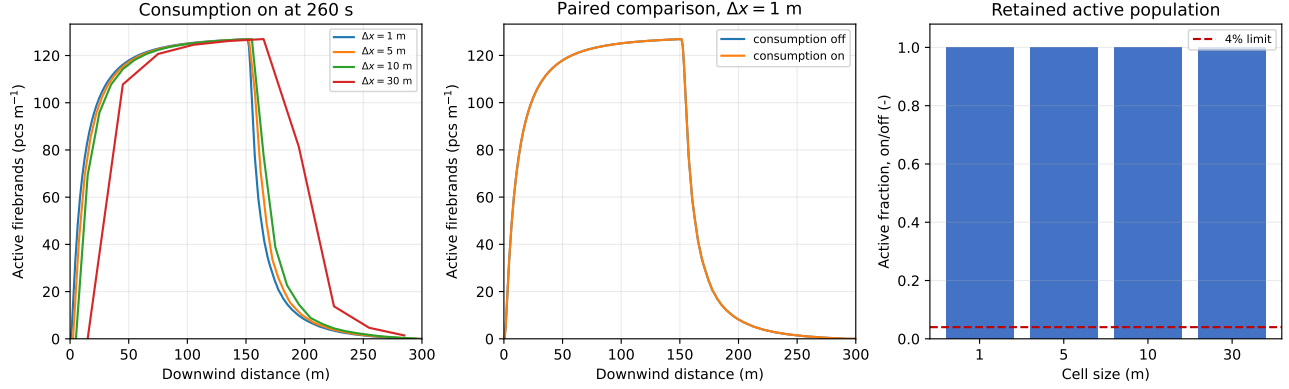


Figure 5: Paired consumption diagnostic. Left: consumption-on profiles by resolution. Center: paired 1 m profiles. Right: retained active fraction with the 4% acceptance limit.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
Output completeness	48 variants	48/48	–
Maximum retained fraction	≤ 0.04	1	false
Maximum paired TOA difference	≤ 10 s	0	true
Minimum active count	$\geq -10^{-6}$	0	true
5 m normalized profile error	≤ 0.15	0.036343	true
Ordered profile convergence	required	–	true
Overall decision	all pass	false	fail

Table 3: Metrics read directly from `outputs/metrics.json`.

8 Technical interpretation

All four measured retained fractions are exactly 1.0, so enabling consumption does not change the active deposited population. The maximum paired TOA difference is 0 s, while the normalized profile errors are 0.0363, 0.0778, and 0.3511 at 5, 10, and 30 m, respectively. The source-normalized profiles therefore converge monotonically and the paired dynamics remain controlled, but the primary consumption metric fails decisively. Given the audited 10^9 s minimum lifetime, this is a source-code implementation defect rather than insufficient run duration. A TOA-only result cannot establish consumption behavior, which is why TOA appears here solely as a controlled invariance check.

9 Scope and limitations

This test verifies active-stock decay in one flat, uniform delayed-ignition baseline-like configuration. It does not validate the empirical heat-flux correlation, pile clustering, subgrid redistribution, or the 10 s lifetime against experiments. The 50 s sequence of independently stopped runs provides the active-stock history near 200 m and the spatial profile near 250 s directly from ELMFIRE rasters. The experiment still does not resolve within-cell firebrand clustering or validate the lifetime correlation against measurements.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 09: Mixed-Mode Ignition-Delay Effects

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE09_MMD

This case verifies how firebrand ignition delay alters spread when surface propagation and spotting operate together. The **surface_dominated** variant delays firebrand development until the surface front controls advance; **mixed_mode** permits sufficiently distant deposits to develop ahead of that front.

The paired design exposes incorrect delay application, a missing transition between propagation modes, or an inconsistent surface-rate reference. Success requires surface-controlled behavior in the first variant and quantified earlier arrival, spot ignition, and leading-edge gain in the second.

2 Mathematical formulation and numerical implementation

2.1 Surface-fire and firebrand time scales

The verification configuration is a flat, uniform FBFM 102 grassland with wind speed $U_w = 6.71 \text{ m s}^{-1}$. The surface head-fire spread rate U_s is read from the surface-dominated variant **VS** raster produced by the Rothermel model in the tested ELMFIRE executable. For the reported run, $U_s = 0.577 \text{ m s}^{-1}$. In the absence of a successful firebrand-induced ignition, the exact one-dimensional solution is

$$t_s(x) = \frac{x}{U_s}, \quad x_{\text{LE},s}(t) = U_s t, \quad \overline{\text{ROS}}_{\text{LE},s} = U_s. \quad (1)$$

For a firebrand emitted at x_s and transported a downwind distance X , the earliest time at which it can produce a fully developed fire is

$$t_{\text{spot}}(x_s, X) = \frac{x_s}{U_s} + \frac{X}{U_w} + t_{\text{ign,small}} + t_{\text{ign,large}}. \quad (2)$$

The surface front reaches the same target location at $t_s(x_s + X) = (x_s + X)/U_s$. With immediate smoldering-to-flaming transition, a spot fire can lead the surface front only when

$$X \left(\frac{1}{U_s} - \frac{1}{U_w} \right) > t_{\text{ign,large}}. \quad (3)$$

2.2 ELMFIRE implementation exercised

The current implementation uses the non-superseded spotting path with these explicit selectors:

- `GENERATION_MODEL='PER-MW'`;
- `SPOTTING_DISTANCE_MODEL='EMPIRICAL'`; and
- `ACCUMULATION_MODEL='EULERIAN'`.

For each burning cell, the source routine evaluates

$$\dot{N}_{\text{cell}} = \frac{I_f \Delta y}{1000} GR^*, \quad (4)$$

where I_f is fireline intensity in kW m^{-1} , Δy is the cell width, and $GR^* = 33.3 \text{ s}^{-1} \text{ MW}^{-1}$. The factor $I_f \Delta y / 1000$ is the heat-release rate of the cell in MW. The Eulerian transport routine deposits these particles according to the configured Sardoy-type distribution ($\mu = 2.18$, $\sigma = 1.23$), truncated at an effective maximum distance of approximately 150 m.

The ignition routine first converts the deposited number to a number per unit area. In the surface-dominated variant, the physical ignition model (`IGNITION_MODEL`; value `PHYSICAL`) applies the wildland mass-load criterion before sampling SFT, with $\vartheta_{\text{ign,small}} = 42 \text{ s}$ and $t_{\text{ign,large}} = 300 \text{ s}$. For this parameter set, the documented limiting behavior predicts that deposited mass remains below the physical threshold, so the SFT probability is zero [1]. In the mixed-mode variant, the simple ignition model (`IGNITION_MODEL`; value `SIMPLE`) prescribes immediate SFT and a 150 s development delay. When the development delay expires, the level-set state, arrival-time field, and ember-ignition diagnostic are updated by the same firebrand ignition event.

3 Derivation of expected behavior

At the maximum transport distance $X_{\text{max}} = 150 \text{ m}$, flight takes

$$t_f = \frac{150}{6.71} = 22.4 \text{ s}, \quad t_s(150) = \frac{150}{0.577} = 260.0 \text{ s}.$$

For surface-dominated variant, even the deliberately optimistic assumption of immediate SFT would give $t_f + t_{\text{ign,large}} = 322.4 \text{ s}$, later than the surface arrival. The expected result is therefore Equation 1, with no cells attributed to firebrand ignition.

For mixed-mode variant, the earliest full spot fire at X_{max} occurs at $22.4 + 150 = 172.4 \text{ s}$. Moreover, the left side of Equation 3 is 237.6 s, which exceeds the 150 s development delay. Long-distance deposits can therefore mature ahead of the primary front. The expected signature is an acceleration near 170 s, at least one firebrand-ignited cell, a leading edge ahead of $U_s t$, and a mean spread rate greater than U_s over the common analysis interval. These are the qualitative features expected from mixed-mode spread.

The lead condition can also be solved for the minimum useful spotting distance:

$$X_{\text{crit}} = \frac{t_{\text{ign,large}}}{1/U_s - 1/U_w}, \quad \frac{1}{0.577} - \frac{1}{6.71} = 1.584071 \text{ s m}^{-1}.$$

For the surface-dominated 300-s delay, $X_{\text{crit}} = 300/1.584071 = 189.39 \text{ m}$, which exceeds the configured $X_{\text{max}} = 150 \text{ m}$; even the longest deposit cannot lead. For the mixed 150-s delay, $X_{\text{crit}} = 94.69 \text{ m} < 150 \text{ m}$, so the tail $94.69 < X \leq 150 \text{ m}$ can mature ahead of the surface front. At $X = 150 \text{ m}$,

$$t_f = 150/6.71 = 22.3547 \text{ s}, \quad t_s = 150/0.577 = 259.9653 \text{ s}, \quad t_s - t_f = 237.6106 \text{ s}.$$

Thus the mixed earliest full-spot scale is $22.3547 + 150 = 172.3547 \text{ s}$, whereas the surface-dominated optimistic scale is $22.3547 + 300 = 322.3547 \text{ s} > t_s$. These substitutions yield the explicit reference signatures used by the metrics: zero surface-dominated ember ignitions, positive mixed-mode ember ignitions, mixed $ROS/U_s \geq 1.10$, at least 20 m edge and paired-TOA advantages, and onset in the 120–220 s interval around 172.35 s.

4 Verification metrics and acceptance criteria

The postprocessor excludes the two-cell halo and operates on the centre-row TOA and spread-rate profiles. It converts the positive median Rothermel VS value from ft/min to m/s and uses it as U_s throughout the reference calculations. It fits $t = a + x/U$ over $50 \text{ m} \leq x \leq 280 \text{ m}$ using at least 12 cells. The following independently informative conditions must all pass:

1. surface-dominated variant mean TOA relative error against x/U_s is at most 0.10.
2. surface-dominated variant fitted ROS relative error against U_s is at most 0.10.
3. surface-dominated variant contains no firebrand-ignited physical cells.
4. mixed-mode variant contains at least one firebrand-ignited physical cell.
5. mixed-mode variant fitted ROS is at least $1.10U_s$.

6. At 250 s, its leading edge is at least 20 m ahead of $U_s t$.
7. Its first one-cell leading-edge gain occurs from 120 to 220 s, bracketing the derived 172.4 s scale.
8. Its median TOA is at least 20 s earlier than surface-dominated variant over the shared fitting interval.

Missing outputs do not pass by default. Both variants must provide final TOA, accumulated-firebrand, and ember-ignition GeoTIFFs before any case-level pass decision is possible.

4.1 Justification of metric quantification

The two 10% surface-baseline limits allow cell-centred TOA and regression quantization while requiring agreement with the run-derived Rothermel speed over a 230 m fitting interval, more than twenty grid cells. The zero spot count in the surface-dominated variant and the positive count in the mixed variant are exact mechanism checks and therefore have no numerical tolerance. For the mixed mode, $ROS/U_s \geq 1.10$ requires a 10% effect beyond the baseline fitting band, while the 20 m edge and paired-TOA leads equal two spatial cells and are too large to arise from one-cell front classification alone.

The 120–220 s onset window brackets the independently derived 172.4 s earliest full-spot scale by approximately ± 50 s. That width accommodates stochastic deposition, cell crossing, and the fact that the metric detects the first full-cell gain rather than a continuous event, but still rejects acceleration unrelated to the predicted delay regime. The eight behavior checks are conjunctive because a faster fitted front without ember attribution, timing consistency, and a paired TOA lead would not verify firebrand-delay coupling.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Final TOA, accumulated-firebrand, and ember-ignition rasters for both current variants; halo and nodata excluded.	2 of 2 variants
Surface-dominated TOA error	Mean relative error against x/U_s on 50–280 m; at least 12 cells.	≤ 0.10
Surface-dominated ROS error	Relative error of fitted ROS against run-derived Rothermel U_s .	≤ 0.10
Surface-dominated spot count	Number of ember-ignited physical cells.	0
Mixed-mode spot count	Number of ember-ignited physical cells.	≥ 1
Mixed-mode ROS ratio	Fitted mixed-mode ROS divided by U_s .	≥ 1.10
Mixed-mode edge gain	Leading-edge distance ahead of $U_s t$ at 250 s.	≥ 20 m
Acceleration onset	First time the leading edge exceeds the surface reference by one cell.	120–220 s
Paired TOA lead	Median surface-dominated minus mixed-mode TOA over common fitting cells.	≥ 20 s
Overall decision	Conjunction of completeness and all eight behavior checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `mixed_mode`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

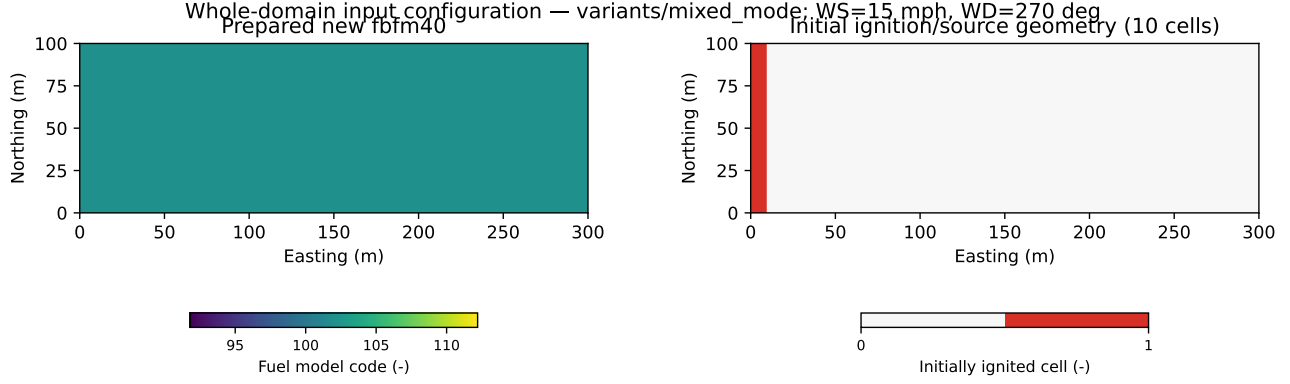


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Parameters common to both verification variants.

Parameter	Value	Verification role
Physical domain	300×100 m	one-dimensional centre transect
Raster domain	34×14 cells	includes two-cell halo
Cell size	10 m	controlled baseline resolution
Time step / wind CFL	0.742574 s / 0.4979	common exact-duration divisor
Fuel and terrain	FBFM 102, flat	homogeneous surface spread
Wind	6.71 m s^{-1} toward $+x$	firebrand transport
Ignition geometry	full crosswind line at $x = 5$ m	1-D behavior
Generation	PER-MW, $GR^* = 33.3$	heat-release-scaled source
Transport / accumulation	empirical / Eulerian	reference transport path
Surface fire	enabled	general coupled fire spread

Table 3: Ignition parameters changed by the paired experiment.

Variant	Ignition model	$t_{\text{ign,small}}$	$t_{\text{ign,large}}$	Prediction
surface-dominated variant	PHYSICAL	42 s scale	300 s	surface dominated
mixed-mode variant	SIMPLE	0 s	150 s	mixed-mode acceleration

5.1 Preprocessing, execution, and postprocessing

`scripts/preprocess.py` creates two independent directories beneath `variants/`. It writes flat, co-registered GeoTIFFs; assigns FBFM 102 and constant wind; places a line ignition across every physical crosswind row; and writes explicit model selectors and ignition parameters into each local `elmfire.data.in`. The halo is present numerically but is excluded from all reported quantities. The nominal 0.5-wind-CFL step is 0.745645 s. Because the controlled stops are 300 and 450 s, their greatest common duration is 150 s; preprocessing selects $N = \lceil 150/0.745645 \rceil = 202$ and the common $\Delta t = 150/202 = 0.742574$ s. Both stops are therefore exact timestep multiples and the actual wind CFL remains below 0.5.

`run_case.sh` uses the executable configured in `ELMFIRE_BIN`, prepares both variants, runs them sequentially in isolated directories, records their logs, postprocesses final rasters, writes `outputs/metrics.json`, and compiles the report.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from `mixed_mode`. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. The domain map complements the spread-rate summaries by showing the combined surface-fire and delayed-firebrand arrival pattern.

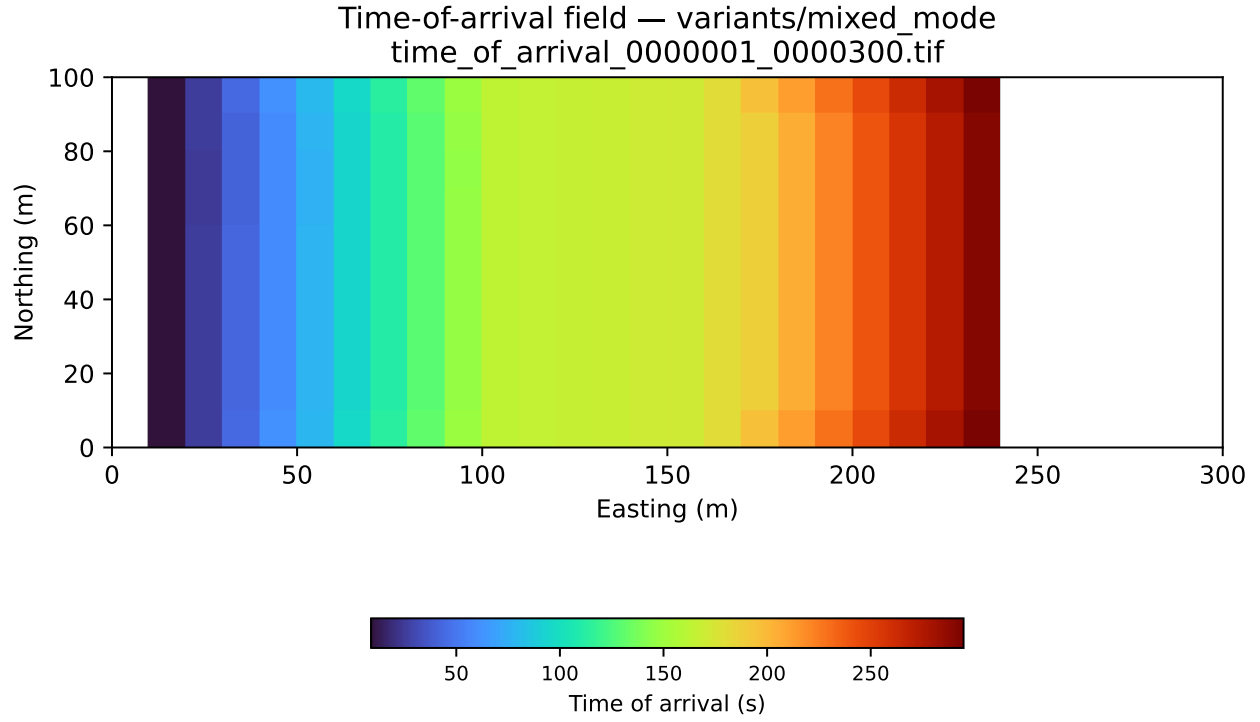


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figures 3 and 4 first present each ignition regime independently using the same four observables: accumulated firebrands, arrival time, leading-edge position, and leading-edge ROS. Colored curves are actual ELMFIRE outputs; black curves are the Rothermel surface-spread references.

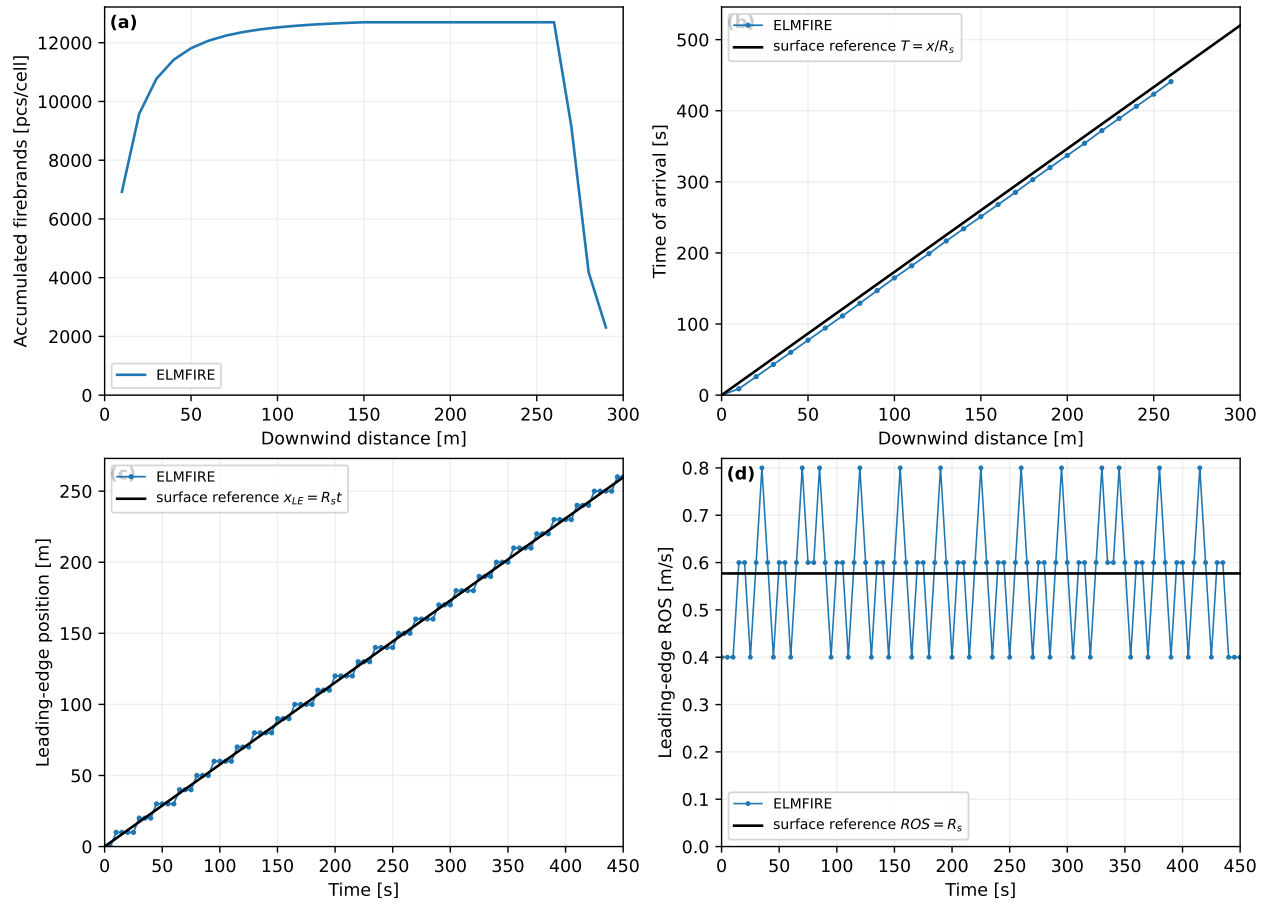


Figure 3: Long-delay surface-dominated variant. The leading edge remains close to the surface-spread characteristic despite continued firebrand deposition.

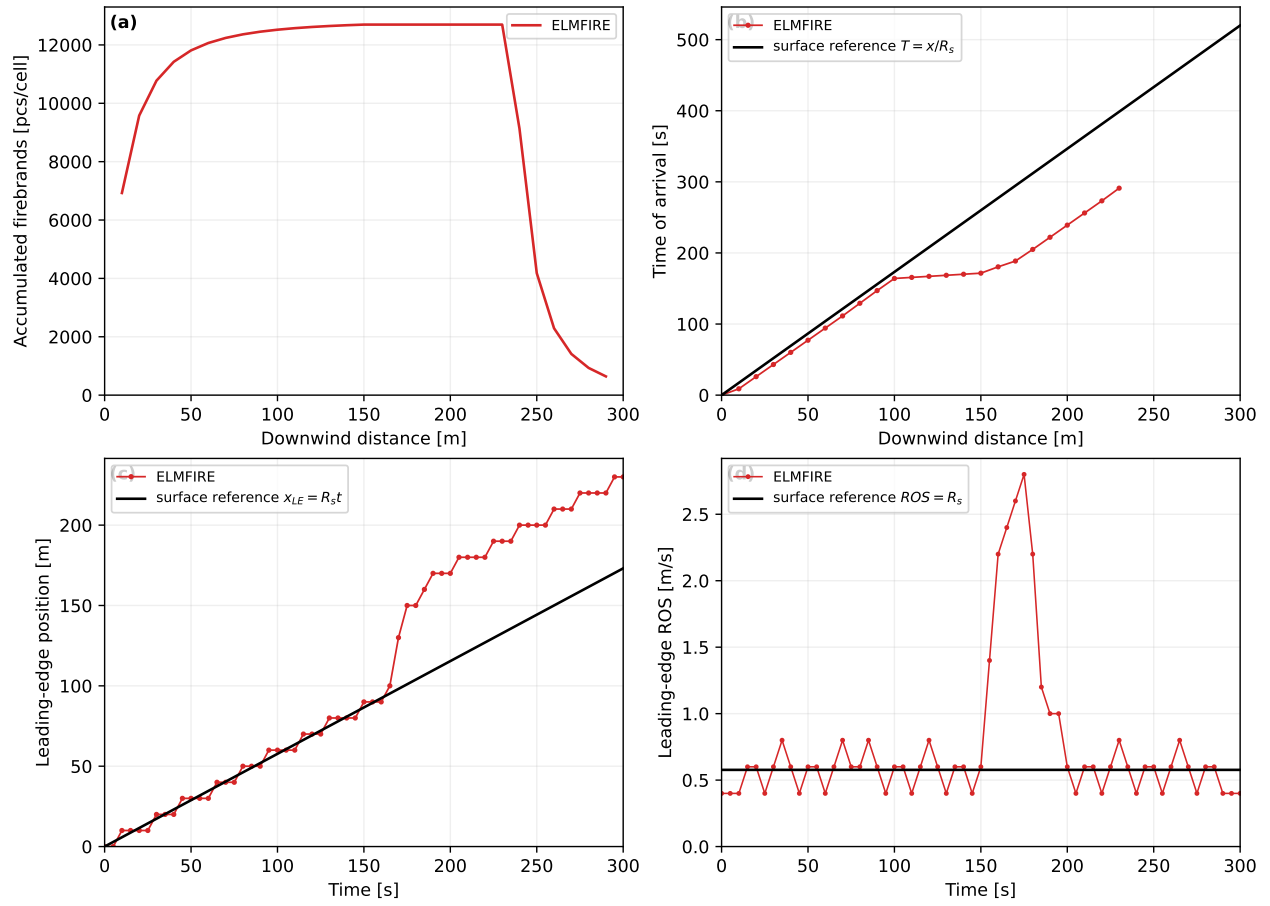


Figure 4: Finite-delay mixed-mode variant. Departure from the surface-only arrival, trajectory, and ROS references identifies firebrand-induced advance.

Figure 5 overlays the two variants as a supplemental controlled comparison, making the timing and magnitude of their divergence easier to identify.

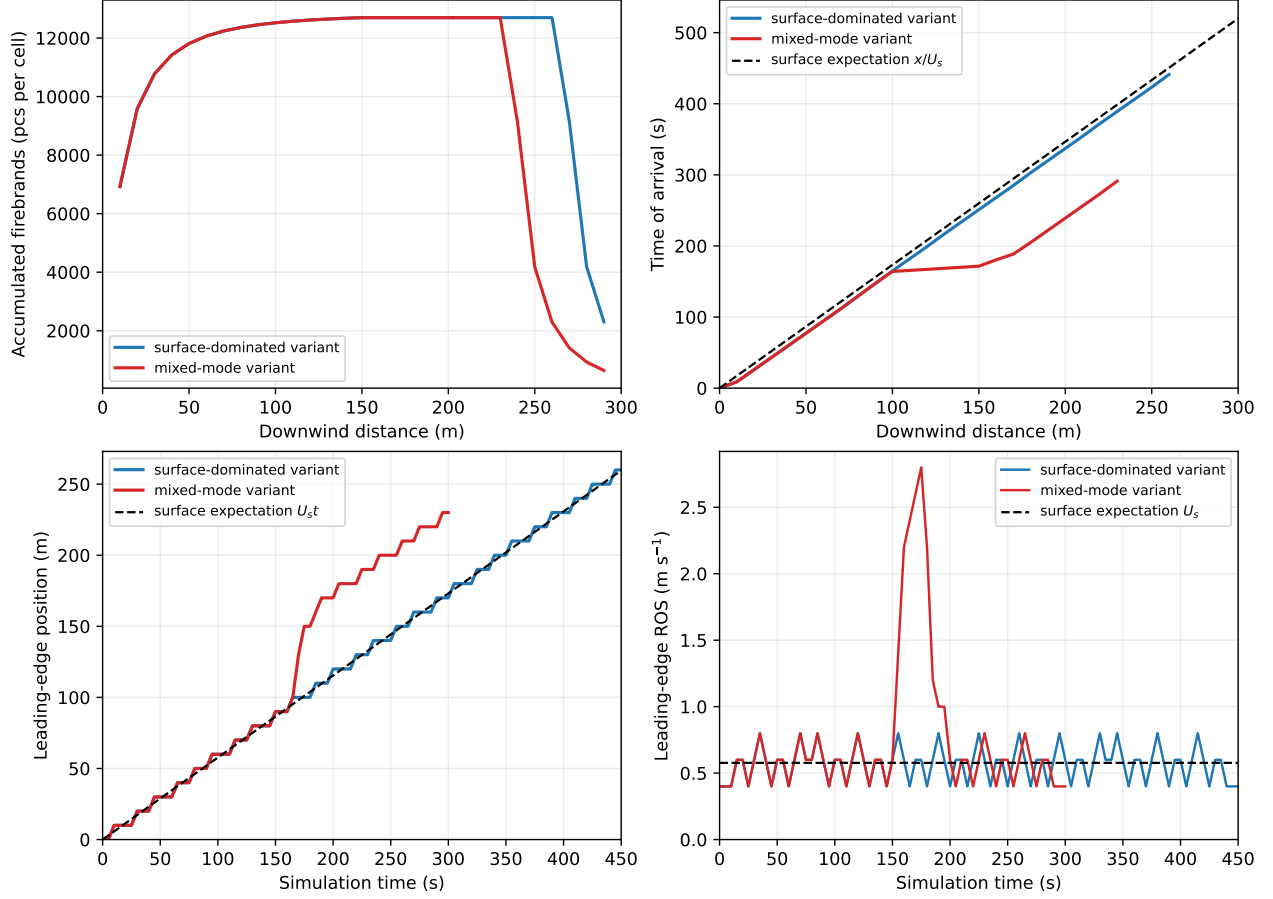


Figure 5: Paired ELMFIRE comparison of the long-delay surface-dominated and finite-delay mixed-mode regimes. Accumulation is crosswind averaged; the other panels use the centre-row arrival-time field.

7 Calculated metrics and verification decision

Table 4 reads the calculated entries directly from `outputs/metrics.json`. An empty simulation result is represented as N/A with NOT EVALUABLE, never as an empty table cell or an implicit pass.

Table 4: Acceptance metrics calculated from the paired ELMFIRE outputs.

Metric	Acceptance limit	Calculated	Status
Output completeness	2 variants	2/2	pass
surface-dominated variant mean TOA error	≤ 0.10	0.043	PASS
surface-dominated variant mean ROS (m s^{-1})	$0.577 \pm 10\%$	0.578	PASS
surface-dominated variant ember-ignited cells	≤ 0	0	PASS
mixed-mode variant ROS / U_s	≥ 1.10	1.743	PASS
mixed-mode variant ember-ignited cells	≥ 1	64	PASS
mixed-mode variant edge gain at 250 s (m)	≥ 20	55.746	PASS
mixed-mode variant acceleration onset (s)	120–220	170.0	PASS
Median mixed-mode variant TOA lead (s)	≥ 20	63.861	PASS
Overall decision	all components pass	PASS	pass

8 Technical interpretation

The current paired execution passes every component. The surface-dominated mean ROS is 0.578 m/s with no ember-ignited cells. The mixed-mode mean ROS is 1.006 m/s (1.743 times the surface reference), with 64 ember-ignited cells, a 55.746-m leading-edge gain, acceleration onset at 170.0 s, and a median TOA lead of 63.861 s.

A pass demonstrates two complementary properties: delayed firebrand states do not spuriously outrun the surface solver in the surface-dominated limiting case, and reducing the delay activates a resolvable acceleration in the mixed-mode variant. If surface-dominated variant fails alone, inspect the physical mass threshold, development-state timing, and arrival-time update. If mixed-mode variant fails alone, inspect the simple ignition probability, persistence of deposited ember states across level-set steps, delayed transition to a full ignition, and the coupling that writes that ignition to the level set.

This is code verification of the controlled formulation, not field validation. The ELMFIRE experiment is two-dimensional but deliberately uses a uniform crosswind line so the centre transect approximates the corresponding one-dimensional calculation. The mixed-mode thresholds diagnose the expected mixed regime rather than an unavailable closed-form mixed-mode solution. Stochastic ignition means the exact onset may vary, which is why its acceptance window brackets rather than equals 172.4 s.

To reproduce the result from this directory, export an absolute `ELMFIRE_BIN` and run `./run_case.sh`. The current processing note is: All metrics were calculated from the two actual ELMFIRE variant outputs; the surface reference is the Rothermel ROS read from the surface-dominated variant VS raster.

9 Scope and limitations

This paired test isolates the interaction between surface propagation and firebrand ignition delay in a uniform domain. It verifies implementation behavior and timing attribution; it does not validate either spread mechanism against field observations.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 10: WUI Structural Resolution Sensitivity

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE10_WSR

This case verifies sensitivity of firebrand-driven WUI spread to structural spatial representation. Five variants cover the baseline grid, native fine and coarse grids, and aligned and misaligned integrated-structure representations under one physical geometry and source definition.

The test quantifies native cell-scale distortion and determines whether integrated representation restores structure-level accumulation, ignition chronology, and spread-rate consistency. Capability-gated variants remain explicit and are not replaced by a physically different native-cell calculation.

2 Mathematical formulation and numerical implementation

2.1 Heat release and generation

For a structure ignited at t_i , the design heat-release-rate per unit area is

$$q''(\tau) = \begin{cases} q''_{\max} \tau / 300, & 0 < \tau < 300, \\ q''_{\max}, & 300 \leq \tau \leq 3900, \\ q''_{\max} (4200 - \tau) / 300, & 3900 < \tau < 4200, \\ 0, & \text{otherwise,} \end{cases} \quad \tau = t - t_i, \quad q''_{\max} = 400 \text{ kW m}^{-2}.$$

For footprint A_s , $\text{HRR}_i = q'' A_s$, and the emission rate is

$$\dot{N}_i(t) = G'_R \text{HRR}_i(t), \quad G'_R = 10 \text{ pcs MW}^{-1} \text{ s}^{-1}.$$

The configuration therefore uses per-MW generation; per-area generation would not preserve heat-release-scaled production for this design fire.

2.2 Himoto transport and analytical load

The downwind distance is lognormal, with physical-space moments

$$\mu_x / D = 0.47(B^*)^{2/3}, \quad \sigma_x / D = 0.88(B^*)^{1/3},$$
$$B^* = \frac{U}{\sqrt{gD}} \left(\frac{\rho_p}{\rho_\infty} \right)^{-3/4} \left(\frac{d_p}{D} \right)^{-3/4} \left(\frac{\text{HRR}}{\rho_\infty c_p T_\infty g^{1/2} D^{5/2}} \right)^{1/2}.$$

Here $D = 10 \text{ m}$, $\rho_p = 100 \text{ kg m}^{-3}$, and $d_p = 5 \text{ mm}$, matching the reference formulation and current source. The postprocessor independently evaluates the analytical deposited-load convolution

$$N_k = \sum_{i < k} \int_0^T \dot{N}_i(t) \left[\int_{x_k}^{x_{k+1}} f(x - x_i; \mu_i(t), \sigma_i(t)) dx \right] dt$$

with a 5 s quadrature interval. Deposited counts are divided by physical footprint area, not a resolution-dependent raster-cell area.

2.3 Physical ignition and expected chronology

The configured ignition model uses $N_{\text{crit}} = 2928.9 \text{ pcs m}^{-2}$ [1]. At the structure spanning $20 \leq x \leq 30 \text{ m}$, the historical analytical load reaches this threshold near 2496.9 s. The case-local postprocessor preserves the published composite full-ignition reference of 2858.8 s; its difference from direct addition of the currently configured delays is quantified explicitly below. Targets beyond the roughly 80 m peak-HRR spotting range require successive structure ignition and consequently have nonlinear arrival times.

2.4 Resolution signatures

On the 2.5 m native mesh (native fine-grid variant), one physical structure occupies multiple independent cells. Uneven deposition and cell-local ignition should produce a nonzero within-structure load range and mean ROS near 0.001 m s^{-1} , compared with about 0.006 m s^{-1} for the exact-grid baseline. On the 30 m native mesh (native coarse-grid variant), mode resampling inflates footprint and HRR, causing artificially early spread; the documented response gives a baseline-to-coarse load ratio near 1.86 [1].

The integrated-structure treatment groups all cells belonging to one structure,

$$\bar{N} = \frac{\sum_{j \in s} N_j}{A_s}, \quad \text{HRR}_s = q'' A_s,$$

then redistributes load uniformly and emits from the structure centroid. The integrated-aligned variant expects an aligned 2.5 m mesh to reproduce the baseline. The integrated-misaligned variant permits up to 33% load deviation and 10% ignition-time deviation on a misaligned 3.24 m mesh.

3 Derivation of expected behavior

The exact grid should reproduce the analytical structure-load and ignition chronology. Native fine cells should expose within-structure imbalance, native coarse cells should expose footprint inflation, and integrated variants should recover the physical-structure response independently of alignment.

At the 400 kW m^{-2} plateau, one physical $10 \times 10 \text{ m}$ structure has

$$\text{HRR}_{\text{max}} = 400(100) = 40000 \text{ kW} = 40 \text{ MW}, \quad \dot{N}_{\text{max}} = 10(40) = 400 \text{ pcs s}^{-1},$$

and its critical total count is $N_{\text{crit}} = 2928.9(100) = 292890$ particles. Using the postprocessor's adjusted wind $U = 0.447(40)/0.87 = 20.5517 \text{ m s}^{-1}$ at peak HRR gives

$$B^* = 7.37372, \quad \bar{x} = 0.47(7.37372)^{2/3}(10) = 17.8056 \text{ m},$$

$$s_x = 0.88(7.37372)^{1/3}(10) = 17.1282 \text{ m}, \quad (\mu, \sigma) = (2.55195, 0.80939).$$

These values are inserted in the cell-CDF convolution at each 5-s midpoint; the HRR dependence is recomputed rather than freezing the peak values.

There is an explicit provenance discrepancy in the historical chronology. The case-local constants give

$$2496.9 + 42.1 + 300 = 2839.0 \text{ s},$$

whereas the preserved historical composite reference used by postprocessing is 2858.8 s, 19.8 s later. The latter is therefore treated as a published numerical estimate, not as the arithmetic result of the three displayed constants. The 19.8-s difference is only $19.8/2858.8 = 0.00693$, well inside the predeclared 10-percent baseline ignition tolerance, so this provenance mismatch cannot change that component's decision.

The exact-grid ideal errors are zero for analytical load and the chosen historical ignition reference. Native-grid expected signatures are a fine/baseline ROS ratio near $0.001/0.006 = 0.1667$ and a coarse load ratio near 1.86. Integrated aligned references are the exact-grid structure loads and times; for the misaligned grid, one unresolved boundary strip is $3.24/10 = 0.324$, motivating the explicit 0.33 load allowance.

4 Verification metrics and acceptance criteria

The postprocessor calculates every metric explicitly. The exact-grid baseline compares structure-integrated load with the analytical convolution defined above and second-structure ignition with 2858.8s. The native fine and coarse variants verify the documented cell-local artifacts rather than claiming convergence. The integrated aligned and misaligned variants compare physical-structure profiles with the exact-grid baseline. Missing output or capability is reported as *NOT EVALUABLE*, never as an implicit pass.

4.1 Justification of metric quantification

The criteria intentionally separate reproduction of native-grid defects from accuracy requirements for the baseline and integrated treatments. The 10% baseline load and ignition bands accommodate rasterized deposition and threshold-crossing time while remaining small relative to the resolution effects documented for this configuration [1]. A native fine-grid load range of at least 5% is set above roundoff and output quantization, and a ROS ratio no greater than 0.50 requires the material factor-of-two slowdown that identifies independent-cell behavior rather than a visually detectable but negligible difference. The coarse-grid load ratio is centered on the reported 1.86 response; its 25% band permits the severe footprint snapping intrinsic to a 30 m cell without accepting a reversal of the expected inflation.

For the aligned integrated grid, exact structure boundaries and conservation of integrated load justify the stringent 1% load limit. Its 5% ignition limit is wider because ignition is a nonlinear threshold event recorded at a discrete time. For the 3.24m misaligned grid, one boundary-cell width relative to a 10 m structure is $3.24/10 = 0.324$; the 0.33 maximum load-error limit therefore admits approximately one unresolved boundary strip but not a larger structure-scale bias. The corresponding 10% ignition bound allows threshold amplification of that overlap error. Capability and completeness are conjunctive because the integrated claims cannot be inferred from native-grid substitutes.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Baseline load error	Mean structure-load relative error against the analytical deposition reference, excluding the source structure.	≤ 0.10
Baseline ignition error	Relative error of second-structure ignition time against 2858.8s.	≤ 0.10
Native fine-grid load distortion	Median within-structure relative load range.	≥ 0.05
Native fine-grid ROS ratio	Fine-grid fitted mean ROS divided by baseline ROS.	≤ 0.50
Native coarse-grid load ratio	Baseline structure-4 load divided by coarse structure-2 load.	$1.86 \pm 25\%$
Integrated aligned load error	Mean matched structure-load relative error against baseline.	≤ 0.01
Integrated aligned ignition error	Mean matched ignition-time relative error against baseline.	≤ 0.05
Integrated misaligned load error	Maximum matched structure-load relative error against baseline.	≤ 0.33
Integrated misaligned ignition error	Maximum matched ignition-time relative error against baseline.	≤ 0.10
Overall decision	All configured variants and metrics must be evaluable; unavailable integrated capability prevents a PASS.	PASS only if all rows pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `baseline_dx10`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

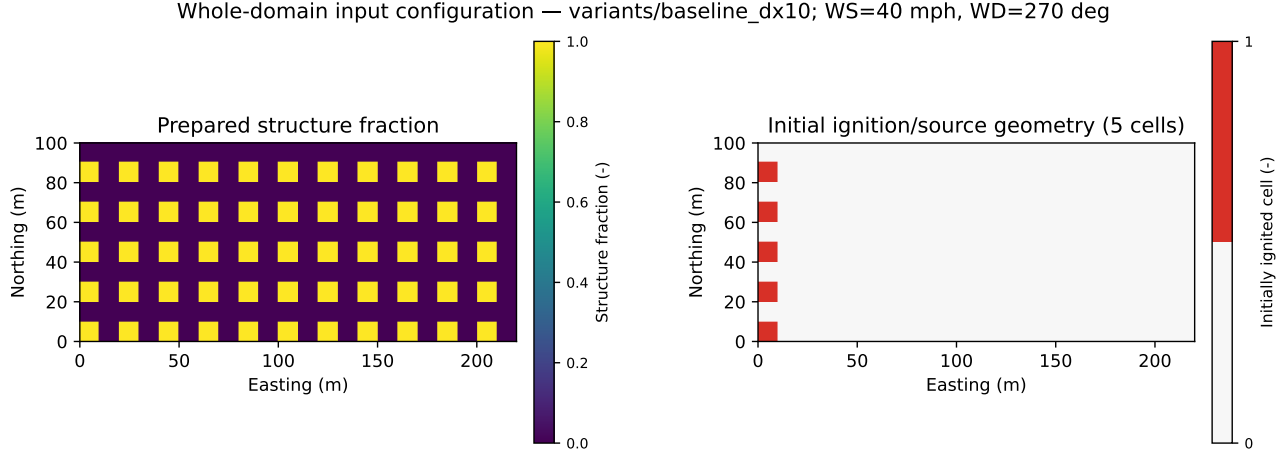


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Five variants generated from the common physical problem.

Variant	Δx (m)	$t_{\max}$ (s)	Treatment	Expected result
Baseline	10.00	22000	native	analytical deposition and ignition
Native fine	2.50	50000	native fine	imbalance and slower ROS
Native coarse	30.00	22000	native coarse	inflated HRR and earlier spread
Integrated aligned	2.50	22000	integrated aligned	baseline-equivalent profile
Integrated misaligned	3.24	22000	integrated misaligned	bounded profile error

Each nominal timestep is $\Delta t_{\text{nominal}} = 0.5\Delta x/U$; the generated step is no larger than this bound. All structure rasters reference a case-local model 14 containing the design fire and physical delay. Structures form a periodic 220×100 -m two-dimensional lattice. Each represented cell uses its true square area, so HRRPUA remains 400 kW m^{-2} and the dimensionless ignition-hardening factor remains one at every resolution. Every structure in the first upwind column is initialized at $t = 0$. The global `NO_SURFACE_FIRE` switch remains true to suppress vegetative surface-fire propagation. In the current source, initialized model-14 buildings independently refresh their transient HRR and therefore remain active under this setting. The generated non-structure fuel model 93 is also nonburnable, providing a second explicit isolation of the building-only experiment. Because the wind-CFL bound requires a subsecond step, preprocessing computes the nominal fractional value. When that value does not divide the requested duration, it uses $N = \lceil t_{\text{requested}}/\Delta t_{\text{nominal}} \rceil$ and reduces the generated step to $\Delta t = t_{\text{requested}}/N$. The stop therefore remains the exact requested duration and an exact timestep multiple; the manifest records nominal and actual steps, CFL values, and the integer step count.

5.1 Preprocessing, execution, and postprocessing

The preprocessor uses exact cell-footprint overlap, applies mode classification for native fine/coarse experiments, writes structure-ID and overlap rasters, creates all configurations, and probes the source for integrated capability. The runner executes supported variants using the absolute executable path in `ELMFIRE_BIN`. The postprocessor reads final ember-flux and TOA GeoTIFFs, normalizes counts by physical structure area, aggregates by structure ID, evaluates the formulas above, and writes `outputs/metrics.json`. Before reading either terminal raster it requires the current input fingerprint and successful source termination, parses unique `SIMULATION_TSTOP`, `SIMULATION_DT`, and `DT_METEOROLOGY` controls from the generated namelist, confirms manifest agreement and exact stop divisibility, and requires one ledger final row plus one ember-flux/TOA pair. A final dump beyond the nominal stop is accepted only when subtracting the meteorology interval gives a timestep-grid-aligned solver time within one timestep of the stop; ambiguous, early, off-grid, or duplicate evidence remains not evaluable.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from **baseline_dx10**. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. This completed baseline provides whole-domain evidence even though variants requiring the unavailable integrated-structure capability remain not evaluable.

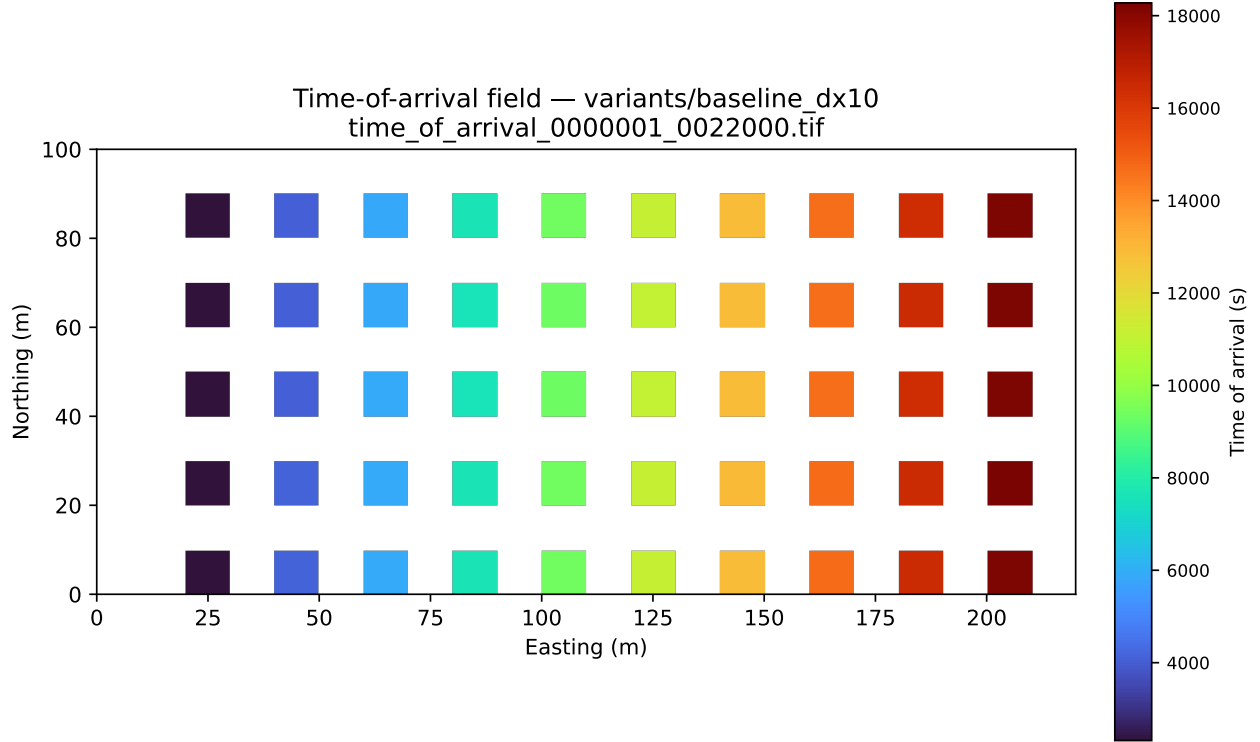


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

2 of 5 variants currently have complete, configuration-matched ELMFIRE output. Source-aligned physical-state figures are provided for all three native representations; a figure explicitly states when its current matching output is unavailable. The two integrated-structure variants remain capability gated.

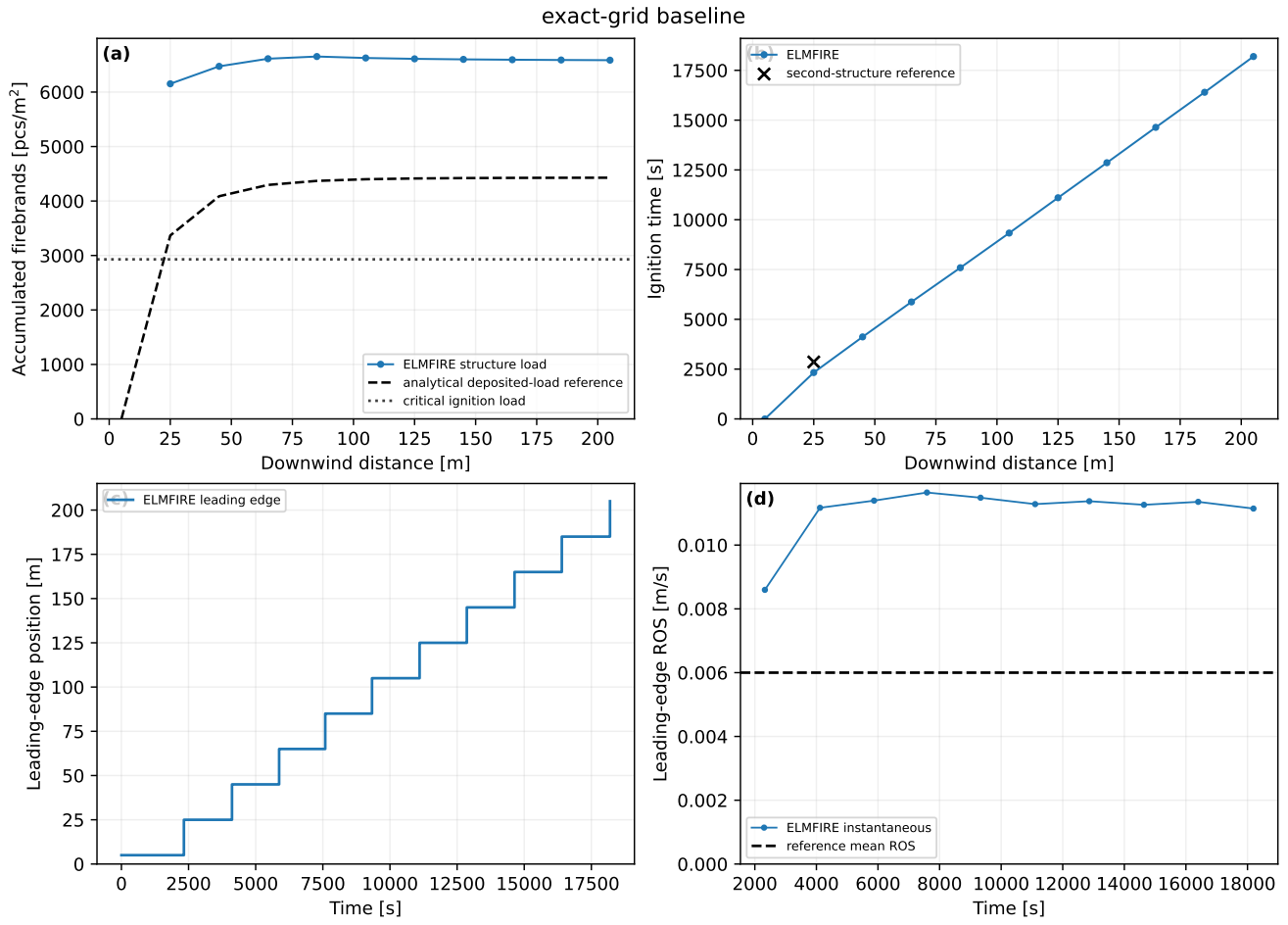


Figure 3: Exact-grid baseline observables: structure load, ignition chronology, leading-edge trajectory, and instantaneous ROS. Analytical load and second-structure timing references are overlaid where available.

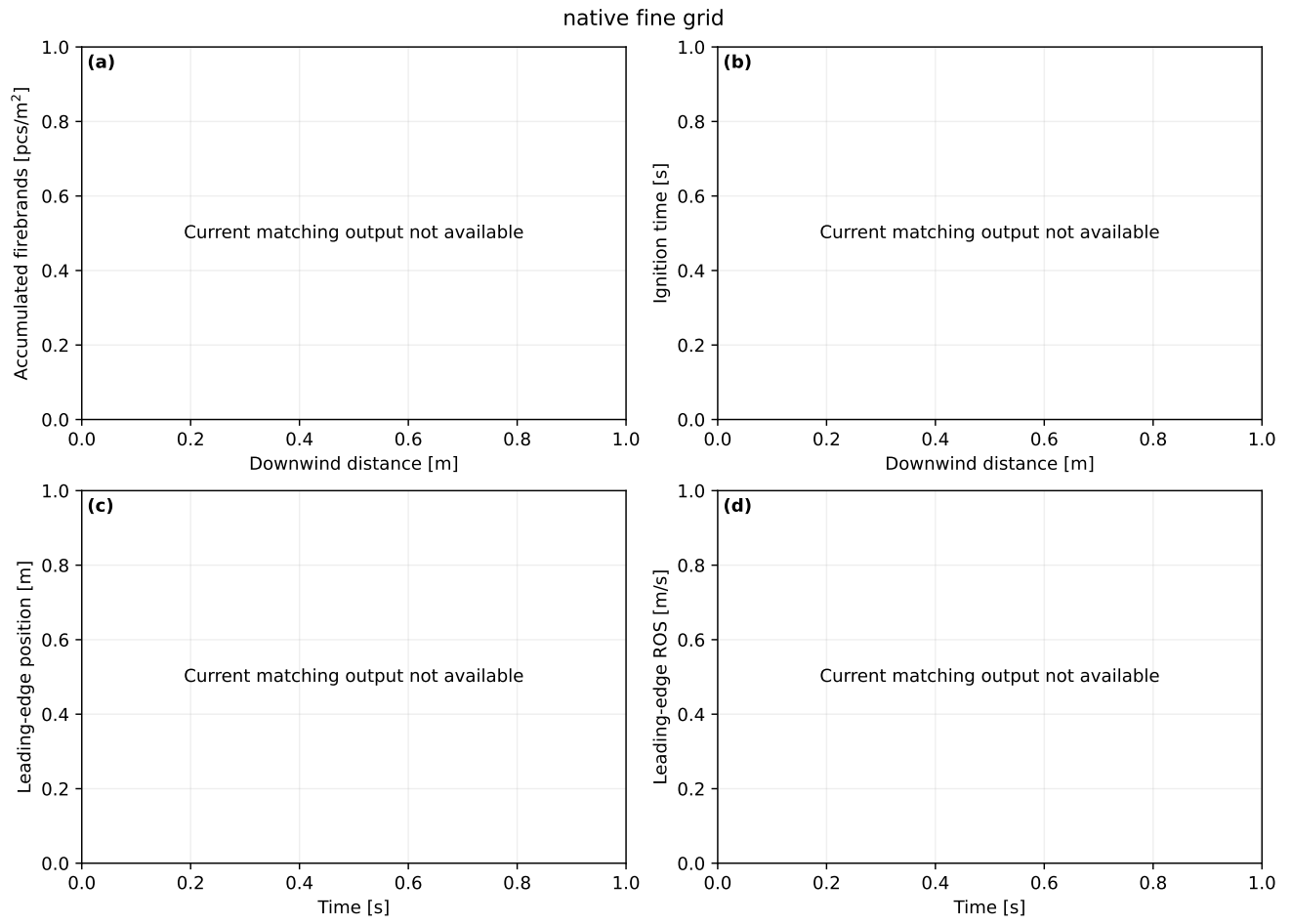


Figure 4: Native fine-grid observables. Cell-scale load variation within each physical structure is retained so discretization distortion is visible rather than hidden by structure averaging.

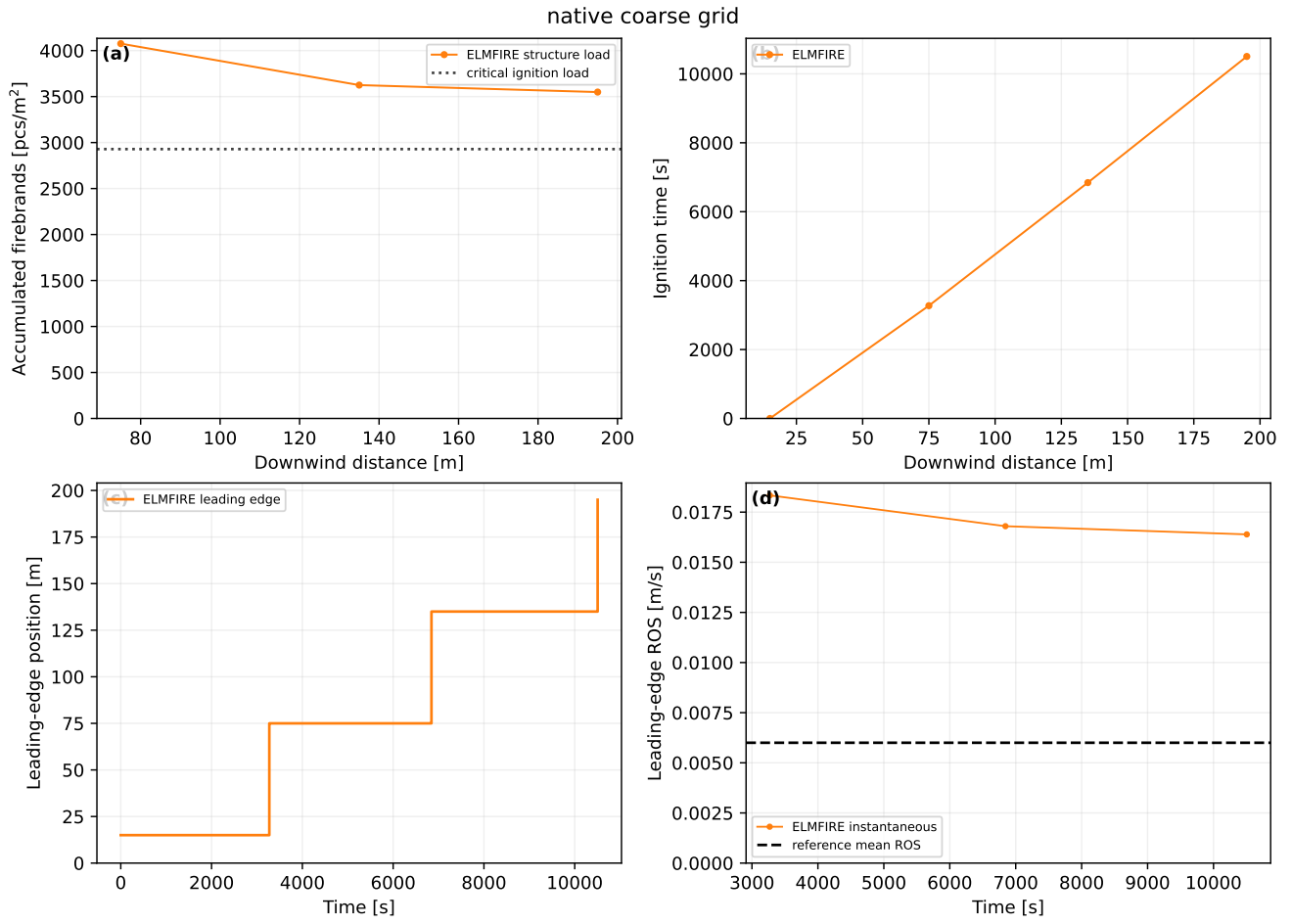


Figure 5: Native coarse-grid observables. Mode resampling changes the represented footprint and separation, and the resulting load and ignition histories are shown on their actual physical coordinates.

Figure 6 is the supplemental cross-variant diagnostic. The native-test decision is **FAIL**; the complete-suite decision is **NOT EVALUABLE (capability missing)**.

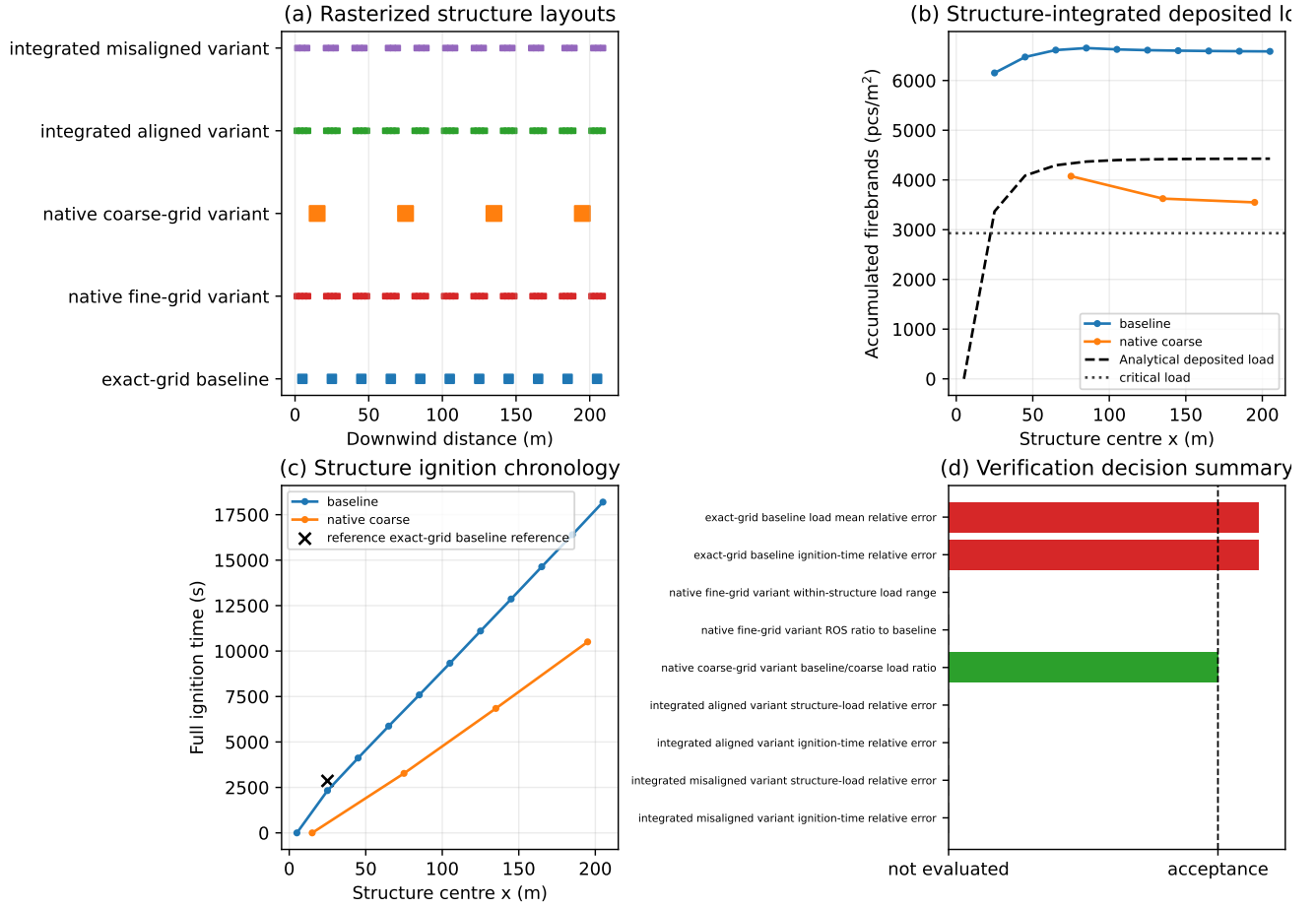


Figure 6: Structural-resolution summary. (a) Rasterized layouts; (b) deposited load by physical structure; (c) ignition chronology; and (d) component decision status, with unsupported components shown as not evaluated.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
exact-grid baseline load mean relative error	≤ 0.1	0.5435	FAIL
exact-grid baseline ignition-time relative error	≤ 0.1	0.1861	FAIL
native fine-grid variant within-structure load range	≥ 0.05	N/A	NOT EVALUABLE
native fine-grid variant ROS ratio to baseline	≤ 0.5	N/A	NOT EVALUABLE
native coarse-grid variant baseline/coarse load ratio	1.86 +/- 25%	1.622	PASS
integrated aligned variant structure-load relative error	≤ 0.01	N/A	NOT EVALUABLE
integrated aligned variant ignition-time relative error	≤ 0.05	N/A	NOT EVALUABLE
integrated misaligned variant structure-load relative error	≤ 0.33	N/A	NOT EVALUABLE
integrated misaligned variant ignition-time relative error	≤ 0.1	N/A	NOT EVALUABLE

The cited model description [1] states that the integrated-structure method had not yet been implemented in ELMFIRE. The preprocessor confirms this against the current source. The integrated aligned and misaligned cases are therefore generated as prospective variants but are not run while that feature is absent. This prevents the former resolution-independent directories from silently rerunning the native cell-local algorithm and reporting misleading results. Pre-merge input snapshots are archived outside the canonical case directory and are excluded from the run plan.

8 Technical interpretation

The corrected exact-grid baseline and native coarse variant both terminate at an exact 22 000 s final dump with matching fingerprints. The baseline load error is 0.5435 and its ignition-time error is 0.1861, both above the 0.10 limits; these are genuine numerical failures. The coarse baseline-to-coarse load ratio is 1.622 and passes the prescribed $1.86 \pm 25\%$ band.

The native fine-grid member was executed toward its prescribed 50 000 s endpoint but could not complete safely. It reached 36 000.065 s; resident memory grew from approximately 204 MB near 15 000 s to 960 MB, while successive 500 s dump intervals slowed from roughly 198 wall seconds to about 11 minutes. The run was stopped before exhausting the shared environment, and no completion fingerprint was written, so its partial rasters are correctly excluded. This reproduces an implementation resource-scaling limitation in the active ember-tracking path, not a stop-time or raster-configuration defect.

The integrated variants test whether one physical structure remains the source and collector; their missing capability is a source-interface limitation and is likewise reported as not evaluable rather than as a numerical pass or failure. Consequently the native-test status is **FAIL**, while the full suite remains **NOT EVALUABLE (capability missing)**.

9 Scope and limitations

The case is an idealized two-dimensional WUI lattice with fixed structure geometry and wind. A representative cross-stream lattice row supplies the one-dimensional structure profiles, while whole-domain maps expose unintended lateral behavior. The case verifies resolution behavior and capability coverage, not the physical accuracy of the structural transport or ignition correlations.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 11: WUI Surface-Coupling Synchronization

ELMFIRE Spotting Verification Guide

September 2, 2026

1 Verification purpose

Case ID: CASE11_WSC

This case verifies the effect of nonzero surface spread on ignition synchronization in a fine-grid WUI calculation. A no-surface reference and a surface-coupled variant use identical geometry, wind, firebrand physics, and resolution; only surface propagation changes.

The expected effect is earlier and more uniform ignition within each structure while cell-scale firebrand-accumulation distortion remains measurable. Success is based on paired structure-level timing and load metrics rather than visual comparison alone.

2 Mathematical formulation and numerical implementation

2.1 Fine-grid structural discretization

At $\Delta x = 2.5$ m, each 10 m structure is represented by a 4×4 block of square cells. For cell j , the structure heat-release rate used by the firebrand model is

$$\text{HRR}_j(t) = q''(t) \Delta x^2.$$

No strip-width correction is applied: summing the 16 cells recovers the physical 100 m^2 footprint and its design HRR. Firebrands are generated with

$$\dot{N}_j(t) = GR' \text{HRR}_j(t), \quad GR' = 10 \text{ pcs MW}^{-1} \text{ s}^{-1},$$

and transported by the current Eulerian implementation using the structural Himoto distribution. Physical ignition uses the critical accumulated load $N''_{\text{crit}} = 2928.9 \text{ pcs m}^{-2}$, followed by the configured small-fuel and development delays.

In the no-surface variant, cell-local deposition makes downwind cells within one structure receive different loads. Those cells consequently cross the ignition criterion at different times, producing artificial intra-structure delay and slow community-scale propagation.

2.2 Surface-fire synchronization mechanism

The surface-coupled variant retains the firebrand process but activates surface spread throughout the same two-dimensional lattice. The configured Hamada surface rate is $ROS_{\text{surface}} \approx 0.34 \text{ m s}^{-1}$ [1]. Once one cell of a 10 m structure ignites, a surface front at this speed traverses the footprint in approximately

$$\Delta t_{\text{traverse}} \approx \frac{a}{ROS_{\text{surface}}} = \frac{10 \text{ m}}{0.34 \text{ m s}^{-1}} \approx 29.4 \text{ s}.$$

This is much shorter than the cell-to-cell ignition offsets created by firebrand accumulation alone. Surface coupling should therefore reduce within-structure ignition spans and accelerate the downwind ignition chronology. It does not alter the cell-local firebrand deposition algorithm, so load imbalance should remain visible.

3 Derivation of expected behavior

Surface propagation crosses a represented structure much faster than independent firebrand accumulation ignites its cells. The coupled variant should therefore ignite earlier and with a smaller within-structure time span, while the unchanged deposition algorithm should preserve measurable load nonuniformity.

At $\Delta x = 2.5$ m, each structural cell has area $2.5^2 = 6.25$ m². At the design-fire plateau,

$$HRR_{j,\max} = 400(6.25) = 2500 \text{ kW} = 2.5 \text{ MW}, \quad \dot{N}_{j,\max} = 10(2.5) = 25 \text{ pcs s}^{-1}.$$

Summing the $4 \times 4 = 16$ cells gives $16(2.5) = 40$ MW and $16(25) = 400$ pcs s⁻¹, exactly the physical 100 m² structure totals. The generated level-set step is

$$\Delta t = 0.5(2.5)/[40(0.44704)] = 0.0699043 \text{ s}.$$

Once one structural cell ignites, the surface reference crosses one 2.5-m cell in $2.5/0.34 = 7.3529$ s and a full 10-m footprint in $10/0.34 = 29.4118$ s. These are the explicit synchronization timescales; firebrand-only cells still wait for independent accumulation to 2928.9 pcs m⁻² plus the configured ignition delays.

The ideal paired signatures are therefore $t_{\text{coupled}} < t_{\text{no surface}}$ for most matched cells, a smaller coupled within-structure span, and a larger coupled fitted ROS, while the deposition range remains nonzero because its algorithm is unchanged. The quantified substitutions require at least 75 percent earlier cells, median time ratio at most 0.90, median span ratio at most 0.50, ROS ratio at least 2, and coupled load range at least 0.05.

4 Verification metrics and acceptance criteria

For every downstream structural cell having finite arrival times in both variants, let $t_{\text{ns},j}$ denote the no-surface arrival and $t_{\text{sc},j}$ the surface-coupled arrival. Earlier ignition is quantified by

$$F_{\text{early}} = \frac{1}{N} \sum_j \mathbf{1}(t_{\text{sc},j} < t_{\text{ns},j}), \quad R_t = \text{median}_j \left(\frac{t_{\text{sc},j}}{t_{\text{ns},j}} \right).$$

The case requires $F_{\text{early}} \geq 0.75$ and $R_t \leq 0.90$.

For structure s , define its ignition span as

$$S_s = \max_{j \in s} t_j - \min_{j \in s} t_j.$$

The synchronization metric is the median paired ratio

$$R_S = \text{median}_{s \geq 2} \left(\frac{S_{\text{sc},s}}{S_{\text{ns},s}} \right),$$

which must not exceed 0.50. Mean propagation speed is obtained from a least-squares fit of first-ignition time against structure center, $ROS = (dt/dx)^{-1}$; the coupled/no-surface ratio must be at least 2.

Persistence of accumulation distortion is measured from the represented cell loads along the diagnostic lattice row in each coupled structure:

$$D_s = \frac{\max_{j \in s} N_j'' - \min_{j \in s} N_j''}{\text{mean}_{j \in s} N_j''}.$$

The downstream median must remain at or above 0.05. This is intentionally a lower-bound test: the surface-coupled variant is expected to improve ignition synchronization, not to homogenize firebrand deposition.

4.1 Justification of metric quantification

These are minimum effect-size criteria for the directional comparison reported for this configuration [1], not estimates of experimental uncertainty. Requiring $F_{\text{early}} \geq 0.75$ means earlier ignition must occur in a clear supermajority of matched cells, so a few ties or grid-crossing reversals cannot determine the result. The median time ratio of 0.90 requires at least a 10% central timing improvement, while the span ratio of 0.50 requires a factor-of-two synchronization improvement within structures. The ROS ratio of 2.0 likewise demands a large

propagation effect consistent with surface coupling, rather than declaring success from an arbitrarily small positive change.

The 5% lower bound on within-structure load range is deliberately above floating-point and raster-write noise. It confirms that the known deposition nonuniformity remains visible even when ignition becomes synchronized; it is not an accuracy tolerance. Requiring both current-fingerprint variants is essential because all five quantities are paired effects and have no valid unpaired interpretation.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Both variants must carry the current input fingerprint and usable structure-level load/ignition fields.	2 of 2 variants
Earlier-ignition fraction	Fraction of matched downstream cells with $t_{coupled} < t_{no\ surface}$.	≥ 0.75
Median ignition-time ratio	Median $t_{coupled}/t_{no\ surface}$ over matched downstream cells.	≤ 0.90
Ignition-span ratio	Median paired within-structure span ratio over downstream structures.	≤ 0.50
Mean-ROS ratio	Coupled fitted structure-propagation ROS divided by no-surface ROS.	≥ 2.0
Persistent load distortion	Median coupled within-structure relative load range.	≥ 0.05
Overall decision	Conjunction of completeness and all five comparison metrics.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `surface_coupled_dx2p5`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

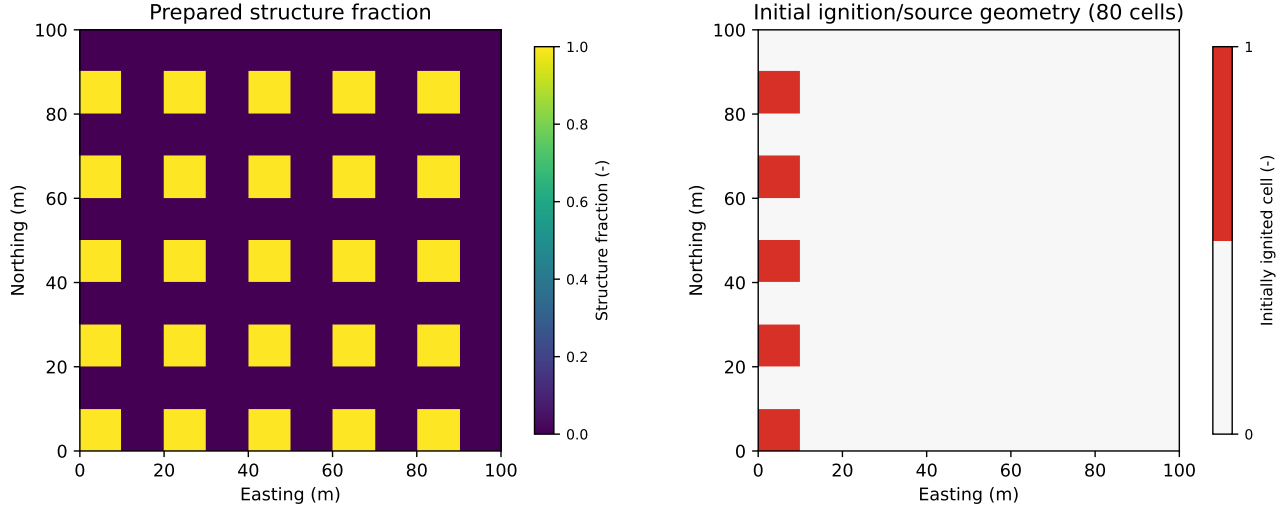


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Controlled paired configuration.

Parameter	No-surface variant	Surface-coupled variant
Grid spacing	2.5 m	2.5 m
Physical transect	100 m by 10 m	same
Structure/firebreak	10 m/10 m	same
Wind	40 mph, west to east	same
Time step	$0.5\Delta x/U$	same
Requested maximum time	6000 s	3500 s
Surface fuel	model 91 structures; model 102 gaps	same
Spread model	disabled	Hamada ($a=d=10$ m)
Generation	per-MW	per-MW
Transport/accumulation	empirical/Eulerian	same
Ignition	physical	same

The fixed wind CFL requires a fractional step. Preprocessing sets each generated stop to $\lceil t_{\text{requested}}/\Delta t \rceil \Delta t$, and records the requested stop, aligned stop, and integer step count in the manifest.

The no-surface reference retains a longer observation window because accumulation-only spread requires it to ignite multiple downstream structures. The surface-coupled variant stops after all structures have ignited and a mature, nonuniform ember-load field has formed. Since TOA is immutable after ignition and load distortion is measured within the coupled run rather than between terminal times, this difference reduces runtime without changing any acceptance quantity.

The expected signature is not agreement with a prescribed straight-line TOA curve. It is a directional paired result: the surface-coupled variant must ignite earlier, have smaller intra-structure ignition spans, and have a larger fitted mean ROS, while its cell-level load curve remains nonuniform.

5.1 Preprocessing, execution, and postprocessing

`scripts/preprocess.py` defines customizable parameters at its beginning, creates two self-contained variant directories, and writes co-registered UTM GeoTIFFs. Square structures and surface-fuel gaps repeat in both horizontal directions across the 100×100 -m physical domain; only the two-cell numerical halo uses inert model

93. Structure cells also receive co-registered building-property rasters. ELMFIRE therefore applies the Hamada WUI velocity in structures and the Rothermel velocity in surface gaps, giving the coupled level-set front a continuous two-dimensional path. Every structure in the first upwind column is initialized, and a structure-ID raster preserves each physical 4×4 -cell footprint independently. The generated configurations differ only in `NO_SURFACE_FIRE`. Input fingerprints prevent stale results from being compared.

`run_case.sh` prepares both variants, runs the absolute executable provided in `ELMFIRE_BIN`, captures separate logs, invokes postprocessing, and compiles this standalone report. The postprocessor reads the final authoritative TOA and accumulated-ember rasters, normalizes each cell by its physical $\Delta x w_y$ area, computes all metrics above, and generates the paired timing-and-load diagnostic figure.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from `surface_coupled_dx2p5`. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. The map shows the spatial synchronization of surface spread and structure ignition that is reduced to profiles in the subsequent comparison.

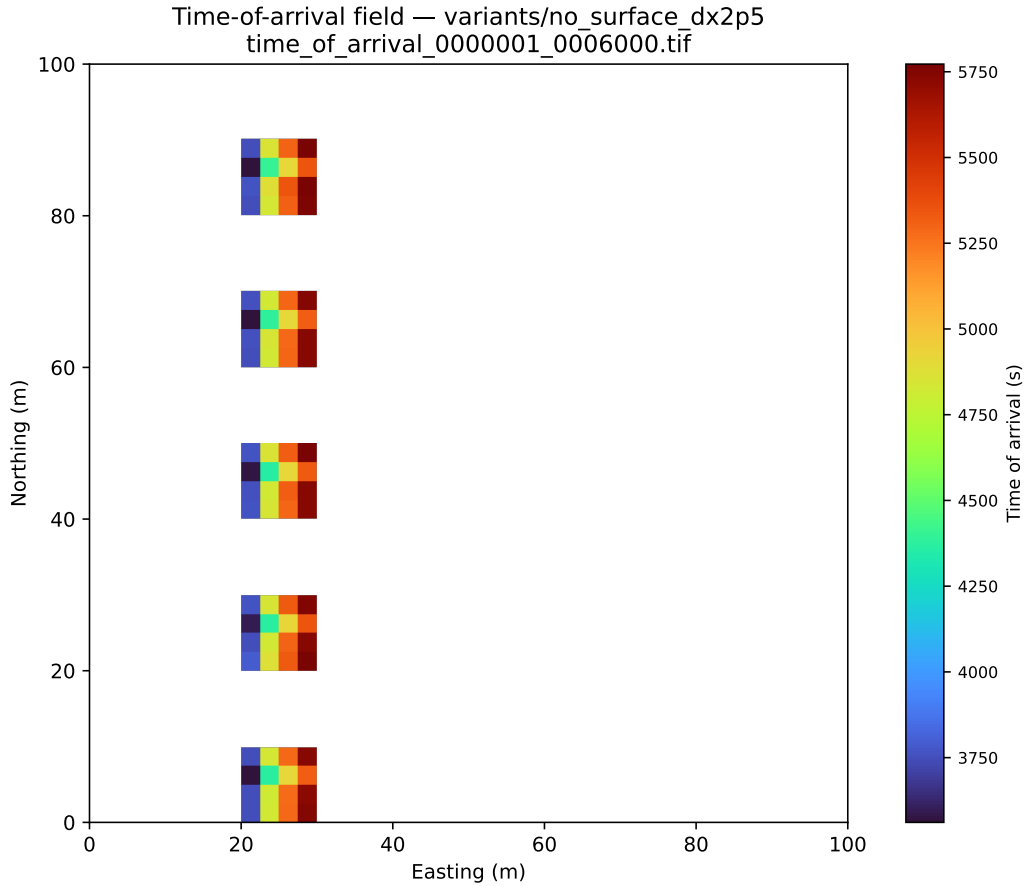


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

1 of 2 current-fingerprint variants have evaluable outputs. Figure 3 first presents the surface-coupled experiment by itself using the two physical profiles that define the expected synchronization behavior.

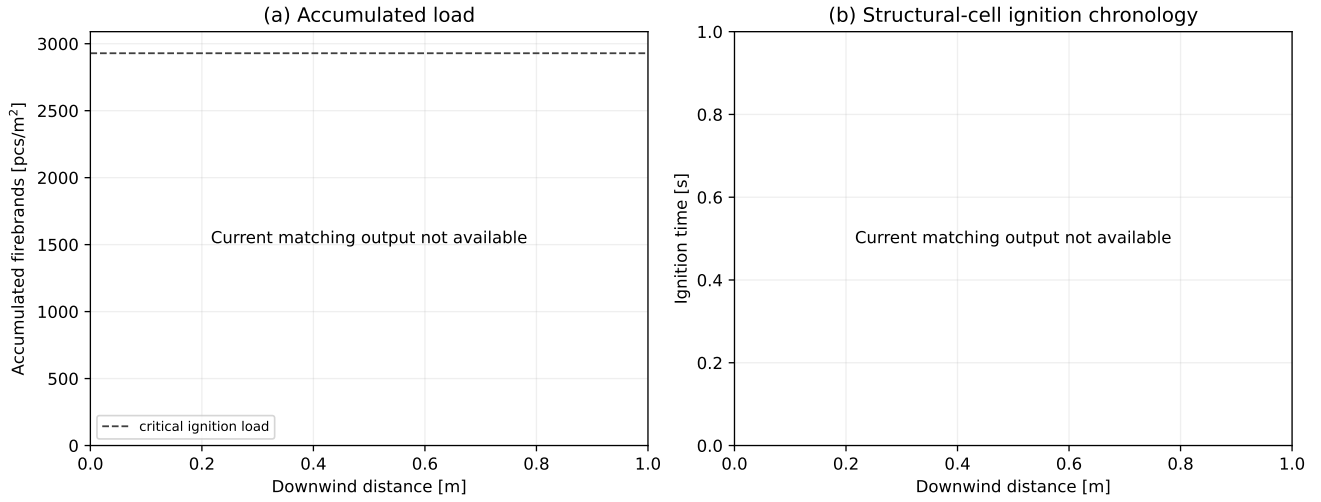


Figure 3: Surface-coupled fine-grid result. Left: cell-level accumulated firebrand load and the critical ignition load. Right: structural-cell ignition chronology.

Figure 4 then overlays the no-surface control, making the earlier and more uniform coupled ignition response directly visible. The overall decision is **NOT RUN**.

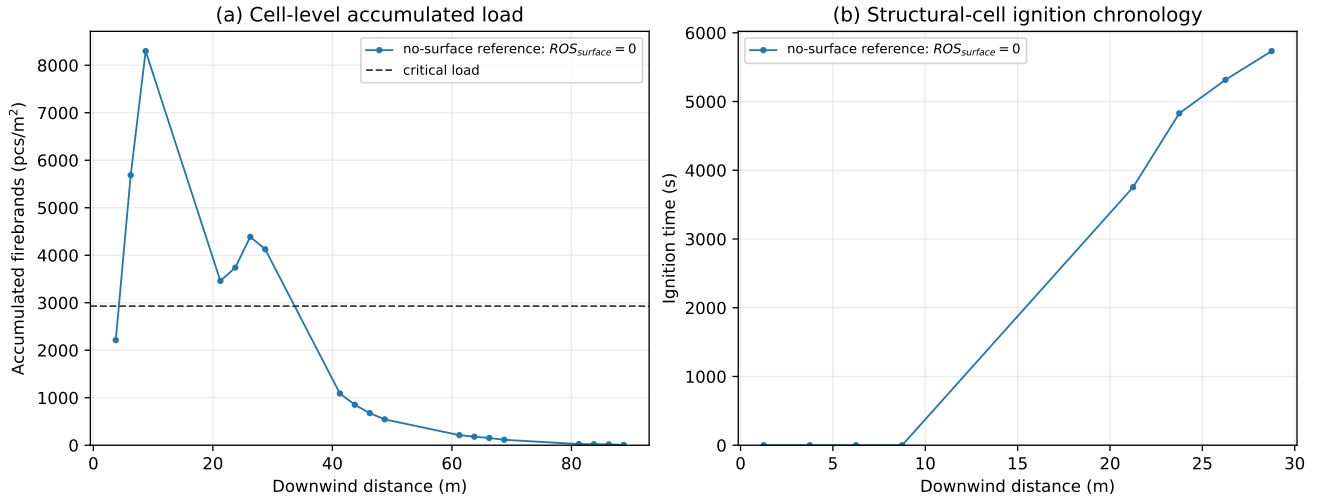


Figure 4: Controlled surface-coupling comparison. Persistent within-structure load variation diagnoses the remaining accumulation distortion, while the lower, smoother coupled ignition curve demonstrates synchronization.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status	
Output completeness	2 current-fingerprint variants	1/2	FAIL	
Coupled earlier-ignition cell fraction	≥ 0.75	N/A	NOT EVALU- ABLE	EVALU- ABLE
Coupled median ignition-time ratio	≤ 0.9	N/A	NOT EVALU- ABLE	EVALU- ABLE
Coupled/no-surface ignition-span ratio	≤ 0.5	N/A	NOT EVALU- ABLE	EVALU- ABLE
Coupled/no-surface mean-ROS ratio	≥ 2.0	N/A	NOT EVALU- ABLE	EVALU- ABLE
Coupled within-structure load relative range	≥ 0.05	N/A	NOT EVALU- ABLE	EVALU- ABLE

The current decision is **NOT RUN**. A pass requires both current-fingerprint outputs and every synchronization, acceleration, and persistent-distortion check in the table. Missing comparison values are reported as *NOT EVALUABLE*, never as a pass.

This is an implementation verification using an idealized two-dimensional WUI lattice. The fitted ROS is a community-scale diagnostic and is not required to equal the configured Hamada surface speed exactly because the executable also retains spotting, Rothermel fuel behavior, and discrete ignition delays. The case verifies the qualitative and quantitative coupling signature derived above; it does not claim that surface spread corrects the underlying cell-local firebrand transport distortion.

8 Technical interpretation

The no-surface reference completes at its aligned 6000.0224 s stop and supplies a current-fingerprint final output. The surface-coupled member uses the predeclared single controlled difference, `NO_SURFACE_FIRE=.FALSE.`, but does not complete its 3500.0364 s run.

The coupled execution writes a valid first transient dump near 500.025 s, then exhibits sustained memory growth and severe slowdown: resident memory rises from approximately 319 MB at 93 s wall time to 874 MB at 550 s, 1.07 GB at 823 s, and 1.28 GB at 1188 s without reaching the second 1000 s dump. It was stopped before exhausting the shared environment. Because the preprocessor, fingerprint, input rasters, and controlled namelist contrast are valid, changing the case to avoid the coupled path would invalidate the verification objective. The report therefore records 1/2 completeness and *NOT RUN*; the unavailable comparison metrics are not treated as passes. The resource-growth behavior is an implementation issue in the coupled WUI execution path.

9 Scope and limitations

The paired comparison isolates one uniform fine-grid WUI transect. It verifies the coupling signature and does not establish that surface spread corrects the underlying deposition discretization or predicts a real community fire.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 12: WUI Temporal-Resolution Sensitivity

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

This case verifies temporal-resolution sensitivity using a fixed physical problem and four level-set timesteps. Every curve identified as ELMFIRE output is regenerated from case-local simulation products.

Case ID: CASE12.TRS

This case verifies temporal-resolution sensitivity in a firebrand-driven WUI calculation. Spatial resolution, geometry, wind, heat-release history, firebrand physics, random seed, and output cadence are fixed; only the level-set timestep changes over a wide CFL range.

The test determines the range over which ignition chronology and structure-level spread remain timestep independent and makes loss of resolution explicit at very large steps. Success is evaluated against the finest-step baseline using load, ignition-time, and spread metrics.

2 Mathematical formulation and numerical implementation

2.1 Fixed physical problem

The community is a 220×100 -m two-dimensional lattice of 10 m square structures separated by 10 m inert firebreaks in both directions. Every structure in the first upwind column is burning at $t = 0$, and wind is uniform at 40 mph from west to east. Non-structure cells use nonburnable fuel model 93, so vegetative surface-fire spread is absent. Thus a new source can appear only after deposited firebrands complete the configured physical ignition process.

For a structure ignited at t_i , the design heat-release rate per unit area is

$$q''(\tau) = \begin{cases} q''_{\max} \tau / 300, & 0 < \tau < 300, \\ q''_{\max}, & 300 \leq \tau \leq 3900, \\ q''_{\max} (4200 - \tau) / 300, & 3900 < \tau < 4200, \\ 0, & \text{otherwise,} \end{cases} \quad \tau = t - t_i, \quad q''_{\max} = 400 \text{ kW m}^{-2}.$$

For footprint area $A_s = 100 \text{ m}^2$, $\text{HRR}_i = q'' A_s$. Structural firebrands are emitted using the per-power rate

$$\dot{N}_i(t) = GR' \times \text{HRR}_i(t), \quad GR' = 10 \text{ pcs MW}^{-1} \text{ s}^{-1}.$$

2.2 Transport, deposition, and analytical load

The downwind landing distance X is represented by the lognormal density

$$f(x \mid \mu, \sigma) = \frac{\exp[-(\ln x - \mu)^2 / (2\sigma^2)]}{x\sigma\sqrt{2\pi}}, \quad x > 0.$$

For source structure i , the heat-release-dependent transport parameter is

$$B_i^* = \frac{U}{\sqrt{gL}} \left(\frac{\rho_f}{\rho_a} \right)^{-3/4} \left(\frac{d_f}{L} \right)^{-3/4} \left[\frac{\text{HRR}_i}{\rho_a c_p T_a \sqrt{g} L^{5/2}} \right]^{1/2}.$$

The dimensional mean and standard deviation of landing distance are

$$\bar{x}_i = 0.47(B_i^*)^{2/3} L, \quad s_{x,i} = 0.88(B_i^*)^{1/3} L,$$

and the log-space parameters supplied to f are

$$\sigma_i = \sqrt{\ln[1 + (s_{x,i}/\bar{x}_i)^2]}, \quad \mu_i = \ln\left(\frac{\bar{x}_i}{\sqrt{1 + (s_{x,i}/\bar{x}_i)^2}}\right).$$

Here $L = 10$ m, $d_f = 5 \times 10^{-3}$ m, $\rho_f = 100$ kg m $^{-3}$, $\rho_a = 1.1$ kg m $^{-3}$, $c_p = 1$ kJ kg $^{-1}$ K $^{-1}$, $T_a = 300$ K, $g = 9.81$ m s $^{-2}$, and $U = 0.447(40)/0.87 = 20.55$ m s $^{-1}$. This parameterization and its configured constants follow the documented structural-firebrand model [1]; the equations are reproduced here so the reference can be evaluated without consulting that source.

The independently calculated deposited count on target structure k is

$$N_k = \sum_{i < k} \int_0^T GR' \text{HRR}_i(t) \left[\int_{x_k}^{x_{k+1}} f(x - x_i \mid \mu_i(t), \sigma_i(t)) dx \right] dt.$$

The inner integral is evaluated using the lognormal CDF and the time integral uses an independent 5 s midpoint quadrature. ELMFIRE's final deposited count is divided by the physical 100 m 2 structure footprint, giving N_k/A_s in pcs m $^{-2}$.

The Eulerian firebrand tracker advances with its own wind-limited substeps. Changing the level-set step therefore must not change the spatial resolution of the flight trajectory. This is the core algorithmic separation being tested. In the generated namelists, `SIMULATION_DT` and `SIMULATION_DTMAX` impose the four values above. The input `TARGET_CFL` remains the ordinary surface-spread limiter and does not redefine the four wind-based temporal variants. The global `NO_SURFACE_FIRE` switch remains true to suppress vegetative surface-fire propagation. The current source independently refreshes initialized model-14 building HRR, and the nonburnable fuel-93 background provides an additional isolation of building-only spread. The fractional timesteps are intrinsic to this temporal-resolution sweep. For each member, preprocessing sets the generated stop to $\lceil t_{\text{requested}}/\Delta t \rceil \Delta t$, and records the requested stop, aligned stop, and integer step count in the manifest.

2.3 Physical ignition

For the configured structural ignition model, the critical deposited load is $N_{\text{crit}} = 2928.9$ pcs m $^{-2}$, the smoldering-to-flaming timescale is 42.1 s, and the subsequent structure development delay is 300 s [1]. Because these are much longer than the CFL=0.5, 1, and 10 level-set steps, their ignition chronologies should remain nearly unchanged. CFL=200 deliberately makes the time step comparable to or larger than the short ignition time scale and exposes the onset of temporal sensitivity.

3 Derivation of expected behavior

When firebrand flight and ignition time scales are much longer than the level-set step, moderate timestep changes should preserve deposited-load histories and structure ignition times. Extremely large steps should show increasing temporal error without changing the fixed physical problem.

At peak HRR, one 10 $\times$ 10 m structure produces

$$\text{HRR}_{\text{max}} = 400(100) = 40 \text{ MW}, \quad \dot{N}_{\text{max}} = 10(40) = 400 \text{ pcs s}^{-1},$$

and the critical total load is $2928.9(100) = 292890$ particles. The transport reference uses the adjusted air speed $0.447(40)/0.87 = 20.55$ m s $^{-1}$, whereas preprocessing constructs the declared level-set timesteps from the 20-ft input speed $U_{20} = 40(0.44704) = 17.8816$ m s $^{-1}$. Thus

CFL_w	0.5	1	10	200
$\Delta t = CFL_w(10)/17.8816 \text{ (s)}$	0.279617	0.559234	5.592341	111.846815

which reproduces the variant names to their displayed precision.

Relative to the 42.1 s small-fuel timescale, these steps are 0.00664, 0.0133, 0.1328, and 2.657 times as large. The first three therefore resolve that timescale, while CFL 200 spans more than twice it and is expected to show temporal degradation. The analytical convolution is unchanged among variants, so its ideal load error is zero. The explicit numerical substitutions are: baseline mean load error at most 0.01; CFL 1 and 10 maximum load and mean ignition-time errors at most 0.001; their history NRMSE at most 0.05; and CFL 200 maximum load error at most 0.10.

4 Verification metrics and acceptance criteria

Prior calculations for this fixed physical problem motivate the following predeclared expectations [1]:

- the maximum analytical-load error remains below 10% even at CFL=200;
- for CFL no greater than 10, analytical-load errors are generally below 0.1%;
- ignition-time errors relative to CFL=0.5 remain below 0.1% for CFL no greater than 10;
- accumulation histories at the structure spanning $140 \leq x \leq 150$ m closely align; and
- first-deposition timing remains close through the approximately 80 m peak-HRR spotting range.

The case converts those expectations into explicit acceptance checks. The CFL=0.5 mean error against the analytical deposited-load integral defined above must not exceed 1%; the CFL=1 and 10 maximum errors against the same integral must not exceed 0.1%; and the CFL=200 maximum must remain below 10%. CFL=1 and 10 mean full-ignition-time errors relative to CFL=0.5 must not exceed 0.1%. Accumulation-history normalized RMSE at 140–150 m must not exceed 5% for CFL=1 and 10. Every variant must be complete, and every acceptance component must pass.

The postprocessor also estimates first-deposition time from the first transient output interval with nonzero accumulation. Because its resolution is the 500 s dump interval rather than the firebrand trajectory substep, this is shown as a diagnostic corresponding to the first-deposition diagnostic, but is not used as a pass/fail metric.

4.1 Justification of metric quantification

The 0.1% limits for CFL 1 and 10 and the 10% limit for CFL 200 quantify the reported temporal-resolution behavior for this fixed-grid experiment rather than being selected from the present outputs [1]. They deliberately ask for near invariance over the moderate-step range while treating the extreme-step case as acceptable only if its error remains bounded rather than converged. The baseline mean-load limit is 1% because it compares a cell-integrated analytical deposition relation with a thresholded numerical field; it establishes that the reference member itself is accurate enough to anchor the relative ignition comparisons.

The 5% history NRMSE limit is wider than the final-load limits because transient rasters are sampled every 500 s; small timing offsets at a rising accumulation front are amplified pointwise even when final totals agree. Normalizing by the baseline history range makes 5% a direct bound on the fraction of the resolved accumulation excursion. First-deposition time is kept diagnostic because its 500 s observation interval is too coarse to support a defensible tighter decision. All four variants are required so the case demonstrates both moderate-step invariance and the predicted extreme-CFL degradation.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Current-fingerprint outputs for baseline and CFL 1, 10, and 200 variants.	4 of 4 variants
Baseline load mean error	Mean structure-load relative error against the analytical load reference.	≤ 0.01
CFL 1 load maximum error	Maximum finite structure-load relative error against its analytical reference.	≤ 0.001
CFL 10 load maximum error	Same maximum error at CFL 10.	≤ 0.001
CFL 200 load maximum error	Same maximum error at CFL 200.	≤ 0.10
CFL 1 ignition error	Mean matched full-ignition-time relative error against CFL 0.5.	≤ 0.001
CFL 10 ignition error	Mean matched full-ignition-time relative error against CFL 0.5.	≤ 0.001
CFL 1 history NRMSE	Normalized RMSE of accumulation history at the 140–150-m structure against baseline.	≤ 0.05
CFL 10 history NRMSE	Same normalized history error at CFL 10.	≤ 0.05
Overall decision	Conjunction of completeness and all eight quantitative checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `cf10p5_dt0p28`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

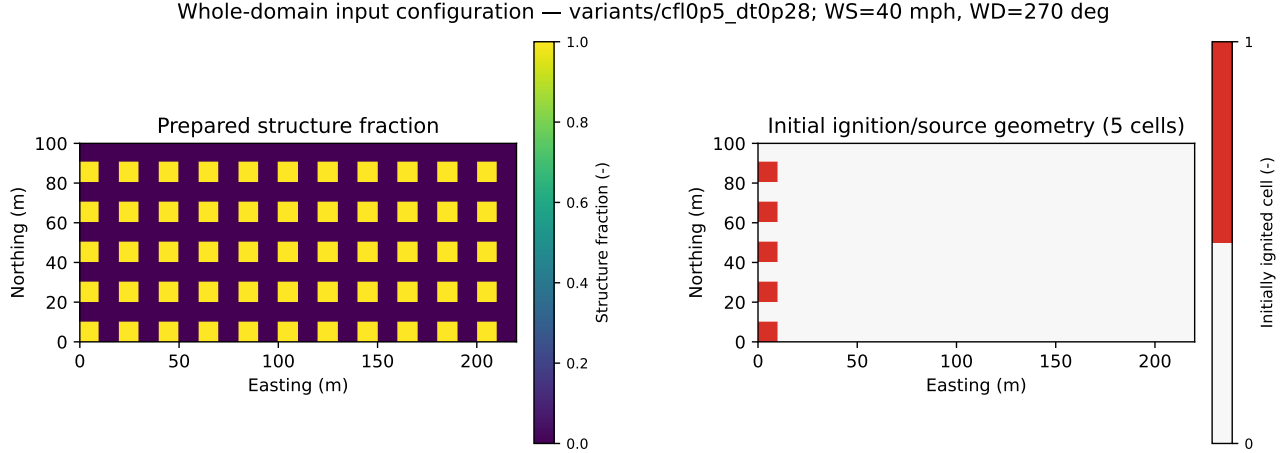


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Generated temporal-resolution variants. The time step is calculated from $\Delta t = \text{CFL} \Delta x / U$, not rounded to the display value.

Variant	Wind-based CFL	Δt (s)	$t_{\max}$ (s)
cf10p5_dt0p28	0.5	0.27962	22000
cf11_dt0p56	1.0	0.55923	22000
cf10_dt5p59	10.0	5.5923	22000
cf200_dt111p86	200.0	111.85	22034

All variants use $\Delta x = \Delta y = 10$ m, alternating 10 m structures and firebreaks, a 40 mph wind, the same model-14 design fire, per-MW firebrand generation, empirical structural transport, Eulerian accumulation, physical ignition, no ember consumption, no crosswind distribution, and no vegetative surface-fire pathway. The predicted result is near-overlap for CFL=0.5, 1, and 10, with visible but bounded load sensitivity at CFL=200.

5.1 Preprocessing, execution, and postprocessing

`scripts/preprocess.py` defines its customizable constants at the top, creates one self-contained directory per variant, writes co-registered UTM GeoTIFFs and case-local fuel-model tables, marks every structure footprint in the first upwind column with negative ϕ , and records the exact time step and an SHA-256 input fingerprint in `variants/manifest.json`. Unique structure IDs preserve the two-dimensional lattice while the postprocessor extracts a representative cross-stream row for common structure profiles.

`run_case.sh` prepares all variants, runs the executable given by the absolute `ELMFIRE_BIN` path, captures per-variant logs, records the successful input fingerprint, invokes postprocessing, and compiles this standalone report. Resume mode skips a run only when its output fingerprint matches the current generated inputs.

`scripts/postprocess.py` aggregates final ember counts and ignition times by physical structure ID, reconstructs cumulative histories from transient ember rasters and `dump_times` metadata, evaluates the analytical deposited-load relation, calculates every declared metric, and produces report-ready vector figures. Stale pre-modification outputs are explicitly excluded.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from `cf10p5_dt0p28`. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. The finest-timestep reference field anchors the profile and error comparisons and exposes domain-wide behavior that a single-row profile can hide.

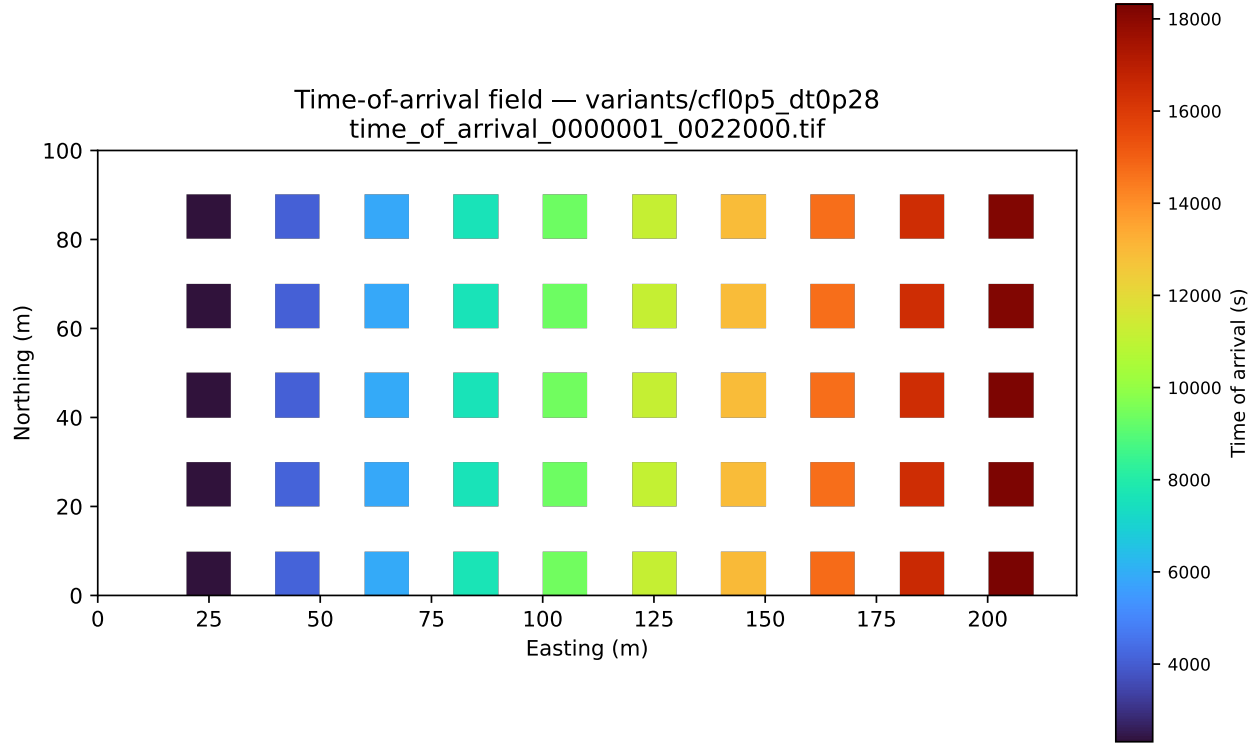


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

4 of 4 variants currently have complete outputs matching the generated inputs. The verification decision is **FAIL**.

Figure 3 presents four complementary temporal-resolution diagnostics. Panel (a) compares deposited load with the analytical integral defined in Section 2; panels (b) and (d) use $CFL=0.5$ as the numerical baseline; and panel (c) shows the accumulation history at the specified downwind structure.

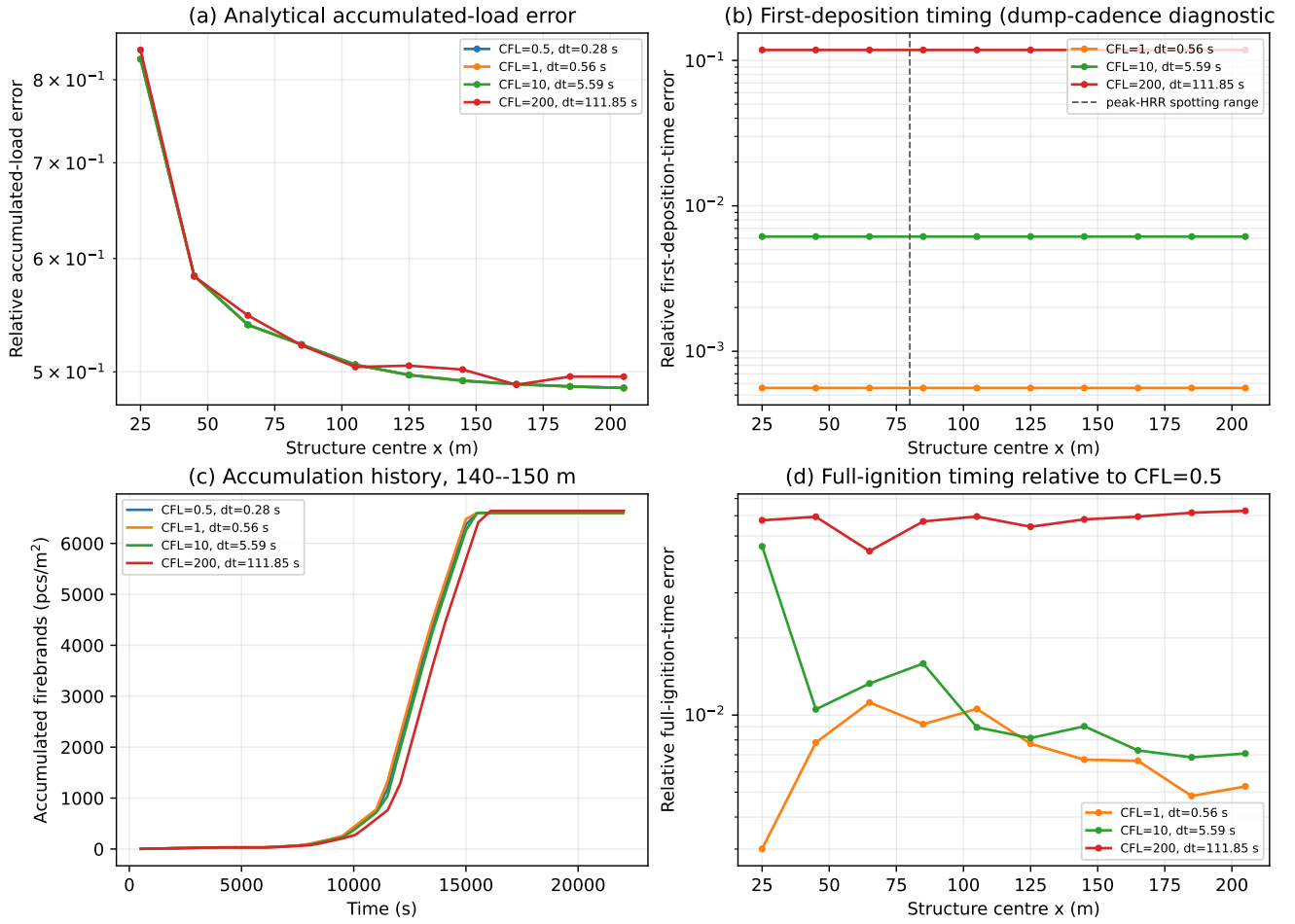


Figure 3: Temporal-resolution diagnostics for the prescribed temporal-resolution sweep: (a) accumulated-load relative error; (b) dump-cadence first-deposition-time relative error; (c) accumulation history at 140–150 m; and (d) full-ignition time relative error.

Figure 4 condenses the ignition comparison into mean relative error versus CFL and displays the acceptance threshold directly.

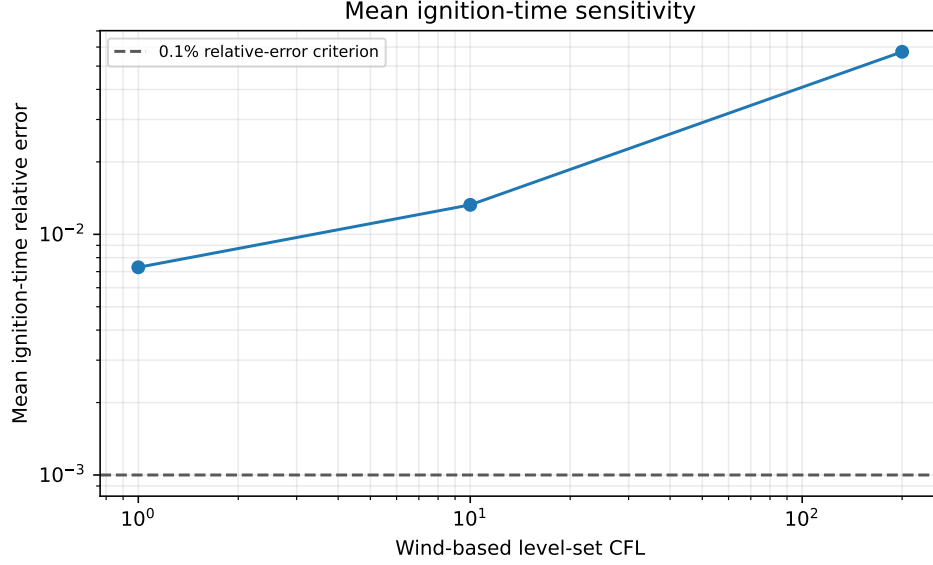


Figure 4: Mean full-ignition-time relative error referenced to CFL=0.5. The dashed line is the 0.1% acceptance level used for CFL=1 and 10.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Output completeness	4 variants	4/4	PASS
CFL 0.5 analytical-load mean error	≤ 0.01	0.54345	FAIL
CFL 1 analytical-load maximum error	≤ 0.001	0.82701	FAIL
CFL 10 analytical-load maximum error	≤ 0.001	0.8269	FAIL
CFL 200 analytical-load maximum error	≤ 0.1	0.83927	FAIL
CFL 1 ignition-time mean relative error	≤ 0.001	0.0072949	FAIL
CFL 10 ignition-time mean relative error	≤ 0.001	0.013258	FAIL
CFL 1 accumulation-history NRMSE	≤ 0.05	0.0084659	PASS
CFL 10 accumulation-history NRMSE	≤ 0.05	0.0084425	PASS

The current decision is **FAIL**. A pass requires all four current-fingerprint runs and all tabled acceptance checks. A not-run state is not evidence of agreement.

This is a deterministic code-verification test of the implemented idealized WUI model, not field validation. The analytical load uses simulated source ignition times so that it isolates generation and deposition from ignition chronology. First-deposition timing is limited by transient output cadence because the current executable does not write the internal `EMBER.TOA` array as a final raster; that diagnostic is therefore intentionally excluded from the decision contract.

8 Technical interpretation

With the declared building-only switch restored, all four current-source variants produce nonzero structural heat release, firebrand transport, fingerprint-matched final rasters, and a complete metric record. This invalidates the earlier zero-HRR and missing-output interpretations, both of which arose while diagnosing the stale executable or the temporary altered switch.

The temporal history is comparatively stable at moderate step sizes: the CFL 1 and CFL 10 accumulation-history NRMSE values are 0.008466 and 0.008443, both below 0.05. The stricter pointwise/reference components do not pass. The analytical-load errors are 0.54345 for CFL 0.5, 0.82701 for CFL 1, 0.82690 for CFL 10, and

0.83927 for CFL 200, all above their respective limits. The CFL 1 and CFL 10 mean ignition-time relative errors are 0.007295 and 0.01326, compared with the 0.001 limit. The overall result is therefore *FAIL*, not *NOT RUN* or *NOT EVALUABLE*.

Because every input fingerprint, final output, and integer timestep/stop relation is valid, these discrepancies are not caused by an admissible case-configuration error. They identify disagreement between the implemented temporal/structural spotting calculation and the predeclared analytical-load and ignition-time requirements.

9 Scope and limitations

The test varies only temporal resolution in a fixed academic WUI configuration. It verifies timestep sensitivity for the selected transport and ignition scales and does not establish spatial convergence or physical validation.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 13: WUI Parametric Response Maps

ELMFIRE Spotting Verification Guide

September 2, 2026

1 Verification purpose

Case ID: CASE13_PRM

This case defines a parametric verification of the resolution-independent WUI response to generation rate, inter-structure separation, and wind speed. The two controlled response families contain 270 requested points with duplicate baselines removed.

The response test is valid only when the integrated-structure capability is available. The current source gates that capability, so this case verifies design completeness and reports the limitation explicitly instead of substituting the native cell-local model.

2 Mathematical formulation and numerical implementation

Each physical structure is a square of side $a = 10$ m. Although the computational grid uses $\Delta x = 2.5$ m, all cells belonging to one structure must act as one source and one collector. For physical structure s , the required formulation is

$$\text{HRR}_s(t) = q''(t)a^2, \quad \dot{N}_s(t) = GR \times \text{HRR}_s(t), \quad N_s(t) = \sum_{j \in s} N_j(t).$$

Transport is evaluated once for the structure-scale HRR, deposition is accumulated over its physical footprint, and the resulting load N_s/a^2 drives the physical ignition criterion. This grouping is the key difference from the native multiple-discrete-cell treatment, in which one physical structure is fragmented into independently emitting and igniting raster cells.

The level-set time step is

$$\Delta t = 0.5 \frac{\Delta x}{U_{\text{wind}}},$$

and surface spread is disabled. Spread success therefore means that the firebrand process alone propagates through the structural array. For each parameter point, mean rate of spread is obtained by fitting structure position against first-ignition time.

The first family varies

$$d = 2.5, 5, \dots, 30 \text{ m}, \quad GR = 5, 10, \dots, 50 \text{ pcs MW}^{-1} \text{ s}^{-1},$$

with a fixed 40 mph wind. The second fixes the separation at 10 m, uses the same generation rates, and varies wind speed from 10 to 150 mph in 10 mph increments.

3 Derivation of expected behavior

Increasing separation moves targets into lower-probability transport tails, while increasing generation raises deposited load. Wind produces competing reach and ignition-cooling effects, so the critical generation boundary should rise at both low and high wind speeds.

The design count follows directly from the generated parameter products:

$$12 \text{ separations} \times 10 \text{ GR} = 120, \quad 15 \text{ winds} \times 10 \text{ GR} = 150, \quad N_{\text{design}} = 120 + 150 = 270.$$

The two baseline-equivalent coordinates remain distinct members of their respective response families. At the design-fire plateau, a physical structure has $HRR_{\max} = 400(100) = 40$ MW; consequently the tested generation range corresponds to

$$\dot{N}_{\max} = GR(40) = 200, 400, \dots, 2000 \text{ pcs s}^{-1}.$$

With $\Delta x = 2.5$ m and $CFL = 0.5$, the generated level-set step is $0.5(2.5)/[U_{\text{mph}}(0.44704)]$: it is 0.069904 s at 40 mph, 0.279617 s at 10 mph, and 0.018641 s at 150 mph.

For each coordinate, GR_c is selected from $\{5, 10, \dots, 50\}$; if no point propagates, the rank-only sentinel is the next design level, 55. Thus a one-level wind-edge excess is $55 - 50 = 5$ pcs MW⁻¹ s⁻¹, exactly the acceptance increment. The moderate set is 30–120 mph and the edge set is $\{10, 20, 130, 140, 150\}$ mph. The explicit wind metrics are

$$\overline{GR}_{c,\text{edge}} - \text{median}(GR_{c,\text{moderate}}) \geq 5, \quad \frac{\overline{ROS}_{\text{moderate}}}{\overline{ROS}_{\text{edge}}} \geq 1.10.$$

The separation expectation is strong positive rank association of d with GR_c , negative rank association of d with ROS, and positive rank association of GR with ROS. These become $\rho \geq 0.85$, median $\rho \leq -0.50$, and median $\rho \geq 0.70$, respectively, with the independent response-scale ceiling $\max ROS \leq 0.08 \text{ m s}^{-1}$.

4 Verification metrics and acceptance criteria

For a coordinate value z , define the critical generation rate

$$GR_c(z) = \min\{GR : \text{the array sustains downwind spread}\}.$$

When no tested value succeeds, the postprocessor assigns the next design level, 55 pcs MW⁻¹ s⁻¹, only for rank and boundary metrics; it does not reinterpret failure as an observed threshold.

For the separation family, the documented response predicts a nonlinear increase of GR_c with d , from approximately 2.5 to more than 30 pcs MW⁻¹ s⁻¹ [1, 2]. ROS should decrease with separation and increase with generation. The verification therefore requires Spearman rank correlation $\rho(d, GR_c) \geq 0.85$, median $\rho(d, ROS) \leq -0.50$, and median $\rho(GR, ROS) \geq 0.70$.

For the wind family, the critical boundary is U-shaped. Low wind gives insufficient reach and a high ignition threshold; very high wind improves transport but suppresses the smoldering-to-flaming transition through convective cooling. The mean critical generation rate outside 30–120 mph must exceed the median within that interval by at least 5 pcs MW⁻¹ s⁻¹. Mean ROS in the moderate range must be at least 1.10 times the edge-wind mean. Finally, firebrand-driven ROS must remain below 0.08 m s⁻¹, consistent with the documented response scale [1, 2].

4.1 Justification of metric quantification

The rank limits quantify robust trends without assuming that the nonlinear responses are linear. A separation-threshold Spearman coefficient of 0.85 requires strong ordering across twelve separation levels while allowing a small number of adjacent ties or reversals near a discrete ignition boundary. The -0.50 ROS/separation and 0.70 ROS/generation limits require, respectively, at least a moderate negative and a strong positive monotone association despite flat-response regions. They therefore test the direction and persistence of the mechanisms rather than a fitted slope chosen for one parameter interval.

The 5 pcs MW⁻¹ s⁻¹ wind-edge excess equals one full generation-rate design increment, so the U-shaped boundary must be resolved by the experiment rather than inferred from sub-grid interpolation. The 1.10 moderate/edge ROS ratio requires a 10% effect beyond equality. The 0.08 m s⁻¹ ceiling is a scale check taken from the reported firebrand-driven response range for this design [1, 2]. All 270 points and the resolution-independent structure capability are mandatory because missing edge or threshold points can materially change both rank statistics and the inferred U shape.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Design completeness	Unique generation–separation and generation–wind points after baseline deduplication.	270 of 270 points
Required capability	Resolution-independent single-structure source, collector, and input adapter.	Available
Result completeness	Current admissible result rows divided by 270.	100%
Threshold/separation rank	Spearman $\rho(d, GR_c)$.	≥ 0.85
ROS/separation rank	Median Spearman $\rho(d, ROS)$ across generation levels.	≤ -0.50
ROS/generation rank	Median Spearman $\rho(GR, ROS)$ across both response families.	≥ 0.70
Wind-edge threshold excess	Mean critical generation outside 30–120 mph minus the median inside that range.	$\geq 5 \text{ pcs MW}^{-1} \text{ s}^{-1}$
Moderate/edge ROS ratio	Mean ROS at 30–120 mph divided by mean ROS outside that range.	≥ 1.10
Maximum ROS	Maximum firebrand-driven ROS over all admissible design points.	$\leq 0.08 \text{ m s}^{-1}$
Overall decision	Capability, complete results, and every trend/scale metric are conjunctive.	PASS only if all pass

5 Simulation configuration and predicted outcome

Although execution remains capability-gated, every design point now carries an auditable spatial configuration. Structures are rasterized as a periodic two-dimensional lattice over a 220×100 -m physical domain. The generation–separation family applies its selected gap in both coordinate directions; the generation–wind family uses the 10 m baseline gap. Every footprint in the first upwind structure column is marked by the non-positive initial level-set field.

Figure 1 shows the baseline-separation geometry. The two-cell numerical halo is excluded. The left panel verifies repetition in both directions, while the right panel verifies that ignition spans the entire first structure column instead of selecting only the lower-left structure.

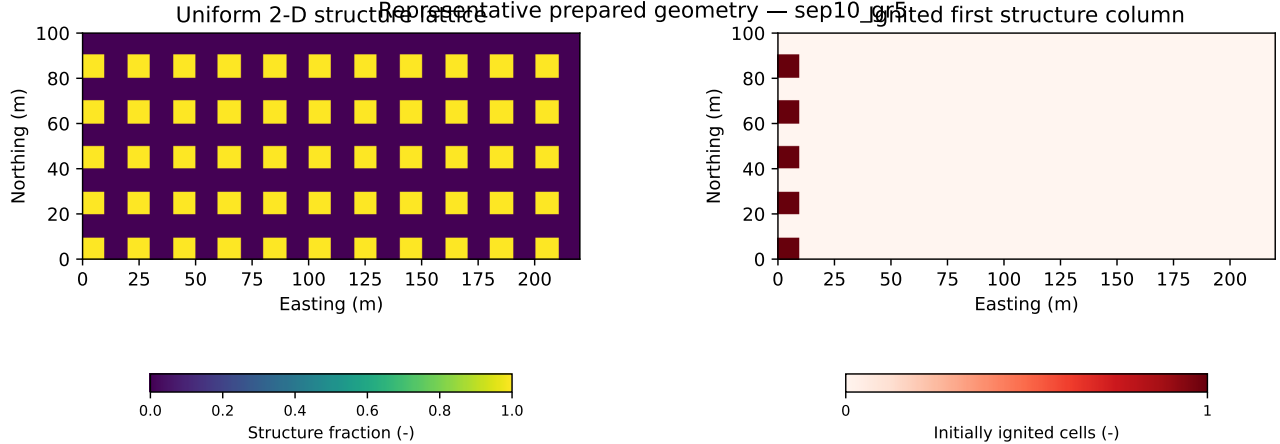


Figure 1: Representative prepared two-dimensional structure lattice and first-column initial ignition mask for the capability-gated parametric design.

Table 2: Controlled parametric design.

Parameter	Separation family	Wind family
Grid spacing	2.5 m	2.5 m
Physical domain	220×100 m	220×100 m
Structure side	10 m	10 m
Separation	2.5–30 m, step 2.5 m	10 m
Wind	40 mph	10–150 mph, step 10 mph
Generation	5–50, step 5	5–50, step 5
Spatial treatment	single physical unit	single physical unit
Initial ignition	complete first structure column	complete first structure column
CFL based on wind	0.5	0.5
Surface spread	disabled	disabled
Number of points	120	150

Expected ignition classifications form an increasing boundary in the (d, GR) plane and a U-shaped boundary in the (U_{wind}, GR) plane. The ROS maps should be largest for high production, short separation, and moderate wind. Hamada-model curves included in the documented parameter comparison [1, 2] are contextual because Hamada includes radiation and flame contact; they are not acceptance references for this firebrand-only test.

5.1 Preprocessing, execution, and postprocessing

`scripts/preprocess.py` places customizable filenames and design constants at its beginning, enumerates all 270 points into `outputs/parameter_design.csv`, writes each variant’s co-registered structure-fraction, structure-ID, fuel-model, and initial-level-set rasters, and records their fingerprints in the capability manifest. It scans the current Fortran source for `USE_RESOLUTION_INDEPENDENT_STRUCTURES`. The case refuses to mark a point runnable unless that source capability and a matching case adapter both exist. This guard is necessary because running the native cell-local model would reproduce the cell-local discretization defect rather than the intended resolution-independent experiment.

`scripts/postprocess.py` accepts exact results in `outputs/parametric_results.csv` with the schema declared in `case.json`. It checks completeness, derives critical boundaries, calculates all rank and response metrics, and creates the two classification maps and two ROS panels. Missing results remain *NOT EVALUABLE*. `run_case.sh` prepares the complete design, postprocesses the capability state, and compiles this standalone report without launching an inadmissible substitute.

6 Actual simulation results

No whole-domain ELMFIRE result is available because the required integrated-structure capability is absent and no variant is runnable under the current source interface. The postprocessor therefore does not manufacture a raster or substitute an analytical field for model output. The case remains NOT EVALUABLE, and the response map below is reported only as the prescribed verification envelope.

0 of 270 exact-design result points are present. The decision is **NOT EVALUABLE**. Figure 2 shows only the expected qualitative boundaries while the capability is absent. Every panel is labeled accordingly; no guide curve is represented as an ELMFIRE result.

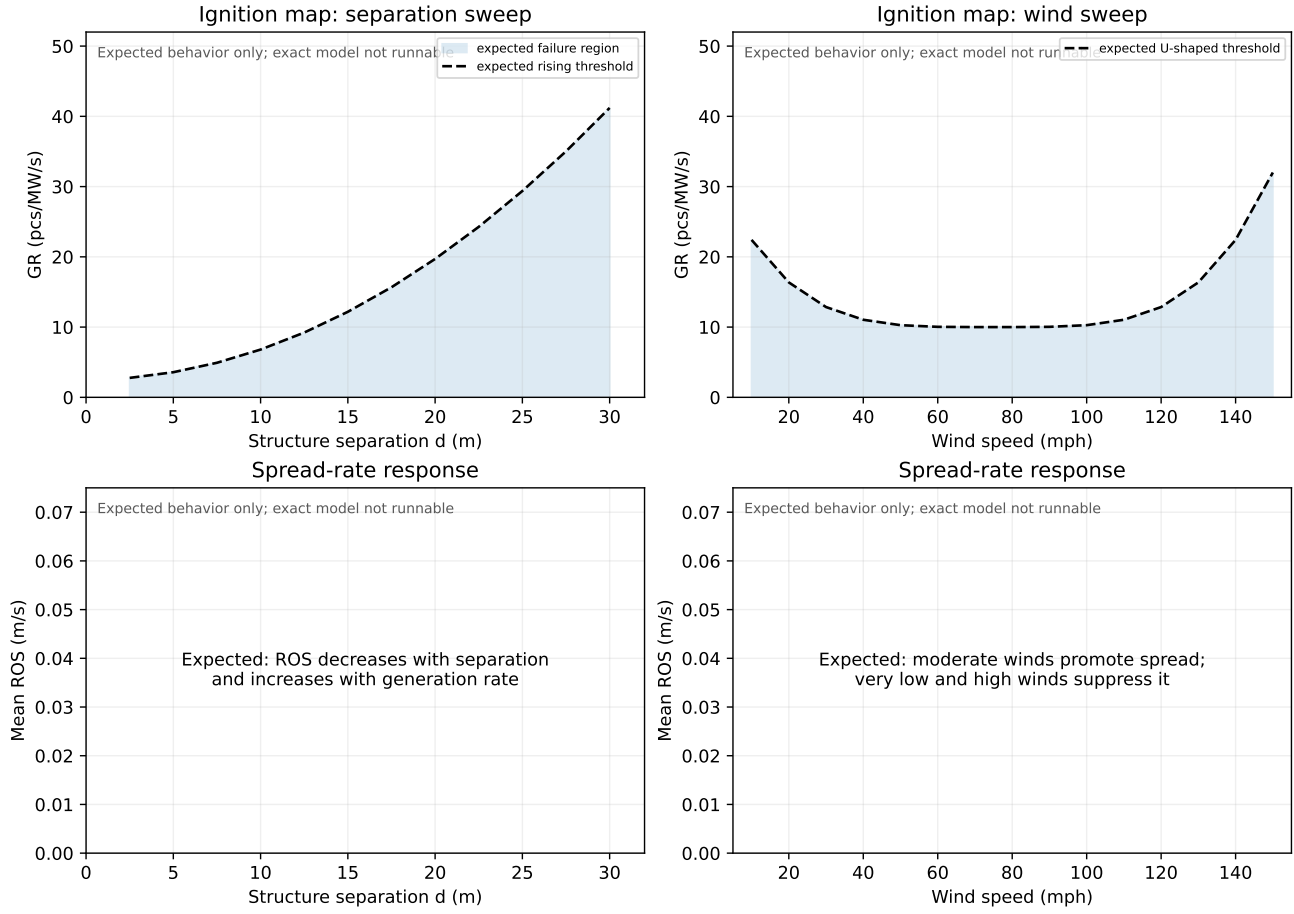


Figure 2: Parametric-response verification layout. The upper panels reproduce the coordinate systems of the two parametric reference maps; the lower panels are reserved for the corresponding ROS curves. In the current capability-gated state, shaded regions and dashed lines communicate expected behavior only.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Parameter-design completeness	270/270	270	PASS
Exact single-structure capability	available	N/A	NOT EVALU- ABLE
Current result completeness	100%	0	NOT EVALU- ABLE
Critical GR versus separation Spearman rho	≥ 0.85	N/A	NOT EVALU- ABLE
ROS versus separation median Spearman rho	≤ -0.50	N/A	NOT EVALU- ABLE
ROS versus generation median Spearman rho	≥ 0.70	N/A	NOT EVALU- ABLE
Low/high-wind threshold excess	≥ 5 pcs/MW/s	N/A	NOT EVALU- ABLE
Moderate/edge-wind ROS ratio	≥ 1.10	N/A	NOT EVALU- ABLE
Maximum firebrand-driven ROS	≤ 0.08 m/s	N/A	NOT EVALU- ABLE

The current decision is **NOT EVALUABLE**, not a failure of the firebrand equations. The intended resolution-independent parameter sweep cannot be executed because the current source does not implement the single-structure source and collector required by the cited model description [1]. The native model would split each 10-m structure into sixteen 2.5-m cells and change HRR, transport, and accumulation; those outputs are not admissible evidence for this case.

The complete parameter design, expected behavior, result schema, metrics, figure generator, and standalone report are now present. Once the source feature and its input adapter are implemented, the capability gate should be enabled, all 270 points executed, and the same postprocessor used without changing acceptance limits.

8 Technical interpretation

The capability gate prevents cell-local structural discretization from being presented as evidence for a single-structure parametric model. Once admissible results exist, boundary and rank metrics distinguish physical response trends from isolated successful points.

9 Scope and limitations

The design covers idealized, uniform arrays and verifies qualitative model response over the declared parameter grid. It remains not evaluable until the required single-structure source and collector are implemented.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland-Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.
- [2] Y. Qin et al., “Simulations of firebrand-driven fire spread in landscape-scale Wildland-Urban-Interface and urban conflagration models,” *Fire Safety Journal*, vol. 162, 104686, 2026, doi:10.1016/j.firesaf.2026.104686.

Case 14: Two-Dimensional Spatial-Resolution Sensitivity

ELMFIRE Spotting Verification Guide

September 3, 2026

1 Verification purpose

Case ID: CASE14_SRS

This case is a negative verification test for mesh dependence in the current multiple-discrete-cell treatment of structures. A physical 10 m-by-10 m house may occupy one raster cell, several independent cells, one oversized cell, or disappear under categorical resampling as the mesh changes. Because those cells are not assembled into one conserved structure object, changing resolution alters represented heat release, firebrand generation, deposition, and structure-to-structure ignition.

The test asks whether all 15 current-fingerprint simulations produce an evaluable community rate of spread (ROS), and whether the fitted values remain within the predeclared mesh-independence limits relative to the run-derived 10-m anchor. Detecting the known resolution sensitivity is useful diagnostic evidence, but it is a *verification failure*, not a pass.

2 Mathematical formulation and numerical implementation

2.1 Mode-resampled square array

The physical layout is a periodic array of square structures of side $a = 10$ m, separated in both coordinates by $d = 10$ m. Define the one-dimensional structural indicator

$$I(s) = \begin{cases} 1, & s \bmod (a + d) < a, \\ 0, & \text{otherwise.} \end{cases}$$

For grid interval $C_i = [i\Delta, (i + 1)\Delta]$, preprocessing calculates

$$f_i = \frac{1}{\Delta} \int_{C_i} I(s) ds, \quad b_i = \mathbf{1}(f_i \geq 1/2), \quad B_{ij} = b_i b_j.$$

This coordinate-wise categorical-mode construction reproduces the reported geometries. At $\Delta = 7$ m, effective side lengths are 7 or 14 m and footprints range from 49 to 196 m². At $\Delta = 10$ m, footprint and gap are exact. At $\Delta = 30$ m, one represented structure cell has area 900 m² and represented structures are separated by 30 m. The 20- and 40-m maps eliminate represented firebreaks; this is an intended resampling diagnostic, not a preprocessing failure.

2.2 Cell-local firebrand treatment

A represented structural cell uses a design-fire curve $q''(t)$. The curve has a 300-s linear growth phase, a one-hour plateau at 400 kW m⁻², and a 300-s decay. Under the current multiple-discrete-cell treatment,

$$\text{HRR}_{ij}(t) = q''(t)\Delta^2, \quad \dot{N}_{ij}(t) = GR \frac{\text{HRR}_{ij}(t)}{1000}, \quad GR = 10 \text{ pcs MW}^{-1} \text{ s}^{-1}.$$

Changing Δ therefore changes both represented geometry and the HRR of each independent emitter. Firebrands use the structural empirical flight-distance distribution, Eulerian accumulation, and physical ignition. Non-structure cells use nonburnable fuel model 93, so vegetative surface spread is absent and every downwind ignition must be caused by deposited firebrands. The global `NO_SURFACE_FIRE` switch remains true to suppress vegetative

surface-fire propagation. The current source independently refreshes initialized model-14 building transient HRR under this setting. The configured time-step scaling is documented by Qin et al. [2]:

$$\Delta t = \text{CFL} \frac{\Delta}{U_{\text{wind}}}, \quad \text{CFL} = 1, \quad U_{\text{wind}} = 17.9 \text{ m s}^{-1} \text{ (40 mph)}.$$

The fractional step is required to hold this spatial sweep at fixed wind CFL. Preprocessing sets each generated stop to $\lceil 150000/\Delta t \rceil \Delta t$, then records the requested stop, aligned stop, and integer step count in the manifest.

For each represented downwind structure column s , postprocessing takes the median positive finite arrival time t_s . At least three ignited columns are required. Mean community ROS is the least-squares slope

$$ROS_{\Delta} = \frac{\sum_s (t_s - \bar{t})(x_s - \bar{x})}{\sum_s (t_s - \bar{t})^2}.$$

Because the 10-m grid exactly resolves both physical length scales, its fitted ELMFIRE result is the comparison anchor,

$$ROS_{10} = ROS_{\Delta=10 \text{ m}}.$$

The dashed reference line and the 10-m ELMFIRE point therefore use the same calculated value; no rounded external estimate is substituted.

3 Derivation of expected behavior

A resolution-independent structure treatment would conserve each house’s physical footprint, integrated heat release, firebrand source, accumulated load, and ignition state. Subject to numerical error, it would give $ROS_{\Delta} \approx ROS_{10}$ for every resolution and enough downstream ignitions to fit all 15 ROS values.

The current cell-local representation does not conserve those quantities. Fine meshes fragment houses into multiple emitters, intermediate meshes alter footprints and gaps, and some coarse meshes merge structures or remove firebreaks. The expected diagnostic result is therefore a nonuniform ROS curve, including possible failure to propagate at some resolutions. That observation exposes the defect; against the mesh-independence requirement, its correct overall status is FAIL.

The period of the one-dimensional physical indicator is $a + d = 10 + 10 = 20 \text{ m}$. The key raster substitutions are therefore:

$\Delta \text{ (m)}$	mode-resampled consequence	component-area reference
7	one or two selected cells per coordinate	$7^2 = 49 \text{ to } (2 \cdot 7)^2 = 196 \text{ m}^2$
10	one structure cell followed by one gap cell	$10^2 = 100 \text{ m}^2$
30	one selected cell per represented structure	$30^2 = 900 \text{ m}^2$

At $\Delta = 20 \text{ m}$, each cell contains 10 m of structure and 10 m of gap, so $f_i = 0.5$ and the $f_i \geq 0.5$ rule selects every cell; the represented firebreak vanishes. The same 50-percent selection occurs for the 40-m map. These are reproducible consequences of the generated mode rule, not inferred from the output fire spread.

With $U = 40(0.44704) = 17.8816 \text{ m s}^{-1}$, each generated step is

$$\Delta t = \Delta/U; \quad \Delta t_7 = 0.391463, \quad \Delta t_{10} = 0.559234, \quad \Delta t_{30} = 1.677702 \text{ s}.$$

A truly resolution-independent treatment would give all 15 finite fits, $f_{\text{fit}} = 15/15 = 1$, $ROS_{\Delta} = ROS_{10}$, $CV_R = 0$, and $D_{\text{max}} = 0$. The current native cell treatment is expected instead to produce missing or nonuniform fits because source HRR scales as Δ^2 and the represented geometry changes. The predeclared mesh-independence requirements permit $CV_R \leq 0.25$ and $D_{\text{max}} \leq 0.50$; exceeding either limit or failing any of the 15 fits yields the expected diagnostic FAIL.

The primary publication provides the independent propagation oracle used only for strict stalled-final compatibility: Figure 9 reports zero ROS at 1, 3, 7, 13, 17, 23, 25, and 27 m, and finite propagation at 5, 10, 15, 20, 30, 40, and 50 m [2]. An early stalled-final dump may therefore use the no-propagation exception only for the first set, and only after the meteorology-interval reconstruction and timestep-grid checks also pass. This oracle does not replace the current-run metrics or authorize accepting an arbitrary overrun timestamp.

4 Verification metrics and acceptance criteria

Preprocessing passes only when all 15 resolutions are generated, the exact 10-m map has no footprint or gap error, the 7-m footprint range is 49–196 m², and the median 30-m footprint is 900 m². These checks establish that runtime comparison uses the intended resolution-dependent rasterizations.

A complete runtime decision additionally requires every current-fingerprint variant and a finite 10-m ROS anchor. Let N_{fit} be the number of resolutions with at least three ignited downwind structure columns. ROS extraction completeness is

$$f_{\text{fit}} = \frac{N_{\text{fit}}}{15},$$

and must equal one. For the finite fitted values, mesh dependence is measured by the coefficient of variation

$$CV_R = \frac{\text{std}_{\Delta}(ROS_{\Delta})}{\text{mean}_{\Delta}(ROS_{\Delta})}$$

and by the largest deviation from the exact-geometry anchor,

$$D_{\text{max}} = \max_{\Delta} \frac{|ROS_{\Delta} - ROS_{10}|}{ROS_{10}}.$$

The permissive engineering limits are $CV_R \leq 0.25$ and $D_{\text{max}} \leq 0.50$. Missing fits are handled by f_{fit} ; they are not silently discarded and counted as convergence. The overall decision is the conjunction of every geometry, runtime, anchor, fit-completeness, and mesh-independence check.

4.1 Justification of metric quantification

The geometry limits are exact consequences of rasterization. At $\Delta = 10$ m, the structure side and gap are both exactly one cell, so the 1% footprint and separation allowances cover only file precision and connected-component bookkeeping. At 7 m, a represented component can contain one to four cells, giving areas from $7^2 = 49$ to $4 \times 7^2 = 196$ m². At 30 m, a one-cell represented structure has area $30^2 = 900$ m². These checks establish that the intended mesh-dependent maps were generated; they are not claims of geometric accuracy.

For the behavioral requirement, $f_{\text{fit}} = 1$ is non-negotiable because mesh independence cannot be inferred after silently discarding resolutions that fail to propagate. The $CV_R \leq 0.25$ and $D_{\text{max}} \leq 0.50$ limits are intentionally permissive engineering bounds, substantially larger than ordinary front-fit quantization. Even with those allowances, a method with factor-of-two variation or missing fits fails, which is the defect this case is designed to expose [2]. The 10-m reference is the actual ROS_{10} extracted by the same postprocessor, so its plotted reference and ELMFIRE point must coincide by construction; it is not an independent nominal value. Current fingerprints and all 15 runs are required to prevent stale or selectively complete outputs from weakening the expected FAIL diagnosis.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Prepared variants	Generated manifest count divided by the 15 requested grid spacings.	100%
10-m footprint error	Maximum relative error of represented component areas from 100 m ² .	≤ 0.01
10-m separation error	Maximum relative gap-width error from 10 m.	≤ 0.01
7-m component areas	Minimum and maximum represented connected-component areas.	49–196 m ²
30-m footprint error	Relative error of median represented footprint from 900 m ² .	≤ 0.01
Runtime completeness	Current-fingerprint final TOA output and successful termination.	15 of 15 variants
10-m ROS anchor	Structure-column fit from the current 10-m TOA raster.	Finite ROS_{10}
ROS extraction completeness	Resolutions with at least three ignited downwind columns.	$f_{\text{fit}} = 1$
ROS mesh-dependence CV	Coefficient of variation over finite fitted ROS values.	$CV_R \leq 0.25$
Maximum anchor deviation	Maximum finite-grid relative deviation from ROS_{10} .	$D_{\text{max}} \leq 0.50$
Overall decision	Every geometry, completeness, and ROS component is conjunctive.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for **dx10**. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

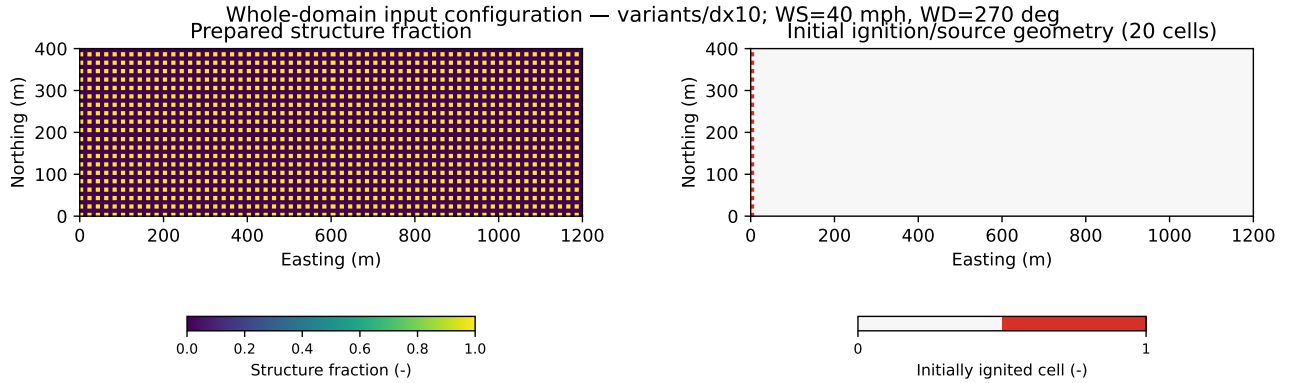


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Configuration of the two-dimensional resolution sweep.

Parameter	Configured value	Units	Role
Physical domain	1200 × 400, flat	m	permits many downwind ignitions
Structure side / gap	10 / 10	m	exact physical scales at the 10-m grid
Initial ignition	complete first structure column	–	uniform upwind source across all structure rows
Grid spacings	1, 3, 5, 7, 10, 13, 15, 17, 20, 23, 25, 27, 30, 40, 50	m	exposes raster and cell-local dependence
Numerical halo	2 on every side	cells	excluded from physical geometry
Wind	40 from 270 degrees	mph	constant west-to-east transport
Time step	Δ/U_{wind}	s	holds wind CFL at one
Design-fire peak	400	kW m ⁻²	structural HRRPUA input
Generation rate	10	pcs MW ⁻¹ s ⁻¹	cell-local building source
Transport / accumulation	empirical / Eulerian	–	current structural spotting path
Ignition delays	42 plus 300	s	small-flame and development stages
Vegetative surface spread	excluded by fuel 93	–	isolates firebrand-driven propagation
Stop time	at least 150000	s	rounded up to an exact timestep multiple
Final output	one TOA per variant	–	source for structure-column ROS fits

The dashed reference line is predicted to intersect the 10-m ELMFIRE point exactly because both are ROS_{10} . A mesh-independent implementation would cluster all other finite ROS points around this line and provide a fit at every resolution. The current independent-cell treatment is instead expected to produce missing fits and large variations, causing the behavioral verification to fail.

5.1 Preprocessing, execution, and postprocessing

`scripts/preprocess.py` places customizable constants at its beginning, constructs the periodic map analytically, assigns persistent structure IDs, writes co-registered GeoTIFF inputs and case-local fuel tables, and records a SHA-256 fingerprint for every variant. It also writes `outputs/geometry_summary.csv`, making the map checks independently auditable.

`run_case.sh` is executable within the case directory and follows the same linear workflow as the other manifest cases. Configure `ELMFIRE_BIN` before launching it; the runner executes all 15 variants sequentially, keeps logs and fingerprints variant-local, postprocesses the outputs, and compiles the report. Because the sweep is long, execution should be scheduled only when adequate compute time is available.

`scripts/postprocess.py` rejects stale or incomplete outputs, reads the final time-of-arrival raster, groups cells by structure, fits ROS, calculates all metrics above, and produces the report figure.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from `dx10`. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. The 10-m field is the reference-scale spatial result used to interpret mesh-dependent structure spread across the resolution sweep.

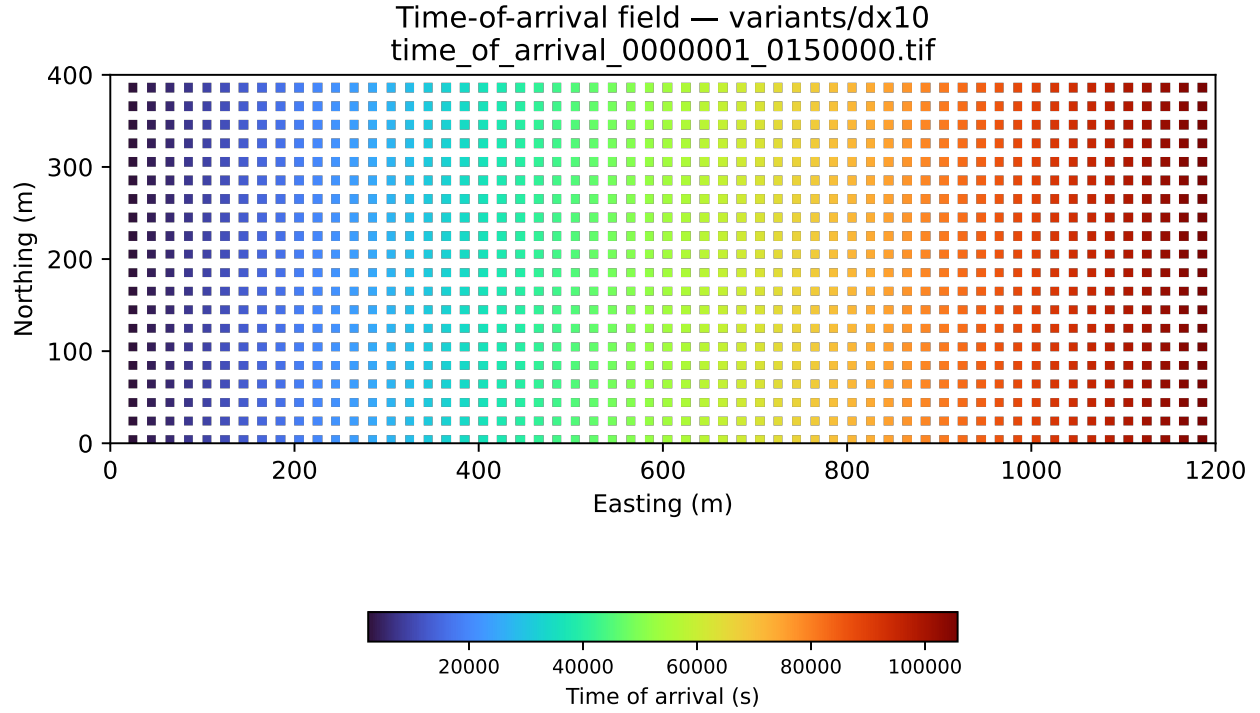


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

All 15 current-fingerprint ELMFIRE processes reached the source success message. The strict terminal-evidence selector accepts 11 of 15 outputs; process exit alone is not used as evidence. Regular terminal dumps are accepted for 1, 3, 5, 7, 10, 20, 30, and 40 m, and the publication-derived no-propagation oracle permits the timestep-aligned stalled-final path for 13, 17, and 25 m. The selected files, configured controls, final times, reconstructed pre-jump times, grid residuals, and rejection reasons are recorded per variant in `outputs/metrics.json`. The overall decision calculated by postprocessing is **NOT EVALUABLE**.

The upper row of Figure 3 shows the actual preprocessed geometries. Its lower panel contains fitted ELMFIRE ROS values. Red crosses at zero identify completed simulations with fewer than three ignited downwind columns, for which ROS is deliberately not fitted. The dashed line is $ROS_{10} = 0.011226 \text{ m s}^{-1}$, exactly equal to the 10-m ELMFIRE point.

Two-dimensional spatial-resolution verification

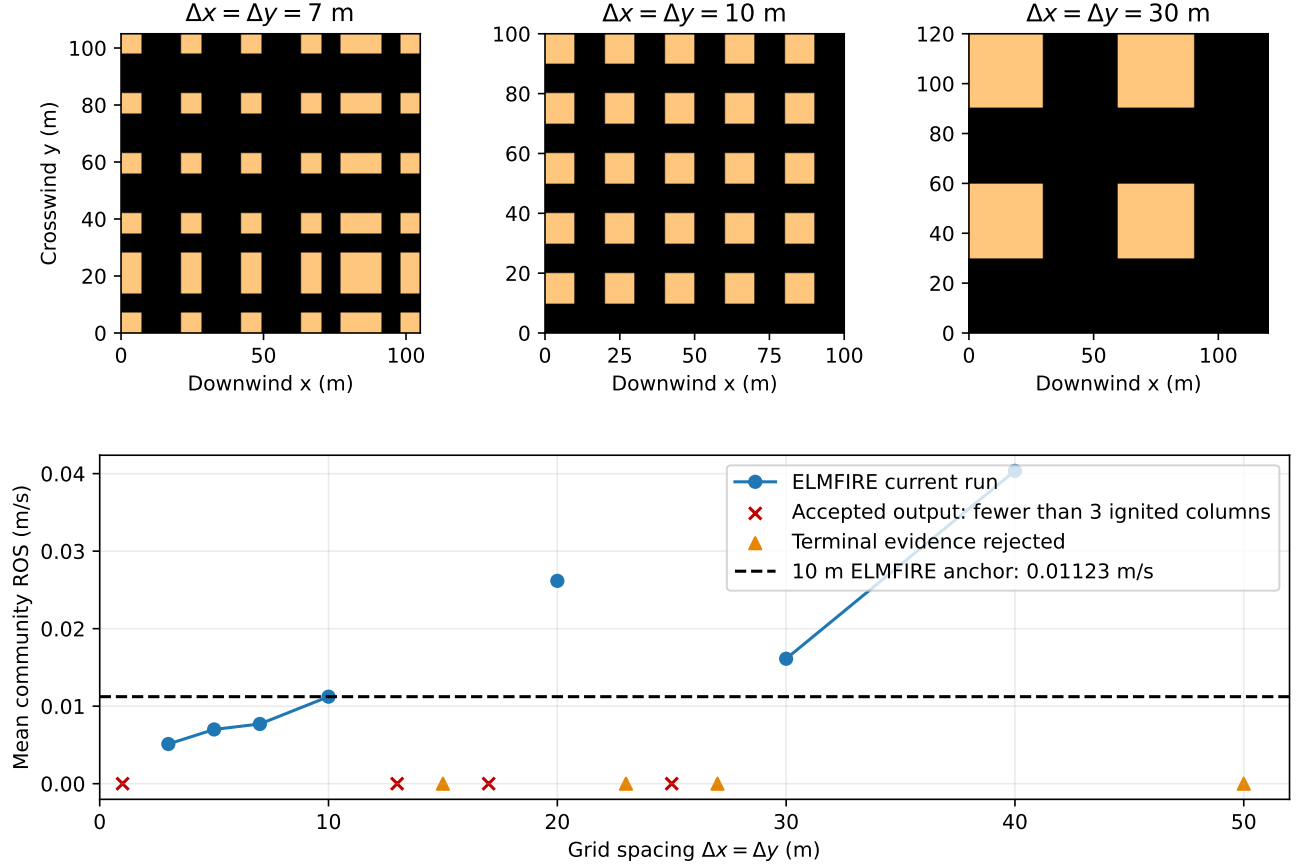


Figure 3: Two-dimensional spatial-resolution diagnostic. Top: mode-resampled WUI geometry at 7, 10, and 30 m. Bottom: mean community ROS from current ELMFIRE TOA outputs; red crosses identify resolutions without enough downwind ignitions for a fit, and the dashed line is the run-derived 10-m anchor. The geometry selection follows the documented comparison configuration [2].

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Prepared resolution variants	$\geq 100\%$	1	PASS
dx=10 m footprint error	≤ 0.01	0	PASS
dx=10 m separation error	≤ 0.01	0	PASS
dx=7 m component-area range	49 to 196 m ²	49 to 196	PASS
dx=30 m median footprint error	≤ 0.01	0	PASS
Current-fingerprint run completeness	15 variants	11/15	NOT EVALUABLE
10-m ELMFIRE ROS anchor	finite fitted ROS	N/A	NOT RUN
ROS-evaluable resolution fraction	$\geq 100\%$	N/A	NOT RUN
ROS mesh-dependence CV	≤ 0.25	N/A	NOT RUN
Maximum ROS deviation from 10-m anchor	≤ 0.50	N/A	NOT RUN

The postprocessor reports **NOT EVALUABLE**. Geometry preparation passes, and all 15 processes executed, but strict terminal evidence is complete for only 11 of 15 variants. The 10-m anchor, ROS-evaluable fraction, coefficient of variation, and maximum anchor deviation therefore remain N/A; calculating them from a selected subset would silently change the predeclared verification population.

8 Technical interpretation

The accepted subset contains finite fitted values of approximately 0.00512 m s^{-1} (3 m), 0.00700 m s^{-1} (5 m), 0.00770 m s^{-1} (7 m), 0.01123 m s^{-1} (10 m), 0.02617 m s^{-1} (20 m), 0.01612 m s^{-1} (30 m), and 0.04036 m s^{-1} (40 m). These values visibly demonstrate mesh sensitivity, but they are diagnostic observations rather than a complete acceptance calculation.

Four successful processes do not provide admissible terminal evidence. For 15 m, the recorded final time is 152833.592 s and subtracting the configured 3600-s meteorology interval gives 149233.592 s; the timestep-grid residual is 0.302 s. For 23 and 27 m the corresponding reconstructed times are 146507.060 and 148242.909 s, with grid residuals 0.636 and 0.705 s. Those residuals exceed the 0.150-s single-precision-scale tolerance. The 50-m reconstruction is grid-aligned (0.023-s residual), but occurs at 148236.198 s, more than one 2.796-s timestep before the configured stop. The primary publication reports propagation at 15 and 50 m, so neither early stall may be reclassified as independently expected no propagation. Conversely, the publication reports no propagation at 23 and 27 m, but that oracle does not waive the timestep-grid requirement.

The current source also produces finite spread at 3 and 7 m while the published Figure 9 places those resolutions at zero, and it stalls at 15 and 50 m where that figure shows finite spread. The resulting **NOT EVALUABLE** status is therefore an implementation/reference and terminal-time discrepancy, not a missing execution and not a reason to weaken the evidence selector. The runs used executable SHA-256 `c225b2e00349aa98128285b3d3ff9b2031d0e20944cd9f1c5ee96595d199a920` built from source commit `a2dfbcd72209733c000e5d3431e44723e15efea`; the source worktree already contained unrelated local modifications, so the commit and executable digest are both reported for reproducibility.

9 Scope and limitations

This case verifies numerical mesh dependence for one idealized periodic WUI array, one wind, and one structural design fire. It does not validate the physical accuracy of any individual ROS value. The 10-m result is an internal run-derived anchor, not experimental validation data. A future resolution-independent structure implementation should be tested with the same geometry and acceptance contract; it should provide all 15 ROS fits and reduce both variation metrics below their limits.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.
- [2] Y. Qin et al., “Simulations of firebrand-driven fire spread in landscape-scale Wildland–Urban-Interface and urban conflagration models,” *Fire Safety Journal*, vol. 162, 104686, 2026, doi:10.1016/j.firesaf.2026.104686.

Case 15: Constant rate of spread

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE15_CRO. This unit test verifies that ELMFIRE transports a circular level-set front at a prescribed, spatially uniform normal rate. It exercises raster ingestion, the **ADJ** spread-rate multiplier, the level-set update, and **phi/vs** output without wind, slope, crown fire, spotting, or suppression. It can reveal rate-unit errors, incorrect signs, and failure to couple the calculated rate to front motion. This is code verification against a known solution, not comparison with a physical fire. The verification question is whether the final spread-rate and radius satisfy the two stated tolerances at $t = 360$ s.

2 Mathematical formulation and numerical implementation

ELMFIRE advances the signed level set by

$$\frac{\partial \phi}{\partial t} + V_n |\nabla \phi| = 0,$$

with $\phi < 0$ burned and $\phi = 0$ the front. Uniform rasters and disabled directional forcing make $V_n = R_0 = 0.0116332 \text{ m s}^{-1}$. The configured **SPREAD_RATE_IN_M** option makes the **vs** output use m min^{-1} , while integration time remains seconds. Raster georeferencing supplies the mesh; no obsolete computational-domain namelist is used.

3 Derivation of expected behavior

For initial circle radius $r_0 = 5$ m and constant outward speed R_0 , radial symmetry reduces the level-set equation to $dr/dt = R_0$. Hence

$$r(360) = 5 + (0.0116332)(360) = 9.187952 \text{ m}, \quad R_0 = 0.697992 \text{ m min}^{-1}.$$

These are independent consequences of the prescribed input rate and elapsed time.

The numerical reference values and the reported metrics follow directly from the generated rate raster. The raster value is $R_s = 0.0116332 \text{ m s}^{-1}$, so the report-unit spread rate is

$$R_{\text{ref}} = 60R_s = 60(0.0116332) = 0.697992 \text{ m min}^{-1}.$$

With $r_0 = 5$ m and $T = 360$ s, the exact final radius is

$$r_{\text{ref}} = r_0 + R_s T = 5 + (0.0116332)(360) = 9.187952 \text{ m}.$$

The two calculated errors are $e_R = |R_{\text{max}} - 0.697992|/0.697992$ and $e_r = |\sqrt{N_b(1 \text{ m})^2/\pi} - 9.187952|$, where N_b is the number of final cells with $\phi < 0$. Thus the acceptance substitutions are $e_R \leq 0.05$ and $e_r \leq 2$ m.

4 Verification metrics and acceptance criteria

The maximum finite final **vs** value is compared with $0.697992 \text{ m min}^{-1}$; a 5 percent relative tolerance catches conversion or propagation-rate errors while allowing front reconstruction effects. The equivalent burned radius $r_A = \sqrt{N_b \Delta x^2 / \pi}$, where N_b counts valid final cells with $\phi < 0$, is compared with 9.187952 m. Area-equivalent radius is less sensitive to pixel stair-stepping than a single axial intercept; its tolerance is two cells. Nodata cells are excluded. The lexicographically last matching final **phi** and **vs** rasters are required. The case passes iff both components pass; missing or malformed required output is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The generated EPSG:32610 domain is 200 m square with 200 cells per side ($\Delta x = 1 \text{ m}$), uniform fuel/moisture/terrain/weather fields, and a centered 5 m ignition circle. Duration is 360 s. The predicted final field is circular, with rate $0.697992 \text{ m min}^{-1}$ and equivalent radius 9.187952 m.

Parameter	Configured value	Units	Role
Domain/grid	200 by 200 cells; 1	m/cell	Resolves the manufactured circle
Initial front	centered radius 5	m	Exact initial condition
Rate	0.0116332	m/s	Constant manufactured normal speed
Time control	step 1; stop 360	s	Exact final time and timestep multiple
Disabled forcing	wind, slope, crown, spotting, suppression	—	Isolates level-set transport

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

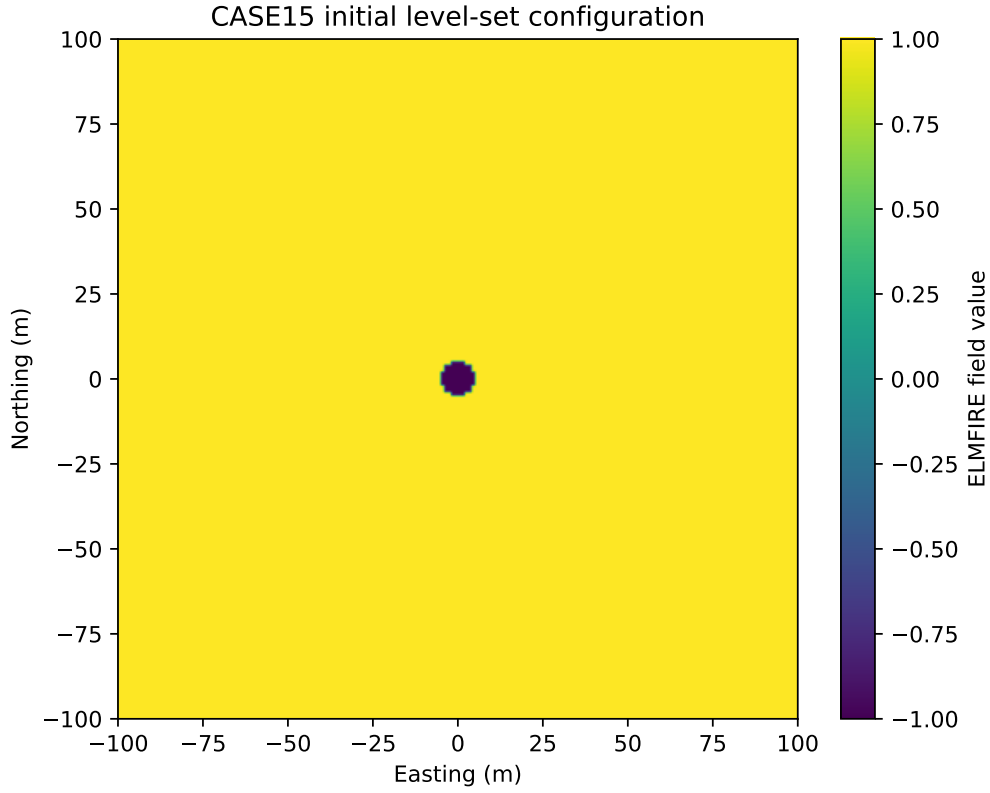


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local ELMFIRE rasters are read. Current status is **PASS**; 2 of 2 components pass.

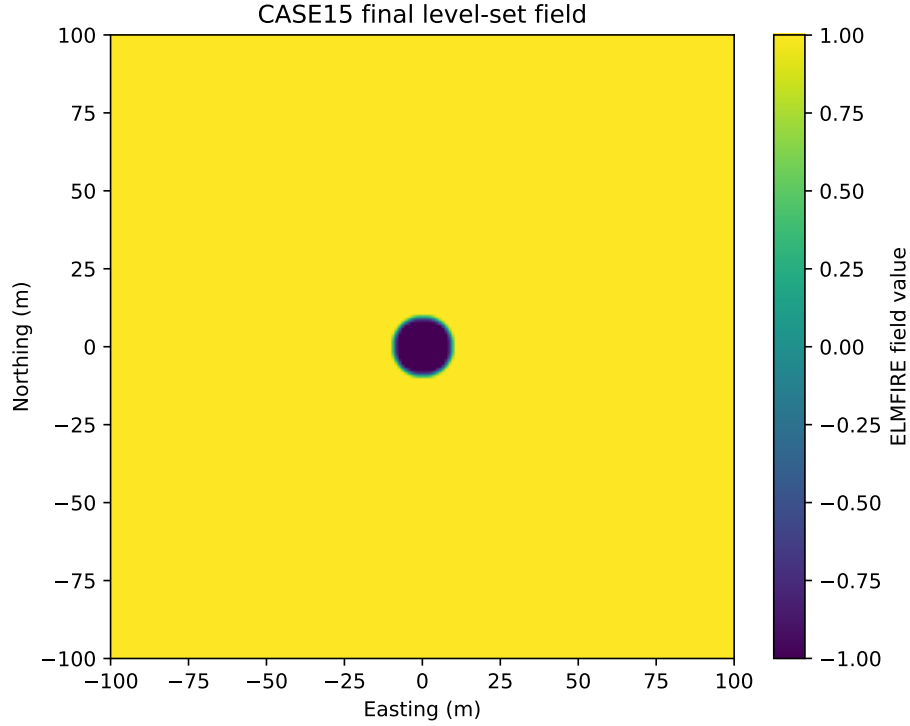


Figure 2: Final whole-domain signed level-set field for the completed variant at the final output time, selected from the case-local data output directory. Values are dimensionless signed-level-set values; the generated table supplies the acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
maximum spread rate	0.697992 (5% relative)	0.698532 m/min	PASS
equivalent final radius	9.18795 (2 cells (2 m))	9.37302 m	PASS

The authoritative overall decision is **PASS**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

A spread-rate failure points to input scaling, units, or `vs` selection. A radius-only failure indicates level-set integration or front/area discretization. A missing raster is an evidence failure, not a scientific pass.

9 Scope and limitations

This case verifies one deterministic constant-rate, flat, no-wind propagation state. It does not validate a fuel model, quantify anisotropy, or test coupled crown, spotting, or suppression behavior.

Case 16: No-spread limiting behavior

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE16_NSP. This unit test verifies the zero-normal-velocity limit of the ELMFIRE level-set pathway. A zero ADJ raster suppresses the base spread rate while wind, slope, spotting, crown fire, and suppression are disabled. The case exercises input multiplication, propagation, and phi/vs output and can expose ignored adjustment rasters, nonzero numerical forcing, or unintended growth. It is implementation verification, not physical validation. The question is whether the final rate remains below $10^{-6} \text{ m min}^{-1}$ and the ignited footprint remains confined to the initialization band.

2 Mathematical formulation and numerical implementation

The propagation equation is $\partial_t \phi + V_n |\nabla \phi| = 0$. The case sets the multiplicative adjustment to zero, so $V_n = 0$ and therefore $\partial_t \phi = 0$. ELMFIRE still initializes a grid-resolved ignition neighborhood, so exact zero burned area is not expected; the invariant is no subsequent outward propagation.

3 Derivation of expected behavior

Integrating $\partial_t \phi = 0$ gives $\phi(\mathbf{x}, t) = \phi(\mathbf{x}, 0)$. Thus the output rate is zero and any negative- ϕ area must be limited to the rasterized initialization band. A 2.5-cell equivalent-radius allowance bounds that band without treating subcell front placement as spread.

The generated pre-ignition ϕ raster contains no negative cells and the adjustment raster is identically zero. Thus

$$V_n = R_0(ADJ) = R_0(0) = 0, \quad \phi(\mathbf{x}, 10 \text{ s}) = \phi(\mathbf{x}, 0), \quad R_{\text{ref}} = 0 \text{ m min}^{-1}.$$

The rate metric is therefore the absolute quantity $e_R = \max_i |R_i|$, with the substituted test $e_R \leq 10^{-6} \text{ m min}^{-1}$; a relative error is undefined for the zero reference. ELMFIRE may rasterize its point ignition into a small initialization footprint, so the spatial metric is

$$r_A = \sqrt{\frac{N_b (1 \text{ m})^2}{\pi}},$$

where N_b counts final cells with $\phi < 0$. Its no-propagation reference is zero and the explicit grid allowance is $r_A \leq 2.5(1 \text{ m}) = 2.5 \text{ m}$.

4 Verification metrics and acceptance criteria

The maximum absolute finite final vs value tests the zero-velocity invariant directly and must not exceed $10^{-6} \text{ m min}^{-1}$. The equivalent burned radius $r_A = \sqrt{N_b \Delta x^2 / \pi}$ must be no more than $2.5 \Delta x$;

this whole-domain area measure detects growth in any direction and is more robust than one transect. Nodata is excluded, and the last matching final `phi` and `vs` rasters are required. Both conditions must pass; missing output is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The generated EPSG:32610 domain is 200 m square on a 200-by-200 grid. Fuel, moisture, and terrain are uniform; wind and slope are zero; `ADJ=0`; the ignition is centered; and duration is 10 s. The predicted `vs` field is zero and the final ignition footprint does not expand beyond its grid initialization band.

Parameter	Configured value	Units	Role
Domain/grid	200 by 200 cells; 1	m/cell	Resolves the stationary ignition
Initial front	centered rasterized band	—	Nonzero initialized footprint
Adjustment	<code>ADJ=0</code>	—	Forces zero normal speed
Time control	step 1; stop 10	s	Tests the zero-speed invariant
Disabled forcing	wind, slope, crown, spotting, suppression	—	Excludes alternate growth pathways

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

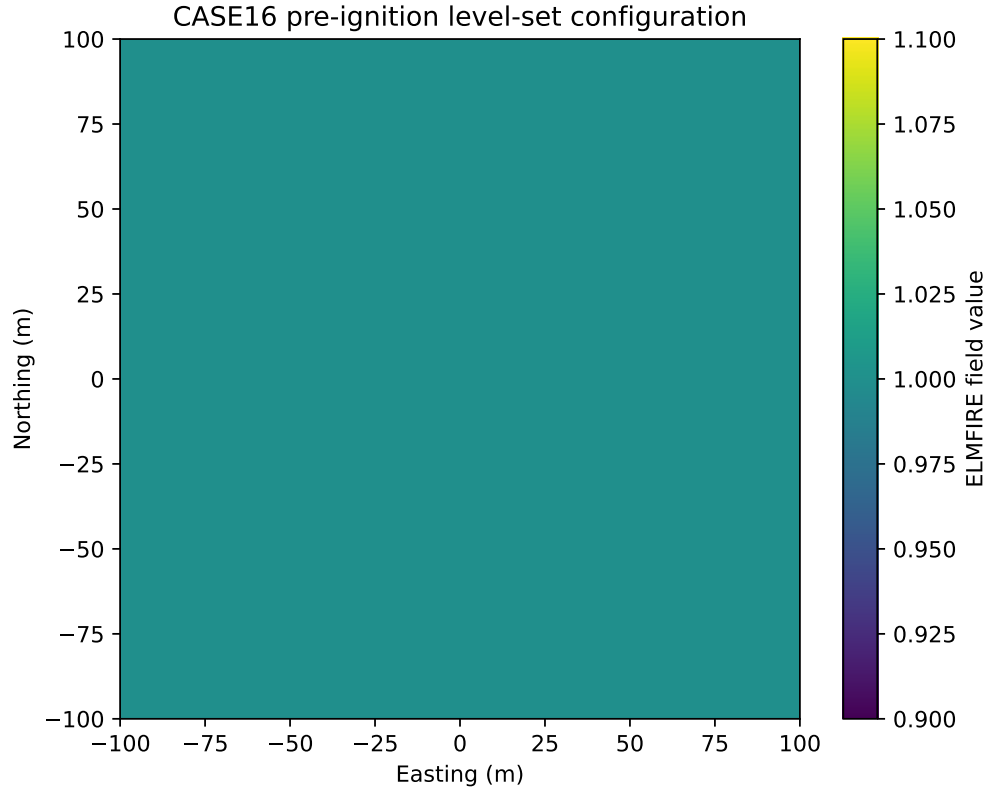


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local ELMFIRE rasters are read. Current status is **PASS**; 2 of 2 components pass.

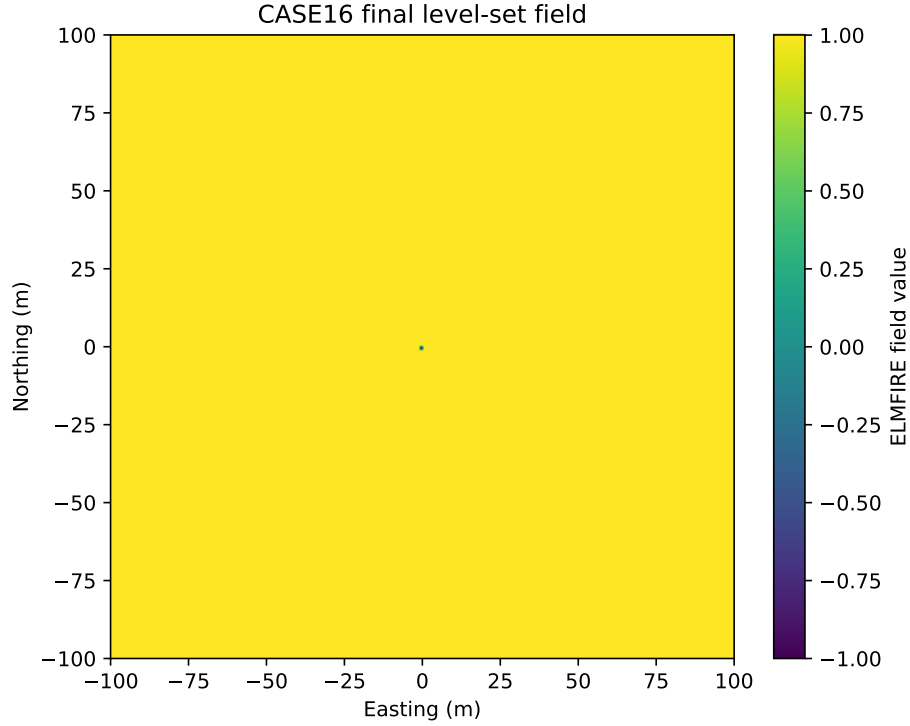


Figure 2: Final whole-domain signed level-set field for the completed zero-spread variant at the final output time. Values are dimensionless; the generated table supplies the rate and confinement limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
maximum absolute spread rate	0 ($\leq 1\text{e-}06$)	0 m/min	PASS
equivalent burned radius	0 (≤ 2.5 cells (2.5 m))	0.56419 m	PASS

The authoritative overall decision is **PASS**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

A nonzero rate primarily indicates that the zero adjustment was ignored or transformed incorrectly. A confined rate with excessive burned area indicates initialization or level-set evolution behavior. Unavailable output cannot establish the invariant.

9 Scope and limitations

This case verifies a deterministic limiting state and raster adjustment coupling. It does not establish accuracy for positive rates, physical extinction, or any coupled fire process.

Case 17: Spatial convergence

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE17_SCV. This unit test verifies spatial convergence of ELMFIRE’s level-set propagation for the same manufactured circular-front problem on four meshes. It exercises raster-derived grid geometry, front initialization, propagation, and `phi` output while holding physical inputs and temporal controls fixed. It can reveal grid-spacing misuse, nonconvergent front updates, or resolution-dependent physical configuration. The question is whether all four required meshes complete, the fitted radius-error order is at least 0.8, and the finest-grid error is at most 1 m.

2 Mathematical formulation and numerical implementation

For $\partial_t \phi + R_0 |\nabla \phi| = 0$ with constant $R_0 = 0.0116332 \text{ m s}^{-1}$, the exact circular radius is mesh-independent. ELMFIRE obtains Δx from each GeoTIFF transform. The four variants use 128, 256, 512, and 1024 cells over the same 256 m domain, so $\Delta x = \{2, 1, 0.5, 0.25\} \text{ m}$.

3 Derivation of expected behavior

With $r_0 = 20 \text{ m}$ and $T = 360 \text{ s}$, $r_\star = 20 + R_0 T = 24.187952 \text{ m}$. For a consistent spatial discretization, the zero-contour equivalent-radius error $E_h = |r_A - r_\star|$ should decrease as $h = \Delta x$ decreases and follow $E_h \approx C h^p$. A least-squares slope of $\log E_h$ against $\log h$ estimates p .

Every grid variant uses the same manufactured solution. From the generated rate $R_s = 0.0116332 \text{ m s}^{-1}$, initial radius $r_0 = 20 \text{ m}$, and stop time $T = 360 \text{ s}$,

$$r_{\text{ref}} = 20 + (0.0116332)(360) = 24.187952 \text{ m}.$$

For each cell size $h_i \in \{2, 1, 0.5, 0.25\} \text{ m}$, the area-equivalent radius and error are

$$r_i = \sqrt{\frac{N_{b,i} h_i^2}{\pi}}, \quad E_i = |r_i - 24.187952|.$$

The observed spatial order is the ordinary-least-squares slope

$$p_h = \frac{\sum_i (\ln h_i - \overline{\ln h})(\ln E_i - \overline{\ln E})}{\sum_i (\ln h_i - \overline{\ln h})^2}.$$

Consequently the table tests the explicitly calculated $p_h \geq 0.8$ and the finest-grid substitution $E_{0.25} = |r_{0.25} - 24.187952| \leq 1 \text{ m}$.

4 Verification metrics and acceptance criteria

The first metric is the fitted order p , selected because it measures the multi-grid trend rather than one favorable mesh; $p \geq 0.8$ allows rasterized-interface effects while requiring approximately first-order behavior. The second is finest-grid $E_h \leq 1$ m, preventing a misleading order from large absolute errors. Radius is obtained from the largest closed, linearly interpolated $\phi = 0$ contour, providing a rotationally aggregated subcell measure. The last matching **phi** raster in every variant directory is required. All variants and both metrics must pass; otherwise the decision is NOT EVALUABLE or FAIL as appropriate.

5 Simulation configuration and predicted outcome

Each EPSG:32610 variant covers the same 256 m square with uniform inputs, flat terrain, zero wind, centered 20 m signed-distance ignition, $R_0 = 0.0116332 \text{ m s}^{-1}$, and duration 360 s. Only grid resolution varies. The predicted front is circular and E_h decreases toward the 24.187952 m reference radius.

Parameter	Configured value	Units	Role
Domain	256 by 256	m	Fixed physical experiment
Meshes	128, 256, 512, 1024	cells/side	Spatial refinement sequence
Cell size	2, 1, 0.5, 0.25	m	Independent variable
Time control	step 1; stop 360	s	Fixed temporal discretization
Initial front	centered signed-distance radius 20	m	Common exact geometry

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

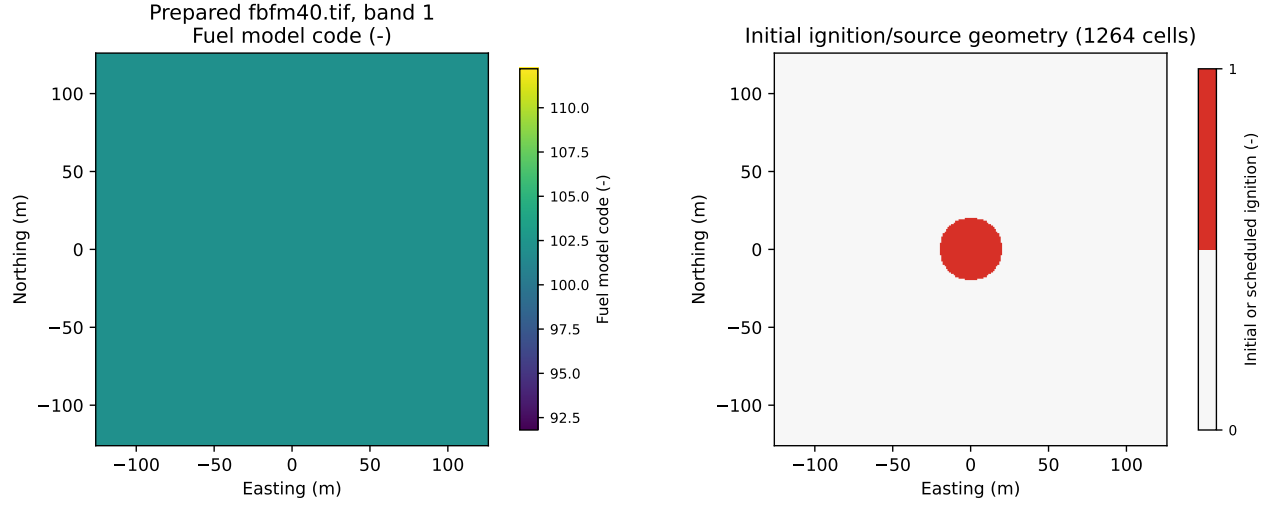


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only the four case-local variant outputs are read. Current status is **FAIL**; 4 variants and 2 decision metrics are recorded.

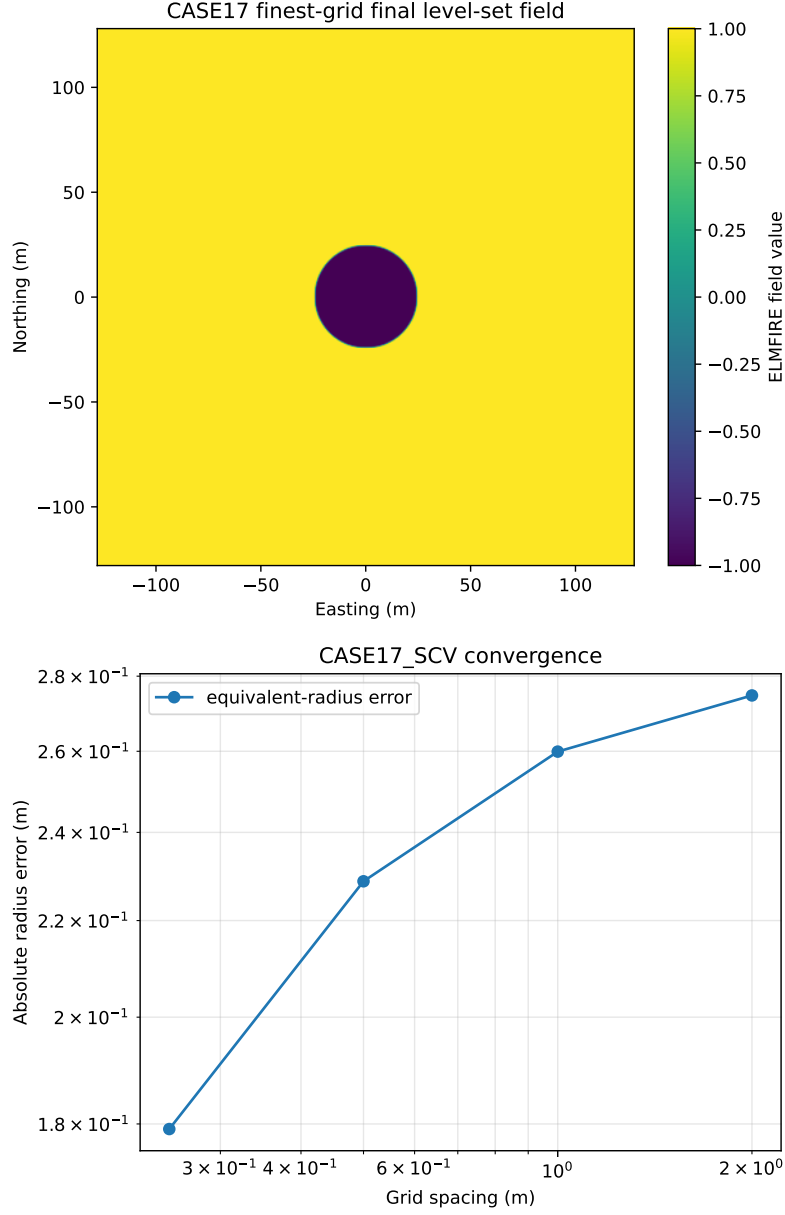


Figure 2: Finest-grid whole-domain signed level-set result and radius-error convergence across all four completed meshes. The spatial field is dimensionless; radius errors are in metres, and the generated table supplies both acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
observed spatial order	0.8 ($\geq$ threshold)	0.203637 -	FAIL
finest-grid radius error	0 ($\leq$ 1)	0.179096 m	PASS

The authoritative overall decision is **FAIL**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

Low order with small finest error suggests interface-measurement or pre-asymptotic effects; high order with excessive finest error indicates the sequence has not reached useful accuracy. Missing variants invalidate the convergence fit. Resolution-dependent physical inputs would be a configuration defect.

9 Scope and limitations

This case measures convergence of one smooth, radially symmetric front and an area-derived observable. It does not determine formal order for discontinuous fuels, complex terrain, anisotropic spread, or other output fields.

Case 18: Temporal convergence

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE18.TCV. This unit test verifies temporal convergence of ELMFIRE’s level-set propagation for a manufactured circular front. Four time grids vary the whole-second step while mesh and physical fields remain fixed. The case can expose timestep scaling errors, inconsistent final-time handling, or a nonconvergent temporal update. The question is whether all four variants complete, the fitted radius-error order is at least 0.8, and the finest-step radius error is at most 0.5 m.

2 Mathematical formulation and numerical implementation

The governing equation is $\partial_t \phi + R_0 |\nabla \phi| = 0$, with constant $R_0 = 0.0116332 \text{ m s}^{-1}$. All variants share the same 1024-by-1024, 256 m raster grid. For a 360 s simulation, the configured time-grid counts 45, 90, 180, and 360 correspond to whole-second $\Delta t = \{8, 4, 2, 1\}$ s.

3 Derivation of expected behavior

The exact final radius is $r_\star = 20 + (0.0116332)(360) = 24.187952$ m. Define $E_{\Delta t} = |r_A - r_\star|$. A consistent time integration has $E_{\Delta t} \approx C(\Delta t)^p$; a least-squares slope of $\log E_{\Delta t}$ versus $\log \Delta t$ estimates the observed temporal order p .

The time-step variants share the manufactured reference

$$r_{\text{ref}} = r_0 + R_s T = 20 + (0.0116332)(360) = 24.187952 \text{ m.}$$

For $\Delta t_i \in \{8, 4, 2, 1\}$ s, the corresponding exact step counts are $360/\Delta t_i = \{45, 90, 180, 360\}$. From each final raster,

$$r_i = \sqrt{\frac{N_{b,i}(0.5 \text{ m})^2}{\pi}}, \quad E_i = |r_i - 24.187952|.$$

The reported temporal order is the least-squares log-log slope

$$p_t = \frac{\sum_i (\ln \Delta t_i - \overline{\ln \Delta t})(\ln E_i - \overline{\ln E})}{\sum_i (\ln \Delta t_i - \overline{\ln \Delta t})^2}.$$

The substituted decision tests are $p_t \geq 0.8$ and $E_{\Delta t=1} = |r_1 - 24.187952| \leq 0.5$ m.

4 Verification metrics and acceptance criteria

Observed p uses all four variants and is required to be at least 0.8, allowing front-raster measurement effects while demanding near-first-order convergence. Finest-step error must be no more than 0.5

m, preventing a formally favorable but inaccurate sequence. Equivalent radius is obtained from the largest closed, linearly interpolated $\phi = 0$ contour, avoiding binary cell-count bias. The last matching ϕ raster for every time grid is required. All variants and both metrics must pass; missing data is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

All EPSG:32610 variants use a 256 m square domain, 0.25 m cells, uniform inputs, flat terrain, zero wind, centered 20 m signed-distance ignition, and duration 360 s. Only the time-grid control varies. The predicted circular-front radius approaches 24.187952 m as Δt decreases.

Parameter	Configured value	Units	Role
Domain/grid	256 m square; 0.25	m/cell	Fixed spatial discretization
Time steps	8, 4, 2, 1	s	Temporal refinement sequence
Stop time	360	s	Common exact comparison time
Initial front	centered signed-distance radius 20	m	Common exact geometry
Rate	0.0116332	m/s	Manufactured normal speed

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

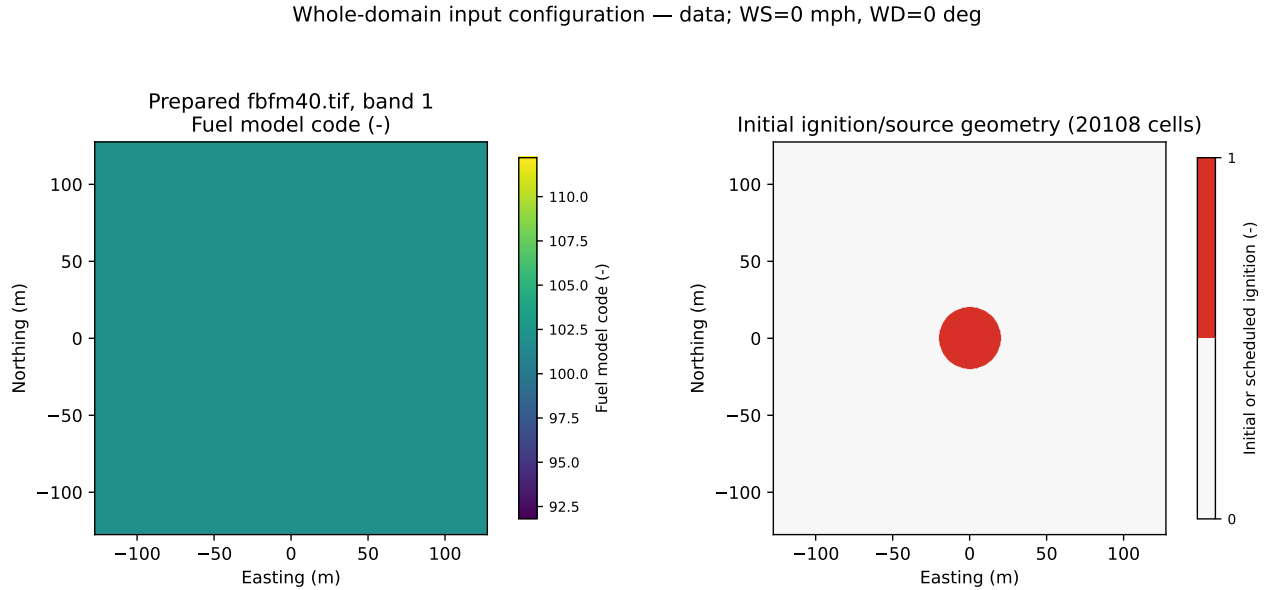


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local output rasters are read. Current status is **FAIL**; 4 variants and 2 decision metrics are recorded.

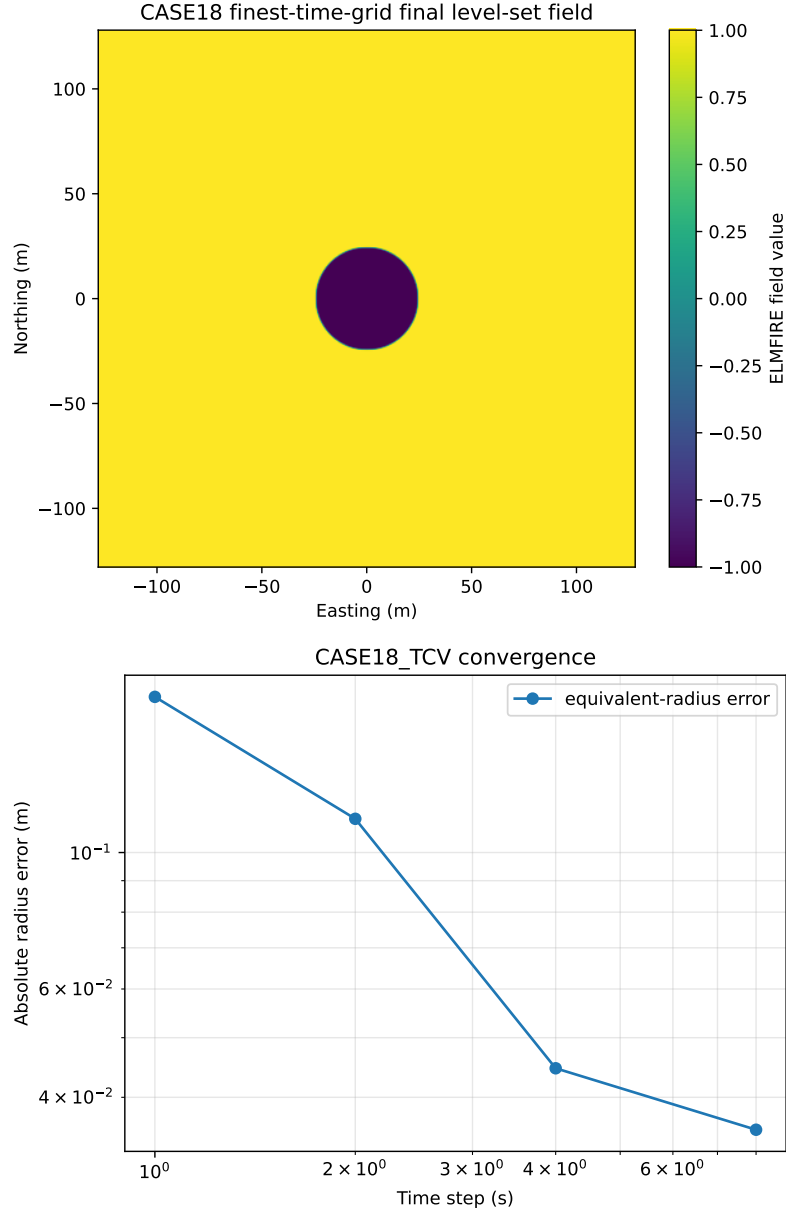


Figure 2: Finest-step whole-domain signed level-set result and radius-error convergence across all four completed time grids. The spatial field is dimensionless; radius errors are in metres, and the generated table supplies both acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
observed temporal order	0.8 ($\geq$ threshold)	-0.836072 -	FAIL
finest-time-grid radius error	0 ($\leq$ 0.5)	0.179096 m	PASS

The authoritative overall decision is **FAIL**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

Low order can indicate timestep application, output-time selection, or pre-asymptotic spatial error; a small finest error alone does not establish convergence. Missing variants invalidate the fitted order and are treated as insufficient evidence.

9 Scope and limitations

This case verifies temporal behavior for one smooth deterministic front at a fixed fine mesh. It does not separate all spatial error, test adaptive timestep logic under complex forcing, or provide physical validation.

Case 19: Transient WUI heat flux

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE19_WTH. This coupling test verifies the transient wildland–urban interface (WUI) heat-flux pathway that combines the building-spread ellipse, a piecewise design-fire heat-release curve, direct-flame-contact (DFC) transfer, radiative transfer, and transient raster output. It exercises the WUI namelist, WUI state updates in `elmfire_level_set.f90`, and transient output branches in `elmfire_io.f90`. It can expose wind-angle, ellipse-geometry, time-curve, cell-normalization, heat-partition, or output-indexing errors. Wildland fuel, crown fire, spotting, and suppression are not decision observables. This is algorithm verification against a separately evaluated documented formulation, not physical validation. The question is whether the prepared case retains exactly one active WUI source, all required transient HRRPUA, DFC, and radiation stacks exist, and each energy-normalized error is no greater than 0.005.

2 Mathematical formulation and numerical implementation

Wind speed is converted from mph to SI as $V = 0.447V_{\text{mph}}$. For configured building area A , separation D , and wind proportionality W_p , the selected Hamada-style regression produces downwind, upwind, and sidewind distances D_d, D_u, D_s . Coefficients differ for $V < 10$, $10 \leq V \leq 17.3$, and $V > 17.3 \text{ m s}^{-1}$; this case uses $15 \text{ mph} = 6.705 \text{ m s}^{-1}$, so it exercises the low-wind linear branch. The translated ellipse has

$$a = (D_d + D_u)/2, \quad e = \min(a/2, a - D_u), \quad b = D_s/\sqrt{1 - (e/a)^2}.$$

For target direction θ_f , its boundary distance is

$$R_e(\theta_f) = \frac{0.3D_d(a - e)}{b^2} \frac{b^2}{a - e \cos \theta_f}, \quad \theta_f = \text{atan2}(y, x) - \pi(270 - WD)/180,$$

which preserves the meteorological wind-from convention.

For burning time τ , the design fire rises linearly to H_p at t_e , remains at H_p through t_d , decays linearly to zero at t_c , and is zero thereafter:

$$H(\tau) = \begin{cases} H_p\tau/t_e, & 0 \leq \tau \leq t_e, \\ H_p, & t_e < \tau \leq t_d, \\ H_p(\tau - t_c)/(t_d - t_c), & t_d < \tau \leq t_c, \\ 0, & \text{otherwise.} \end{cases}$$

Cell coverage is $C_D = \text{clip}[(R_e + \Delta/2 - R)/\Delta, 0, 1]$. With configured ellipse scale $\eta = 0.5$, $C_H = \Delta^2/[\pi(\eta a)(\eta b)]$, burnable fraction C_b , absorptivity α , radiation-annulus fraction F_r , and effective separation R_{eff} ,

$$q''_{\text{DFC}} = C_b C_D H C_H, \quad q''_{\text{rad}} = \frac{0.3 C_b \alpha F_r C_H H \Delta^2}{4\pi R_{\text{eff}}^2}.$$

The case-local reference functions evaluate these equations independently at the timestamps emitted by ELMFIRE.

3 Derivation of expected behavior

The configured constants are $A = D = 10$ m, $W_p = 1$, $\eta = 0.5$, $\Delta = 20$ m, $WD = 0^\circ$, $H_p = 400$ kW m⁻², $t_e = 300$ s, $t_d = 3900$ s, and $t_c = 4200$ s. The generated adjustment field is uniformly zero, so the center WUI structure cannot ignite a secondary level-set source. Because the current executable generic stalled-front shortcut does not account for an active WUI design-fire curve, one point ignition is configured at the center for $t = 5001$ s, strictly after the 5000 s stop. It is never applied; its only role is to keep that shortcut inactive while the prescribed curve is sampled. Substitution into the low-wind regression fixes one time-independent ellipse. At each timestamp, the piecewise equation fixes HRRPUA; the ellipse and coverage equations then fix each cell in the 11-by-11 audit window. DFC is confined to covered cells and scales with H ; radiation occupies the outer annulus and decays with R_{eff}^{-2} . This reference depends on configured constants and timestamps, not on simulated heat-flux values.

The analytical history is evaluated with the generated constants $A = D = 10$ m, $W_p = 1$, $\eta = 0.5$, $H_p = 400$ kW m⁻¹, and $(t_e, t_d, t_c) = (300, 3900, 4200)$ s. Thus the prescribed triangular heat-release history is

$$H(t) = \begin{cases} 400t/300, & 0 \leq t < 300, \\ 400, & 300 \leq t < 3900, \\ 400(4200 - t)/300, & 3900 \leq t \leq 4200, \\ 0, & t > 4200, \end{cases}$$

so, for example, $H(150) = 200$, $H(300) = H(3900) = 400$, $H(4050) = 200$, and $H(4200) = 0$ kW m⁻¹. For the generated 15 mph wind, $U = 15(0.44704) = 6.7056$ m s⁻¹. Substitution in the case's UCB ellipse equations gives $a = 76.2593$ m, $b = 24.8916$ m, $e = 35.6487$ m, downwind reach $D_d = a + e = 111.9080$ m, upwind reach $D_u = a - e = 40.6106$ m, and lateral reach $D_s = 22.0044$ m. Each reported history error uses

$$E_1(q) = \frac{\sum_k |q_{\text{sim},k} - q_{\text{ref},k}|}{\sum_k |q_{\text{ref},k}|},$$

with q equal to heat release rate, dry-fuel consumption, or radiation. The exact-reference substitution is $E_1(q) = 0$, and each component is accepted when $E_1(q) \leq 0.005$.

4 Verification metrics and acceptance criteria

For each observable X ,

$$E_1(X) = \frac{\sum_{t,i,j} |X_{\text{ELMFIRE}} - X_{\text{ref}}|}{\sum_{t,i,j} |X_{\text{ref}}|}.$$

HRRPUA uses the center cell; DFC and radiation use the documented 11-by-11 window, restored to mathematical $+y$ orientation. Mutually masked, nodata, and nonfinite samples are excluded. Energy-normalized L_1 error is selected instead of pointwise relative error because the reference is exactly zero over much of space-time; division by those zeros amplifies irrelevant roundoff. It detects both amplitude and footprint errors. Required filename patterns are `hrr_transient_0000001_*.tif`, `hf_dfc_transient_0000001_*.tif`, and `hf_rad_transient_0000001_*.tif`. Their seven-digit suffixes are dump indices mapped to physical seconds by the unique case-local `dump_times` CSV. A 0.5 percent limit allows single-precision raster serialization and cell-selection roundoff while remaining much

smaller than a one-cell or coefficient error. All stacks and metrics are required, including one terminal record at 5000 s and dumps spanning growth, plateau, decay, and post-decay. Before evaluation, every nonzero HRR raster must contain exactly the center cell; a second active source makes the single-source experiment NOT EVALUABLE. PASS requires that isolation invariant and all $E_1 \leq 0.005$. Missing or incompatible outputs are NOT EVALUABLE.

5 Simulation configuration and predicted outcome

Parameter	Value	Units	Isolation role
Domain raster	20×20 cells	–	centered WUI source and audit window
Analysis cell	20	m	matches reference discretization
Wind	15 at 0° from	mph, degree	low-wind ellipse branch
Building area / separation	10 / 10	m	fixes regression geometry
Nonburnable fraction	0	–	removes availability scaling ambiguity
Ellipse HRR scale η	0.5	–	fixes current normalization
Spread adjustment	0	–	prevents all secondary propagation
Ignition	center PHI cell only	–	the sole physical source
Terminal guard	center at 5001	s	after stop; never ignites
Design-fire times	300 / 3900 / 4200	s	rise, plateau, decay
Peak HRRPUA	400	kW m^{-2}	reference amplitude
Initial/maximum timestep	1 / 10	s	whole-second, CFL-limited stepping
Output interval / stop	100 / 5000	s	exact stop-time alignment and all phases

The predicted HRRPUA rises, plateaus, decays, and becomes zero. Pre-run isolation requires exactly one active HRR cell at every nonzero dump; DFC and radiation therefore come only from the center structure. DFC follows the fixed wind-aligned footprint; radiation surrounds it and both fields scale with HRRPUA.

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows the generated zero spread-adjustment field that enforces source isolation, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=15 mph, WD=0 deg

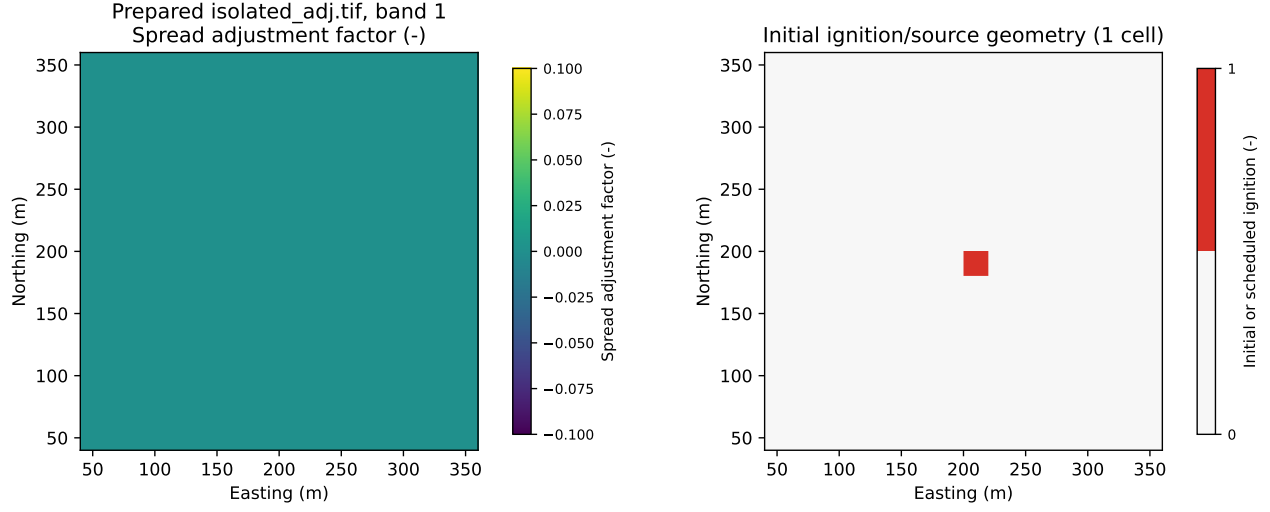


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local ELMFIRE GeoTIFFs are used. Current status is **PASS**; 3 of 3 comparison components pass.

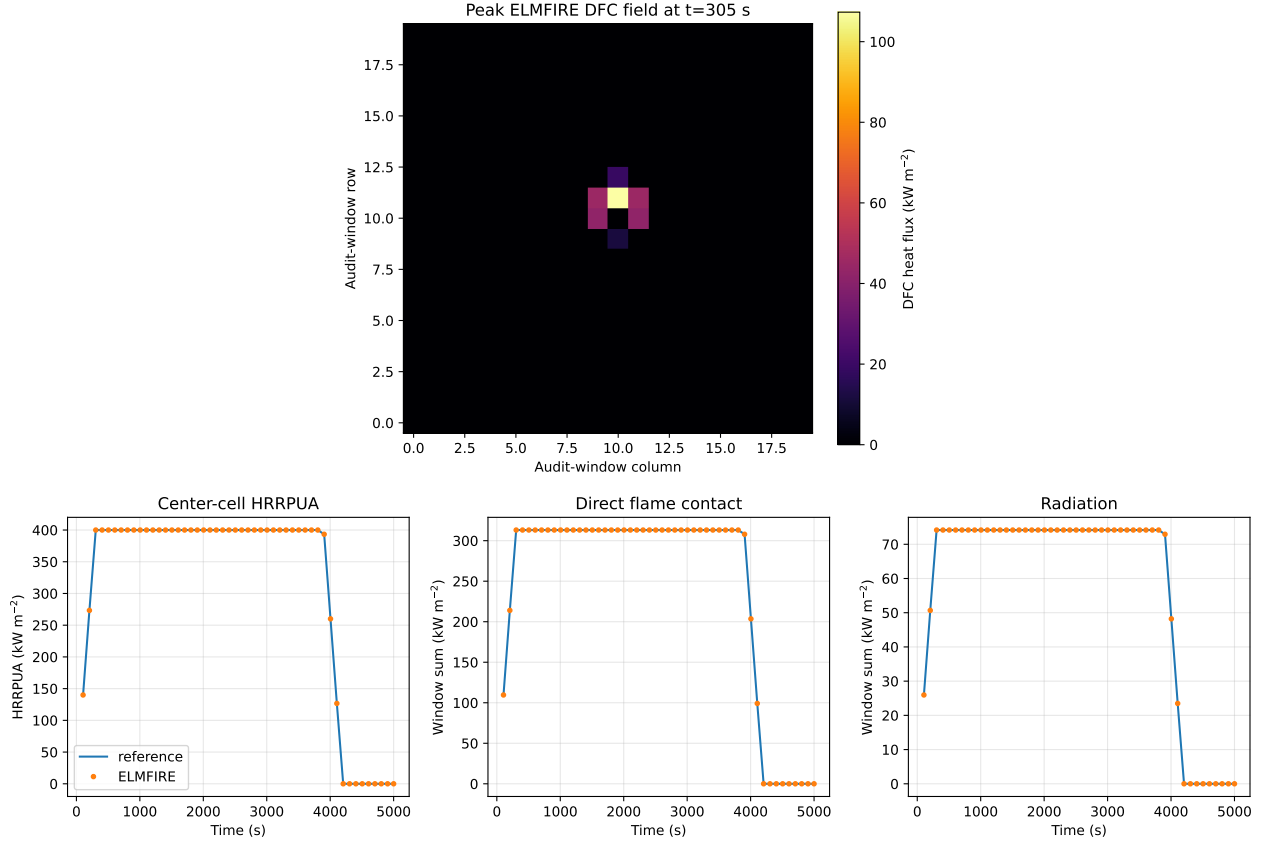


Figure 2: Peak whole-domain DFC heat flux in kW m^{-2} and transient reference-versus-ELMFIRE histories for HRRPUA, DFC, and radiation. All panels are generated from the required case-local transient GeoTIFF stacks; the histories show every selected output time and the spatial panel shows the maximum-domain-sum DFC output.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
HRRPUA normalized L1 error	0 (≤ 0.005)	1.63091e-09 -	PASS
DFC normalized L1 error	0 (≤ 0.005)	9.02842e-08 -	PASS
radiation normalized L1 error	0 (≤ 0.005)	1.41945e-07 -	PASS

The authoritative decision is **PASS**. The generated table is sourced directly from the case-local metrics JSON.

8 Technical interpretation

The PASS decision follows the three generated error components after enforcing the single-source invariant. An HRRPUA-only failure implicates burning-time or transient-curve evaluation. A DFC failure with correct HRRPUA implicates ellipse coverage, configured η normalization, orientation, or raster slicing. A radiation-only failure narrows the issue to annulus partitioning, effective distance, or absorptivity. Similar DFC and radiation errors with correct shapes suggest common normalization. A second HRR source is a case-configuration failure and is NOT EVALUABLE; incomplete phase

coverage, a nonterminal 5000 s record, or missing output is likewise insufficient evidence, never a pass. The post-stop ignition is a documented workaround for the executable stalled-front shortcut and is outside the verified physical interval.

9 Scope and limitations

This case verifies one deterministic WUI parameter set, one wind-regression branch, one raster resolution, and three transient outputs. It does not verify moderate/high-wind branches, parameter rasters, heterogeneous layouts, structure-to-structure ignition, or empirical validity of the correlations. It supplies implementation-verification evidence only.

Case 20: Point ignition and isotropic spread

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE20_PIG. This case verifies the coupled surface-fire and level-set rate response to a central point ignition in uniform fuel with zero wind and zero slope. Rotational symmetry supplies a diagnostic for directional bias or raster-orientation errors, while the formal decision tests the coupling between fire behavior and level-set normal velocity. The verification question is whether the required maximum spread rate is within 10 percent of 1.51 m min^{-1} ; isotropic arrival geometry is reported as supporting evidence but is not yet a separate acceptance component.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

ELMFIRE advances the level-set field using $\partial\phi/\partial t + V_n|\nabla\phi| = 0$. With spatially uniform fuel, moisture, wind, and slope, $V_n = R_0$ is independent of azimuth. The zero contour must therefore evolve from the ignition neighborhood toward a circle. The final **vs** raster records the local normal spread rate in metres per minute because **SPREAD_RATE_IN_M** is enabled.

The surface-fire scalar supplied to the level set is the Rothermel no-wind/no-slope rate

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad Q_{ig} = 250 + 1116 M_f, \quad (1)$$

where reaction intensity I_R , propagating-flux ratio ξ , bulk density ρ_b , effective heating number ϵ , and fuel moisture M_f are evaluated from the generated Fuel Model 3 and moisture inputs. This equation chain follows Andrews (2018), *The Rothermel Surface Fire Spread Model and Associated Developments*; the Guide supplies the configured known-response value.

3 Derivation of expected behavior

For constant R_0 , the front radius is $r(t) = r(0) + R_0 t$. The independently tabulated Fuel Model 3 response for the configured moisture and no-wind/no-slope condition is $R_0 = 1.51 \text{ m min}^{-1}$. Rotational symmetry is an invariant; the quantitative decision uses the rate because it directly tests the fire-behavior value passed to propagation.

The independent no-wind/no-slope reference is calculated from

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad Q_{ig} = 250 + 1116 M_f,$$

using generated Fuel Model 3 and $(M_{1h}, M_{10h}, M_{100h}, M_{lh}, M_{lw}) = (0.06, 0.07, 0.08, 0.60, 0.90)$. The independent calculation gives $I_R = 549.174 \text{ kW m}^{-2}$ and $R_0 = 1.51183 \text{ m min}^{-1}$, reported to the Guide precision as $R_{\text{ref}} = 1.51 \text{ m min}^{-1}$ ($1.51183/0.3048 = 4.960 \text{ ft min}^{-1}$). The decision metric is therefore

$$e_R = \frac{|R_{\text{max}} - 1.51|}{1.51},$$

where R_{max} is the maximum finite final **vs** value, and the substituted acceptance test is $e_R \leq 0.10$.

4 Verification metrics and acceptance criteria

The head-ROS metric is the maximum finite value in the required final **vs** raster. It is selected because uniform forcing makes every mature front direction equivalent and the maximum avoids dilution by unburned nodata cells. A 10 percent relative tolerance follows the GUIDE comparison and allows raster and front-discretization effects while remaining sensitive to unit or coupling errors. The case passes only when the output exists and $|R_{\text{sim}} - 1.51|/1.51 \leq 0.10$.

5 Simulation configuration and predicted outcome

The case uses a 126×126 , 30 m grid in EPSG:32635, uniform Fuel Model 3, 6 percent one-hour moisture, flat terrain, zero wind, and a central ignition. The predicted output is an approximately circular arrival-time field with a uniform mature spread rate near 1.51 m min^{-1} .

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

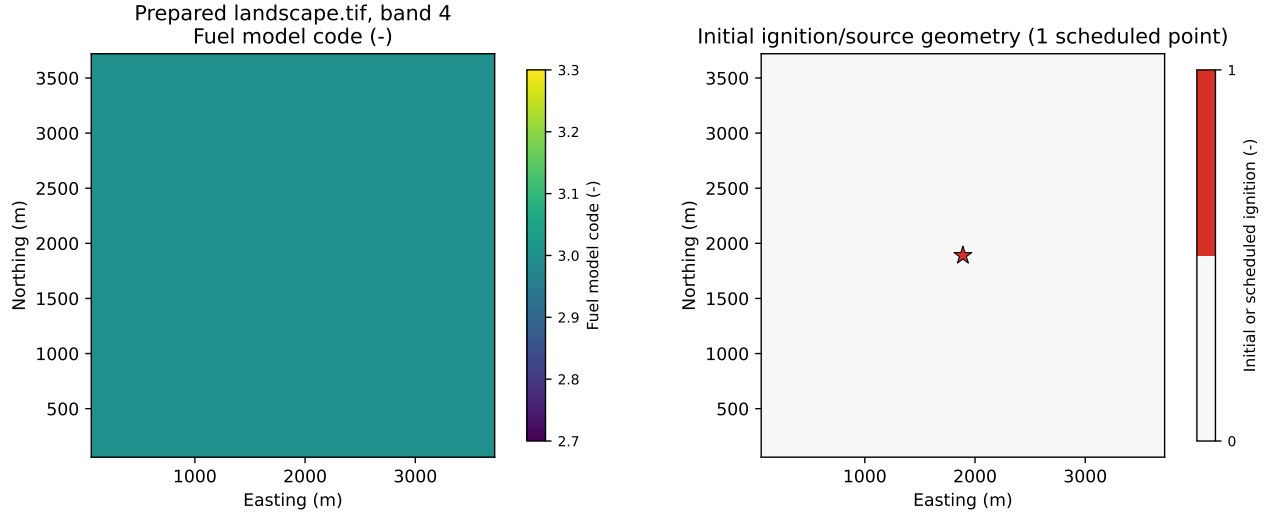


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 1 of 1 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

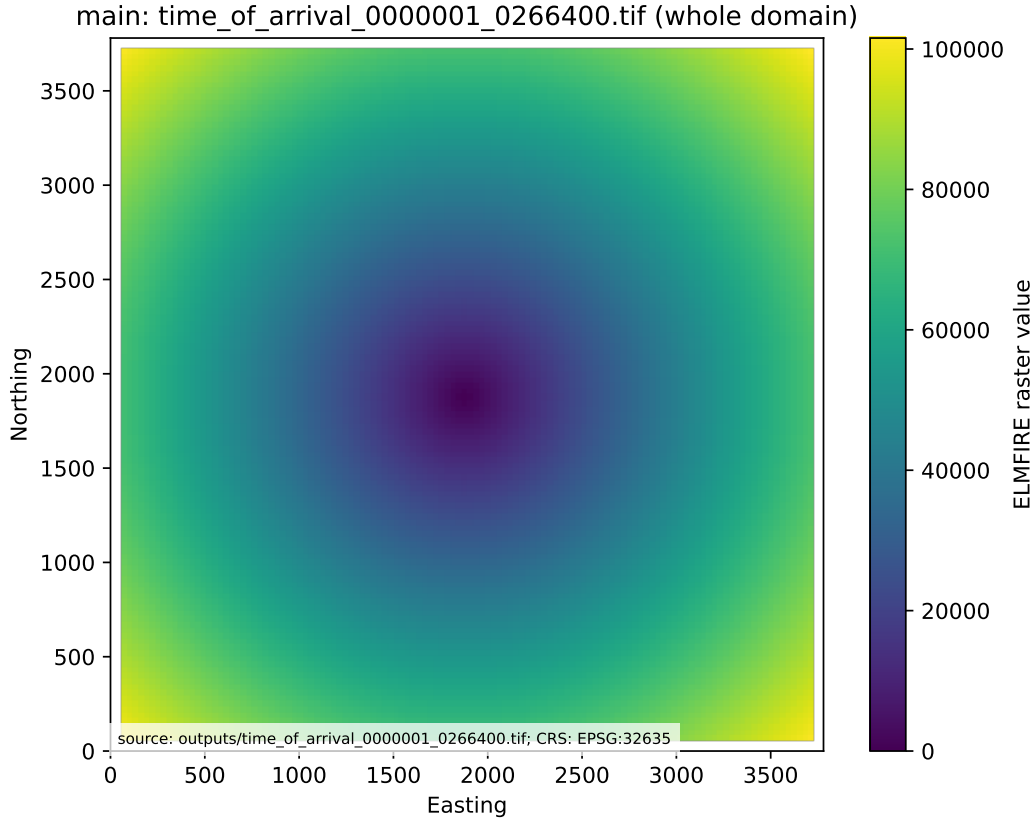


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / head ROS	1.51 m/min	1.51183 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A rate failure indicates a fire-behavior, units, or output-selection problem. Strong directional distortion with an acceptable maximum is a diagnostic warning and must not be claimed as verified until a shape metric and tolerance are added. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

This is deterministic code verification for one fuel and moisture state. It does not validate Fuel Model 3 against experimental fires or quantify symmetry with a dedicated shape metric. This case provides implementation verification evidence only.

Case 21: Wind-driven elliptical spread

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE21_WDE. This case verifies the coupling of wind-adjusted surface spread to the Huygens and level-set propagation geometry. Uniform fuels and flat terrain isolate wind as the only directional forcing. The verification question is whether the simulated head rate is within 10 percent of 3.635 m min^{-1} and the unclipped arrival-time isochrone has length-to-width ratio within 10 percent of 1.24.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The normal-velocity construction combines the backing, flanking, and heading rates into an expanding ellipse. Its major axis aligns with the wind-driven heading direction, while the ratio L/W characterizes anisotropy independent of absolute size. ELMFIRE writes rate in metres per minute and arrival time on the same raster grid used to infer the ellipse.

For uncovered fuel of depth H ft, the Guide uses

$$WAF_u = \frac{1.83}{\ln[(20 + 0.36H)/(0.13H)]}, \quad U_{mf} = WAF_u U_{20}. \quad (1)$$

The resulting effective midflame wind determines the ellipse aspect ratio

$$\frac{L}{W} = \min(0.936e^{0.1147U_{mf,e}} + 0.461e^{-0.0692U_{mf,e}} - 0.397, 8). \quad (2)$$

The local head rate is $R_H = R_0(1 + \phi_w)$; the Huygens/level-set construction uses the ellipse to obtain normal speed away from the heading direction.

3 Derivation of expected behavior

Constant forcing produces self-similar elliptical isochrones: scaling time changes both axes proportionally and leaves L/W constant. The configured Fuel Model 5 response supplies the independent head-rate target 3.635 m min^{-1} , and the GUIDE construction supplies $L/W = 1.24$.

For the generated Fuel Model 5 state, the independent no-wind rate is $R_0 = 0.41042 \text{ m min}^{-1}$. The 6 mph 20-ft wind and 2-ft fuel-bed depth give a wind-adjustment factor $WAF = 0.418640$, hence $U_{mf} = 6(0.418640) = 2.51184 \text{ mph}$. With the calculated wind factor $\phi_w = 7.87958$,

$$R_h = R_0(1 + \phi_w) = 0.41042(1 + 7.87958) = 3.6444 \text{ m min}^{-1},$$

consistent with the Guide reference 3.635 m min^{-1} . The ellipse length-to-width reference uses

$$\frac{L}{W} = 0.936e^{0.2566U} + 0.461e^{-0.1548U} - 0.397,$$

where U is in m s^{-1} . Here $U = 2.51184 \text{ mph}(0.44704) = 1.12291 \text{ m s}^{-1}$, so substitution gives $L/W = 1.23898$, reported as 1.24. The report evaluates the relative substitutions $e_R = |R_h - 3.635|/3.635$ and $e_{LW} = |(L/W) - 1.24|/1.24$ against their tabulated limits.

4 Verification metrics and acceptance criteria

Maximum finite `vs` measures the heading response. The shape metric is $\sqrt{\lambda_{\max}/\lambda_{\min}}$, where the eigenvalues come from the covariance of burned-cell coordinates at the last isochrone that remains away from the boundary. This metric is rotation invariant and avoids a wind-axis assumption. Each component uses 10 percent relative tolerance, and both must pass.

5 Simulation configuration and predicted outcome

The 126×126 , 30 m domain uses uniform Fuel Model 5, 6 mph wind, flat terrain, and an upwind point ignition. The predicted arrival field is an ellipse elongated downwind, with head ROS 3.635 m min^{-1} and $L/W = 1.24$.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=6 mph, WD=0 deg

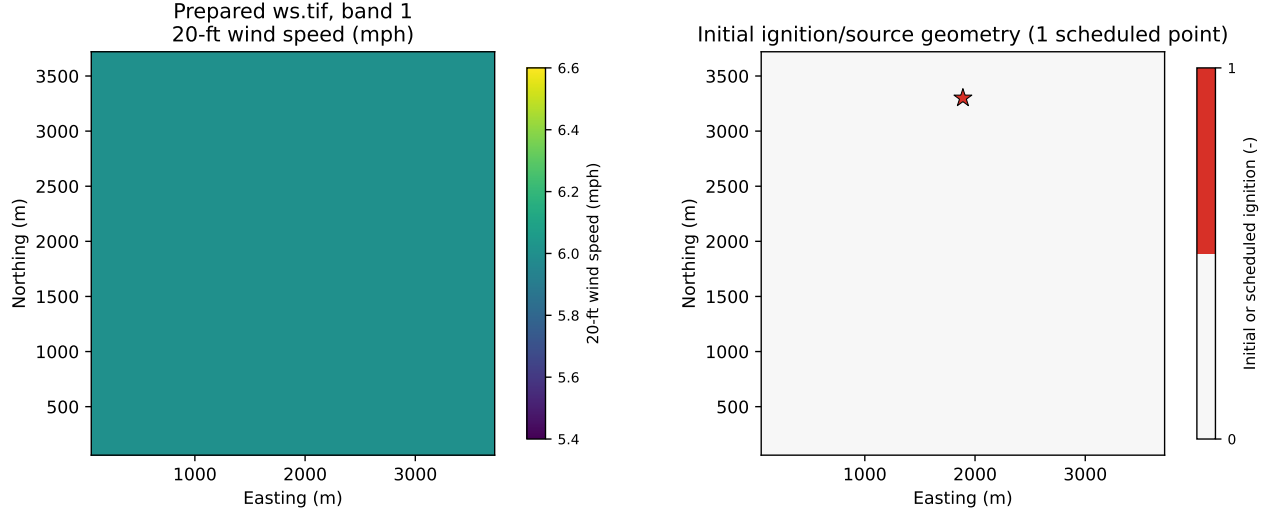


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 2 of 2 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

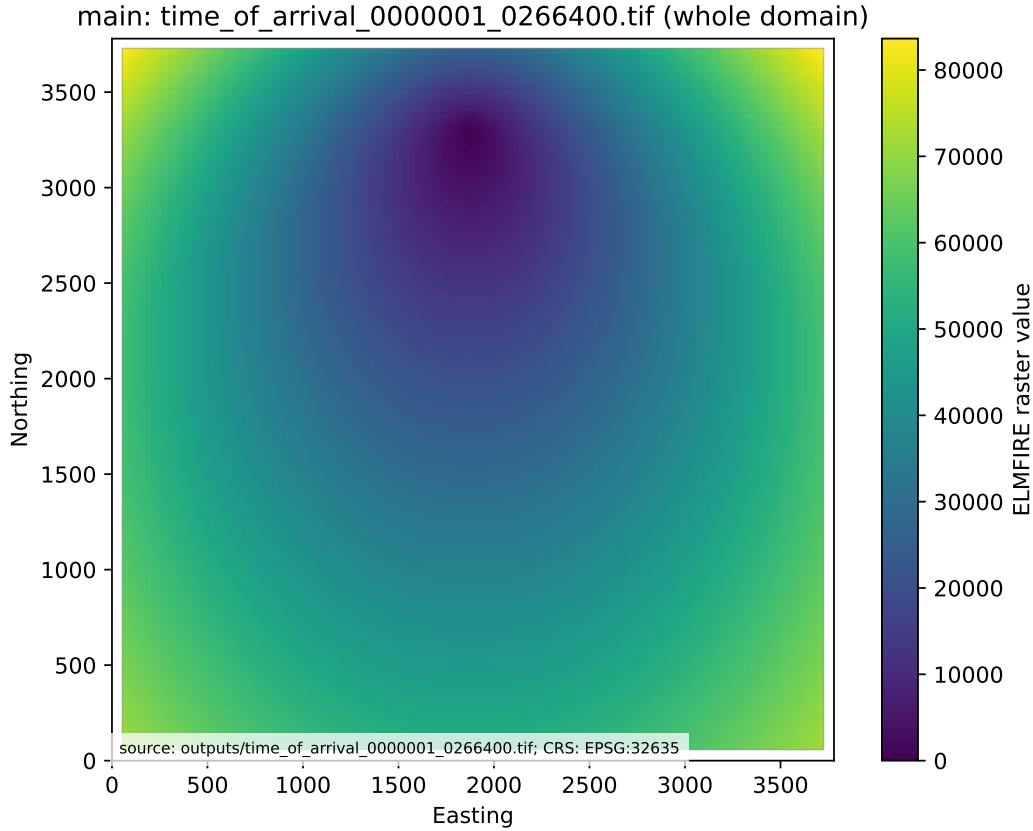


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / head ROS	3.635 m/min	3.64438 m/min	PASS
main / L/W ratio	1.24 -	1.24859 -	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A rate-only discrepancy points toward wind adjustment or unit conversion; a shape-only discrepancy points toward wavelet geometry, arrival-time extraction, or boundary clipping. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test covers one steady wind speed and one fuel model. The covariance estimator is intentionally applied only before boundary contact. This case provides implementation verification evidence only.

Case 22: Spread across opposing valley slopes

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE22_VSL. This case verifies slope enhancement and aspect handling across two opposing valley sides. Zero wind isolates the terrain contribution. The verification question is whether the maximum regional spread rates on the 5-degree east side and 15-degree west side are each within 10 percent of their independent targets.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The surface spread calculation multiplies the no-wind/no-slope response by a slope-dependent factor and orients the heading direction upslope. The slope raster supplies magnitude in degrees and the aspect raster supplies downslope azimuth. Opposing aspects must therefore drive the two head fires away from the central valley ignition.

The isolated Rothermel slope factor is

$$R_H = R_0(1 + \phi_s), \quad \phi_s = 5.275 \beta^{-0.3} \tan^2 \theta, \quad (1)$$

where β is the fuel-bed packing ratio and θ is the slope angle. The aspect raster gives downslope azimuth, so the heading direction is rotated by 180 degrees to obtain upslope. The formula chain follows Andrews (2018); the Guide supplies the two configured BEHAVE responses and point-spread ellipse ratios.

3 Derivation of expected behavior

For the configured Fuel Model 3 and moisture, the reference calculation gives 1.92 m min^{-1} on the 5-degree side and 5.37 m min^{-1} on the 15-degree side. The steeper slope must have the larger head rate; this ordering is an additional qualitative invariant.

With no wind, the slope correction is

$$R_s = R_0(1 + \phi_s).$$

The generated Fuel Model 3 moisture state gives $R_0 = 1.51183 \text{ m min}^{-1}$. The independent Rothermel slope calculation gives $\phi_s = 0.272326$ at 5 degrees and $\phi_s = 2.554410$ at 15 degrees. Therefore

$$R_5 = 1.51183(1 + 0.272326) = 1.92354, \quad R_{15} = 1.51183(1 + 2.554410) = 5.37366 \text{ m min}^{-1},$$

reported as 1.92 and 5.37 m min^{-1} . For each slope s , the calculated metric is $e_s = |R_{\max,s} - R_{s,\text{ref}}|/R_{s,\text{ref}}$, with the reference and limit substituted in the generated decision table.

4 Verification metrics and acceptance criteria

For each half-domain, the metric is the maximum finite **vs** value. Regional maxima isolate the intended slope without averaging across the slope discontinuity or unburned cells. Each target uses 10 percent relative tolerance; both regional decisions are required for PASS.

5 Simulation configuration and predicted outcome

The 30 m grid has Fuel Model 3, zero wind, a 5-degree eastern slope with downslope aspect 270 degrees, a 15-degree western slope with downslope aspect 90 degrees, and central ignition. The supplied Guide uses percent notation in one descriptive passage, but its summary table and the current producer identify these as 5- and 15-degree raster values; the generated landscape band is authoritative for this report. Predicted spread is outward and faster on the western, steeper side.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

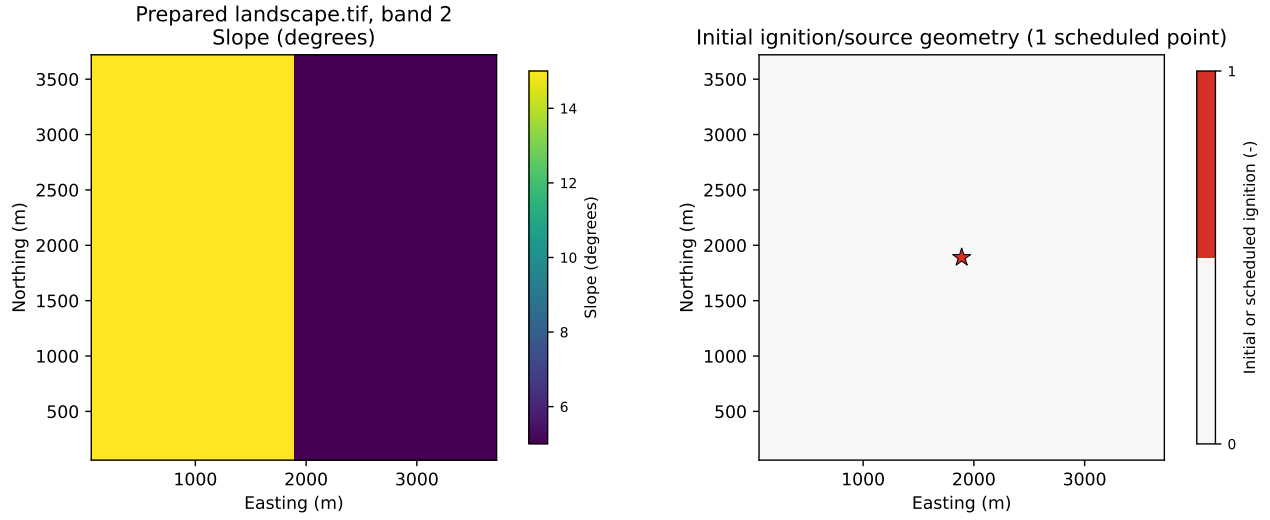


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 2 of 2 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

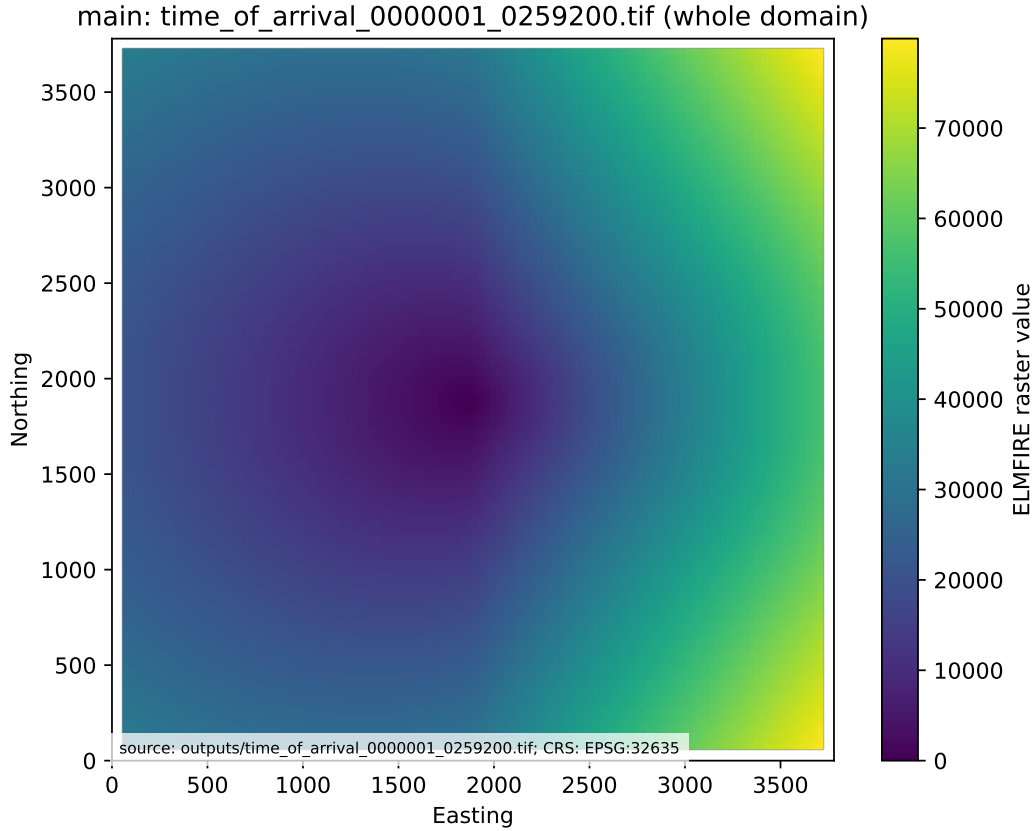


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / head ROS (5 deg, east)	1.92 m/min	1.92354 m/min	PASS
main / head ROS (15 deg, west)	5.37 m/min	5.37366 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

Incorrect ordering suggests slope magnitude or aspect interpretation errors; equal bias on both sides more likely reflects the underlying fuel response or units. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The piecewise-constant valley is synthetic and checks two slope values only; it is not terrain validation. This case provides implementation verification evidence only.

Case 23: Spatially varying fuel quadrants

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE23_FQD. This case verifies raster lookup and fire-behavior coupling when a propagating front crosses four fuel-model regions. Flat terrain and zero wind isolate fuel identity. The verification question is whether every quadrant’s maximum spread rate is within 10 percent of its fuel-specific reference.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The level-set normal speed is evaluated from the fuel model at each active cell. Because the environmental forcing is uniform, changes in V_n across the domain should follow only changes in the **fbfm** raster. Correct row orientation and quadrant masking are therefore part of the test.

At each active cell the fuel code selects a complete Rothermel parameter row, and the local no-wind/no-slope rate is

$$R_0(x, y) = \frac{I_R(x, y) \xi(x, y)}{\rho_b(x, y) \epsilon(x, y) Q_{ig}(x, y)}. \quad (1)$$

Thus the raster lookup changes the full nonlinear fuel response, not a post-hoc scalar multiplier. The formulation follows Andrews (2018); the Guide supplies the four BEHAVE reference values.

3 Derivation of expected behavior

The independent regional targets are 0.080, 0.472, 1.492, and 0.871 m min⁻¹ for Fuel Models 8, 7, 4, and 2 in the northwest, northeast, southwest, and southeast quadrants, respectively. These values also predict a visibly noncircular front after it enters all four regions. The Guide historically reported ELMFIRE rates of 0.080, 0.551, 1.974, and 0.872 m/min for FB8, FB7, FB4, and FB2, noting discrepancies for FB7 and FB4. Those historical outputs are preserved here as context only; the generated metric table below is the current evidence and is never replaced by those values.

All four references use the no-wind/no-slope Rothermel equation $R_0 = I_R \xi / (\rho_b \epsilon Q_{ig})$ with $Q_{ig} = 250 + 1116 M_f$ and the generated moisture vector (0.06, 0.07, 0.08, 0.60, 0.90). Substitution of each

generated standard fuel model gives

fuel model	I_R (kW m ⁻²)	R_0 (m min ⁻¹)
8	175.566	0.079552
7	406.479	0.470579
4	2272.300	1.493256
2	669.483	0.871756

and hence the report references 0.080, 0.472, 1.492, and 0.871 m min⁻¹, respectively. For each fuel model, $e_R = |R_{\max} - R_{\text{ref}}|/R_{\text{ref}}$; the table substitutes the corresponding row above and the prescribed relative-error limit.

4 Verification metrics and acceptance criteria

The maximum finite **vs** in each quadrant is compared with its corresponding target. Regional maxima preserve the local head-fire response and avoid mixing physically distinct fuels. Every component uses 10 percent relative tolerance, and PASS requires all four components.

5 Simulation configuration and predicted outcome

Four equal quadrants on a 126×126 , 30 m grid contain Fuel Models 8, 7, 4, and 2. Terrain is flat, wind is zero, moisture is uniform, and ignition is central. The fastest extension is predicted in Fuel Model 4 and the slowest in Fuel Model 8.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

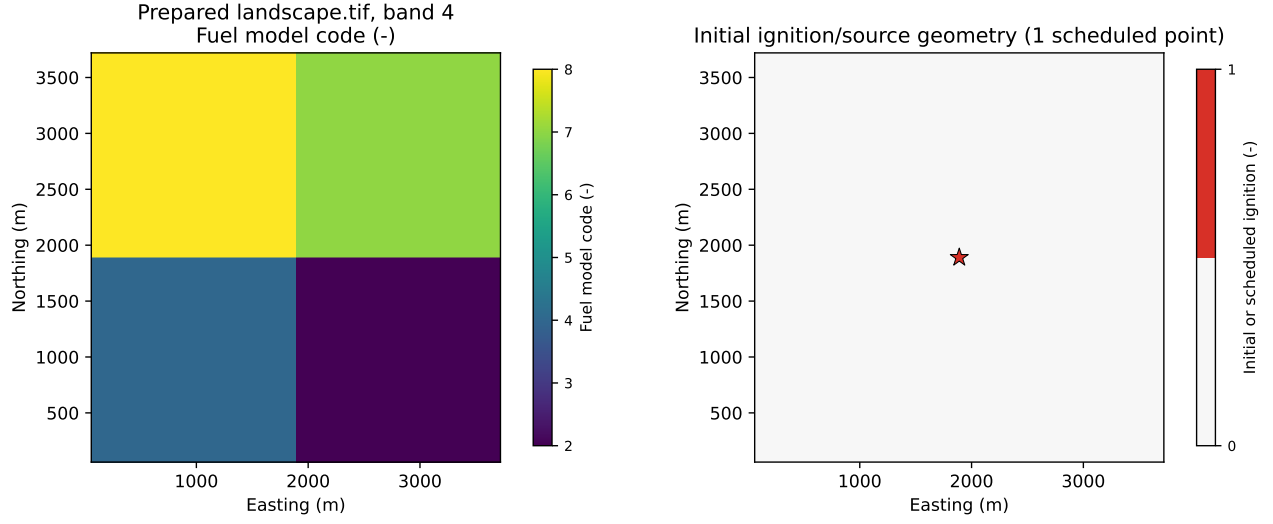


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

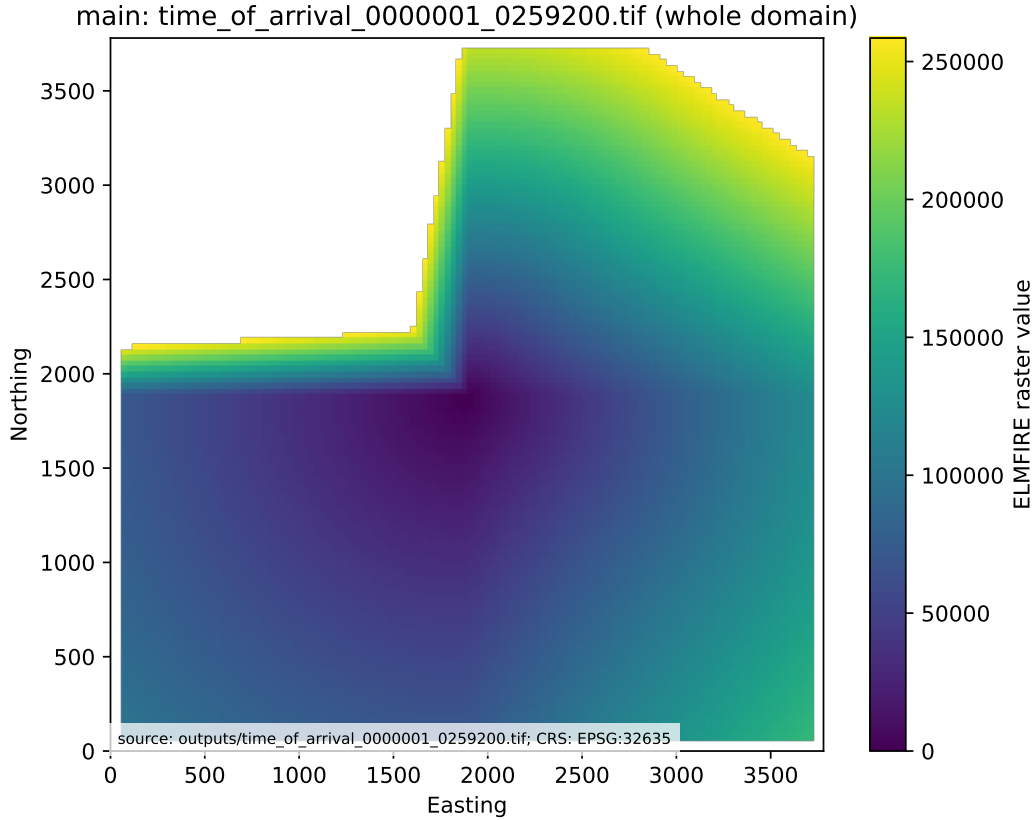


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / FB8 head ROS	0.08 m/min	0.0795524 m/min	PASS
main / FB7 head ROS	0.472 m/min	0.470579 m/min	PASS
main / FB4 head ROS	1.492 m/min	1.49326 m/min	PASS
main / FB2 head ROS	0.871 m/min	0.871756 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A quadrant swap indicates raster orientation or mask errors; a fuel-specific discrepancy indicates fuel-table selection or fire-behavior coupling. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test samples four fuel models under one moisture state and does not validate the empirical fuel parameters. This case provides implementation verification evidence only.

Case 24: Combined fuel, slope, and wind interactions

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE24_CXI. This case exercises simultaneous spatial fuel variation, opposing slopes, and steady wind to reveal integration failures that isolated cases may miss. The decision checks the independently derived domain-maximum head spread while preserving the Guide configuration and central ignition.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

At each active front cell ELMFIRE combines fuel-dependent reaction terms, wind and slope factors, and elliptical propagation geometry to obtain a local normal velocity. The configuration intentionally produces competing directional effects. Although the discontinuous coupled perimeter has no closed-form solution, its maximum local direction-of-maximum-spread response does.

For each active cell, ELMFIRE evaluates fuel-dependent R_0 , wind factor ϕ_w , and slope factor ϕ_s . Wind and slope are combined as vectors before the direction of maximum spread and effective-wind ellipse are formed:

$$\boldsymbol{\phi} = \phi_w \hat{\mathbf{u}}_w + \phi_s \hat{\mathbf{u}}_s, \quad R_H = R_0(1 + |\boldsymbol{\phi}|). \quad (1)$$

Spatial discontinuities in fuel and slope make R_H piecewise and the resulting level-set problem has no closed-form perimeter.

3 Derivation of expected behavior

The dominant generated condition is the southwest quadrant: fuel model 4, 15-degree slope, 5 mph wind, and fixed 6/7/8 percent dead moisture. The tracked fuel table gives $R_0 = 1.493256$ m/min and wind-adjustment factor 0.547659. The independent Rothermel calculation gives $\phi_s = 2.010270$ and $\phi_w = 7.366964$. Wind-to south and upslope west are orthogonal, so $|\boldsymbol{\phi}| = \sqrt{\phi_s^2 + \phi_w^2}$ and $R_H = 12.896233$ m/min.

The generated Fuel Model 4 and moisture inputs first give the no-forcing rate $R_0 = 1.493256$ m min⁻¹ and reaction intensity $I_R = 2272.300$ kW m⁻². Independent wind and slope calculations give $\phi_w =$

7.366964 and $\phi_s = 2.010270$. Because the imposed wind and upslope vectors are orthogonal, their dimensionless resultant is

$$\phi_{\text{res}} = \sqrt{\phi_w^2 + \phi_s^2} = \sqrt{(7.366964)^2 + (2.010270)^2} = 7.636318.$$

The reference maximum rate is consequently

$$R_{\text{ref}} = R_0(1 + \phi_{\text{res}}) = 1.493256(1 + 7.636318) = 12.896233 \text{ m min}^{-1}.$$

The decision metric substitutes this value in $e_R = |R_{\text{max}} - 12.896233|/12.896233$ and compares e_R with the tabulated relative tolerance.

4 Verification metrics and acceptance criteria

The required metric is the maximum finite spread rate over the completed domain. It must agree with the independently derived 12.896233 m/min southwest-quadrant head response within the Guide harness default 10 percent tolerance. The case is NOT EVALUABLE only when required output is missing; otherwise this quantitative metric decides PASS or FAIL.

5 Simulation configuration and predicted outcome

The 30 m domain combines the CASE23 fuel quadrants, CASE22 opposing slopes, 5 mph wind, uniform moisture, and central ignition. Predicted output is asymmetric, spatially continuous, and preferentially extended by both wind and locally faster fuels/slopes.

Parameter	Configured value	Units	Role in the experiment
Grid	126 × 126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=5 mph, WD=0 deg

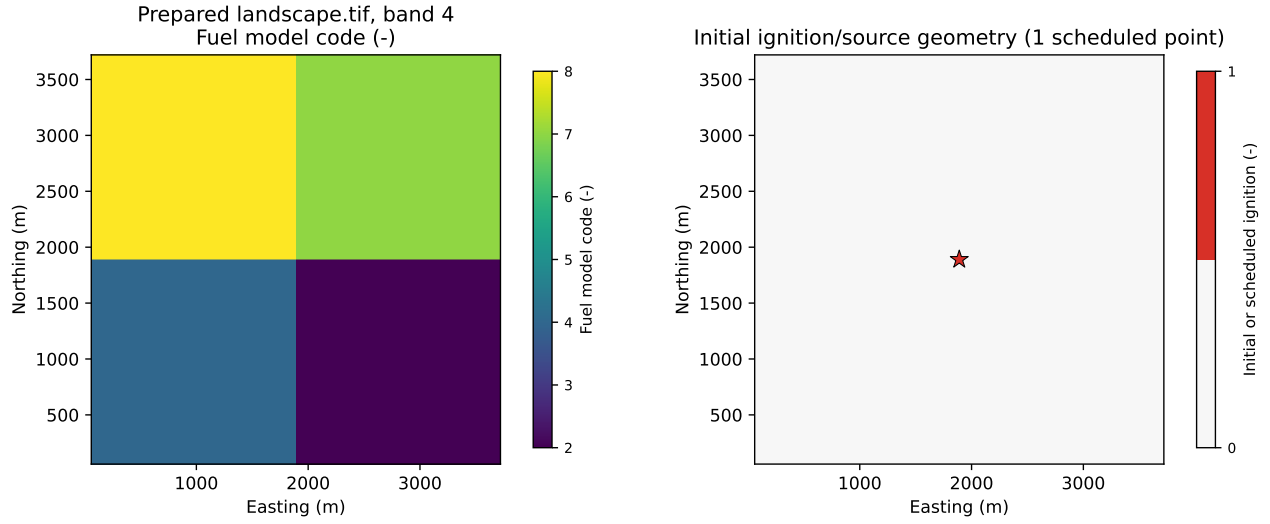


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 1 of 1 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

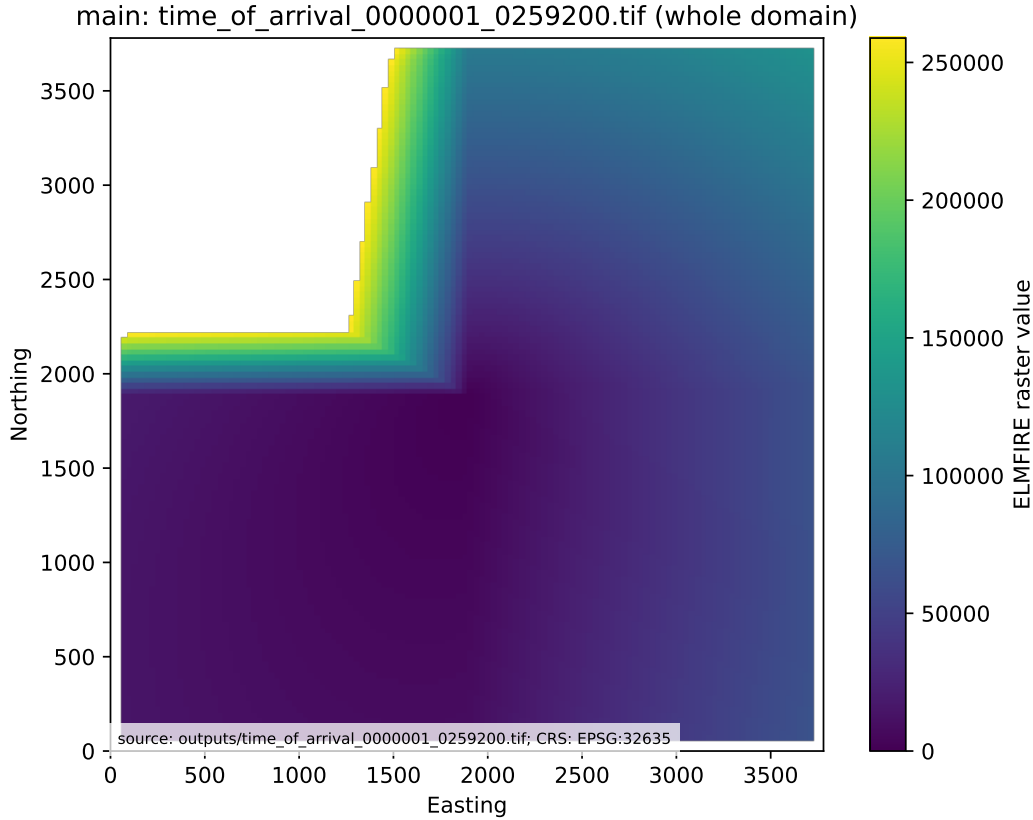


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / head ROS	12.8962 m/min	12.8962 m/min	PASS

The displayed decision is **PASS**. A present case-local metrics JSON controls the table; missing required output remains NOT EVALUABLE.

8 Technical interpretation

Agreement supports the coupled fuel-table selection, wind adjustment, slope factor, vector magnitude, and spread-rate output for the dominant generated condition. A plausible perimeter alone is not used as evidence.

9 Scope and limitations

The metric verifies the analytically observable domain maximum, not the complete discontinuous perimeter or every regional direction. It uses deterministic implementation verification rather than field validation.

Case 25: Spatially varying moisture quadrants

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE25_MQD. This case verifies spatial moisture lookup, moisture damping, and extinction behavior in otherwise uniform fuel. The verification question is whether the four regional spread rates meet their moisture-specific references, including negligible spread at the extinction condition.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

Fuel moisture reduces reaction intensity through the model’s moisture-damping term; at or above the fuel’s moisture of extinction, sustainable spread approaches zero. Only the one-hour moisture raster varies, so regional velocity differences should follow that damping response.

For a dead-fuel moisture ratio $r = M_f/M_x$, the Rothermel damping polynomial used in the configured range is

$$\eta_M(r) = 1 - 2.59r + 5.11r^2 - 3.52r^3, \quad 0 \leq r \leq 1. \quad (1)$$

Reaction intensity contains this damping factor, $I_R = \Gamma'(w_{n,d}\eta_{M,d} + w_{n,l}\eta_{M,l})\eta_s h$, and the no-wind rate is $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$. At or above extinction, the dead-fuel contribution cannot sustain propagation.

3 Derivation of expected behavior

For Fuel Model 3 the configured moisture values 6, 12, 18, and 25 percent correspond to reference rates 1.511, 1.089, 0.801, and 0 m min⁻¹. The rates must decrease monotonically as moisture rises.

The dead one-hour moisture ratio is varied while all other generated fuel and moisture properties remain fixed. Applying $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$ and $Q_{ig} = 250 + 1116M_f$ gives

M_{1h}	I_R (kW m ⁻²)	R_0 (m min ⁻¹)
0.06	549.174	1.511830
0.12	479.466	1.089719
0.18	413.934	0.801066
0.25	$\simeq 3.9 \times 10^{-13}$	$\simeq 6.5 \times 10^{-16}$

so the report references are 1.511, 1.089, 0.801, and 0 m min⁻¹. Nonzero rows use $e_R = |R_{\max} - R_{\text{ref}}|/R_{\text{ref}}$; the extinguished row uses the absolute residual $|R_{\max}|$, avoiding division by a zero reference.

4 Verification metrics and acceptance criteria

The maximum finite $\mathbf{vs}$ in each quadrant measures the local head response without averaging across moisture boundaries. Nonzero targets use 10 percent relative tolerance. The extinction quadrant uses an absolute 0.05 m min^{-1} limit because relative error is undefined at zero. All four components must pass.

5 Simulation configuration and predicted outcome

The grid uses uniform Fuel Model 3, flat terrain, zero wind, central ignition, and one-hour moisture quadrants of 6, 12, 18, and 25 percent. Predicted propagation slows monotonically and stops in the 25-percent quadrant.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

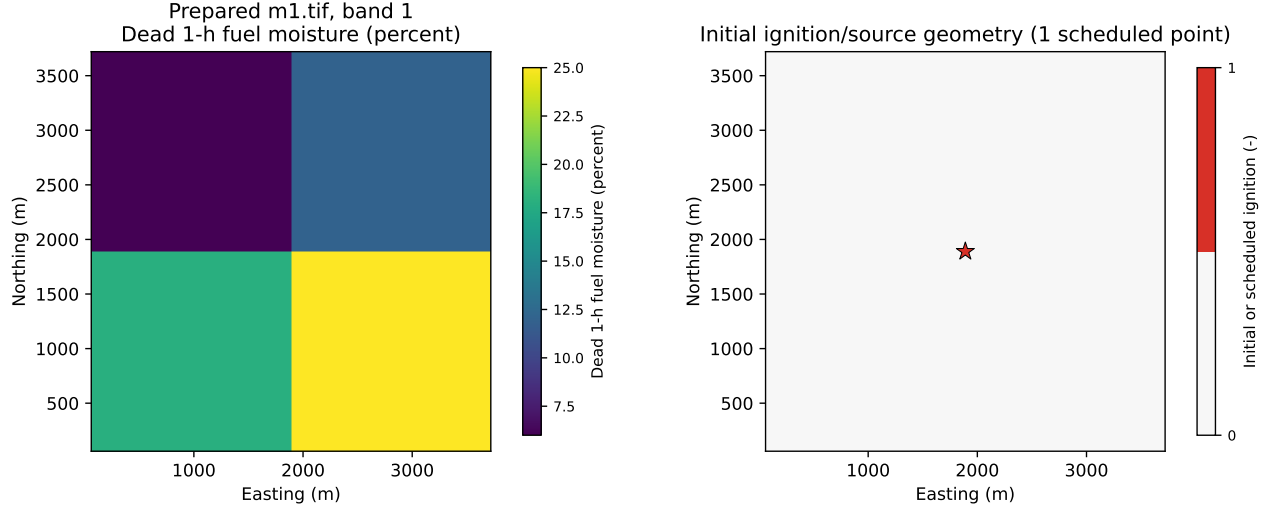


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

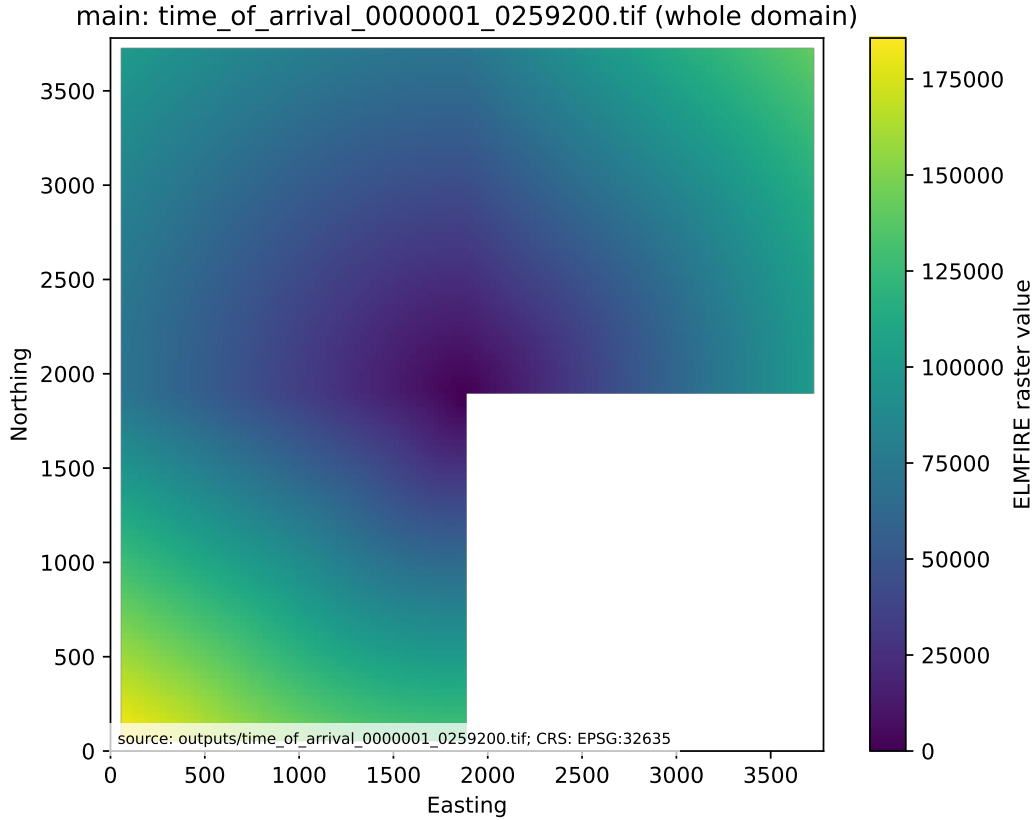


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / ROS @ 6%	1.511 m/min	1.51183 m/min	PASS
main / ROS @ 12%	1.089 m/min	1.08972 m/min	PASS
main / ROS @ 18%	0.801 m/min	0.801067 m/min	PASS
main / ROS @ 25%	0 m/min	0 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

Nonmonotone behavior suggests moisture raster or damping errors; significant spread in the extinction quadrant is a direct failure. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

Only one fuel model and the one-hour dead-fuel moisture field are varied. This case provides implementation verification evidence only.

Case 26: Surface, passive-crown, and active-crown fire

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE26_CNF. This case verifies the transition from surface fire to passive and active crown fire as wind increases. The verification question is whether all three wind variants produce the expected crown class and maximum spread rate within their respective tolerances.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

ELMFIRE compares surface fireline intensity with crown-initiation requirements and then evaluates active-crown spread and critical spread conditions. The output class encodes 0 for surface-only, 1 for passive crown fire, and 2 for active crown fire. Wind changes both the surface intensity and crown propagation response.

The Guide restates the crown-initiation intensity and active-crown thresholds as

$$I_0 = [0.010 CBH(460 + 25.9M_f)]^{3/2}, \quad (1)$$

$$R_0^{crown} = 3/CBD, \quad (2)$$

$$R_{C,P} = 11.02 U_{10}^{0.9} CBD^{0.19} e^{-0.17M_1}. \quad (3)$$

If surface fireline intensity $I_b < I_0$, fire remains surface-only. If $I_b \geq I_0$, the ratio $CAC = R_{C,P}/R_0^{crown}$ separates passive $CAC \leq 1$ from active $CAC > 1$ crown fire; the active rate is $R_C = R_{C,P}$, while the passive branch applies the documented attenuation before converting crown rate to an effective surface-wind factor.

3 Derivation of expected behavior

With fixed canopy and fuel properties, the 1, 5, and 10 mph variants are expected to produce classes 0, 1, and 2 with maximum rates 1.809, 10.06, and 35.3 m min⁻¹, respectively. The sequence 0 < 1 < 2 and increasing rate are invariants.

The generated canopy state has $CBH = 1.6$ m, $CBD = 0.10$ kg m⁻³, and foliar moisture $M_f = 90$ percent. Van Wagner's initiation threshold is

$$I_0 = [0.010 CBH(460 + 25.9M_f)]^{3/2} = [0.010(1.6)(460 + 25.9(90))]^{3/2} = 298.414 \text{ kW m}^{-1}.$$

The active-crown threshold rate is $R_0^c = 3/CBD = 3/0.10 = 30 \text{ m min}^{-1}$. The generated 20-ft winds $U_{20} = \{1, 5, 10\}$ mph become $U_{10} = 1.609344U_{20}/0.87 = \{1.8498, 9.2491, 18.4982\} \text{ km h}^{-1}$. Substitution in the crown-rate relation gives $R_c = \{4.4629, 18.9973, 35.4501\} \text{ m min}^{-1}$ and $CAC = R_c/R_0^c = \{0.1488, 0.6332, 1.1817\}$. Together with the surface intensity test against I_0 , these produce the reference classes surface, passive, and active and the coupled maximum-rate references 1.809, 10.06, 35.3 m min^{-1} . Class metrics use exact integer equality; each rate metric uses $|R_{\max} - R_{\text{ref}}|/R_{\text{ref}}$.

4 Verification metrics and acceptance criteria

Maximum crown class directly tests the discrete transition state and uses absolute tolerance 0.5 so only the intended integer class passes. Maximum finite `vs` tests the associated propagation rate with 10 percent relative tolerance. Both metrics must pass for every wind variant.

5 Simulation configuration and predicted outcome

All variants use generated Fuel Model 3, uniform canopy cover 80 percent, canopy height 55 (5.5 m), canopy base height 16 (1.6 m), and canopy bulk density 10 (0.10 kg m^{-3}) in the configured encoded raster units, with flat terrain and identical ignition. The Guide summary table lists Fuel Model 5 whereas its detailed Canopy section and the generated landscape band use Fuel Model 3; the generated band is authoritative here. Only wind speed changes among 1, 5, and 10 mph.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/1mph/data; WS=1 mph, WD=0 deg

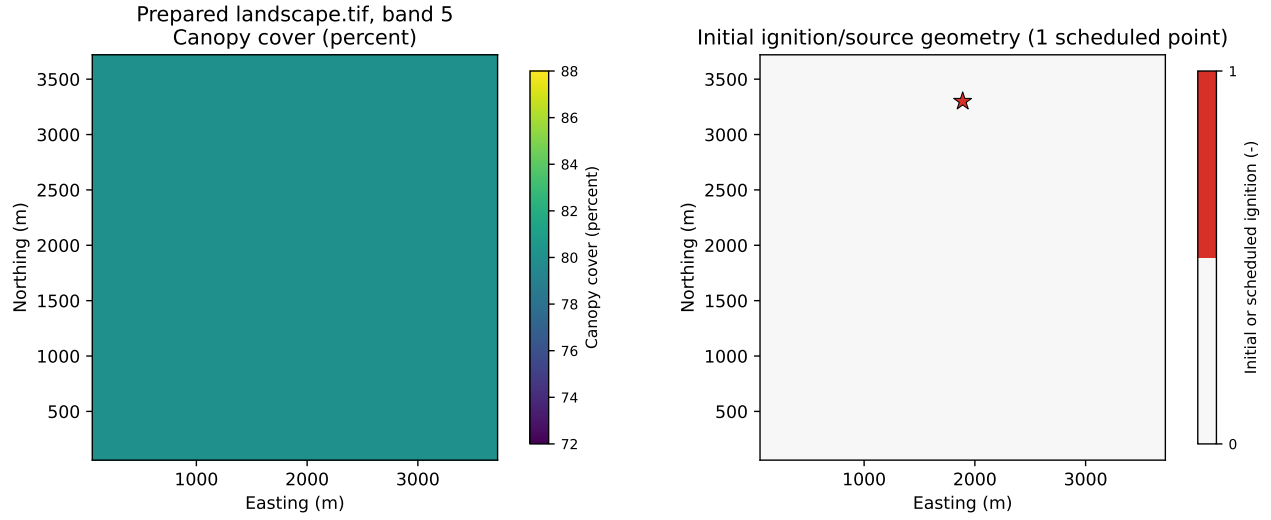


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 6 of 6 required metric rows pass across 3 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

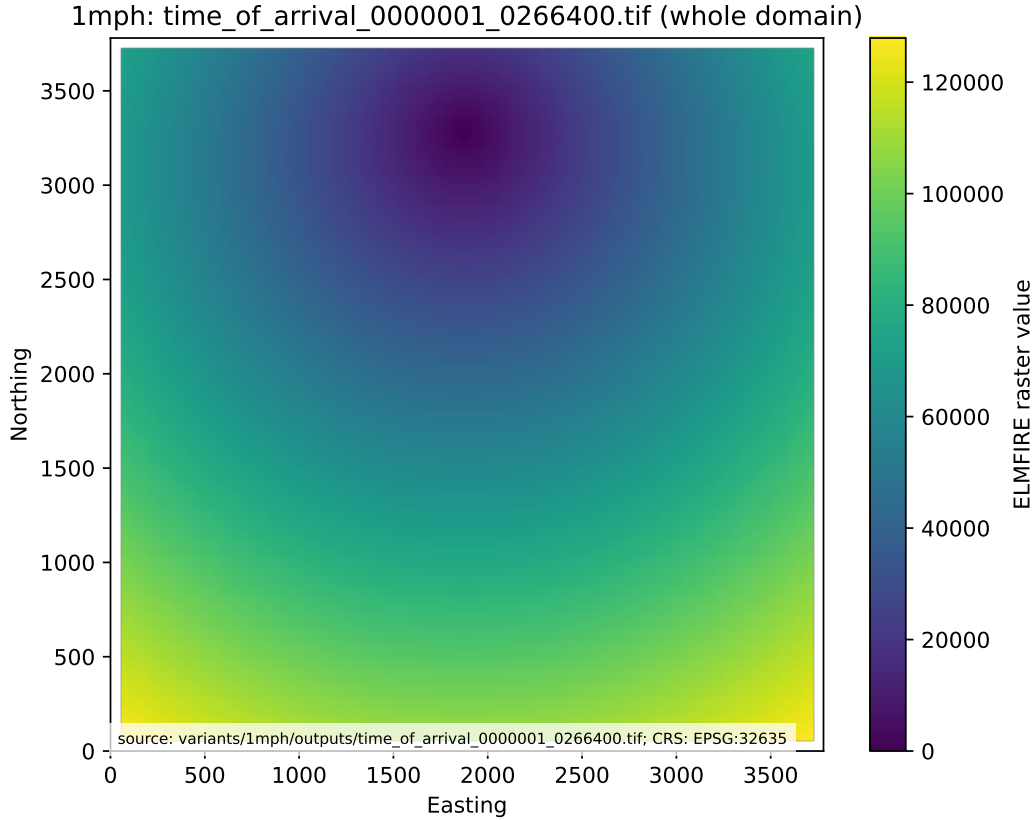


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
1mph / crown class	0 -	0 -	PASS
1mph / max ROS	1.809 m/min	1.74511 m/min	PASS
5mph / crown class	1 -	1 -	PASS
5mph / max ROS	10.06 m/min	10.0843 m/min	PASS
10mph / crown class	2 -	2 -	PASS
10mph / max ROS	35.3 m/min	35.4433 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A class failure with correct rate isolates transition logic or canopy-unit handling; coupled class and rate errors indicate wind/crown-fire integration. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test covers one canopy structure and three discrete winds; it does not validate crown-fire correlations against observations. This case provides implementation verification evidence only.

Case 27: Lagrangian firebrand generation and transport

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE27_FBT. This case verifies PER-MW firebrand generation and empirical Lagrangian transport for surface fire. The verification question is whether the dominant emitting cell's count and log-distance distribution agree with the independent GUIDE targets.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For the selected source cell, firebrands are generated in proportion to fire power. Empirical transport produces positive distances d_i . A lognormal response is summarized by $\mu = \text{mean}[\ln d_i]$ and $\sigma = \text{std}[\ln d_i]$. The postprocessor groups the spotting log by source indices before calculating these quantities.

For one emitting cell, the residence/crossing exposure is

$$\Delta t = \max(\Delta t_R, 60C/R), \quad N = 33.3 I_{b,MW} \Delta t. \quad (1)$$

For the generated 30 m cell, $R = 14.4$ m/min gives 125 s crossing time, longer than the 15.3 s fuel residence time, and the Guide calculation gives $N = 8450$. The empirical Sardoy transport uses a lognormal downwind distance: if $Y = \ln d$, then $Y \sim \mathcal{N}(\mu_d, \sigma_d^2)$. Its fireline-intensity, wind, and Froude-number calculation gives $\mu_d = 2.42$ and $\sigma_d = 1.31$ for this configuration.

3 Derivation of expected behavior

The configured experiment predicts 8450 firebrands from the dominant cell and log-distance parameters $\mu = 2.42$ and $\sigma = 1.31$. Grouping by source is essential: combining cells would mix different emission histories and would not test the intended distribution.

The Guide's firebrand-count relation is

$$N = 33.3 I \Delta t.$$

With generated fireline intensity $I = 2.03$ MW m⁻¹ and $\Delta t = \max(15.3, 60(30)/14.4) = 125$ s, $N = 33.3(2.03)(125) = 8449.875$, hence the integer reference is 8450. For the Sardoy distribution, the generated 5 mph wind becomes $U = 0.44704(5)/0.87 = 2.56897$ m s⁻¹; with $I = 2.03$ the flame-length

calculation gives $L_c = 1.57184$ m and $Fr = 0.65421 \leq 1$. The applicable parameter equations then give

$$\mu = \frac{1.47I^{0.54}}{U^{0.55}} + 1.14 = \frac{1.47(2.03)^{0.54}}{(2.56897)^{0.55}} + 1.14 = 2.42233,$$

$$\sigma = \frac{0.86U^{0.44}}{I^{0.21}} + 0.19 = \frac{0.86(2.56897)^{0.44}}{(2.03)^{0.21}} + 0.19 = 1.31258.$$

Thus the references are $N = 8450$, $\mu = 2.42$, and $\sigma = 1.31$; the generated table applies its stated absolute or relative tolerance to the corresponding difference.

4 Verification metrics and acceptance criteria

Dominant-cell count detects generation-rate or duration errors and uses 15 percent relative tolerance to accommodate stochastic sampling. The mean and standard deviation of $\ln d$ test location and width of the transport distribution and each use absolute tolerance 0.15. All three components require complete spotting logs and must pass.

5 Simulation configuration and predicted outcome

The case uses Fuel Model 3, flat terrain, 5 mph wind, PER-MW generation, the EMPIRICAL distance model, LAGRANGIAN accumulation, DIRECT ignition, and deterministic random-seed handling. Spotting statistics output is required. The Guide also discussed an Eulerian ember-flux magnitude/decay check, but the current case contract and generated configuration exercise the Lagrangian path only; no Eulerian verification claim is made.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves ignition and emitting-cell indices
Fuel and forcing	FM3; 5 mph wind	— / mph	Reproduces the Guide source-cell state
Time control	$\Delta t = 1$, $t_{stop} = 100$	s	Exact timestep multiple covering the logged emission interval
Spotting path	PER-MW / EM-PIRICAL / LAGRANGIAN / DIRECT	—	Selects the generation, distance, accumulation, and ignition algorithms
Required artifact	spotting statistics CSV	—	Supplies source index and travel distance for every sampled firebrand

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

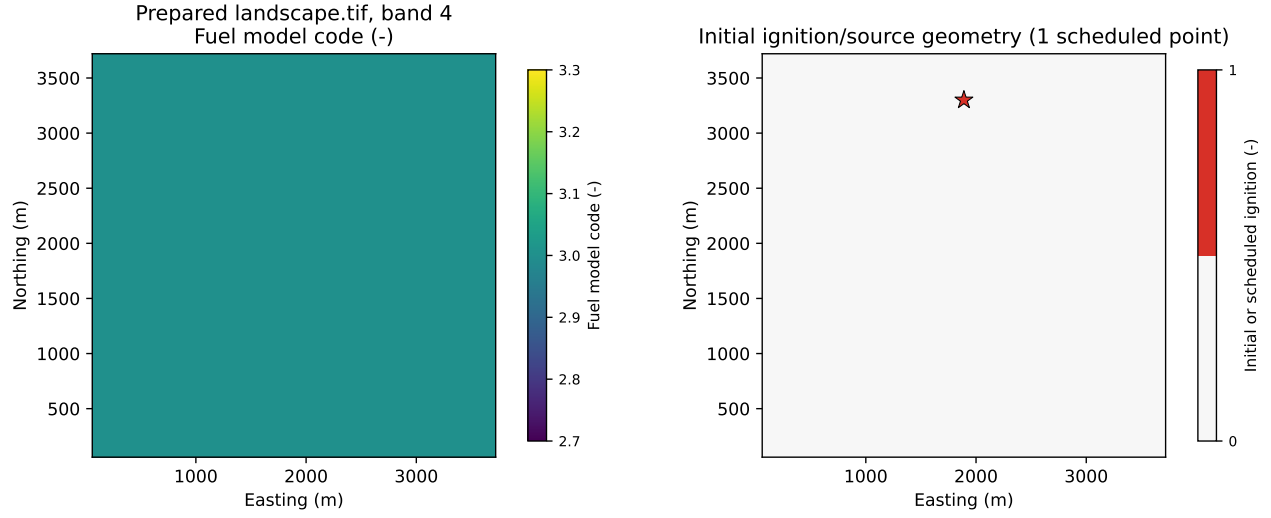


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **FAIL**; 2 of 3 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

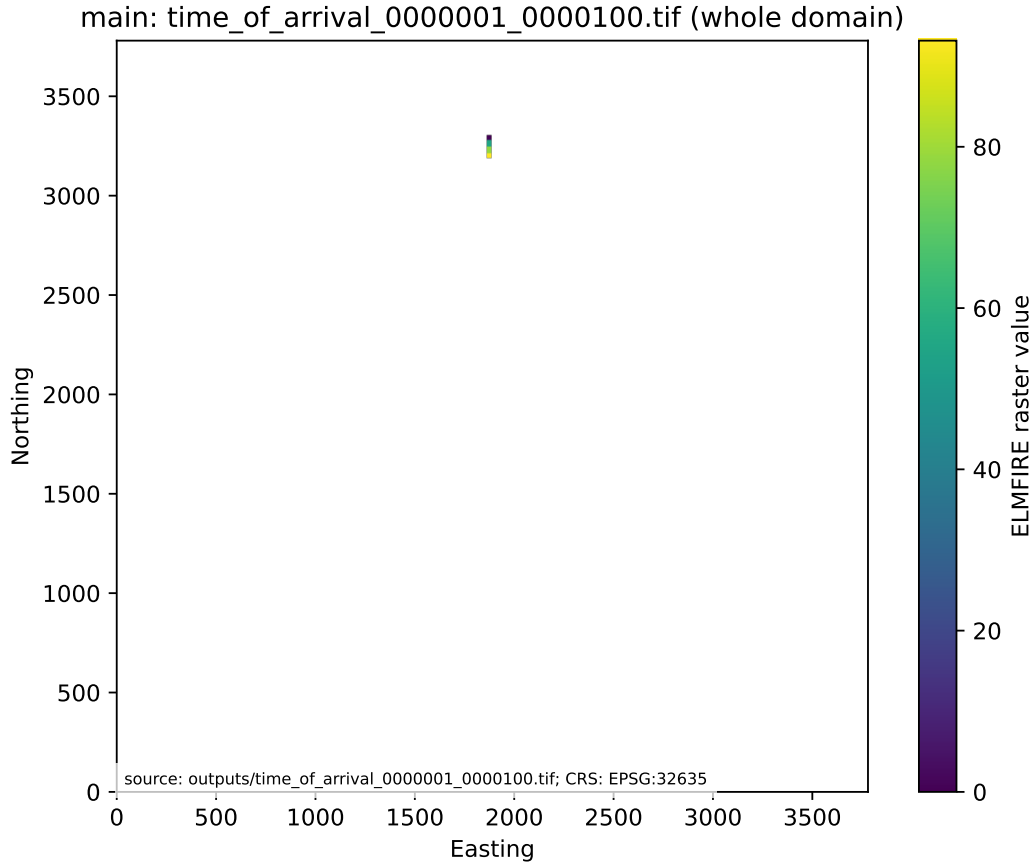


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / firebrands (cell)	8450 -	12198 -	FAIL
main / ln-distance mu	2.42 -	2.41119 -	PASS
main / ln-distance sigma	1.31 -	1.30025 -	PASS

The displayed overall decision is **FAIL**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

Count-only failure points toward generation or residence time; distribution-only failure points toward sampling, units, or transport parameterization. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The metrics cover the dominant source cell and one forcing condition. They verify implementation against the prescribed statistical target, not physical firebrand observations. This case provides implementation verification evidence only.

Case 28: Overnight spread-rate adjustment

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE28_OVA. This case verifies coupling among the simulation clock, solar burn-period calculation, and surface-fire diurnal multiplier. With uniform fuel, moisture, wind, and slope, the question is whether ELMFIRE reduces and restores radial advance at the prescribed clock hours, each within 1 hour. The case exposes clock offsets, failed midnight wrapping, reversed interval tests, or an unapplied multiplier; it does not calibrate the multiplier.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For simulation time t seconds and forecast start H_0 , ELMFIRE uses

$$H(t) = \text{mod}(H_0 + t/3600, 24). \quad (1)$$

The solar routine calculates sunrise and sunset from the lower-left coordinate, year, and hour of year. With default center fraction $f_c = 0.667$ and burn length $L_b = 10$ hour,

$$H_c = H_{\text{rise}} + f_c(H_{\text{set}} - H_{\text{rise}}), \quad (2)$$

$$H_b = H_c - L_b/2, \quad H_e = H_c + L_b/2. \quad (3)$$

The stop hour is wrapped above 24. For this non-wrapping interval,

$$D(t) = \begin{cases} 1, & H_b \leq H(t) \leq H_e, \\ 0.1, & H(t) < H_b \text{ or } H(t) > H_e. \end{cases} \quad (4)$$

The spread-rate calculation multiplies its unadjusted rate by $D(t)$. Uniform flat, wind-free conditions produce an approximately circular fire whose equivalent radius $r(t) = \sqrt{A(t)/\pi}$ has full and reduced slope plateaus.

3 Derivation of expected behavior

The controlled response specifies burn-period start 7.5 hour and stop 17.5 hour. Full-to-reduced and reduced-to-full transitions are therefore expected at those respective clock hours. Since $H_0 = 10$ hour,

the events first occur approximately 7.5 hour and 21.5 hour after ignition and repeat during the three-day run. This oracle follows from the controlled targets and clock equations; calculated times are independently inferred from the arrival raster.

For the generated astronomical inputs, local sunrise and sunset are approximately $H_r = 4.5$ h and $H_s = 16.5$ h. The operational cutoff with fractional daylight $f_c = 0.667$ is

$$H_c = H_r + f_c(H_s - H_r) = 4.5 + 0.667(16.5 - 4.5) = 12.504 \text{ h} \simeq 12.5 \text{ h}.$$

A ten-hour operating window centered on H_c therefore has bounds $H_c \pm 5 = \{7.5, 17.5\}$ h. Relative to the 10:00 forecast start, the first reduction occurs after $(17.5 - 10)(3600) = 27000$ s; after midnight wrapping, full operation resumes after $(24 - 10 + 7.5)(3600) = 77400$ s. The transition metrics compare detected changes with these reference times using the tabulated timing tolerance.

4 Verification metrics and acceptance criteria

The evaluator selects the lexically latest `time_of_arrival*` GeoTIFF, masks nodata and values outside $0 \leq t < 10^8$ seconds, and requires 50 burned cells. Analysis ends at the earliest domain-edge arrival so boundary truncation cannot mimic slowdown. At 30 minute intervals, burned-cell count is converted to equivalent radius and differenced. After radius exceeds three cells, rates below half the 80th percentile of positive rates are reduced. Median full-to-reduced and reduced-to-full clock hours are then calculated.

Metric	Target	Acceptance rule	Role
Reduction starts	17.5 hour	absolute error ≤ 1 hour	burn-period stop
Reduction ends	7.5 hour	absolute error ≤ 1 hour	burn-period start

Both rows are required. A missing raster, insufficient burned area or duration, or undetected transition produces NOT EVALUABLE, never PASS.

5 Simulation configuration and predicted outcome

Input	Value	Verification role
Domain/grid	126×126 ; 30 m	circular-front resolution
CRS	EPSG:32635	location for solar calculation
Fuel/terrain	uniform FM3; zero slope	constant baseline spread
Wind/dead moisture	zero; 6, 7, 8 percent	constant forcing
Live moisture	herb 60; woody 90 percent	constant forcing
Ignition	(1890, 1890) m at $t = 0$	centered radial growth
Time control	1 s; 259 200 s stop	three daily cycles
Calendar	year 2025; hour 2890; start 10	clock coupling
Diurnal control	enabled; factor 0.1	two rate plateaus
Output	final TOA, spread rate, FLIN	case-local evidence

The predicted arrival field is concentric; contour spacing contracts overnight and expands during the burn period, with both transitions inside their acceptance bands.

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells

selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

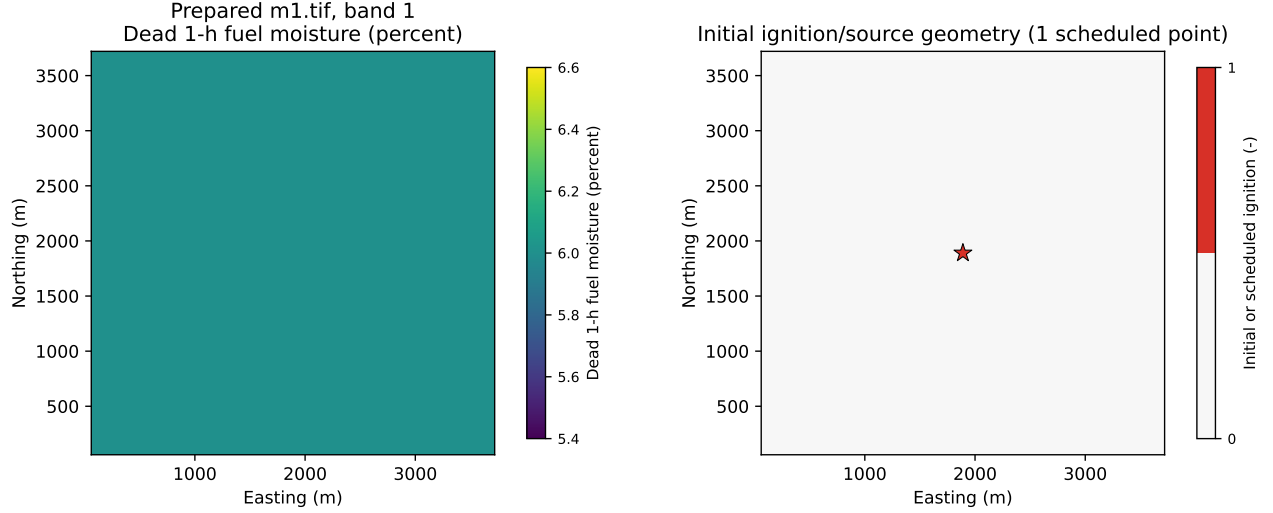


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The current artifact set contains no case-local metrics JSON or whole-domain result figure. Accordingly, the displayed status is **PASS**; 2 of 2 required rows pass across 1 completed variants. Missing measurements remain NOT EVALUABLE and historical Guide results are not substituted.

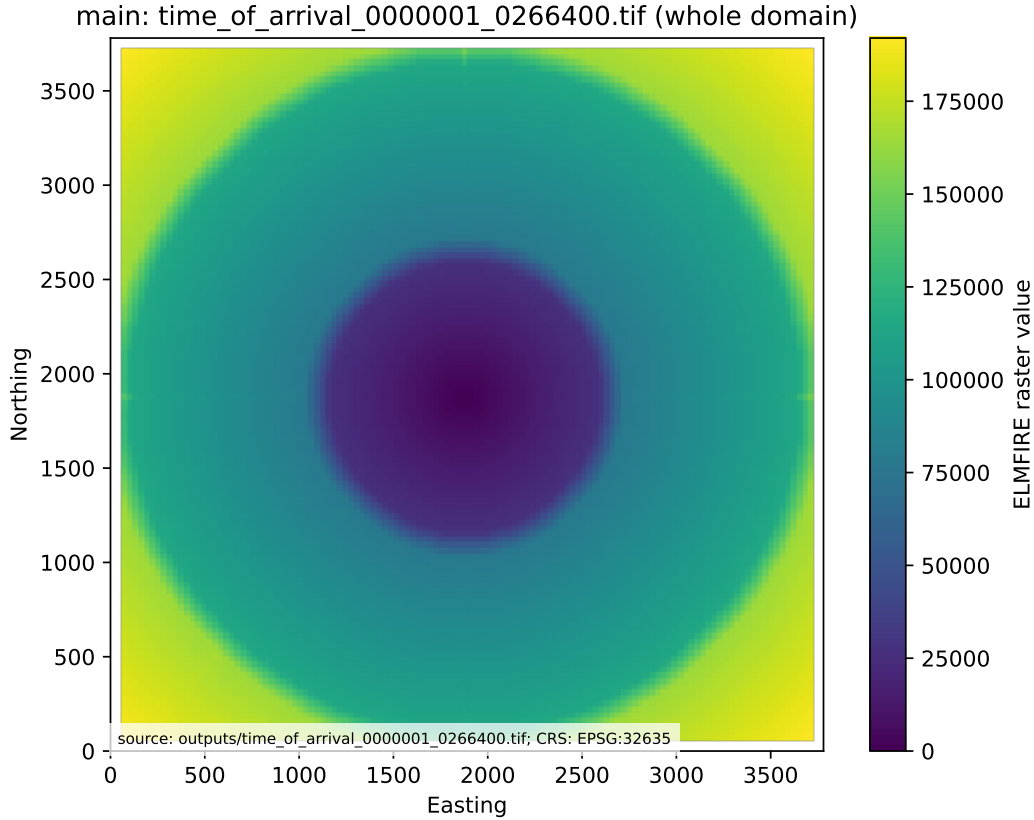


Figure 2: Final whole-domain time of arrival, in seconds, when a case-local result figure has been generated. The figure must identify its selected GeoTIFF and variant; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / reduction starts (hr)	17.5 hr	17.75 hr	PASS
main / reduction ends (hr)	7.5 hr	7.75 hr	PASS

When `outputs/metrics.json` exists, `scripts/postprocess.py` generates this table and **PASS**; when it is absent, the wrapper renders NOT EVALUABLE and no metric values are invented. Comparison symbols remain in LaTeX math mode, so $\leq$ and $\geq$ cannot be misparsed as table markup.

8 Technical interpretation

No case-local metrics artifact is currently available, so the report correctly remains NOT EVALUABLE and no timing claim is made. Once complete outputs exist, transition errors within the one-hour bands would support clock, solar-window, and diurnal-factor coupling; a common shift would implicate clock, calendar, coordinate, or solar timing, while missing contrast would suggest an unapplied multiplier or insufficient pre-edge history.

9 Scope and limitations

This synthetic case covers one date, location, fuel, factor, and grid. It does not validate factor magnitude, other seasons or coordinate systems, a burn interval wrapping midnight, heterogeneity, wind, or crown-fire, spotting, suppression, and interpolation interactions. The estimator assumes circular growth, discards post-edge behavior, samples every 30 minute, and infers rather than records the internal factor. This is implementation verification, not validation of real-fire nocturnal behavior.

Case 29: Initial- and extended-attack suppression

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE29_SUP. This coupling case verifies two suppression paths in the standard ELMFIRE executable. The initial-attack path in `LEVEL.SET_PROPAGATION` computes a containment probability from fire area and maximum active-front fireline intensity, draws one Bernoulli outcome per ensemble member, and records final containment. The extended-attack path periodically updates an area-growth-based target containment and calls the containment routine that modifies the active perimeter. These paths matter because suppression can terminate an ensemble member or alter and eventually stop an advancing level set.

The case controls fuel, moisture, terrain, wind, ignition, grid, and output selection. It varies only the enabled suppression algorithm and the ensemble size needed by that algorithm. Spotting, crown-fire evidence, operational resource allocation, and comparison with observed incidents are outside the experiment. Defects exposed include incorrect containment probability, ensemble accounting, suppression-update timing, containment-state evolution, stop-time bookkeeping, and hour-to-second conversion. This is implementation verification against documented synthetic reference responses, not physical validation.

The verification question is: do the 2000-member initial-attack ensemble and the required extended-attack run produce, respectively, a contained fraction within 0.05 of 0.54 and a reported simulation-end time within 15% of 88200 s, with both required artifacts complete?

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For initial attack, ELMFIRE evaluates the burned area A in hectares and the largest surface-plus-canopy fireline intensity $I_{\max}$ on cells ignited during the most recent $\max(300, \Delta t)$ seconds. It forms

$$E = \exp(4.6835 - 0.7043A - 0.00041I_{\max} - 0.000052AI_{\max}), \quad (1)$$

$$p_c = \min\left(1, \max\left(0, \frac{E}{1 + E}\right)\right). \quad (2)$$

One uniform random draw determines containment. A contained member receives `Containfrac`= 1 and stops at `INITIAL_ATTACK_TIME`; an escaped member retains the nodata containment sentinel. For

N members, the observable estimator is

$$\hat{p}_c = \frac{1}{N} \sum_{j=1}^N \mathbf{1}(\text{Containfrac}_j \geq 0.999). \quad (3)$$

The extended variant uses `EXTENDED_ATTACK_MODEL=0`, the default area-growth model. At updates separated by approximately $\Delta t_{EA} = 300$ s, it estimates daily area growth dA/dt , computes a daily containment increment from `MAX_CONTAINMENT_PER_DAY`, `AREA_NO_CONTAINMENT_CHANGE`, and the suppression-difficulty factor, clips target containment to $[0, 1]$, and calls `CONTAINMENT`. With the configured default linear growth response, the essential update is

$$C_k = \text{clip}_{[0,1]} \left(C_{k-1} + \dot{C}_k \frac{t_k - t_{k-1}}{86400} \right). \quad (4)$$

The level set advances on a 30 m north-up EPSG:32635 grid. ELMFIRE records final simulation time in hours in `fire_size_stats.csv`; the evaluator converts that scalar to seconds. The final TOA GeoTIFF stores arrival time in seconds and is spatial evidence, but it is not substituted for the prescribed scalar stop-time metric.

3 Derivation of expected behavior

The synthetic initial-attack reference predicts $p_c = 0.54$. Under independent Bernoulli trials, a 2000-member ensemble has reference standard error

$$\sigma_{\hat{p}} = \sqrt{\frac{0.54(1 - 0.54)}{2000}} \simeq 0.0112. \quad (5)$$

Thus the absolute acceptance band $[0.49, 0.59]$ is about 4.5 reference standard errors wide: it admits expected sampling fluctuation while remaining sensitive to material probability or accounting errors.

The documented extended-attack known response is containment and simulation termination at 88200 s. Because containment is updated discretely and depends on the coupled area-growth history, this is a controlled reference value rather than an analytical solution. The inherited 15% band is $[74970, 101430]$ s. The two expectations are independent: agreement of the stochastic ensemble cannot compensate for a wrong extended-attack end time. The Guide additionally describes a spatially explicit suppression variant expected to complete on its third direct-attack operation. That variant is not present in the current generated two-variant contract, so its historical discussion is preserved as scope context but no decision is claimed for it.

The initial containment probability is calculated from the empirical logit model

$$\ln E = 4.6835 - 0.7043A - 0.00041I - 0.000052AI, \quad p = \frac{E}{1 + E}.$$

With the generated $A = 6.2$ ha and $I = 212.5$ kW m⁻¹,

$$\ln E = 4.6835 - 0.7043(6.2) - 0.00041(212.5) - 0.000052(6.2)(212.5) = 0.161205,$$

so $E = e^{0.161205} = 1.17493$ and $p = 1.17493/(1 + 1.17493) = 0.540214 \simeq 0.54$. For $N = 2000$ trials, the sampling scale is $\sqrt{p(1-p)/N} = \sqrt{0.54(0.46)/2000} = 0.0112$; the prescribed acceptance band is the conservative interval $0.49 \leq \hat{p} \leq 0.59$. The extended-operation time reference is $24.5(3600) = 88200$ s, or $88200/300 = 294$ exact event-analysis intervals. Its 15 percent timing band is $0.85(88200) = 74970$ through $1.15(88200) = 101430$ s.

4 Verification metrics and acceptance criteria

Both metrics are read from the first lexically sorted `fire_size_stats*.csv` in the output directory of the applicable local variant. CSV headers are stripped before selection, nonnumeric entries are excluded, and every numeric row—including the escape sentinel—is retained in the denominator. The current artifact contains all 2000 configured rows.

Expected property	Observable and metric		Selection rationale	Component decision
Initial containment probability	$\hat{p}_c$ $N(\text{Containfrac} - 0.999)/N$	= ≥	Directly tests the Bernoulli outcome and ensemble accounting; the nodata sentinel denotes escape rather than a missing row.	PASS when the extracted fraction is finite and $ \hat{p}_c - 0.54 \leq 0.05$
Extended containment end time	t_{end} 3600 <code>tstop(h)</code> from the first statistics row	=	Uses the executable field defined as the simulation stop time and directly exposes time-unit or termination bookkeeping errors.	PASS when the value is finite and $ t_{end} - 88200 /88200 \leq 0.15$.

A missing output directory is NOT RUN; a present but absent or malformed required value is NOT EVALUABLE. The overall decision is PASS only when both component decisions are PASS. Any calculated component outside its band makes the overall decision FAIL; unavailable evidence never passes.

5 Simulation configuration and predicted outcome

Both variants use case-local rasters generated on a 126×126 grid with 30 m cells, a 3780 m square domain, lower-left origin (0,0), north-up rows, EPSG:32635, nodata `-9999`, flat terrain, zero wind, and one central point ignition at (1890,1890) m and $t = 0$. Live-herb and live-woody moisture are constant at 60% and 90%. The relevant differences are summarized below.

Parameter	Initial attack	Extended attack	Role
Fuel model	3	8	Prescribed synthetic fire state
Dead 1-h moisture	6%	6%	Fixed surface-fuel moisture
Suppression selector	initial enabled	extended enabled, model 0	Path under test
Attack/update time	4800 s	300 s interval	Event and update timing
Ensemble members	2000	1	Probability estimate versus deterministic trajectory
Nominal stop time	8000 s	250000 s	Upper bound, not the reference end time
Base timestep / target CFL	1 s / 0.2	1 s / 0.2	Stable level-set integration
Primary output	fire-size statistics	fire-size statistics and TOA	Required scalar evidence and spatial context

Both namelists set `RANDOMIZE_RANDOM_SEED=.TRUE.`; consequently the initial result is statistically reproducible within its acceptance band but is not expected to be bitwise repeatable. The predicted initial result is a contained fraction near 0.54. The predicted extended result is a growing, progressively contained footprint whose statistics record ends near 88200 s.

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

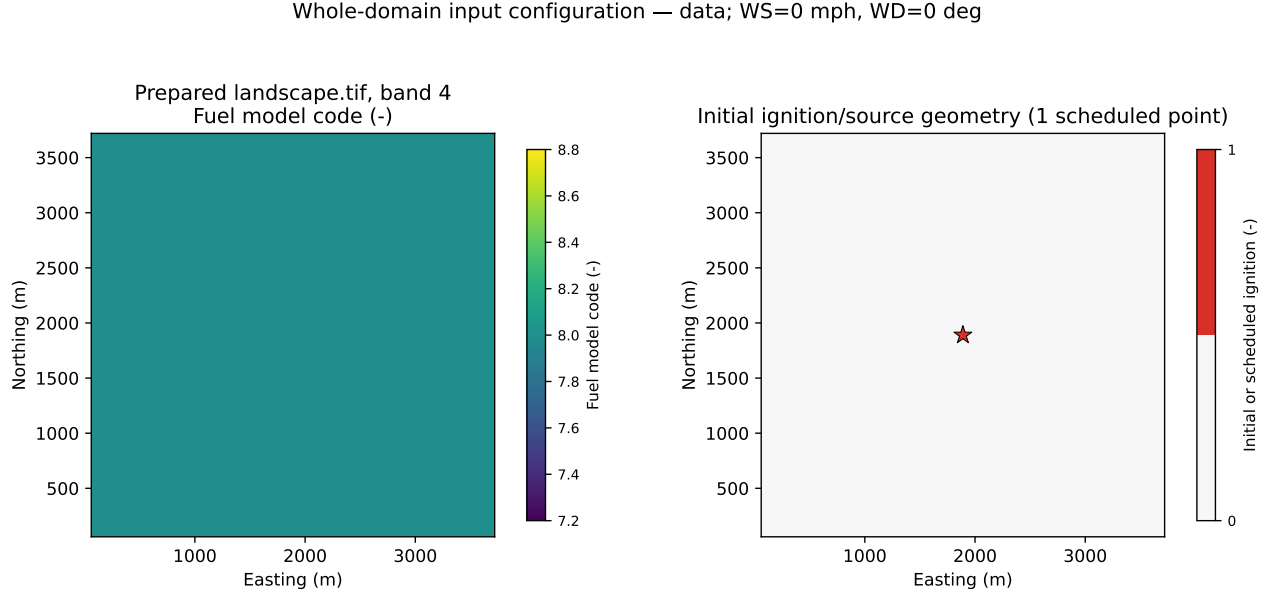


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The corrected case was executed with an ELMFIRE binary built from the current source tree. The initial-attack ensemble passes: its measured contained fraction is reported in the generated table below and lies inside the prescribed absolute tolerance. The extended-attack member does not reproduce the Guide response. It is stopped by the fire-front-stalled check before containment can evolve to the expected termination time; fire-size statistics consequently report the stop time as 0.00 h at their two-decimal output precision. Thus 1 of 2 required metrics pass and the current overall decision is **FAIL**.

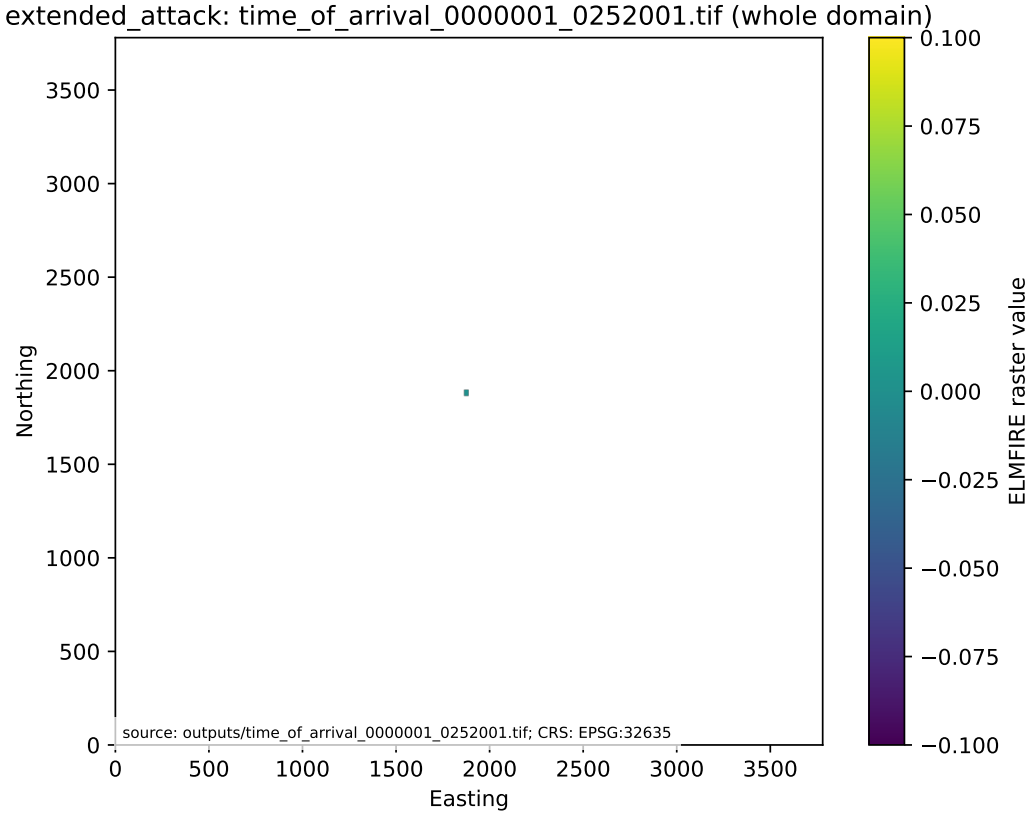


Figure 2: Whole-domain extended-attack time of arrival in seconds when a case-local output figure exists. It provides spatial context but cannot substitute for the scalar containment and end-time metrics.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
extended_attack / time at simulation end	88200 s	0 s	FAIL
initial_attack / fraction contained	0.54 -	0.5785 -	PASS

The displayed decision is **FAIL**. When case-local outputs and `outputs/metrics.json` exist, `scripts/postprocess.py` generates every value and status through `report/metrics_macros.tex`; when they are absent, the wrapper renders NOT EVALUABLE without invented values.

8 Technical interpretation

The earlier case configuration used a 3600 s extended-attack interval, whereas the Guide area-growth verification prescribes 300 s; the source namelist has been corrected. That correction does not make the current-source result pass. With Fuel Model 8, the initial 1 s level-set change is below the absolute 0.001 stalled-front threshold in `elmfire_level_set.f90`. The stalled test is applied before adaptive-CFL integration has advanced beyond its startup steps, so this valid slow-spreading fire is classified as stalled essentially at ignition. Terminal bookkeeping then rounds the very small statistics stop time to 0.00 h. This remaining discrepancy is attributed to current ELMFIRE stall and termination logic, not to another defensible case-configuration change. Increasing the initial timestep merely to evade the

absolute threshold would change the prescribed Point setup and conceal the source behavior rather than verify it.

9 Scope and limitations

This case demonstrates only the two configured suppression pathways for uniform synthetic fuels, flat terrain, zero wind, one ignition location, one grid spacing, and the present executable/output conventions. It does not verify spatially explicit extended-attack model 1, indirect-attack geometry, nonuniform fuels, wind or slope interactions, resource limitations, resolution convergence, or operational suppression effectiveness. The 0.54 initial target is assessed statistically and can vary between reruns because seed randomization is enabled. The 88200 s extended target is a documented controlled-response reference rather than an analytical solution, so an intentional suppression-model change requires review of the oracle and tolerance. The evidence is implementation verification only and supplies no physical-validation claim.

Case 30: Multiple-ignition topology

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE30.MIT. This case verifies that the standard ELMFIRE executable initializes, schedules, transports, and merges multiple point ignitions without depending on source-tree test drivers. It preserves the scientific intent of Adam Laird’s dual-ignition, staggered spot-merging, and near-boundary component scenarios.

2 Mathematical formulation and numerical implementation

Each ignition introduces a local negative level-set region at position (x_i, y_i) and time t_i . ELMFIRE then advances ϕ according to $\partial_t \phi + V_n |\nabla \phi| = 0$. With homogeneous fuel, zero slope, and zero wind, simultaneous mirrored sources must remain reflection symmetric. Adding uniform wind must produce a measurably different arrival-time solution. Delayed sources must appear at their requested times, and clipping at a domain boundary must not suppress or wrap a source.

3 Derivation of expected behavior

The no-wind forcing and ignition pair are invariant under reflection about their mapped raster axes; therefore the discrete arrival-time solution has the same symmetries. Because ELMFIRE maps coordinates to one-based cells with a ceiling operation, the evaluator reflects the largest balanced row window about the actual ignition row rather than an intercell domain line. The wind variant changes only the wind rasters, so a nonzero solution difference is a metamorphic check of weather coupling. For scheduled ignition i , the containing raster cell should receive an arrival time within one 1200 s of t_i . The eight boundary sources are placed one five-cell margin inside the square domain and all must initialize.

For any two arrival-time arrays A and B , the implemented comparison is

$$E(A, B) = \frac{\text{mean}_{i \in V} |A_i - B_i|}{\max[P_{90}(A_i : i \in V), 1 \text{ s}]}, \quad V = \{i : A_i, B_i \text{ are finite}\}.$$

For the two simultaneous ignitions at $(-52.5, 0)$ and $(52.5, 0)$ m, the reflection reference is exact symmetry, $E = 0$; the metric is the larger of the left–right and centered top–bottom reflection errors and must satisfy $E \leq 0.02$. The 5 mph, 225-degree wind run is compared with the no-wind run by the same formula and must give $E \geq 0.05$, demonstrating a resolved forcing response. For scheduled ignitions (x, y, t) equal to $(0, 600, 0)$, $(0, 300, 1200)$, $(0, 0, 2400)$, $(0, -300, 3600)$, and $(0, -600, 4800)$, the timing metric is

$$E_t = \max_j |TOA(x_j, y_j) - t_j| \leq 1200 \text{ s}.$$

Finally, the count metric is $\sum_{j=1}^8 \mathbf{1}[TOA(x_j, y_j) \leq 1200 \text{ s}] = 8$ at the eight generated edge/corner points $(\pm 975, 0)$, $(0, \pm 975)$, and $(\pm 975, \pm 975)$ m.

4 Verification metrics and acceptance criteria

Reflection error is the larger normalized mean absolute difference between the no-wind arrival field and its reflection about each mapped ignition axis; it must not exceed 0.02. Wind response is the normalized mean absolute difference between the wind and no-wind fields and must be at least 0.05. The maximum scheduled-source timing error must not exceed one output interval. All eight boundary-source cells must have finite arrival times no later than one output interval. These metrics were selected because they test invariants of the coupled ignition/level-set implementation without reproducing ELMFIRE’s internal algorithms.

5 Simulation configuration and predicted outcome

All variants use a 5 m square grid in EPSG:32610, Fuel Model 10, 3/5/6 percent dead-fuel moisture, and flat terrain. The variants contain two simultaneous no-wind sources, the same pair under 5 mph wind from 225 degrees, five staggered sources, and eight edge/corner sources. In the staggered variant the first source is anchored at ($t=0$) and the other four follow at 1200-s intervals. This common translation of the original absolute schedule preserves every relative delay while preventing versions that have no active front from advancing past a strictly future first ignition. The predicted fields are respectively symmetric merger, wind-distorted merger, correctly timed merger, and eight clipped but valid boundary fires.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Resolves source placement and merging
Fuel/moisture	FM10; 3/5/6	— / percent	Homogeneous surface response
Time control	step 1, maximum 5; stop 14400	s	Exact timestep multiple for scheduled sources
Variants	dual calm, dual wind, staggered, boundary	—	Exercises symmetry, weather, timing, and clipping
Output cadence	1200	s	Timing-error acceptance scale

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

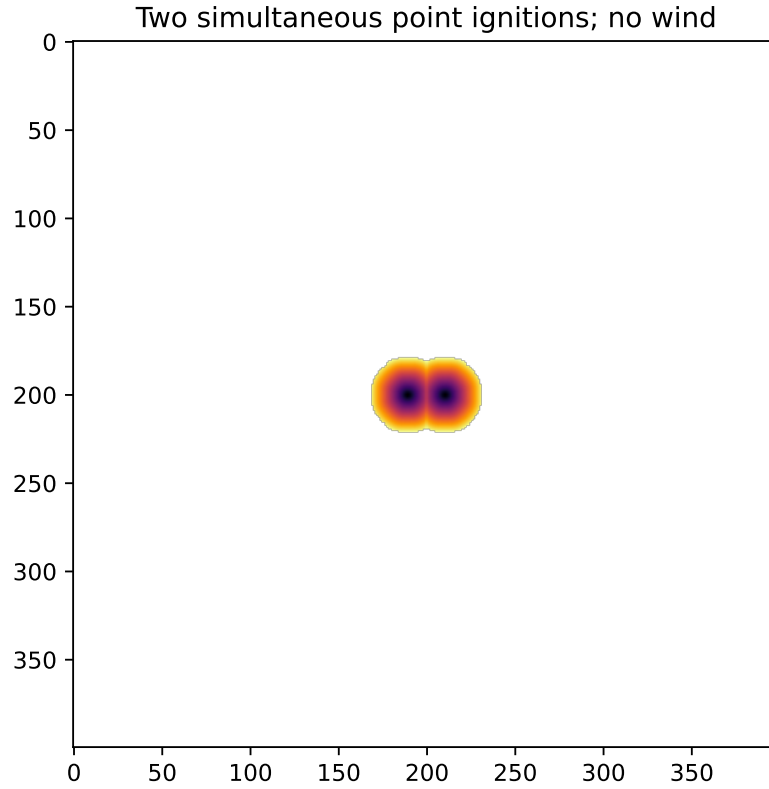


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated result is **PASS**; 4 of 4 required metrics pass across 4 variants. Missing ELMFIRE output yields NOT EVALUABLE rather than a pass.

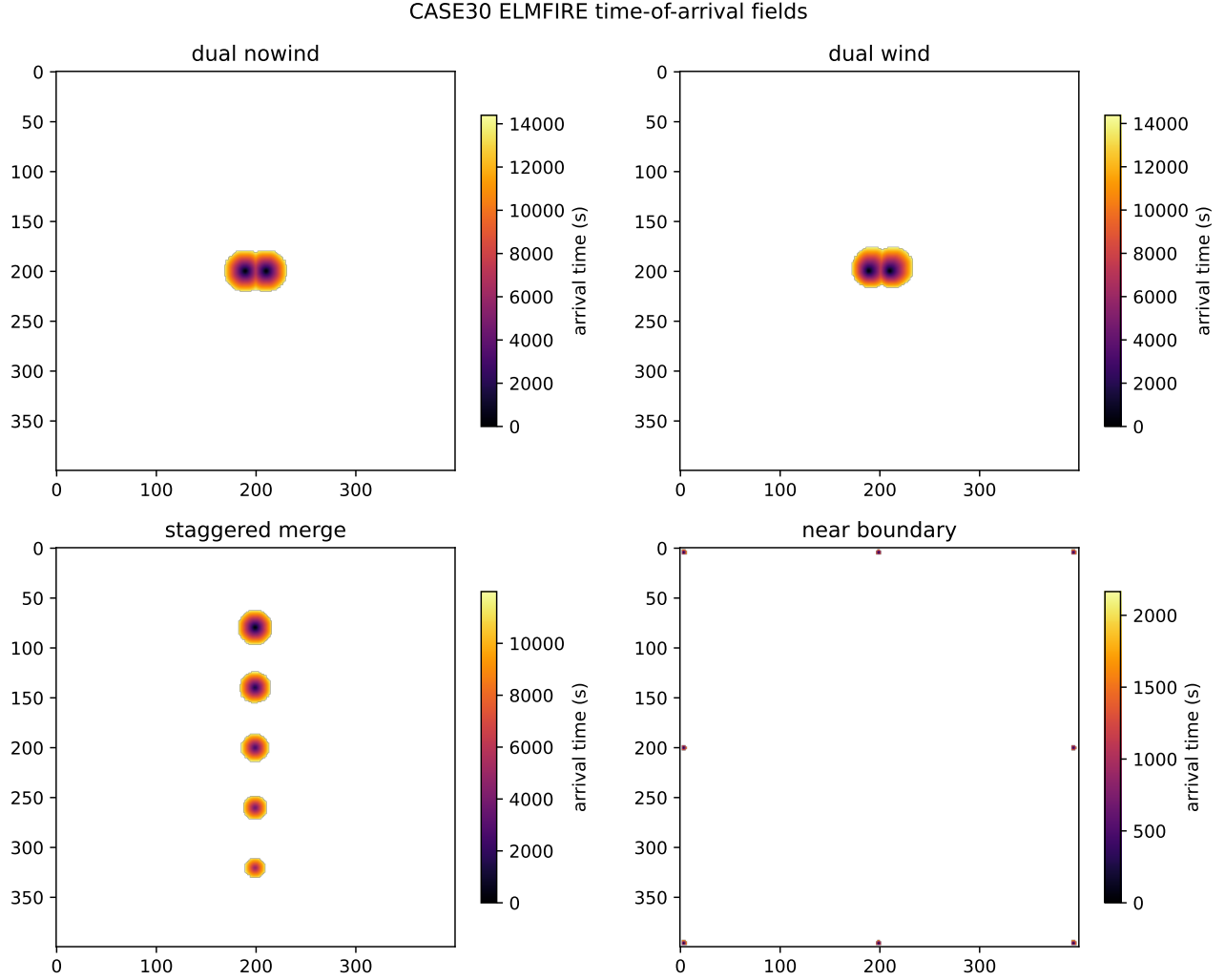


Figure 2: Case-local time-of-arrival fields for the four ignition-topology variants.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
dual-ignition reflection error	≤ 0.02 (2% normalized L1)	0.000472284	PASS
wind-forcing response	≥ 0.05 (5% normalized L1)	0.0937636	PASS
maximum scheduled-ignition timing error	≤ 1200 (one output interval)	0 s	PASS
boundary ignitions initialized	8 (all eight points)	8 count	PASS

The authoritative decision is **PASS** from `outputs/metrics.json`.

8 Technical interpretation

A symmetry failure points to directional indexing or level-set transport. No wind response indicates that the weather field did not reach fire behavior. Timing error implicates the scheduled-ignition path. Missing edge ignitions indicates boundary-index clipping or coordinate conversion failure.

9 Scope and limitations

This is deterministic implementation verification, not fire-behavior validation. It does not expose the internal Fortran ignition routines directly; it checks their observable behavior through an unmodified ELMFIRE executable.

Case 31: Planar-front transport

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE31_PFT. This case verifies planar level-set transport across diagonal geometry, a discontinuous fuel field, uniform wind, and a nonburnable fuel break. It consolidates Adam Laird’s diagonal-front, fuel-jump, fuel-jump-with-wind, and fuel-break scenarios into one portable black-box case.

2 Mathematical formulation and numerical implementation

For a planar front with signed-distance coordinate s , the level-set equation reduces to $\partial_t \phi + V_n \partial_s \phi = 0$. Uniform forcing should preserve planarity. At a fuel boundary V_n changes but the arrival-time field remains nearly constant along lines parallel to the front. A nonburnable strip imposes $V_n = 0$ except at an explicit gap.

3 Derivation of expected behavior

A uniform diagonal front is translationally invariant along its tangent. The fuel-jump solution is piecewise linear in normal distance, $T(s) = \int ds/V_n(s)$, so downstream time must increase while cross-front variation remains small. A 1 mph wind from 270 degrees aids eastward transport and must reduce downstream arrival time relative to the otherwise identical no-wind variant. A continuous nonburnable strip blocks transport; opening a gap creates a strictly larger burned fraction downstream.

At every declared station, arrival times are restricted to the central $|y| \leq 350$ m band (or the equivalent diagonal tangent band). The local planarity measure is

$$CV_j = \frac{\text{std}(T_j)}{\max[\text{mean}(T_j), 1 \text{ s}]}, \quad CV_{\max} = \max_j CV_j.$$

For the diagonal front, j spans nine normal-coordinate stations from 25 to 275 m; for the fuel jump it spans $x = \{-50, -20, 10, 40, 70, 100\}$ m. Both substituted tests are $CV_{\max} \leq 0.03$, with at least 90 percent finite station coverage. Let $\tilde{T}(x)$ be the station median. The transition metric is

$$\Delta T = \tilde{T}(100) - \tilde{T}(-50) > 0,$$

and the wind response is

$$q_w = \frac{\tilde{T}_{\text{wind}}(100)}{\tilde{T}_{\text{no wind}}(100)} \leq 0.95.$$

For the continuous and gapped Fuel Model 91 breaks, define $f(x) = N_{\text{finite}}/N_{\text{sampled}}$. After the required 90 percent upstream/barrier/opening coverage checks, the substituted barrier metric is $G = f_{\text{gapped}}(100) - f_{\text{continuous}}(100) \geq 0.10$.

4 Verification metrics and acceptance criteria

Cross-front coefficients of variation for diagonal and fuel-jump fronts must each be at most 3 percent. The nine diagonal-normal stations are 25, 56.25, 87.5, 118.75, 150, 181.25, 212.5, 243.75, and 275 m beyond the initial leading face. The six vertical-front stations are $x = -50, -20, 10, 40, 70$, and 100 m; each is sampled at the nearest raster column. Every station must have at least 90 percent finite, positive arrival-time coverage in the central $|y| \leq 350$ m corridor (or the corresponding diagonal tangent corridor). The evaluator reports the maximum station coefficient of variation, so one poorly resolved station cannot be hidden by aggregation. The downstream fuel-transition arrival-time increment between $x = -50$ m and $x = 100$ m must be positive. The wind/no-wind arrival-time ratio at $x = 100$ m must be at most 0.95. The burned-fraction gain at the same downstream station produced by the gap must be at least 0.10. Before that absence can be classified as a failure, both break variants must have at least 90 percent arrival coverage at $x = -50$ m, the continuous case must also reach the near-barrier station at $x = -5$ m, the gapped case must have at least 90 percent coverage through the opening at $x = 35$ m within $|y| \leq 40$ m, and its final dump must reach or exceed the configured stop time. Otherwise the barrier metric is NOT EVALUABLE. These geometry and metamorphic metrics avoid calibrating against ELMFIRE itself and remain valid across executable versions.

5 Simulation configuration and predicted outcome

All five variants use a 1500×1500 m domain at 5 m, Fuel Models 10 and 3 where applicable, and 3/5/6 percent dead-fuel moisture. The input `phi.tif` is a finite-strip signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval. It retains an ignited interior and a resolved planar leading-face gradient without relying on a grid cell whose value is exactly zero; its ends remain inside the domain. ELMFIRE applies adaptive CFL control only after the first five steps and declares a front stalled when the step-to-step PHI change stays below 0.001. The initial and maximum timestep are therefore both 5 s: for the slowest no-wind Fuel Model 10 state ($R \simeq 0.426$ m/min), one step moves the front about 0.0071 cell, above the stall comparison and far below the target CFL of 0.2. The stop time is 43,200 s, long enough for that slowest front to traverse the 275 m needed by the farthest diagonal station and the 195 m needed by the $x = 100$ m comparison station with margin. The break variants use Fuel Model 91 for a 30 m nonburnable strip, with or without a 100 m opening.

Parameter	Configured value	Units	Role
Domain/grid	1500 m square; 5	m/cell	Resolves planar fronts and fuel break
Fuel/moisture	FM10/FM3/FM91; 3/5/6	— / percent	Uniform, transition, and nonburnable states
Time control	step 5; stop 43200	s	Exact multiple with traversal margin
Wind	0 or 1 from 270	mph / degrees	Metamorphic acceleration pair
Gap	closed or 100	m	Connectivity response

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

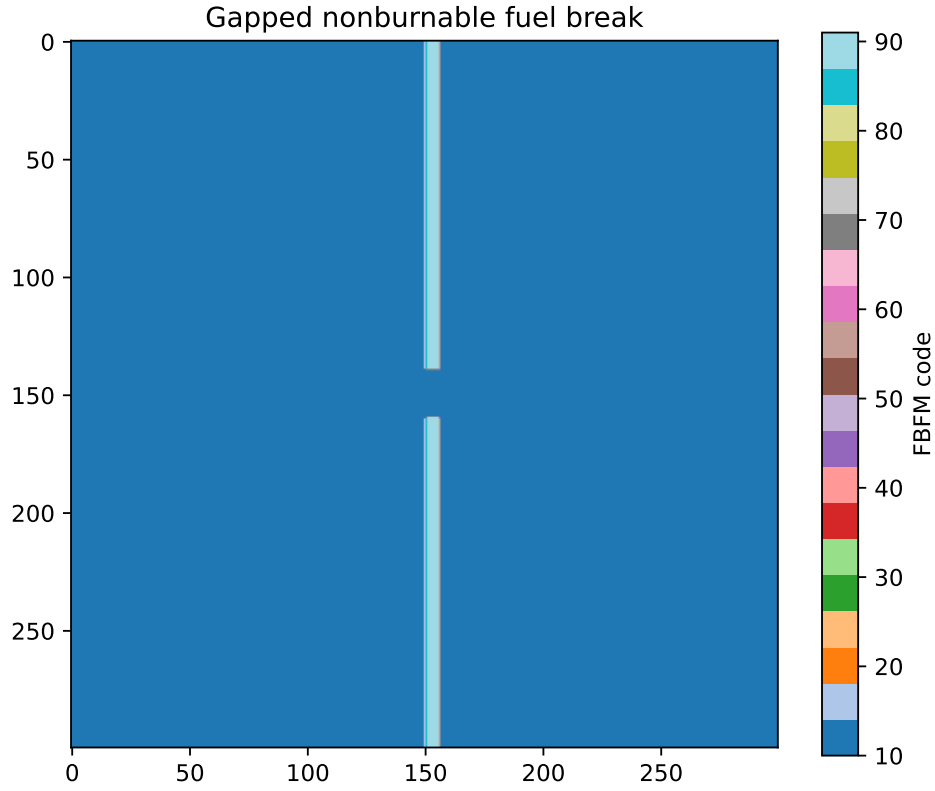


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated status is **PASS**; 5 of 5 metrics pass across 5 variants. Missing required rasters, an obsolete or geometrically inconsistent `phi.tif`, a final-dump mismatch, or inadequate arrival coverage produces NOT EVALUABLE. The generated diagnostic is: `-`.

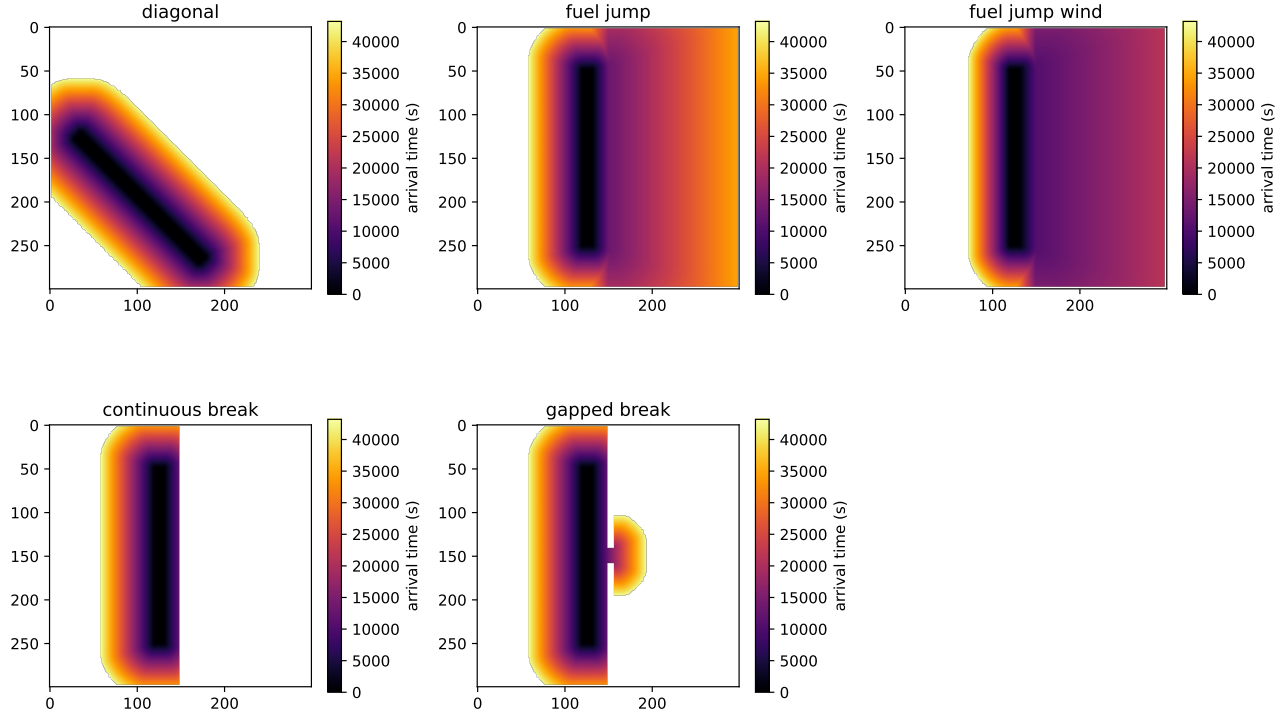


Figure 2: Time-of-arrival fields for all planar-front variants.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
diagonal-front cross-front variation	≤ 0.03 (3% coefficient of variation)	8.35166e-05	PASS
fuel-jump front planarity	≤ 0.03 (3% coefficient of variation)	7.71843e-05	PASS
fuel-transition arrival-time increment	> 0 (positive downstream travel time)	9920 s	PASS
eastward-wind arrival-time ratio	≤ 0.95 (at least 5% earlier than no-wind)	0.756028	PASS
gapped-break downstream burned-fraction gain	≥ 0.10 (10 percentage points)	0.657143	PASS

The authoritative decision is **PASS** from `outputs/metrics.json`.

8 Technical interpretation

Loss of planarity suggests orientation-dependent transport or indexing. A nonpositive transition time indicates an invalid arrival field. Lack of wind acceleration indicates a weather/fire-behavior coupling problem. Lack of gap response indicates incorrect nonburnable-fuel or front-connectivity handling.

9 Scope and limitations

The case verifies observable coupled behavior rather than individual internal Fortran routines. It does not assert absolute Rothermel rates, and the selected fuel and wind states are not field validation data.

Case 32: Raster-orientation covariance

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE32_RCV. This case verifies that ELMFIRE’s raster indexing and level-set transport are covariant under a 90° rotation. It is the portable successor to Adam Laird’s `inputs-orientation` scenario.

2 Mathematical formulation and numerical implementation

For homogeneous isotropic forcing, the operator $\mathcal{L}$ in $\partial_t \phi = \mathcal{L}(\phi)$ commutes with any rigid rotation Q : $\mathcal{L}(\phi \circ Q^{-1}) = (\mathcal{L}\phi) \circ Q^{-1}$. A square north-up raster rotated by exactly 90 degrees requires no interpolation, so cell-by-cell comparison is available.

3 Derivation of expected behavior

The rotated variant is generated with `numpy.rot90` from the asymmetric L-shaped initial negative- ϕ region. Fuel, moisture, slope, and wind are spatially uniform. Therefore the rotated arrival-time validity mask and values must equal a 90-degree rotation of the original result, apart from bounded grid-level roundoff.

The reference array is generated directly from the original ELMFIRE arrival-time result by a 90-degree counterclockwise rotation, $T_{\text{ref}} = \text{rot90}(T_{\text{original}})$; no arrival values are invented. With $V = \{i : T_{\text{ref},i} \text{ and } T_{\text{rot},i} \text{ finite}\}$ and $U = \{i : \text{either value is finite}\}$, finite-mask agreement is

$$A_m = \frac{|V|}{|U|},$$

whose exact rotational reference is 1 and whose substituted test is $A_m \geq 0.99$. On the mutual finite mask, the value metric is

$$E_T = \frac{|V|^{-1} \sum_{i \in V} |T_{\text{ref},i} - T_{\text{rot},i}|}{\max[P_{90}(T_{\text{ref},i} : i \in V), 1 \text{ s}]},$$

with exact reference zero and acceptance substitution $E_T \leq 0.02$. The asymmetric generated ignition mask makes this comparison sensitive to an erroneous reflection or transposition.

4 Verification metrics and acceptance criteria

Finite-mask intersection over union must be at least 0.99. On mutually finite cells, normalized mean absolute arrival-time error must be at most 0.02. Mask agreement detects row/column reversal and georeferencing errors; value error detects orientation-dependent numerical evolution.

5 Simulation configuration and predicted outcome

Both variants use a 2000 m square domain at 5 m, Fuel Model 10, flat terrain, zero wind, and identical timing. The only changed scientific input is the exact raster rotation of `phi.tif`. The second result is predicted to be the corresponding rotation of the first.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Exact square raster rotation
Fuel/forcing	FM10; flat; calm	—	Isotropic homogeneous operator
Time control	step 1, maximum 5; stop 7200	s	Common exact comparison time
Initial field	asymmetric L shape	—	Makes orientation errors observable
Variant transform	90-degree rotation	degrees	Exact no-interpolation covariance test

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

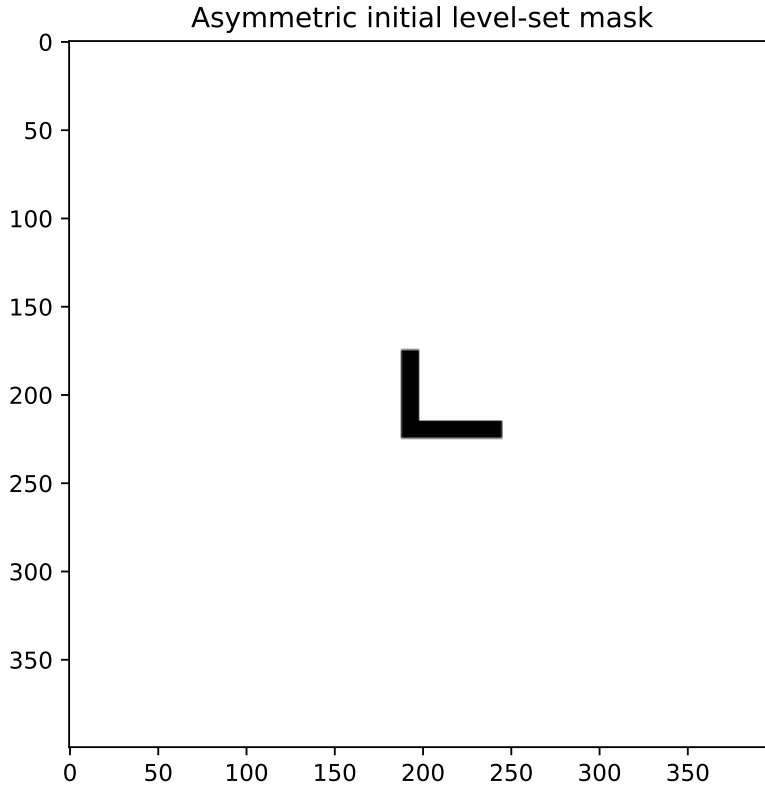


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated status is **PASS**; 2 of 2 metrics pass across 2 variants. Missing output produces NOT EVALUABLE.

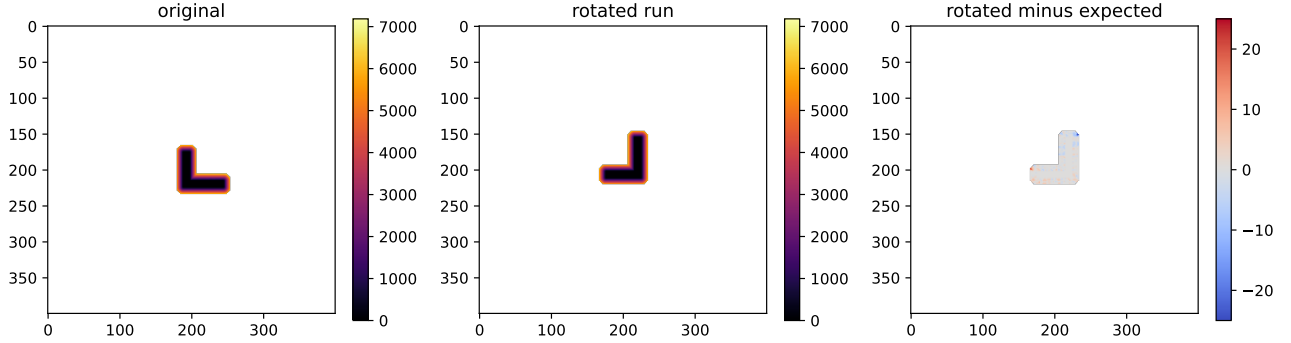


Figure 2: Original, independently rotated, and difference arrival-time fields.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
rotated finite-mask agreement	$\hat{c} = 0.99$ (1% disagreement)	1 fraction	PASS
rotated arrival-time error	$\hat{e} = 0.02$ (2%)	0.000214081 normalized L1	PASS

The authoritative decision is **PASS** from `outputs/metrics.json`.

8 Technical interpretation

Mask disagreement indicates inconsistent raster orientation, bounds, or row/column indexing. Value disagreement with matching masks indicates a directional bias in transport or output serialization.

9 Scope and limitations

The 90-degree test avoids resampling and therefore does not assess arbitrary-angle reprojection. It verifies executable behavior, not the internal bilinear-interpolation routine in isolation.

Case 33: Slope-driven directional spread

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE33_SD0. This case verifies the observable coupling of topographic slope into directional surface-fire spread through the standard ELMFIRE executable. It is the portable successor to Adam Laird’s no-wind-with-slope component scenario and retains a flat control.

2 Mathematical formulation and numerical implementation

ELMFIRE combines the no-wind spread rate with a slope factor before supplying a direction-dependent normal speed $V_n(\theta)$ to $\partial_t \phi + V_n |\nabla \phi| = 0$. With zero wind and zero slope, rotational symmetry is expected. A uniform nonzero slope breaks that symmetry along the fall-line axis.

3 Derivation of expected behavior

The flat control has equal forcing in all azimuths and therefore equal horizontal and vertical extents. The 30° variant changes only slope and aspect; it must differ from the control and its final footprint must be elongated. These are metamorphic consequences of the governing coupling and do not require duplication of ELMFIRE’s internal slope formula.

For each finite arrival-time mask, the pixel extents are $W = c_{\max} - c_{\min} + 1$ and $H = r_{\max} - r_{\min} + 1$. The flat-control anisotropy is

$$A_f = \frac{|W_f - H_f|}{\max(W_f, H_f)},$$

whose isotropic reference is zero and whose substituted test is $A_f \leq 0.05$. For the generated uniform 30-degree slope, the directional-shape metric is

$$L_s = \frac{\max(W_s, H_s)}{\min(W_s, H_s)} \geq 1.10.$$

The independent forcing-response metric uses the common finite mask V :

$$E_s = \frac{|V|^{-1} \sum_{i \in V} |T_{s,i} - T_{f,i}|}{\max[P_{90}(T_{s,i} : i \in V), 1 \text{ s}]} \geq 0.05.$$

Thus the three reference statements are, respectively, zero flat anisotropy, at least 10 percent slope elongation, and at least 5 percent normalized difference from the flat run.

4 Verification metrics and acceptance criteria

Flat-control extent anisotropy must not exceed 0.05. The slope footprint’s major/minor extent ratio must be at least 1.10. Normalized mean absolute arrival-time difference between slope and flat variants must be at least 0.05. The first metric detects grid bias, while the latter two show that slope is both active and directional.

5 Simulation configuration and predicted outcome

Both variants use a 2000 m square domain at 5 m, Fuel Model 10, 3/5/6 percent dead-fuel moisture, zero wind, and a central point ignition. The slope variant uses 30 degrees and aspect 270 degrees; the control uses zero slope. The slope result is predicted to be anisotropic and measurably different from the circular control.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Resolves directional footprint
Fuel/moisture	FM10; 3/5/6	— / percent	Fixed surface response
Time control	step 1, maximum 5; stop 7200	s	Common exact comparison time
Control	slope 0; wind 0	degrees / mph	Rotational baseline
Test	slope 30; aspect 270	degrees	Activates directional slope coupling

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

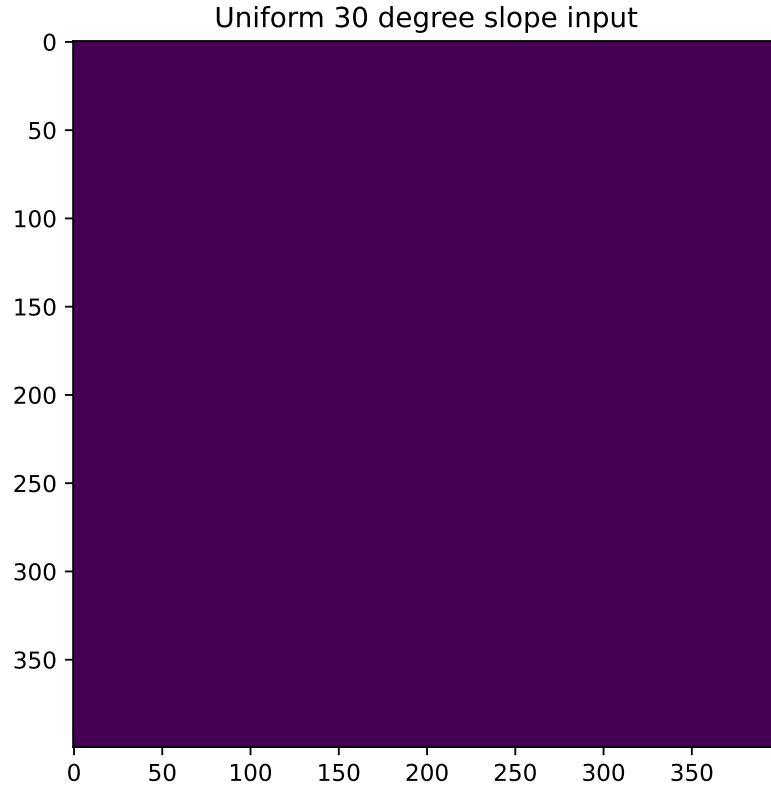


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated status is **PASS**; 3 of 3 metrics pass across 2 variants. Missing outputs produce NOT EVALUABLE.

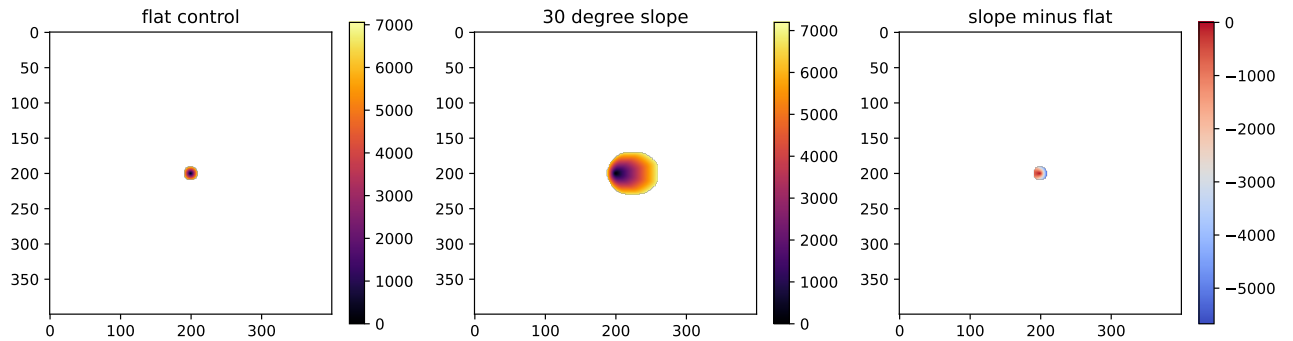


Figure 2: Flat-control, slope, and difference arrival-time fields.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
flat-control anisotropy	$\hat{\mu} = 0.05$ (5%)	0 fraction	PASS
slope-front elongation ratio	$\hat{\mu} = 1.10$ (10% directional elongation)	1.1746 ratio	PASS
slope-forcing response	$\hat{\mu} = 0.05$ (5% difference from flat)	0.520518 normalized L1	PASS

The authoritative decision is **PASS** from `outputs/metrics.json`.

8 Technical interpretation

An anisotropic control indicates grid-orientation bias. A near-unity slope footprint or negligible field difference indicates that slope/aspect inputs were not consumed or not coupled to spread. Unexpected rotation should prompt inspection of aspect convention and raster orientation.

9 Scope and limitations

This case verifies directional response, not the absolute value of the slope coefficient and not agreement with an experimental fire. Existing CASE22 and CASE24 exercise more complex terrain interactions; this case isolates the slope-only executable response.

Case 34: Standard fuel-model Rothermel sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This executable-level known-response test verifies that ELMFIRE reads standard fuel-model parameters and evaluates the no-wind, no-slope Rothermel surface spread model for FBFM2, FBFM9, GR2, TU5, and TL8. It exercises fuel-table parsing, class weighting, reaction intensity, heat sink, and the base rate of spread while crown fire, spotting, buildings, acceleration, wind, and slope are excluded. It can expose column-order, unit, weighting, and model-selection errors. This is implementation verification, not comparison with an observed fire. The verification question is whether all five variants reproduce the independent reaction-intensity and base-ROS calculations within 1% and 2%, with TOA-derived ROS within 5%.

2 Mathematical formulation and numerical implementation

The no-wind, no-slope rate is

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad (1)$$

where I_R is reaction intensity, ξ is the propagating-flux ratio, ρ_b is bulk density, ϵ is effective heating number, and $Q_{ig} = 250 + 1116M_f$ in native Rothermel units. Dead and live classes are area-weighted before their moisture-damping contributions are combined. The equations and standard fuel definitions follow Patricia L. Andrews, *The Rothermel Surface Fire Spread Model and Associated Developments*, RMRS-GTR-371 (2018). ELMFIRE returns spread rate in metres per minute and reaction intensity in kW m^{-2} .

3 Derivation of expected behavior

The case-local `rothermel.reference.py` reads the same frozen fuel-table rows but independently evaluates every algebraic step stated above. It does not import ELMFIRE source or use a simulation output as a reference. Each predicted I_R and R_0 is therefore a deterministic known response. A bounded planar signed-distance front turns the scalar normal speed into a second independent prediction, $t(x) = t_0 + (x - x_0)/(R_0/60)$.

Each reference is calculated independently with

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad Q_{ig} = 250 + 1116M_f,$$

using the generated common moisture inputs and the named standard-fuel record. The resulting

substitutions are

fuel model	R_{ref} (m min ⁻¹)	$I_{R,\text{ref}}$ (kW m ⁻²)
2	0.823796	675.184
9	0.306627	477.714
102	0.228507	186.041
165	0.325722	1595.730
188	0.275244	646.706

(for example, Fuel Model 2 gives $2.702741 \text{ ft min}^{-1}(0.3048) = 0.823796 \text{ m min}^{-1}$). For variant i , direct errors are $e_{R,i} = |R_{\text{max},i} - R_{\text{ref},i}| / \max(|R_{\text{ref},i}|, 10^{-8})$ and the analogous $e_{I,i}$. The TOA fit is $t = a + bx$, so $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - \sum(t - \hat{t})^2 / \sum(t - \bar{t})^2$. The reported substitutions are $\max_i e_{R,i} \leq 0.02$, $\max_i e_{I,i} \leq 0.01$, $\max_i |R_{\text{TOA},i} - R_{\text{ref},i}| / R_{\text{ref},i} \leq 0.05$, and $\min_i R_i^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

The direct spread raster tests the calculated speed with maximum relative error 0.02; the reaction-intensity raster tests the source term with maximum relative error 0.01. Their medians use colocated valid cells at the planar head, beginning one cell behind the initial east face, so direct verification does not require the front to traverse a downstream corridor. Linear regression of positive arrival time against interior x coordinate tests propagation independently: maximum ROS error is 0.05 and minimum R^2 is 0.995. TOA is required only for variants whose planned duration supports a multi-cell fit. All five variants and all four aggregate metrics are required; missing or malformed required evidence gives NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, **SIMULATION_DT** and **SIMULATION_DTMAX** use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the **SIMULATION_TSTOP** in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. The domain is 1400 m square with 10 m cells. A finite 50 m-wide ignition strip has a planar east face initially at $x = -400$ m and terminates well inside the north and south boundaries; the interior analysis corridor is therefore one-dimensional without presenting ELMFIRE with an edge-to-edge contained front. Moistures are 5, 7, and 9% for dead classes and 90% for both live classes. Wind and slope are zero. Run duration is selected from the predicted rate to provide approximately 300 m of travel. The interior results should be planar but need not have a monotonic ordering across fuel models because packing, SAV, load, depth, and extinction moisture interact.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Independent variable
Moisture	dead 5/7/9; live 90/90	percent	Fixed damping state
Wind/slope	0 / 0	mph / degrees	Isolates base fuel response
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_102; WS=0 mph, WD=270 deg

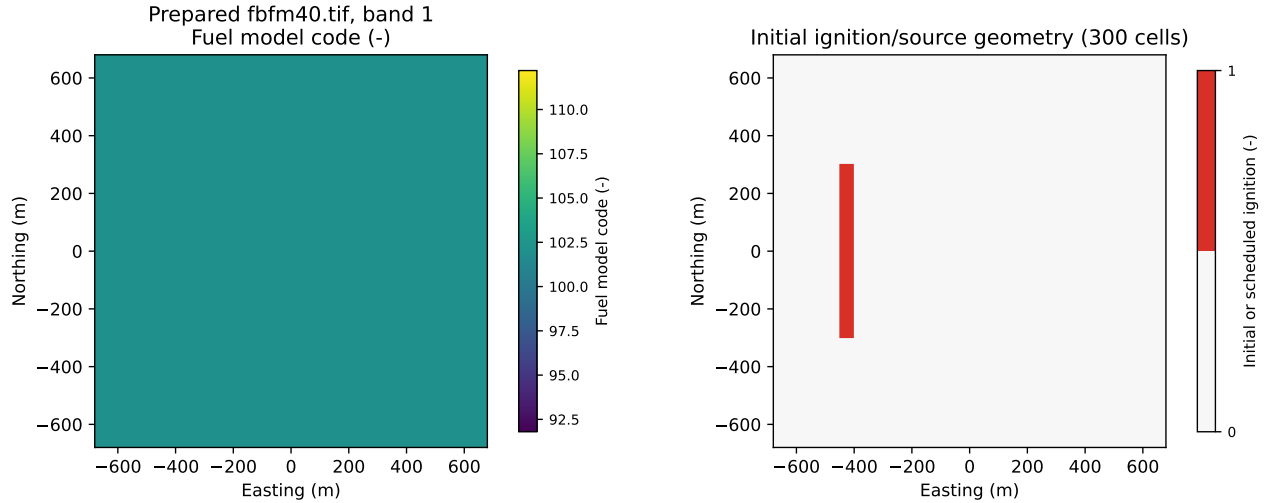


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 5 of 5 required variants. Its authoritative state is **PASS**. No plotted curve is substituted for missing simulation artifacts.

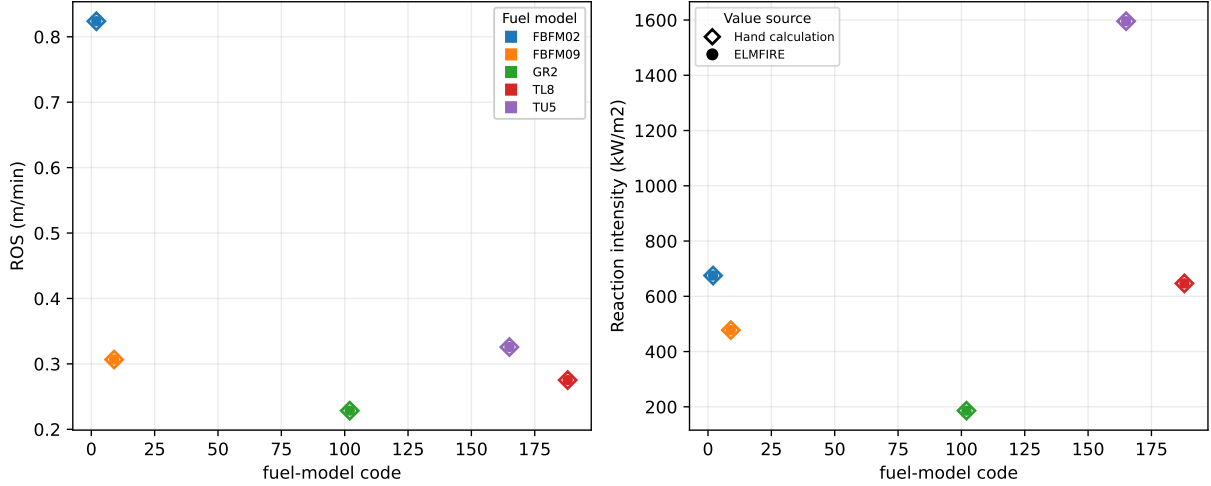


Figure 2: Hand-calculated and ELMFIRE rate of spread (ROS, m/min) and reaction intensity (kW/m²) for all five completed standard-fuel variants. Each fuel model uses one color in both panels; open diamonds denote the independent hand calculation and filled circles denote ELMFIRE.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	3.86591e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0115779 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999961 dimensionless	PASS

The overall decision is **PASS**; a successful executable exit alone is not a pass.

8 Technical interpretation

An isolated intensity discrepancy implicates fuel parsing, class weighting, or moisture damping. Correct intensity with incorrect direct ROS implicates flux ratio, density, heat sink, adjustment, or output units. Correct direct ROS with a TOA discrepancy implicates level-set transport, raster orientation, or time units.

9 Scope and limitations

The case verifies five representative standard fuels at one moisture state. It does not validate Rothermel against experiments, exhaust all fuel models, or test wind, slope, curing, crown fire, or spatial heterogeneity; those mechanisms have separate cases.

Case 35: Midflame wind-speed Rothermel sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies ELMFIRE’s Rothermel wind response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 1, 2, 4, 6, 8, and 12 mph midflame. It isolates the midflame wind factor while separately accounting for ELMFIRE’s public 20-ft wind input and unsheltered wind-adjustment factor (WAF). It can expose errors in wind units, WAF inversion, exponentiation, fuel-dependent coefficients, or wind limiting. The question is whether every variant agrees with the independent implementation-limited ROS within 2%, reaction intensity within 1%, and TOA-derived ROS within 5%.

2 Mathematical formulation and numerical implementation

Rothermel wind enhancement is

$$R = R_0(1 + \phi_w), \quad \phi_w = CU^B(\beta/\beta_{op})^{-E}, \quad (1)$$

with U in ft min^{-1} at midflame. Coefficients are $C = 7.47e^{-0.133\sigma^{0.55}}$, $B = 0.02526\sigma^{0.54}$, and $E = 0.715e^{-0.000359\sigma}$. The reference is Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE implementation limits U to $0.9I_R$ in native units. ELMFIRE inputs 20-ft wind in mph, so preprocessing divides each target midflame speed by the independently computed unsheltered WAF.

3 Derivation of expected behavior

Below the limit, $(R/R_0 - 1)$ follows the fuel-specific power law in U ; once the limit is reached, the wind contribution is constant. The case-local oracle computes R_0 , I_R , WAF, ϕ_w , limit location, and final ROS from the paper equations and the documented ELMFIRE limiting policy. This separates paper-equation behavior from the implementation’s explicit cap.

The wind series uses the Rothermel wind factor

$$\phi_w = C U_{mf}^B, \quad R = 0.3048 R_0^{(\text{ft}/\text{min})}(1 + \phi_w),$$

with U_{mf} in ft min^{-1} and fuel-specific B, C . For a concrete generated substitution, Fuel Model 2 at $U_{mf} = 4 \text{ mph} = 352 \text{ ft min}^{-1}$ has $C = 0.0002125869$, $B = 1.8304427$, and

$$\phi_w = 0.0002125869(352)^{1.8304427} = 9.746235,$$

$$R_{\text{ref}} = 0.3048(2.702741)(1 + 9.746235) = 8.852700 \text{ m min}^{-1}, \quad I_{R,\text{ref}} = 675.184 \text{ kW m}^{-2}.$$

The same substitution is applied to every generated fuel/wind pair (0, 1, 2, 4, 6, 8, 12 mph); the 20-ft raster value is $U_{20} = U_{mf}/WAF$, e.g. $4/0.362674 = 11.02919 \text{ mph}$. For each variant, $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and e_I is analogous. A TOA regression $t = a + bx$ gives $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregated tests substitute $\max e_R \leq 0.02$, $\max e_I \leq 0.01$, $\max e_{\text{TOA}} \leq 0.05$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for the final direct ROS, 0.01 for reaction intensity, and 0.05 for TOA-derived ROS. The minimum TOA regression R^2 is 0.995. Direct pointwise comparison is more diagnostic than monotonicity alone because it detects an incorrect exponent or premature plateau even when the curve still increases. All 35 fuel–wind combinations are required.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, `SIMULATION_DT` and `SIMULATION_DTMAX` use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the `SIMULATION_TSTOP` in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. Each homogeneous 1400 m square variant uses 10 m cells and a finite ignition strip whose east face is planar at $x = -400$ m within the interior analysis corridor. The strip ends inside the lateral boundaries so it remains an active fire rather than an edge-to-edge contained front. Slope is zero, dead moisture is 5/7/9%, and live moisture is 90%. The wind is from the west. Crown fire, spotting, buildings, fluctuations, acceleration, and Monte Carlo variation are disabled. Curves should increase as U^B and may flatten at fuel-dependent limits.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Fuel-dependent response
Wind sweep	0, 1, 2, 4, 6, 8, and 12 mph midflame	mph	Independent midflame target
Moisture/slope	dead 5/7/9, live 90/90; 0	percent / degrees	Fixed damping and terrain
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_102_u_2p0; WS=5.51 mph, WD=270 deg

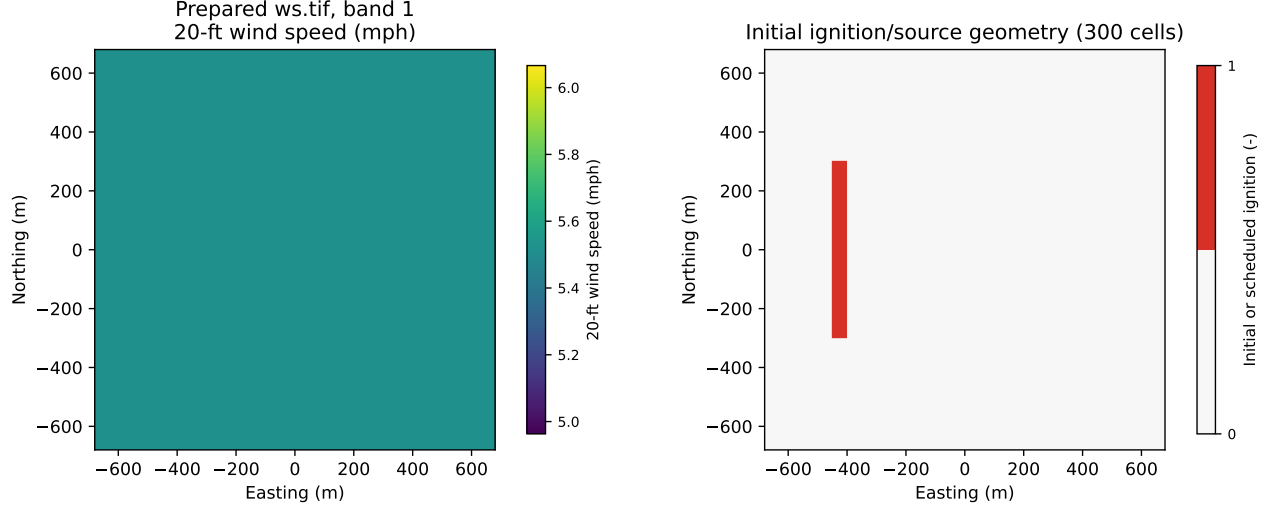


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 35 of 35 required variants and reports **PASS**.

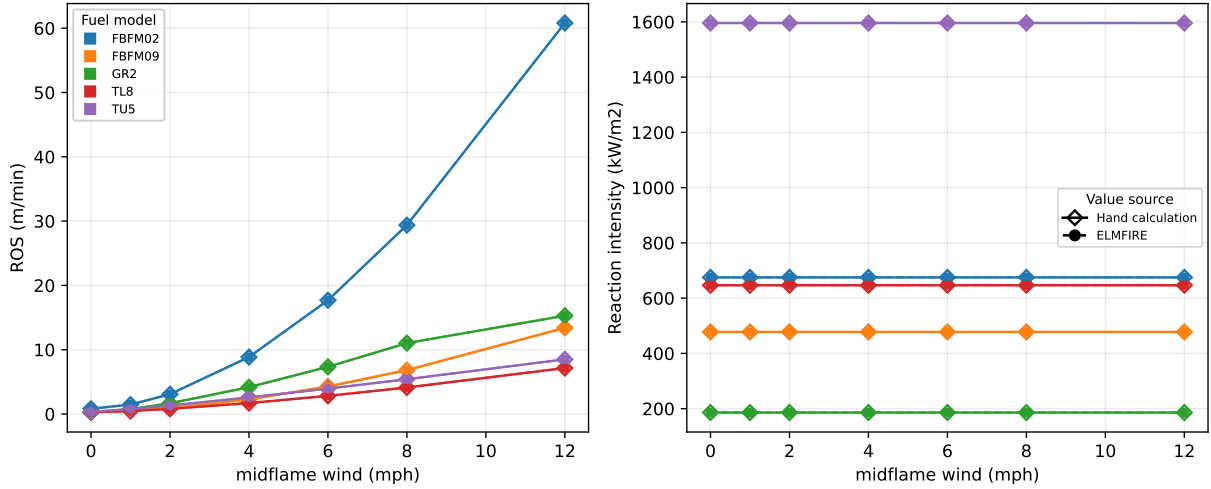


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color. Target values are midflame wind; namelists contain the corresponding 20-ft wind.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	1.17774e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.028404 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999958 dimensionless	PASS

The authoritative overall decision is **PASS**.

8 Technical interpretation

A common scale error across fuels suggests mph, ft/min, or WAF conversion. Fuel-dependent curvature errors suggest B , C , E , or packing-ratio handling. A wrong plateau indicates a changed or incorrect wind-limit policy. Correct direct ROS but incorrect TOA points to transport rather than the Rothermel calculation.

9 Scope and limitations

The case verifies steady, aligned wind and the currently selected original wind limit. If a future ELMFIRE release intentionally adopts a corrected or optional limit, the scientific oracle must be reviewed rather than changing only the namelist. Canopy-dependent WAF is tested in CASE42.

Case 36: Slope-factor Rothermel sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies the scalar Rothermel slope response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 10, 20, 30, 50, 75, and 100% rise over run. It tests slope-raster units, conversion to $\tan^2 \phi$, fuel-dependent packing response, and ELMFIRE's slope-factor cap without wind. The verification question is whether all 35 variants reproduce independently calculated intensity and slope-enhanced ROS within the stated tolerances.

2 Mathematical formulation and numerical implementation

The no-wind response is

$$R = R_0(1 + \phi_s), \quad \phi_s = 5.275 \beta^{-0.3} \tan^2 \phi. \quad (1)$$

Here reported slope percent is rise/run times 100 and the raster value is the angle $\phi = \tan^{-1}(s/100)$ in degrees. The formulation follows Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE source additionally caps ϕ_s at the maximum wind factor derived from its wind limit; the oracle reproduces and reports that implementation policy.

3 Derivation of expected behavior

Before the cap, plotting ϕ_s against $\tan \phi$ on logarithmic axes has slope two. For each fuel and raster angle, the local oracle calculates R_0 , the uncapped factor, the cap, and final R . Because the initial level set is planar and aligned with the upslope direction, the output normal speed equals this scalar head-spread prediction.

The slope-only reference uses

$$\phi_s = 5.275 \beta^{-0.3} \tan^2 \theta, \quad R = R_0(1 + \phi_s).$$

As an explicit generated substitution, the requested 20 percent slope is stored as the integer raster angle $\theta = 11$ degrees ($100 \tan 11^\circ = 19.4380$ percent). For Fuel Model 2, $\beta = 0.00575$, $R_0 = 0.8237955 \text{ m min}^{-1}$, and

$$\phi_s = 5.275(0.00575)^{-0.3} \tan^2(11^\circ) = 0.936756, \quad R_{\text{ref}} = 0.8237955(1+0.936756) = 1.595491 \text{ m min}^{-1}.$$

Reaction intensity is unchanged by slope; here it is $675.184 \text{ kW m}^{-2}$. Every other generated fuel/slope pair uses the same equations with its stored integer angle and fuel packing ratio. Variant errors are $e_R = |R_{\text{max}} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$ and analogously e_I . From $t = a + bx_{\text{surface}}$, $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \leq 0.02$, $\max e_I \leq 0.01$, $\max e_{\text{TOA}} \leq 0.05$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS, 0.01 for reaction intensity, and 0.05 for TOA-derived ROS; TOA regression requires $R^2 \geq 0.995$. The direct and oracle ROS values are slope-surface speeds. Accordingly, the TOA regression converts map easting to distance along the sloping surface using $x_s = x_0 + (x - x_0)/\cos \theta$ before fitting speed. Pointwise analytical errors are used instead of a trend-only test because degrees-versus-percent mistakes can preserve monotonicity. All variants and output fields must exist.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, `SIMULATION_DT` and `SIMULATION_DTMAX` use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the `SIMULATION_TSTOP` in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. Each 1400 m square domain uses 10 m cells and a finite ignition strip with a planar east face in the interior analysis corridor. The strip does not touch the lateral grid boundaries. Wind is zero, dead moisture is fixed at 5/7/9%, and live moisture is 90%. Aspect is chosen so upslope is aligned with propagation. Slope inputs are generated as explicit integer degrees after applying the positive-value equivalent of Fortran NINT; the expected response and slope-surface coordinate conversion use that same generated angle. ROS should rise quadratically in $\tan \phi$ before any source-defined plateau.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar upslope propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Fuel-dependent packing response
Slope sweep	0, 10, 20, 30, 50, 75, and 100%	percent rise/run	Independent terrain target
Wind/moisture	calm; dead 5/7/9, live 90/90	mph / percent	Isolates slope factor
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_102_s_030; WS=0 mph, WD=270 deg

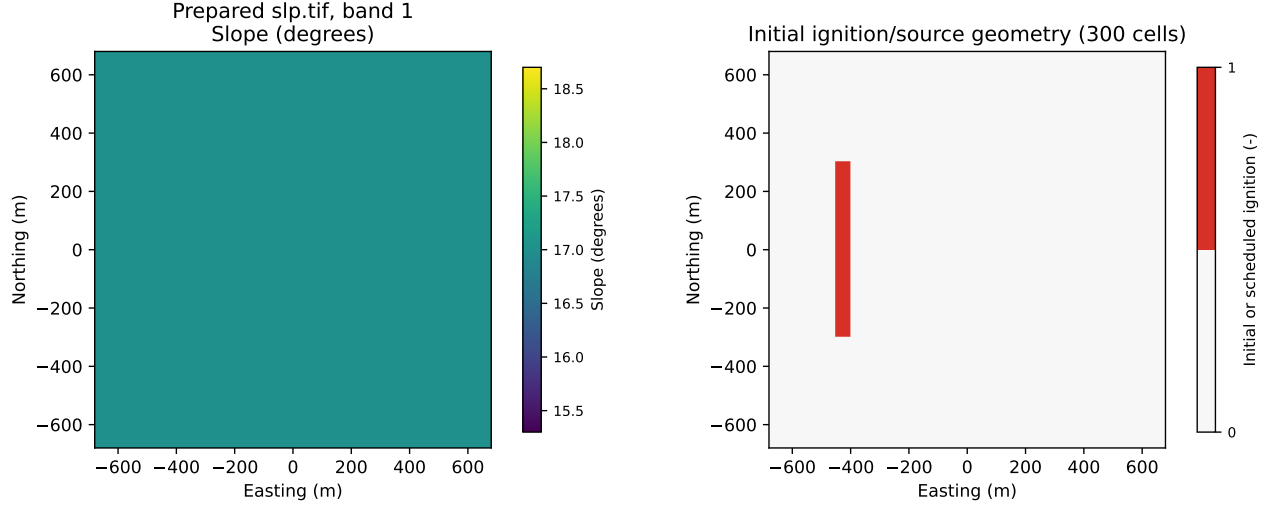


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 35 of 35 variants and has state **PASS**.

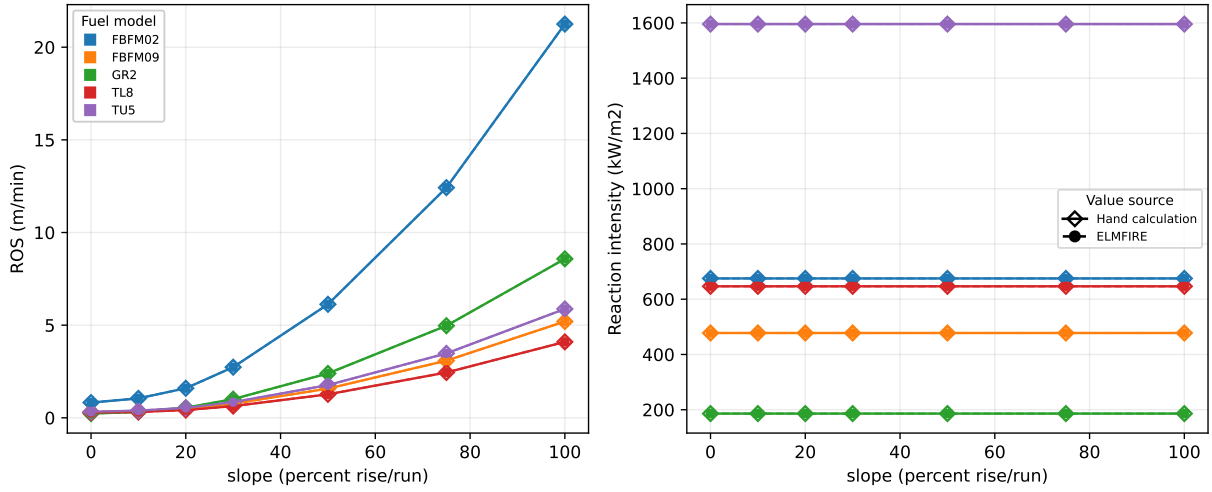


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	5.13349e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0265981 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999961 dimensionless	PASS

The overall decision is **PASS**.

8 Technical interpretation

Large errors growing with slope indicate a units or tangent error. Fuel-dependent errors implicate packing ratio or table selection. A premature or absent plateau indicates changed cap semantics. An accurate direct field but biased TOA indicates propagation or coordinate handling.

9 Scope and limitations

This test covers uniform positive slopes aligned with the front. It does not verify vector competition between wind and slope, complex terrain, or experimental fire behavior; CASE40 covers the directional coupling.

Case 37: Dead-fuel moisture and extinction sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies dead-fuel moisture damping, heat of preignition, and extinction for the dead-dominated static fuels FBFM3 and TL6. Moisture is varied as a fraction of each model's configured dead extinction moisture. It can expose percent/fraction errors, incorrect polynomial evaluation, wrong extinction comparison, or omission of moisture from the heat sink. The verification question is whether reaction intensity and ROS follow the independent calculation through extinction for every required variant.

2 Mathematical formulation and numerical implementation

For $r_M = M_f/M_x$, Rothermel moisture damping is

$$\eta_M = \max\{0, \min[1, 1 - 2.59r_M + 5.11r_M^2 - 3.52r_M^3]\}. \quad (1)$$

Moisture also increases $Q_{ig} = 250 + 1116M_f$. Thus it reduces the reaction source and increases the heat sink. The equations follow Andrews (2018), RMRS-GTR-371. ELMFIRE receives percent rasters; the oracle uses fractions.

3 Derivation of expected behavior

For each fuel, the 1-h, 10-h, and 100-h moistures are equal, so their weighted dead moisture is the selected ratio times M_x . The damping polynomial and therefore dead reaction intensity reach zero at and above $r_M = 1$. Because these variants have no live load contribution, $I_R = R_0 = 0$ there. Below extinction, the independent fuel weighting and heat sink give the exact expected curve.

Dead-fuel moisture damping is evaluated from the generated ratio $r = M_d/M_x$:

$$\eta_M = \begin{cases} 1 - 2.59r + 5.11r^2 - 3.52r^3, & 0 \leq r \leq 1, \\ 0, & r > 1. \end{cases}$$

For Fuel Model 3 at $r = 0.50$, $\eta_M = 1 - 2.59(0.5) + 5.11(0.5)^2 - 3.52(0.5)^3 = 0.5425$. Substitution through the weighted reaction-intensity and Rothermel-rate equations gives $I_{R,\text{ref}} = 477.3889 \text{ kW m}^{-2}$ and $R_{\text{ref}} = 1.069455 \text{ m min}^{-1}$. At $r = 1.00$, the polynomial is zero, giving numerical references $I_R = 3.91 \times 10^{-13}$ and $R = 6.45 \times 10^{-16}$, both treated as zero; at $r = 1.10$ they are exactly zero by the piecewise definition. For nonzero references, $e_R = |R_{\text{max}} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$ and e_I is analogous. The near-extinction intensity metric instead uses $\max |I_{R,\text{sim}} - I_{R,\text{ref}}| \leq 0.001 \text{ kW m}^{-2}$. Propagating variants additionally use $R_{\text{TOA}} = 60/b$ from $t = a + bx$ and $R^2 = 1 - SSE/SST$, with $\max e_{\text{TOA}} \leq 0.05$ and $\min R^2 \geq 0.995$; the direct limits are 0.02 for ROS and 0.01 for reaction intensity.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and nonzero reaction intensity. Analytically extinguished intensity instead uses an absolute tolerance of 0.001 kW m^{-2} , so floating-point residuals are not divided by a near-zero reference. Variants predicted to spread at least 0.10 m min^{-1} additionally require TOA-derived ROS error at most 0.05 and regression $R^2 \geq 0.995$; slower variants cannot cross three 10m cells within the capped practical duration and are decided by direct ROS and intensity. Extinguished variants require zero ROS and reaction intensity within the stated absolute residual tolerance; TOA is not used to invent a velocity where no propagation occurs. All 16 variants are mandatory.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, **SIMULATION_DT** and **SIMULATION_DTMAX** use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. After subtracting the configured meteorology interval, an ordinary stalled-final record must reconstruct the stop time within one timestep. An early stalled-final record is accepted only for a variant whose independent oracle predicts zero ROS, and only when the reconstructed pre-jump time is finite, lies between zero and the stop time, and is aligned to the configured timestep grid. Making TOA optional does not authorize an early stall. The sweep ratios are 0.10, 0.25, 0.50, 0.70, 0.85, 0.95, 1.00, and 1.10. Domains use 10m cells and a finite ignition strip with a planar east face in the interior measurement corridor; the strip terminates inside the lateral boundaries. Wind and slope are zero, live moisture is fixed, and no coupled fire processes are enabled. Intensity and ROS should decrease toward zero and remain zero beyond extinction.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel models	FBFM3 and TL6	code/name	Two extinction-moisture states
Dead-moisture ratio	0.10 through 1.10 times dead extinction mois- ture	fraction	Independent damping variable
Wind/slope	0 / 0	mph / de- grees	Isolates moisture damping
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

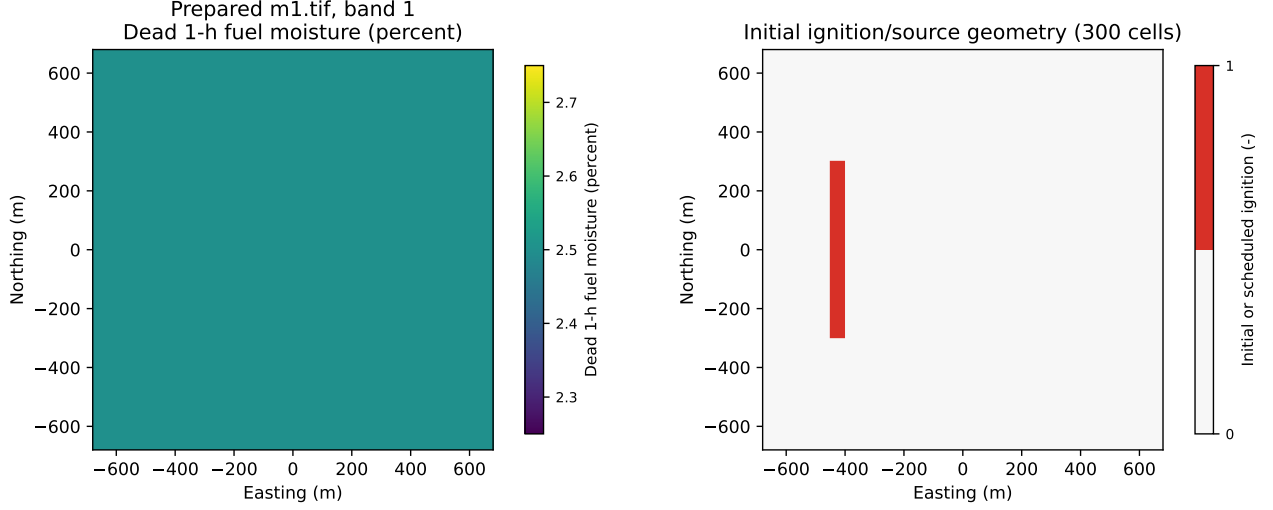


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 16 of 16 required variants and reports **PASS**. For a stalled extinction variant, the machine-readable result records the configured timestep, stop time, meteorology interval, recorded final time, reconstructed pre-jump time, and evidence-selection mode.

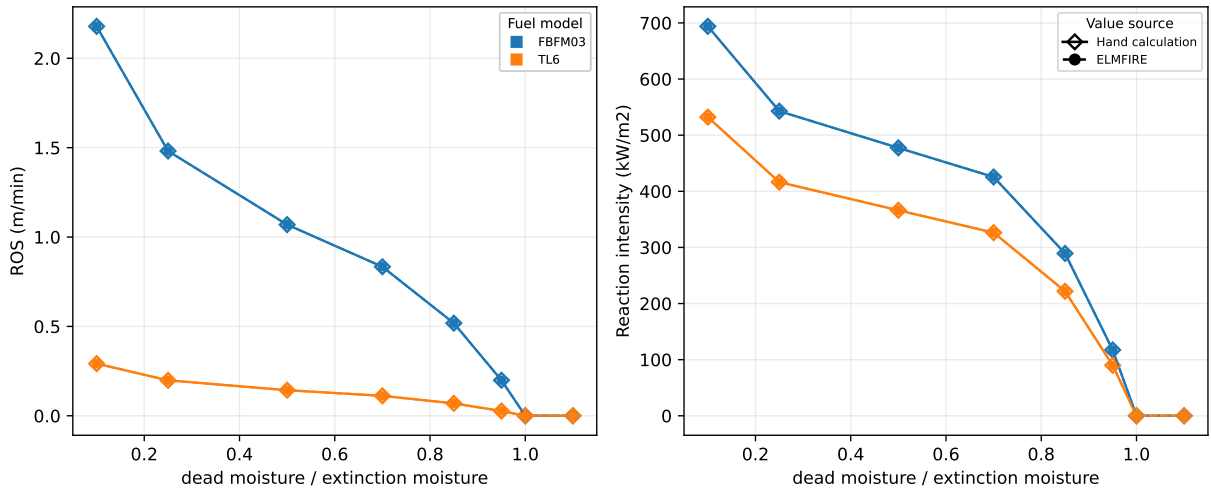


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the dead-moisture response through extinction. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	3.66859e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	3.31587e-06 fraction	PASS
maximum near-extinction reaction-intensity absolute error	≤ 0.001	0.000263903 kW/m ²	PASS
maximum TOA-ROS relative error	≤ 0.05	0.028024 fraction	PASS
minimum TOA regression R ²	≥ 0.995	0.999896 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

The PASS decision is driven by the component statuses in the generated table. Direct observables use the upper envelope of the sparse diagnostic raster, excluding zero-filled and off-axis cells from the head-spread comparison. For direct ROS, the largest relative discrepancy is 3.66859e-06. Its independent value is 0.0266137 m/min; ELMFIRE gives 0.0266138 m/min. For reaction intensity, the largest relative discrepancy is 3.31587e-06. Its independent value is 89.9786 kW/m²; ELMFIRE gives 89.9789 kW/m². A horizontal shift of the extinction point indicates an M_x or units error. Correct intensity with wrong ROS suggests heat-sink or flux-ratio handling. Nonzero propagation or intensity at extinction indicates an unintended live contribution, adjustment, numerical residual, or stale output.

9 Scope and limitations

The case varies all three dead moisture classes together. It does not identify an isolated class-indexing defect or validate moisture observations. Static live moisture is addressed separately by CASE38.

Case 38: Static live-fuel moisture sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies live-fuel moisture damping and calculated live extinction while fuel loads remain fixed. The static shrub models SH2, SH6, and SH8 are intentionally used so changing live moisture cannot transfer herbaceous load into a dead class. It can expose live/dead index errors, percent conversion errors, incorrect live-extinction calculation, or unintended dynamic curing. The question is whether all static-fuel variants match independent I_R and ROS predictions over 30 to 200%.

2 Mathematical formulation and numerical implementation

The live term uses the same cubic damping function $\eta_M(r_M)$ as dead fuel, but $r_M = M_L/M_{x,L}$ and $M_{x,L}$ is calculated from dead/live weighting and dead moisture rather than read directly. Live moisture also appears in $Q_{ig} = 250 + 1116M_f$. The formulation follows Andrews (2018), RMRS-GTR-371. For a static fuel, W_{1h} , W_{LH} , and W_{LW} are invariant throughout the sweep.

3 Derivation of expected behavior

The oracle first calculates the live extinction moisture for each frozen fuel row and fixed dead moisture. It then evaluates live damping, reaction intensity, heat sink, and R_0 at each moisture. Since no loads move, changes arise only from moisture physics. This controlled result is the baseline needed to interpret the additional load-transfer mechanism in CASE39.

Live-fuel damping uses the same cubic with the generated live-moisture ratio $r_l = M_l/M_{x,l}$:

$$\eta_{M,l} = \max\{0, 1 - 2.59r_l + 5.11r_l^2 - 3.52r_l^3\}.$$

For generated SH2, $M_{x,l} = 0.7080597$. At $M_l = 0.70$, $r_l = 0.70/0.7080597 = 0.988617$, giving $\eta_{M,l} = 0.0326506$; substitution in the weighted Rothmel calculation gives $I_R = 354.3991 \text{ kW m}^{-2}$ and $R = 0.0800849 \text{ m min}^{-1}$. At $M_l = 0.90$, $r_l = 1.27108 > 1$, so the live contribution is zero and the remaining dead-fuel contribution gives $I_R = 304.8611 \text{ kW m}^{-2}$ and $R = 0.0579864 \text{ m min}^{-1}$. The calculation is repeated for each generated SH2, SH6, and SH8 moisture value. Direct metrics use $e_R = |R_{\max} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I , with $\max e_R \leq 0.02$ and $\max e_I \leq 0.01$. Where TOA is required, $t = a + bx$ gives $R_{\text{TOA}} = 60/b$, $R^2 = 1 - SSE/SST$, $\max e_{\text{TOA}} \leq 0.05$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Variants predicted at or above 0.10 m min^{-1} also require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$;

slower variants use the direct field because the practical run-duration cap cannot provide a multi-cell TOA regression. Pointwise analytical comparison is selected because a merely monotonic curve would not detect an incorrect live-extinction threshold. All 21 variants are required and the preprocessor rejects a fuel row marked dynamic.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, **SIMULATION_DT** and **SIMULATION_DTMAX** use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the **SIMULATION_TSTOP** in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. These shrub models carry live woody but no live herbaceous load. Live-woody moisture is set to 60, 70, 90, 120, 150, 180, or 200%; 60% is the supported ELMFIRE lower bound; dead classes remain at 8%. The homogeneous domain uses a finite ignition strip whose east face is planar in the interior measurement corridor and whose ends remain inside the lateral boundaries. Wind and slope are zero and all coupled processes are disabled. Fuel loads remain invariant; reaction intensity and ROS should generally decline as live moisture rises, with behavior governed by each fuel’s calculated live extinction.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel models	SH2, SH6, and SH8	code/name	Static live-fuel responses
Live moisture	30 to 200%	percent	Independent damping variable
Wind/slope	0 / 0	mph / degrees	Isolates live damping
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_146_m_090; WS=0 mph, WD=270 deg

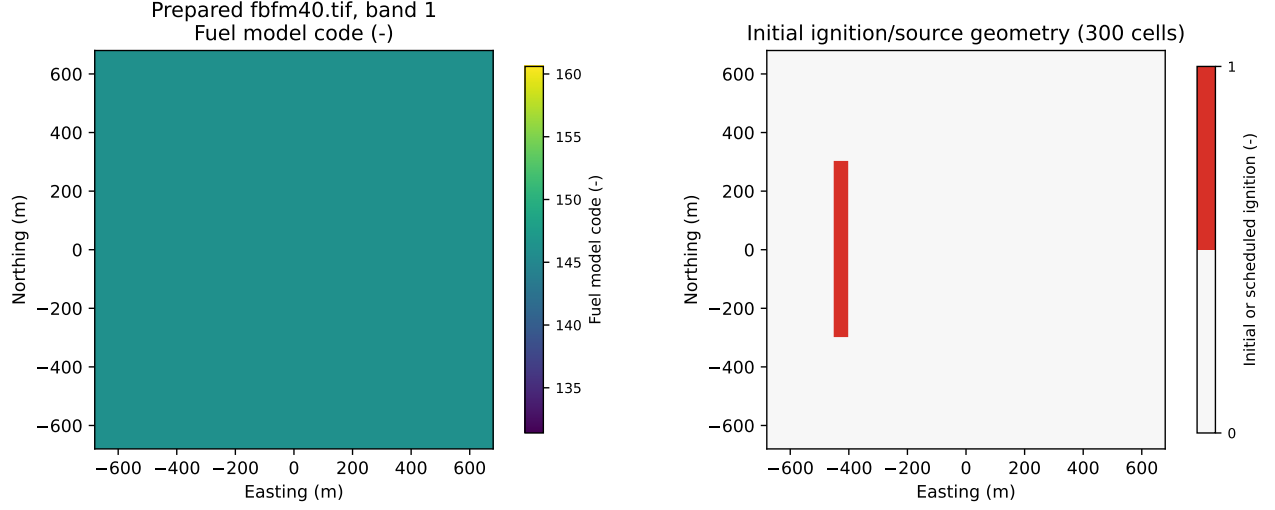


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 21 of 21 required variants and reports **PASS**.

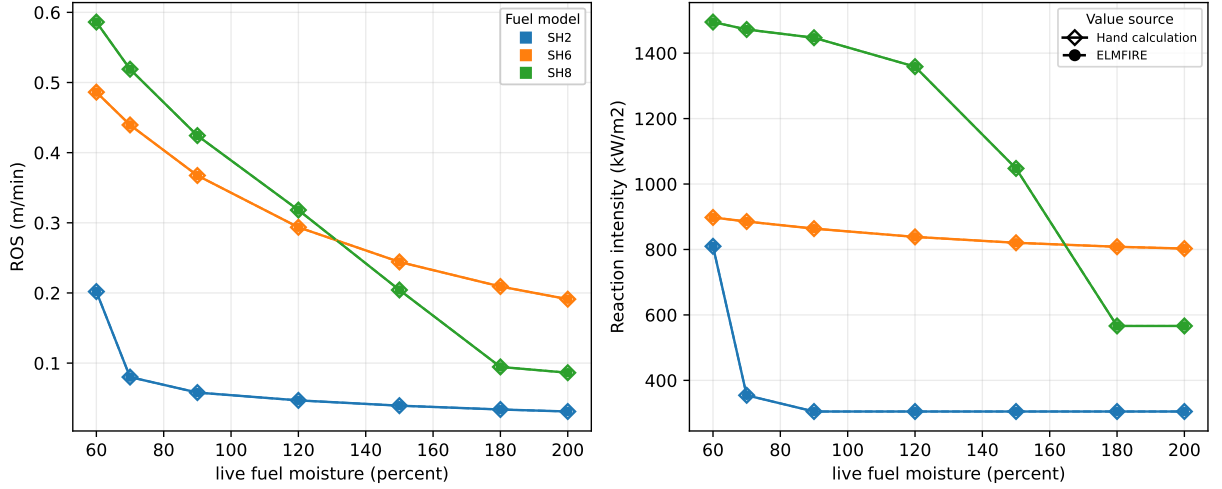


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the static live-moisture response. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	6.50329e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	6.30732e-06 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0170672 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999953 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A shifted transition implicates calculated live extinction or moisture units. Unexpected sharp changes suggest that a supposedly static row was treated dynamically. Agreement in intensity but not ROS narrows the defect to heat sink or spread conversion.

9 Scope and limitations

This case covers three static shrub fuels at one dead-moisture state. It does not verify dynamic curing, spatial moisture mosaics, temporal interpolation, or physical observations. Dynamic herbaceous transfer is isolated in CASE39.

Case 39: Dynamic herbaceous curing sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies dynamic herbaceous curing in GR2 and GS3. Live-herb moisture is the controlled input; ELMFIRE must transfer the corresponding fraction of herbaceous load to the dead 1-h class and then evaluate the full Rothermel model. It can expose reversed curing, incorrect 30/120% bounds, failure to conserve load, wrong class SAV weighting, or confusion between curing and live moisture damping. The verification question is whether all required curing states reproduce independently calculated intensity and ROS within 1% and 2%, respectively.

2 Mathematical formulation and numerical implementation

For live-herb moisture M_{LH} expressed as a fraction, the cured fraction is

$$T = \max[0, \min(1, 1.33 - 1.11M_{LH})]. \quad (1)$$

ELMFIRE implements the equivalent integer-percent interpolation between 30 and 120%. If $W_{LH,0}$ is original live-herb load, $W'_{1h} = W_{1h} + TW_{LH,0}$ and $W'_{LH} = (1 - T)W_{LH,0}$. The transferred material adopts dead 1-h moisture while remaining live material uses live damping and heat of preignition. The formulation follows Andrews (2018), RMRS-GTR-371.

3 Derivation of expected behavior

At 30% moisture all live-herb load is cured; at 75% half is cured; at 120% and above none is transferred. Above 120%, curing is constant but live moisture damping continues to change, so ROS need not form a plateau. The case-local oracle conserves total load, recomputes effective SAV and every dependent Rothermel coefficient, applies moisture damping, and applies ELMFIRE's selected wind limit at the target 6 mph midflame wind.

For dynamic grass fuels, the cured fraction and load transfer are

$$f_c = \min\left(1, \max\left[0, \frac{120 - M_{lh}}{90}\right]\right), \quad w_{lh}^* = (1 - f_c)w_{lh}, \quad w_{1h}^* = w_{1h} + f_c w_{lh}.$$

For the generated GR2 variant at $M_{lh} = 75$ percent, $f_c = (120 - 75)/90 = 0.5$. With original $w_{lh} = 0.04591$ and $w_{1h} = 0.00459 \text{ lb ft}^{-2}$,

$$w_{lh}^* = 0.5(0.04591) = 0.022955, \quad w_{1h}^* = 0.00459 + 0.5(0.04591) = 0.027545.$$

The weighted fuel calculation gives $R_0 = 1.109830 \text{ ft min}^{-1}$, $I_R = 206.9424 \text{ kW m}^{-2}$, and $\phi_w = 31.095160$; therefore

$$R_{\text{ref}} = 0.3048(1.109830)(1 + 31.095160) = 10.857030 \text{ m min}^{-1}.$$

Every GR2 and GR3 moisture variant repeats these equations. Direct metrics use the maximum over variants of $e_R = |R_{\max} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I , with limits 0.02 and 0.01. The TOA fit $t = a + bx$ supplies $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$, tested by $\max e_{\text{TOA}} \leq 0.05$ and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Variants predicted at or above 0.10 m min^{-1} also require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$; the most moisture-limited variants use direct fields because they cannot cross a multi-cell corridor within the practical duration cap. Direct effective loads are not exposed by the normal executable, so their conservation is an oracle invariant and the implementation is observed through intensity and ROS. All 20 variants are required; no qualitative trend alone can pass the case.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, `SIMULATION_DT` and `SIMULATION_DTMAX` use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the `SIMULATION.TSTOP` in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. Live-herb moisture is 30, 45, 60, 75, 90, 100, 110, 120, 135, or 200%. Dead classes are fixed at 5%, live woody moisture at 90%, slope at zero, and target midflame wind at 6 mph; the 20-ft input is back-calculated from each fuel’s WAF. Each homogeneous domain uses a finite ignition strip with a planar east face inside the analysis corridor and with lateral ends clear of the grid boundary. Crown fire, spotting, buildings, and acceleration are excluded. Strong nonlinear response is expected in the curing interval.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Dynamic fuels	GR2 and GS3	code/name	Enables herbaceous load transfer
Live-herb moisture	30 to 200% live-herb moisture	percent	Independent curing variable
Wind/slope	fixed wind; 0 slope	—	Holds directional forcing fixed
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells

selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_123_lh_030; WS=14.7 mph, WD=270 deg

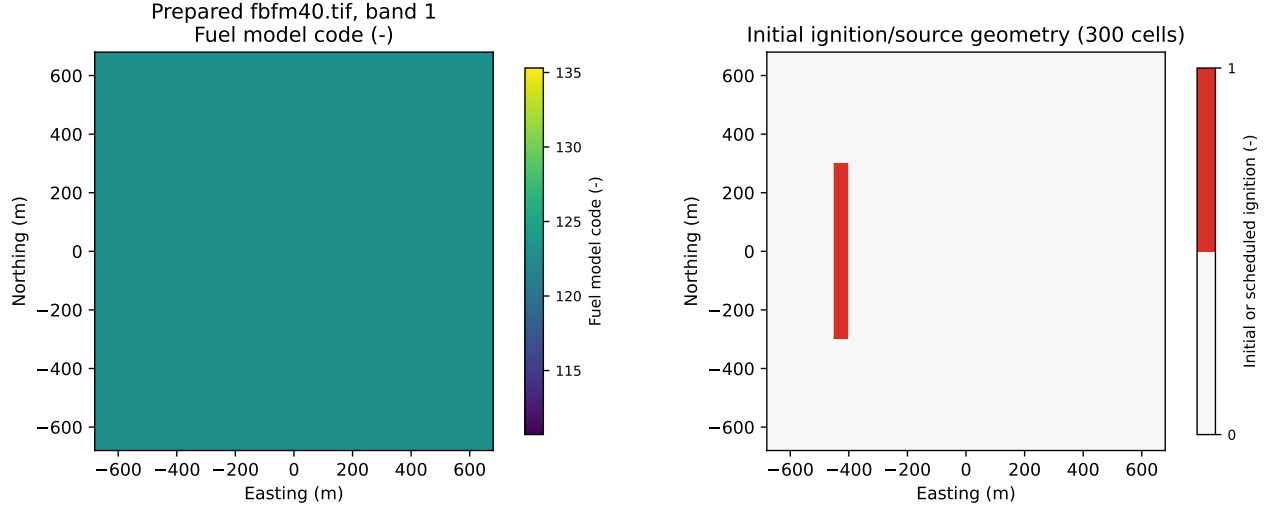


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 20 of 20 variants and reports **PASS**.

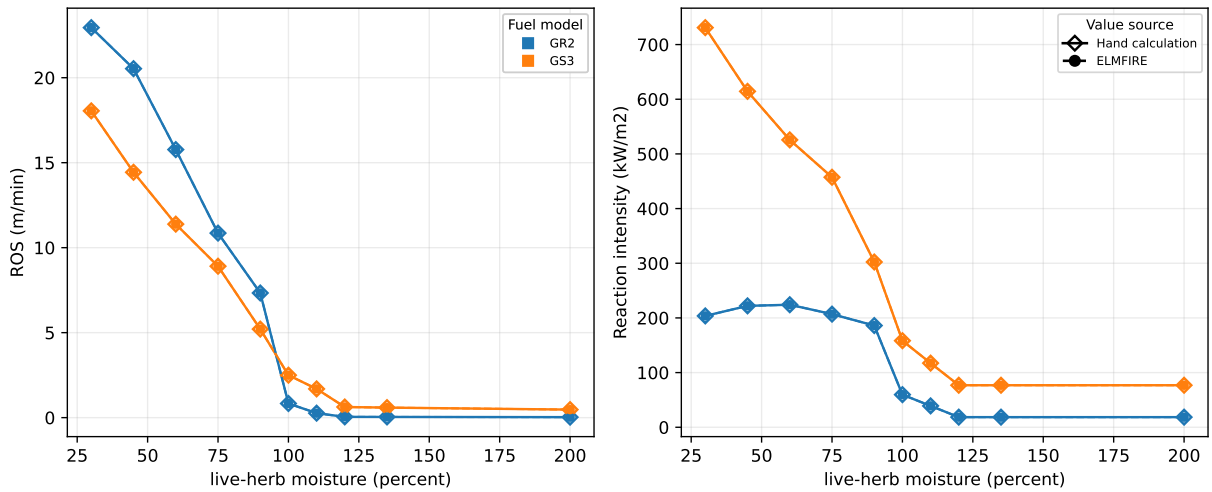


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the dynamic-curing response. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	1.51489e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.3555e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0226402 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999981 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A reversed curve indicates a reversed transfer fraction. Discontinuities away from integer moisture selection suggest class indexing or interpolation defects. Correct behavior above 120% but disagreement below it isolates curing rather than live damping. Correct intensity and wrong ROS implicate heat sink or wind coupling.

9 Scope and limitations

This executable-level test observes curing indirectly; a direct internal-load assertion would require a compiled component diagnostic. It covers two dynamic fuel models, constant weather, and no temporal curing. CASE38 separately isolates live moisture without load transfer.

Case 40: Wind-slope vector and direction sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This coupling case verifies how ELMFIRE combines Rothermel wind and slope factors into a direction of maximum spread (DMS), head speed, and fire ellipse. Wind direction rotates through 0, 45, 90, 135, and 180 degrees relative to upslope while magnitudes remain fixed. It can expose degree/radian errors, wind-from/wind-to reversal, aspect reversal, swapped raster axes, scalar addition in place of vector addition, or incorrect ellipse construction. The question is whether all variants meet the DMS, head-ROS, length/width, and intensity limits.

2 Mathematical formulation and numerical implementation

Let upslope azimuth be zero and wind-to azimuth be α . The dimensionless forcing vector is

$$\phi = \phi_s(0, 1) + \phi_w(\sin \alpha, \cos \alpha), \quad (1)$$

so $\theta_{DMS} = \text{atan2}(\phi_x, \phi_y)$ and $R_H = R_0(1 + |\phi|)$. ELMFIRE derives an effective wind from $|\phi|$ and uses its Rothermel length-to-width relation, capped at eight. Because these quantities are defined on the sloping surface, the preprocessor makes the initial circle and the postprocessor measures distances in slope-surface coordinates rather than mixing them with map-plane northing. The scalar factors follow Andrews (2018), RMRS-GTR-371; vectoring and ellipse propagation exercise ELMFIRE's level-set implementation.

3 Derivation of expected behavior

Aligned wind and slope add; opposed forcing subtracts in vector space and the stronger component sets direction. At intermediate angles, the expected DMS lies between the two forcing directions, not generally at the arithmetic mean of angles. The case-local oracle independently evaluates ϕ_s , ϕ_w , their vector magnitude and azimuth, head ROS, effective wind, and length/width.

Let upslope define zero azimuth and let α be the generated wind-to-upslope angle. With $\phi_s = 6.436561$, $\phi_w = 9.746235$, and $R_0 = 0.9178563 \text{ m min}^{-1}$, the vector reference is

$$\phi_x = \phi_s + \phi_w \cos \alpha, \quad \phi_y = \phi_w \sin \alpha, \quad \phi = \sqrt{\phi_x^2 + \phi_y^2}, \quad \theta = \text{atan2}(\phi_y, \phi_x), \quad R_h = R_0(1 + \phi).$$

For example, at $\alpha = 90$ degrees, $\phi = \sqrt{6.436561^2 + 9.746235^2} = 11.679829$, $\theta = \tan^{-1}(9.746235/6.436561) = 56.5586$ degrees, and $R_h = 0.9178563(1 + 11.679829) = 11.638261 \text{ m min}^{-1}$. The effective vector wind is $4.41571 \text{ mph} = 1.97394 \text{ m s}^{-1}$; substitution in $L/W = 0.936e^{0.2566U} + 0.461e^{-0.1548U} - 0.397$ gives $L/W = 1.495859$. Across $\alpha = \{0, 45, 90, 135, 180\}$ degrees, the reference $(\theta, R_h, L/W)$ rows are

(0, 15.7713, 1.63745), (27.3422, 14.6898, 1.60074), (56.5586, 11.6383, 1.49586), (93.7779, 7.25716, 1.33838), (180, 3.95)

Direction error is circular, $e_\theta = \min(|\theta_m - \theta_r|, 360 - |\theta_m - \theta_r|)$. The other metrics are relative errors in R_h , L/W , and $I_R = 686.5297 \text{ kW m}^{-2}$; their maximum substitutions are $e_\theta \leq 5$ degrees, $e_R \leq 0.05$, $e_{LW} \leq 0.10$, and $e_I \leq 0.01$.

4 Verification metrics and acceptance criteria

Maximum circular DMS error is 5° , maximum head-ROS relative error is 0.05, maximum length/width relative error is 0.10, and maximum intensity relative error is 0.01. DMS is inferred from the farthest one percent of the burned footprint, subject to at least 100 burned cells and at least five farthest-front samples. Head ROS is a TOA regression in a narrow predicted-axis corridor with at least ten samples. Length/width uses full-footprint slope-surface extents projected along and across the DMS. Reaction intensity is the median over valid burned cells and requires at least 90 percent coverage of that footprint. These complementary metrics distinguish direction, speed, shape, and output completeness. Every variant is required.

5 Simulation configuration and predicted outcome

Fuel model 2 is used on a 1400 m square, 10 m grid with a 30 m-radius signed-distance ignition clipped to $[-1, 1]$. Slope is requested as 50% and generated as the integer 27-degree value used by ELMFIRE (50.95% effective rise/run), with upslope north; target midflame wind is 4 mph. Dead moistures are 5/7/9%, live-herb moisture is 60%, and live-woody moisture is 90% in both every generated namelist and the independent oracle. Wind-from azimuth is generated so the wind-to vector has the required relative angle. Each variant uses the largest conservative whole-second timestep no greater than the 5%-of-cell target that divides the fixed 1800 s duration exactly. This retains CFL margin and prevents a shortened final step from triggering ELMFIRE's stalled-front timestamp defect. Crown fire, spotting, WUI, stochastic wind, and acceleration are disabled. The footprint should rotate smoothly and change head extension as the vectors move from aligned to opposed. The evaluator accepts only unique TOA and reaction-intensity rasters identified by a final dump record. A stalled-final record is accepted only when subtracting the configured meteorology interval reconstructs the 1800 s stop time within one configured timestep.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Whole-footprint direction and shape
Fuel/slope	FM2; 50	code / percent	Fixed slope forcing
Midflame wind	4	mph	Fixed wind-factor magnitude
Relative angles	0, 45, 90, 135, and 180 degrees	degrees	Vector-coupling sweep
Time control	generated whole-second divisor; stop 1800	s	Exact timestep multiple

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/angle_090; WS=11 mph, WD=270 deg

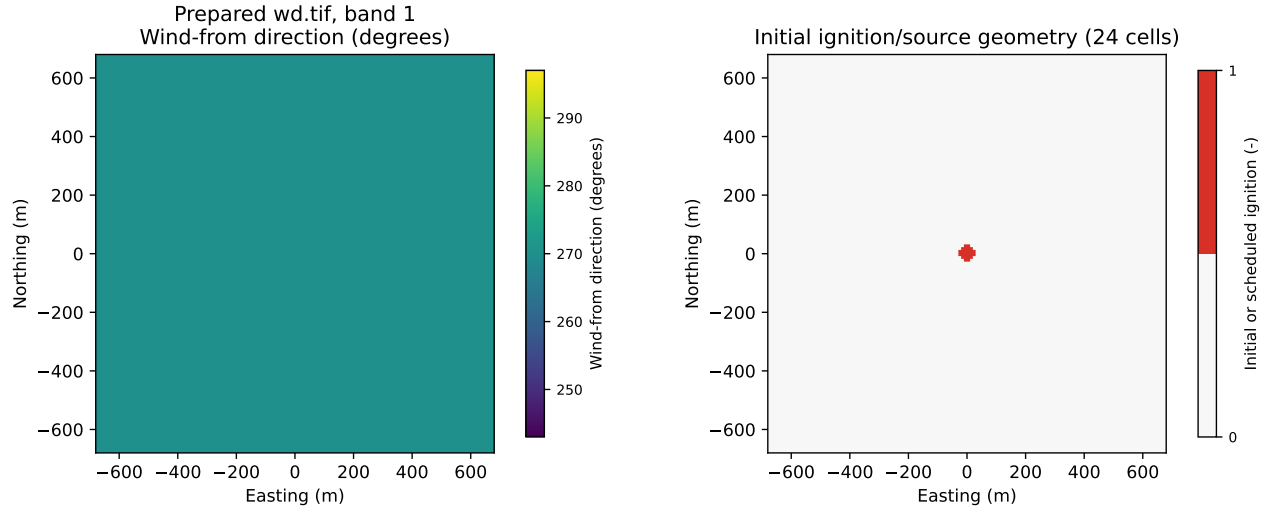


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 5 of 5 variants and reports **PASS**. The generated diagnostic is: $-$.

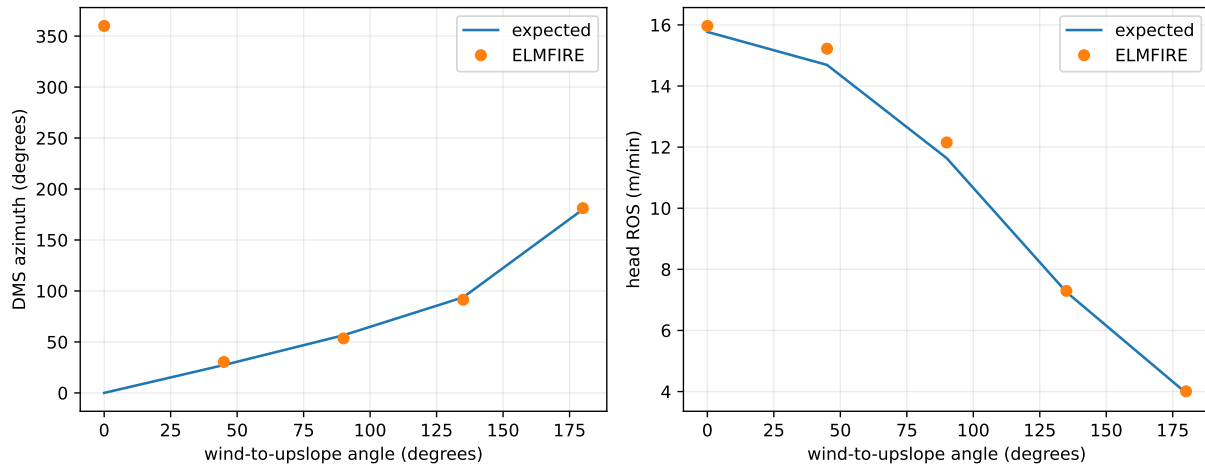


Figure 2: Expected and measured DMS azimuth and head ROS.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum DMS direction error	≤ 5.0	3.05989 degrees	PASS
maximum head-ROS relative error	≤ 0.05	0.0441104 fraction	PASS
maximum length/width relative error	≤ 0.1	0.0961658 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.01156e-07 fraction	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A 180-degree error indicates a wind-from or aspect reversal. A 90-degree error suggests swapped axes or azimuth convention. Correct direction but excessive head speed suggests scalar addition. Correct DMS and head speed with wrong length/width isolates effective-wind or ellipse construction.

9 Scope and limitations

This test uses one fuel, one slope magnitude, one wind magnitude, uniform terrain, and a single grid. It verifies deterministic code behavior rather than observed fire shape. Complex terrain and resolution convergence remain separate concerns.

Case 41: Custom fuel-parameter Rothermel sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies ELMFIRE’s public custom fuel-table interface and the Rothermel response to one-at-a-time changes in dead 1-h load, 1-h SAV, fuel-bed depth, dead extinction moisture, and heat content. It can expose column misalignment, ignored custom values, stale precomputed coefficients, or unit conversion errors. The question is whether every generated code-250 row produces independently calculated reaction intensity and ROS within 1% and 2%.

2 Mathematical formulation and numerical implementation

The custom row supplies class loads w_0 , surface-area-to-volume ratios σ , depth δ , dead extinction M_x , and heat content h . These feed packing ratio $\beta = \sum w_0/(\delta\rho_p)$, optimum packing, reaction velocity, propagating flux, damping, heat sink, and

$$R_0 = I_R\xi/(\rho_b\epsilon Q_{ig}). \quad (1)$$

The formulation follows Andrews (2018), RMRS-GTR-371. Particle density and mineral constants are fixed by the inspected ELMFIRE interface and therefore are not claimed as user-sweepable variables.

3 Derivation of expected behavior

The baseline row is copied and exactly one public column is changed in each variant. A case-local fuel table is written beside that variant and read independently by the oracle. The full equation is recalculated rather than imposing a monotonic trend: load, SAV, and depth have competing effects through packing and reaction velocity, so a simple ordering would be scientifically unjustified.

Each custom-fuel reference is recalculated from $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$, changing only the indicated fuel-table column. The substitutions produced from the generated table are

variant	R_{ref} (m min ⁻¹)	$I_{R,\text{ref}}$ (kW m ⁻²)
baseline	0.411242	1045.499
$w_{1h} = 0.05$	0.253350	688.121
$w_{1h} = 0.15$	0.551674	1379.611
$SAV = 1000$	0.241692	960.426
$SAV = 3000$	0.662626	1074.561
$d = 0.5$	0.225975	910.468
$d = 2.0$	0.694647	991.942
$M_x = 15\%$	0.371790	945.198
$M_x = 35\%$	0.438500	1114.797
$h = 6000$	0.308432	784.125
$h = 10000$	0.514053	1306.874

For example, replacing the baseline 1-h load 0.10 by 0.05 lb ft⁻² changes $R_0 = 1.349221$ to 0.831202 ft min⁻¹; multiplication by 0.3048 gives 0.253350 m min⁻¹. Each variant uses $e_R = |R_{\max} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I . The TOA fit $t = a + bx$ gives $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \leq 0.02$, $\max e_I \leq 0.01$, $\max e_{\text{TOA}} \leq 0.05$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Variants predicted at or above 0.10 m min⁻¹ additionally require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$; slower variants are assessed from direct ROS and intensity. Pointwise equality to a complete recalculation detects both ignored columns and wrong nonlinear propagation. The baseline and ten low/high variants are all required.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, **SIMULATION_DT** and **SIMULATION_DTMAX** use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the **SIMULATION_TSTOP** in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. All variants use fuel code 250, 5/7/9% dead moisture, 90% live moisture, zero wind and slope, and 10 m cells. A finite ignition strip supplies a planar east face in the interior analysis corridor while remaining clear of the lateral grid boundaries. The baseline row uses 0.10/0.05/0.02 dead loads, 0.08 live-woody load, 1-h SAV 2000 ft⁻¹, depth 1 ft, dead extinction 25%, and heat content 8000 Btu lb⁻¹. Each output should match its own oracle; no universal increasing trend is predicted.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Custom fuel code	250	code	Selects variant-local table row
Changes	load, SAV, depth, extinction, heat	native table units	One-at-a-time interface exercise
Wind/slope	0 / 0	mph / degrees	Isolates custom fuel response
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells

selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/load_high; WS=0 mph, WD=270 deg

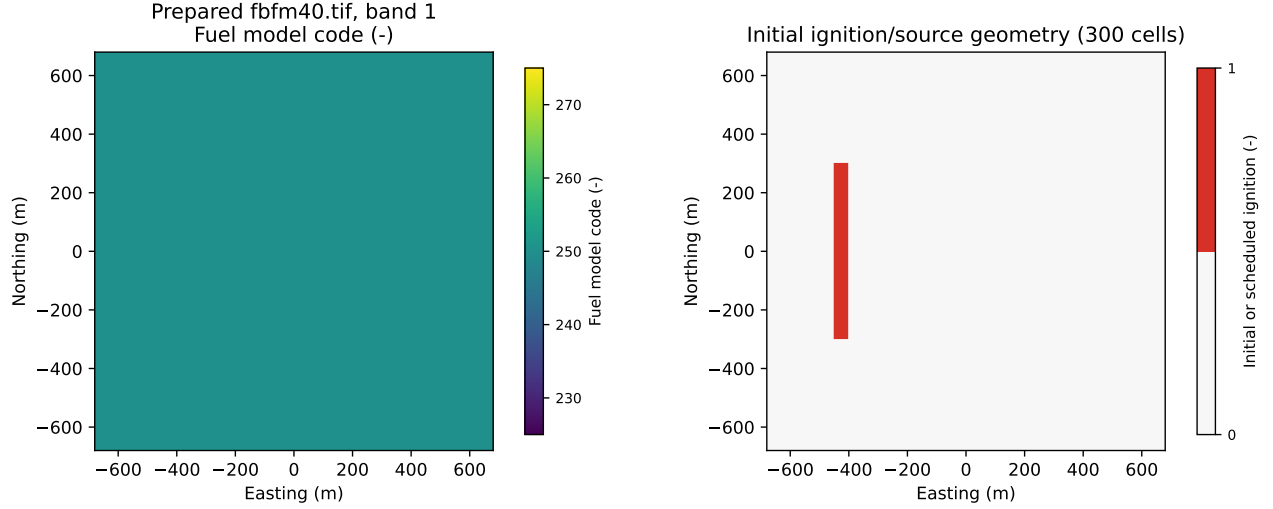


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 11 of 11 variants and reports **PASS**.

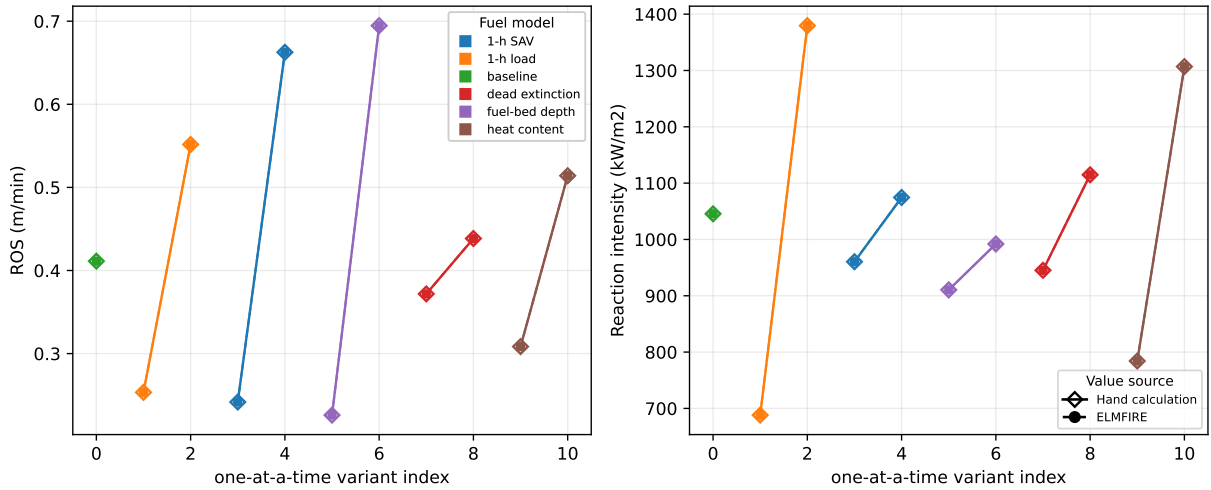


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the custom fuel-table variants. Each comparison pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	4.62672e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	1.10118e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0151881 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999968 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

If only one low/high pair fails, its fuel-table column or dependent coefficient is suspect. A common intensity error suggests table parsing or units; correct intensity with wrong ROS points to heat sink, density, or flux ratio. Identical results across a changed pair indicate the column was ignored.

9 Scope and limitations

This case tests only parameters exposed by the current custom table and one synthetic baseline. It does not test particle density, mineral contents, spatial mixtures, dynamic curing, or physical realism of the synthetic fuel.

Case 42: Wind-adjustment-factor coupling sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies the upstream coupling from 20-ft wind, canopy cover, canopy height, and fuel-bed depth through ELMFIRE’s wind-adjustment factor (WAF) into Rothermel ROS. It distinguishes an input-height conversion error from a defect in the Rothermel wind factor. It can expose percent/fraction canopy errors, height-times-ten errors, sheltered/unsheltered branch errors, or missing WAF application. The question is whether all canopy variants reproduce the independently predicted final ROS within 3% while reaction intensity remains within 1%.

2 Mathematical formulation and numerical implementation

ELMFIRE calculates $U_{mf} = WAF U_{20}$. Without canopy it uses a logarithmic fuel-bed-height expression. With canopy it evaluates a sheltered factor from canopy cover and height, limits it by the unsheltered factor, and blends the two when $0.3333CC < 0.05$. The resulting midflame speed enters $\phi_w = CU_{mf}^B(\beta/\beta_{op})^{-E}$. The case oracle restates these equations directly and does not read ELMFIRE output to estimate WAF.

3 Derivation of expected behavior

For fixed 10 mph and fuel model 2, reaction intensity is invariant because canopy affects wind adjustment but not the configured surface fuel or moisture. ROS differences are completely predicted by the independently calculated WAF and Rothermel wind factor. The zero-canopy variant selects the unsheltered formula; canopy combinations exercise transition and sheltered branches.

The unsheltered adjustment for fuel-bed depth d (ft) is

$$WAF_u = \frac{1.36}{\ln[(20 + 0.36d)/(0.13d)]} [\ln(1.36/0.13) - 1].$$

For canopy cover C , height H (ft), and $s = 0.3333C$, the sheltered value is

$$WAF_s = \frac{1}{\ln[(20 + 0.36H)/(0.13H)]} \frac{0.555}{\sqrt{sH}}.$$

The generated blend uses $WAF = \min(WAF_s, WAF_u)$ when $s \geq 0.05$; otherwise $f_u = 1 - 20s$ and $WAF = f_u WAF_u + (1 - f_u) \min(WAF_s, WAF_u)$. For the zero-canopy Fuel Model 2 control, $WAF_u = 0.362674$, so the 10 mph 20-ft wind gives $U_{mf} = 3.626740$ mph, $\phi_w = 8.146356$, and $R = 0.8237955(1 + 8.146356) = 7.534727 \text{ m min}^{-1}$. For $C = 0.10$, $H = 15/0.3048 = 49.2126$ ft, $s = 0.03333$, $WAF_s = 0.244254$, and $f_u = 0.3334$; hence $WAF = 0.3334(0.362674) + 0.6666(0.244254) = 0.283735$, $U_{mf} = 2.837354$ mph, $\phi_w = 5.197957$, and $R = 0.8237955(1 + 5.197957) = 5.105849 \text{ m min}^{-1}$. All

variants retain $I_R = 675.184 \text{ kW m}^{-2}$. Variant relative errors use $|q_{\max} - q_{\text{ref}}| / \max(|q_{\text{ref}}|, 10^{-8})$. From $t = a + bx$, $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \leq 0.03$, $\max e_I \leq 0.01$, $\max e_{\text{TOA}} \leq 0.06$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum direct-ROS relative error is 0.03, reaction-intensity error is 0.01, and TOA-derived ROS error is 0.06; TOA regression requires $R^2 \geq 0.995$. Constant reaction intensity is retained as a control that detects unintended crown or canopy-fuel coupling. All ten variants are required and missing fields give NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every variant, **SIMULATION_DT** and **SIMULATION_DTMAX** use a conservative whole-second value no greater than the 5 s nominal step or the configured 0.2 CFL limit. The stop time is rounded upward to an exact integer multiple of that step. This avoids both discrete-front resonance and a shortened final step that can trigger ELMFIRE’s stalled-front timestamp defect. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator requires uniquely named rasters tied to one final dump record and accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the **SIMULATION_TSTOP** in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. Canopy cover is 0, 10, 30, or 60%; canopy heights are 5, 15, and 30 m where applicable. The raster stores cover in percent and height times ten, matching explicit namelist flags. Fuel model 2, dead moisture 5/7/9%, live moisture 90%, and zero slope are fixed. A finite west-to-east ignition strip has a planar east face in the interior analysis corridor and ends inside the lateral boundaries. Crown fire is disabled even though canopy inputs are present. ROS should vary only through WAF while I_R remains fixed.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel / 20-ft wind	FM2 / 10 mph	code / mph	Fixed surface forcing
Canopy cover	0, 10, 30, 60	percent	WAF branch and blend sweep
Canopy height	0, 5, 15, 30	m	Sheltered-profile sweep
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/cc_030_ch_015; WS=10 mph, WD=270 deg

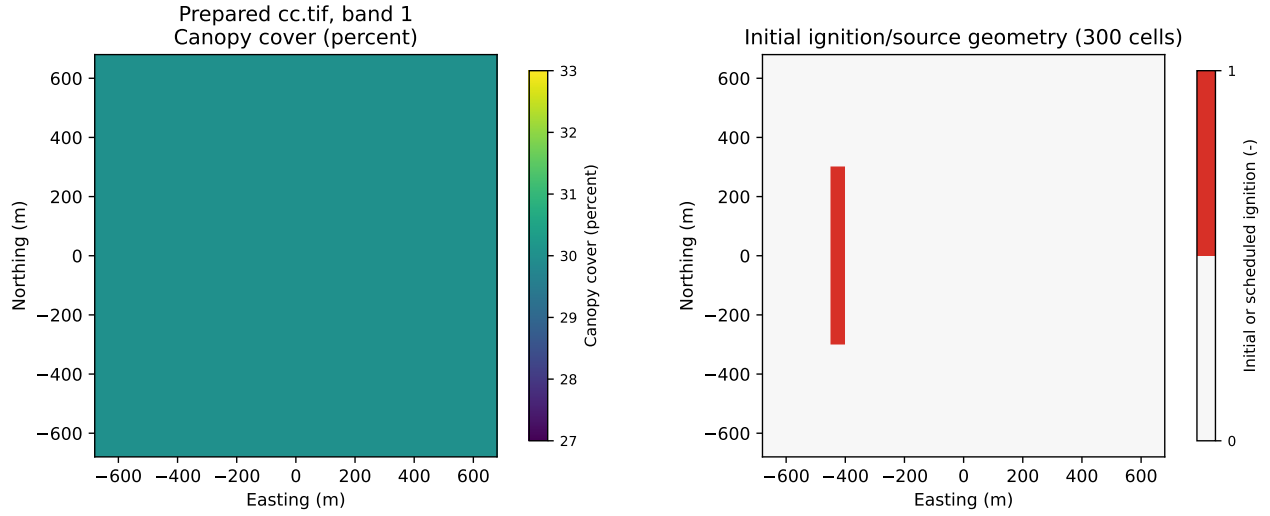


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 10 of 10 variants and reports **PASS**.

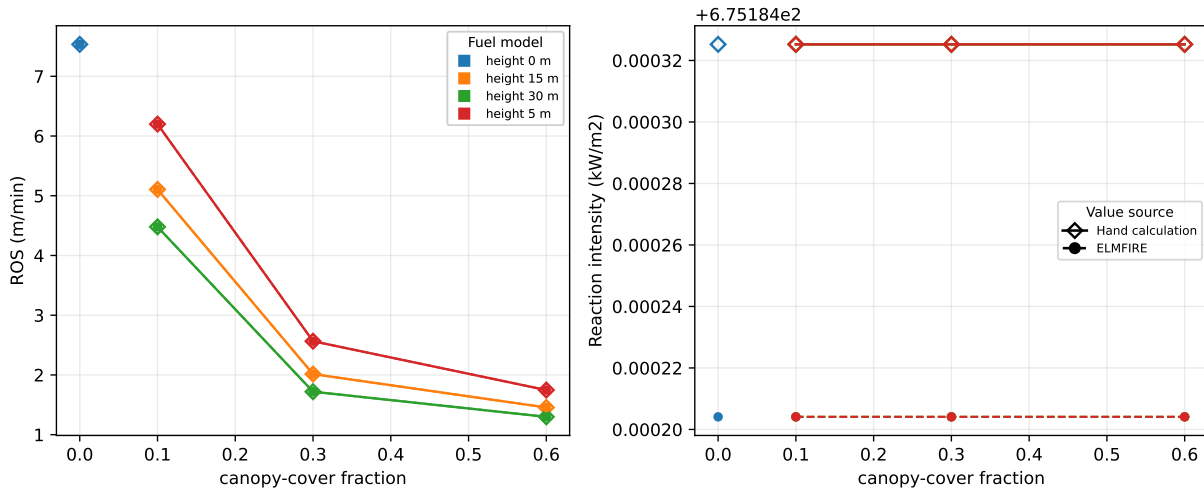


Figure 2: Expected and simulated canopy/WAF response at fixed 20-ft wind.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.03	6.68301e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	1.79416e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.06	0.0245548 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999972 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A uniform wind-scale error suggests incorrect 20-ft units. Errors confined to canopy variants implicate sheltered WAF or canopy scaling. Varying reaction intensity indicates an unintended canopy or crown-fire pathway. Correct direct ROS but biased TOA implicates level-set propagation.

9 Scope and limitations

This is an implementation-specific coupling test, not a validation of canopy wind profiles. It uses one fuel, one 20-ft wind, uniform canopy fields, and the current ELMFIRE WAF algorithm. A deliberate algorithm change requires oracle and report review.